

REPRESENTATION THEORY

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ABSTRACT. It's a lecture note I typed for "Representation theory" taught by Emanuel Scheidegger, in spring 2022. This note mainly follows the blackboard-writing of Prof. I also add some details and my understandings in it.

In this course, we will cover the following aspects:

- (1) Representation of finite groups.
- (2) Symmetric functions.
- (3) Lie groups and Lie algebra.
- (4) Representations of complex semisimple Lie algebra.
- (5) Representations of compact Lie groups.

Attention: there may be a considerable number of mistakes in this note, and that's all my fault, since I still have too many problems to work out.

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1. INTRODUCTION AND OVERVIEW

Group theory is the study of symmetries of a mathematics object. This is the point of view of geometry: given a geometry object X , what is its group of symmetries?

But representation theory reverse this question, given a group G , what object X does it act on? Here we pay more attention on linear action, i.e. X is a vector space.

We can compare with manifolds, since every abstract manifold can be embedded into $\mathbb{R}^n$, every abstract group can be embedded into S_n , according to Cayley's theorem as follows

Theorem 1.1. Any finite group of order n is isomorphic to a subgroup of the symmetric group S_n .

In this course, we are interested in the following groups:

1. finite group, in particular symmetric group, Coxeters groups.
2. Lie groups over $\mathbb{R}$ and $\mathbb{C}$.

And representation theory is a very useful tool, one of the most important applications is the classification of finite simple groups, all kinds of finite simple groups are listed as follows

1. cyclic groups C_p for prime p
2. alternating groups $A_n, n \geq 5$
3. 16 simple groups of Lie type
4. 26 sporadic groups

Among those sporadic groups, the largest one is the monster M , with order $|M| \sim 8 \cdot 10^{53}$, but the number of irreducible representations is only 194. As we will see, all irreducible representations of one group will reflect all information about it, so it's possible for us to learn the properties of monster group, by using its irreducible representations.

It's also worth mentioning that there is a crazy conjecture about monster group, called Monstrous Moonlight conjecture, proven by Borchers in 1992, and he got his Fields medal in 1998.

Part 1. Representation of finite group

2. BASIC DEFINITIONS AND IRREDUCIBILITY

2.1. Basic Definitions.

Definition 2.1.1 (representation). Let G be a finite group, V is a finite-dimensional vector space over k . A representation of G on V is a group homomorphism $\rho : G \rightarrow \text{GL}(V)$.

Notation 2.1. We say V is a representation of G and often write gv instead of $\rho(g)v$, we also say that G acts on V .

Remark 2.1.1. We give following remarks:

1. ρ equips V with the G -module structure. Conversely, a G -module structure on a vector space gives us a representation of G . They are the same thing in different languages.
2. We will mostly work with $k = \mathbb{C}$. More generally, V can be finite-dimensional R -module for a commutative ring with 1.
3. Let $B = (e_1, \dots, e_n)$ be a basis of V , for $\varphi \in \text{End}_k V$, write $\varphi e_i = \sum a_{ji} e_j$, and let $A = (a_{ij}) \in M_n(k)$. If ρ is a representation, the $\rho_B(g)$ is the matrix of $\rho(g)$ with respect to B . Then $g \rightarrow \rho_B(g)$ is a homomorphism from G to $\text{GL}(n, k)$, called the matrix representation.

Definition 2.1.2 (morphism of representation). Let V, W be two representations of finite group G . A linear map $\varphi : V \rightarrow W$ is a morphism of representation of G if the following diagram commutes for all $g \in G$.

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \downarrow g & & \downarrow g \\ V & \xrightarrow{\varphi} & W \end{array}$$

Definition 2.1.3 (quotient representation.). Let $\rho : G \rightarrow \text{GL}(V)$ be a representation. A subrepresentation of V is a vector subspace W of V , such that $\rho(g)W \subseteq W, \forall g \in G$. For a subrepresentation W , the map $\rho(g)(v + W) := \rho(g)v + W$ defines a representation of G on V/W , called the quotient representation.

Lemma 2.1.1. For a map of representation $\varphi : V \rightarrow W$, the kernel of φ is a subrepresentation of V , image and cokernel of φ are subrepresentations of W .

Proof. Trivial. □

By some standard linear algebra methods, we can construct new representations from old ones:

Lemma 2.1.2. Let $\rho : G \rightarrow \text{GL}(V), \sigma : G \rightarrow \text{GL}(W)$ be two representations of G , then

1. $\rho \oplus \sigma : G \rightarrow \text{GL}(V \oplus W), g(v \oplus w) = gv \oplus gw$

2. $\rho \otimes \sigma : G \rightarrow \text{GL}(V \otimes W), g(v \otimes w) = gv \otimes gw$
3. $\rho^{\otimes n} : G \rightarrow \text{GL}(V^{\otimes n}), g(v^{\otimes n}) = (gv)^{\otimes n}$
4. $\wedge^n \rho : G \rightarrow \text{GL}(\wedge V^n), g(v_1 \wedge \cdots \wedge v_n) = gv_1 \wedge \cdots \wedge gv_n$
5. $\text{Sym}^n \rho : G \rightarrow \text{GL}(\text{Sym}^n V), g(v_1 \otimes \cdots \otimes v_n) = gv_1 \otimes \cdots \otimes gv_n$
6. $\rho^\vee : G \rightarrow \text{GL}(V^\vee), \rho^\vee(g) = (\rho(g)^t)^{-1}$
7. $\rho_{V,W} : G \rightarrow \text{Hom}(V, W), (\rho(g)\varphi)(v) = \rho(g)\varphi(\rho(g^{-1})v)$

are representations of G .

Proof. Routines □

Lemma 2.1.3. Let V, W be two representations of G . Then we have the following isomorphism

$$\text{Hom}_G(V, W) \cong \text{Hom}(V, W)^G = G\text{-invariants of } \text{Hom}(V, W)$$

Lemma 2.1.4. The following are isomorphisms of representations U, V, W of G

1. $\text{Hom}(V, W) \cong V^\vee \otimes W$
2. $V \otimes (U \oplus W) \cong V \otimes U \oplus V \otimes W$
3. $\wedge^k(V \oplus W) \cong \bigoplus_{a+b=k} \wedge^a V \otimes \wedge^b W$
4. $\wedge^k(V^\vee) \cong (\wedge^k V)^\vee$
5. $\wedge^k(V^\vee) \cong \wedge^{n-k} V \otimes \det V^\vee$, where $n = \dim V, \det V = \wedge^n V$.

Definition 2.1.4 (group action). Let G be a group and X be a set. A group action of G on X is a map $\sigma : G \rightarrow \text{Aut}(X)$, such that

1. $\sigma(g)x \in X, \forall x \in X$
2. $\sigma(gh)x = \sigma(g)\sigma(h)x, \forall x \in X$
3. $\sigma(e)x = x, \forall x \in X$

If we have such a group action, we can construct many useful representations

Example 2.1.1 (permutation representation). Let V be a finite-dimensional over $\mathbb{C}$ with basis X , and G acts on X via σ , we define $R_X : G \rightarrow \text{GL}(V)$ as follows

$$R_X(g)\left(\sum_{x \in X} a_x e_x\right) = \sum_{x \in X} a_x e_{\sigma(g)x}$$

Here R_X is called permutation representation.

And the following examples are based on above one.

Example 2.1.2 (regular representation). Choose X to be G considered as a set, and G acts on G by left multiply, then $R = R_G$ is called regular representation, in this case V is denoted by $k[G]$, called group algebra.

Example 2.1.3 (alternating representation). Let V be the group algebra of G , and consider the map $\rho : G \rightarrow \text{GL}(V)$ defined as follows

$$\rho(g)\left(\sum_{x \in X} a_x e_x\right) = \sum_{x \in X} \text{sgn}(\sigma(g))a_x e_{\sigma(g)x}$$

is called the alternating representation.

Example 2.1.4 (coset representation). Let H be subgroup of G , and $X = \{g_1, \dots, g_n\}$ be a complete set of representatives of G/H , G acts on X by $g(g_iH) = gg_iH$. In this case, R_X is called the coset representation of G with respect to H .

Now we consider some concrete examples which we will use later.

Example 2.1.5. Consider $G = S_n$ and $X = \{1, 2, \dots, n\}$. Let $V = \mathbb{C}X$, and $W = \mathbb{C}(e_1 + \dots + e_n) \subset V$. Consider the permutation representation R_X , then it's easy to see that $R_X|_W$ is trivial representation.

Example 2.1.6. Regular representation for $X = \{1, 2, 3\}$, we can write down explicitly as follows

$$\begin{aligned} R(1) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & R((12)) &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & R((13)) &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \\ R((23)) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & R((132)) &= \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & R((123)) &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \end{aligned}$$

Example 2.1.7. A 2-dimension representation of S_3 : the symmetry of triangle, denoted by V

$$\begin{aligned} V(1) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & V((12)) &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & V((13)) &= \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix} \\ V((23)) &= \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}, & V((132)) &= \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, & V((123)) &= \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \end{aligned}$$

2.2. Irreducibility.

Definition 2.2.1 (irreducible). A representation of V is called irreducible if there is no subrepresentation W of V .

Definition 2.2.2 (indecomposable). A representation of V is called indecomposable if it can not be written as a direct sum of two nonzero subrepresentation.

Remark 2.2.1. Clearly, from definition we have a irreducible representation must be indecomposable. But when we consider complex representation, the irreducibility and indecomposability coincides, and that's Maschke's theorem.

Theorem 2.2.1 (Maschke's theorem). Let V be a representation of a finite group of $\mathbb{C}$, $W \subseteq V$ is a subrepresentation, then there is a complementary invariant subrepresentation W' of G , such that $V = W \oplus W'$.

Remark 2.2.2. Maschke theorem still holds when $\text{char } k \nmid |G|$

Remark 2.2.3. Any continuous representation of a compact group has this property, but group $(\mathbb{R}, +)$ does not, consider $a \mapsto \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}$ which fixes the x -axis, but there is no complementary subspace.

Lemma 2.2.1 (Schur lemma). Let V, W be irreducible representations of finite group G , and $\varphi \in \text{Hom}_G(V, W)$, then

1. either φ is isomorphism, or $\varphi = 0$
2. If $V = W$, then $\varphi = \lambda I, \lambda \in \mathbb{C}$

Proposition 2.2.1. Let $\rho : G \rightarrow \text{GL}(V)$ be representation of finite group, then there is a unique decomposition

$$V = \bigoplus_{i=1}^N V_i^{a_i}$$

where V_i is distinct irreducible representations.

2.3. Representation of abelian groups and S_3 .

2.3.1. *Representation of abelian groups.* Let $\rho : G \rightarrow \text{GL}(V)$ be any representation, then map $\rho(g) : V \rightarrow V$ is in general not a morphism of representations, i.e. for $h \in G$,

$$\rho(g)(hv) \neq h(\rho(g)v)$$

In fact, we can prove $\rho(g) \in \text{End}_G V$ if and only if $g \in Z(G)$. So if G is abelian, then any $\rho(g)$ is a morphism of representations.

Now let V be an irreducible representation. By Schur lemma, every $g \in G$ acts on V by a scalar multiple of identity, so every subspace of V is invariant, thus V must be one dimensional.

Proposition 2.3.1. Let G be a finite abelian group, then every irreducible representation of G is 1-dimensional.

Remark 2.3.1. The converse statement also holds, see Corollary 2.21.

Definition 2.3.1 (dual group). Let G be a finite group, then $G^\vee = \text{Hom}_G(G, \mathbb{C}^*)$ is called the dual group.

Corollary 2.1. Let G be a finite abelian group, then $\text{Irr } G \xrightarrow{1:1} G^\vee$

Proof. By the Remark 1.27, if G is abelian, then $G = Z(G)$, then $\rho(g) \in \text{End}_G V = \mathbb{C}^*, \forall g \in G$ and $V \in \text{Irr}(G)$. $\square$

2.3.2. *Representation of S_3 .* For S_3 , we have already seen the following representations:

1. trivial representation U , with dimension 1.
2. alternating representation U' , with dimension 1.
3. the regular representation R , with dimension 3.
4. the symmetric of the triangle V , with dimension 2.

And we also note that R has a 1-dimensional subrepresentation $V' = \mathbb{C}(e_1 + e_2 + e_3)$, in fact, it's a trivial representation, hence it is isomorphic to U .

Consider the complementary subspace of V' in R , denoted by $V'' = \{(v_1, v_2, v_2) \in V \mid v_1 + v_2 + v_2 = 0\}$, we can choose a basis $(\omega, 1, \omega^2), (1, \omega, \omega^2)$, where $\omega^3 = 1$.

Now, let W be an arbitrary representation of S_3 , consider $\mathbb{Z}/3\mathbb{Z} = \langle \sigma \rangle \subset S_3$, and decompose W into

$$W = \bigoplus_{i=1}^3 V_i^{\oplus a_i}, \quad V_i = \mathbb{C}v_i, \sigma v_i = \omega^i v_i$$

Let $\tau \in S_3$ be a transposition, such that

$$S_3 = \langle \sigma, \tau \rangle / (\tau \sigma \tau = \sigma^2)$$

then

$$\sigma(\tau v_i) = \tau(\sigma^2 v_i) = \tau(\omega^{2i} v_i) = \omega^{2i} \tau v_i$$

3. CHARACTER THEORY

In this section, G denotes a finite group.

Definition 3.0.1 (character). Let $\rho : G \rightarrow \text{GL}(V)$ be a representation, $\chi_V : G \rightarrow \mathbb{C}, g \mapsto \chi_V(g) = \text{tr}(\rho(g))$ is a character of ρ .

Remark 3.0.1. In fact, χ_V is a class function, i.e.

$$\chi_V \in \mathcal{C}_G = \{f : G \rightarrow \mathbb{C} \mid f|_K = \text{constant}, \forall K \in \text{Conj}(G)\}$$

The dimension of $\mathcal{C}_G = |\text{Conj}(G)|$, and we have the following isomorphism

$$\mathcal{C}_G \cong \mathbb{Z}(\mathbb{C}[G])$$

defined by

$$f \mapsto \sum_{g \in G} f(g)g$$

Proposition 3.0.1. Let V, W be representations of G , then

1. $\chi_{V \oplus W} = \chi_V + \chi_W$
2. $\chi_{V \otimes W} = \chi_V \chi_W$
3. $\chi_{V^\vee} = \overline{\chi_V}$
4. $\chi_{\text{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2))$
5. $\chi_{\wedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$

Proof. Note that $\{\lambda_i \lambda_j \mid i \leq j\}, \{\lambda_i \lambda_j \mid i < j\}$ are the eigenvalues of g on $\text{Sym}^2 V, \wedge^2 V$ respectively, then

$$\begin{aligned} \sum_{i \leq j} \lambda_i \lambda_j &= \frac{1}{2} \left(\sum_{i,j} \lambda_i \lambda_j + \sum_i \lambda_i^2 \right) \\ \sum_{i < j} \lambda_i \lambda_j &= \frac{1}{2} \left(\sum_{i,j} \lambda_i \lambda_j - \sum_i \lambda_i^2 \right) \end{aligned}$$

□

Theorem 3.0.1 (The fixed point formula). Let X be a finite set with an action by V , and V the permutation representation. Let $X^g = \{x \in X \mid gx = x\}, g \in G$. Then $\chi_V(g) = |X^g|$

Proof. Since $\text{Aut}(X) \cong S_{|X|}$, the matrix A representing $\rho(g)$ is a permutation matrix: if $ge_{x_i} = e_{x_j}$ for some $x_i, x_j \in X$, then

$$A_{ik} = \begin{cases} 1, & k = j \\ 0, & \text{otherwise} \end{cases}$$

Then, if $x_i \in X^g$, then $ge_{x_i} = e_{gx_i} = e_{x_i}$, that is $A_{ii} = 1$, so

$$\text{tr}(\rho(g)) = \sum_{i: x_i \in X^g} A_{ii} = \sum_{i: x_i \in X^g} 1 = |X^g|$$

□

Definition 3.0.2 (character table). The character table of G is a table with the conjugacy classes listed a cross, the irreducible representations listed on the left.

Example 3.0.1. Character table for S_3

	1	(12)	(123)
trivial U	1	1	1
alternating U'	1	-1	1
standard V	2	0	-1
permutation P	3	1	0

Use the propsuberty of character, if $W = U^{\oplus a} \oplus U'^{\oplus b} \oplus V^{\oplus c}$, then

$$\chi_W = a\chi_U + b\chi_{U'} + c\chi_V$$

Note that $\chi_U, \chi_{U'}, \chi_V$ is independent, so we can express character of any representation of S_3 in terms of $\chi_U, \chi_{U'}, \chi_V$. For example, we can decompose

$$\chi_{V \otimes V} = (4, 0, 1) = (2, 0, -1) + (1, 1, 1) + (1, -1, 1)$$

Later we will see if two representations have the same character, then they are isomorphic to each other, that is Corollary 2.12. Admitting this fact, we have

$$V \otimes V \cong U \oplus U' \oplus V$$

So we can decompose any representation of S_3 in the above method, if we know what does its character look like.

Remark 3.0.2. Note that different groups can have identical character tables, e.g., dihedral group

$$D_{4n} = \langle a, b \mid a^2 = b^{2n} = (ab)^2 = e \rangle$$

and quaternionic group

$$Q_{4n} = \langle a, b \mid a^2 = b^{2n}, (ab)^2 = e \rangle$$

have the same character table.

Remark 3.0.3. Nevertheless, characters can characterize the group G : order of G , order of all its normal subgroups, whether G is simple or not.

Proposition 3.0.2. Let V be a representation of G . The map $\varphi = \frac{1}{|G|} \sum_{g \in G} g \in \text{End } V$ as a projection from V to $V^G = \{v \in V \mid gv = v, \forall g \in G\}$

Proof. Let $w \in W$, $v = \varphi(w) = \frac{1}{|G|} \sum_{g \in G} gw$, then for any $h \in G$, we have

$$hv = \frac{1}{|G|} \sum_{g \in G} hgw = \frac{1}{|G|} \sum_{g \in G} gw = v$$

So $\text{im } \varphi \subset V^G$; Conversely, if $v \in V^G$, then $\varphi(v) = \frac{1}{|G|} \sum_{g \in G} gv = v$, this implies $V^G \subset \text{im } \varphi$. Moreover, $\varphi \circ \varphi = \varphi$. $\square$

Definition 3.0.3. We let $(\alpha, \beta) = \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)}\beta(g)$ denote a Hermitian inner product on $\mathcal{C}_G$.

Theorem 3.0.2 (First orthogonality relation). Let $V, W \in \text{Irr}(G)$, then

$$(\chi_V, \chi_W) = \begin{cases} 1, & V \cong W \\ 0, & \text{otherwise} \end{cases}$$

Proof. If V, W are irreducible representations, then Schur's lemma implies

$$\dim \text{Hom}(V, W)^G = \dim \text{Hom}_G(V, W) = \begin{cases} 1, & V \cong W \\ 0, & \text{otherwise} \end{cases}$$

However, $\chi_{\text{Hom}(V, W)} = \chi_{V^\vee \otimes W} = \chi_{V^\vee} \chi_W = \overline{\chi_V} \chi_W$. Let $\varphi = \frac{1}{|G|} \sum_{g \in G} g \in \text{End}(\text{Hom}(V, W))$, then we have

$$\begin{aligned} \dim \text{Hom}(V, W)^G &= \text{tr}_{\text{Hom}(V, W)^G} \varphi = \frac{1}{|G|} \sum_{g \in G} \text{tr}_{\text{Hom}(V, W)}(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\text{Hom}(V, W)}(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V}(g) \chi_W(g) \end{aligned}$$

□

Corollary 3.1. Any representation of a finite group G is determined by its character up to isomorphism, i.e. $V \cong W \iff \chi_V(g) = \chi_W(g), \forall g \in G$.

Corollary 3.2. If $V = \bigoplus_i V_i^{\oplus a_i}$, V_i are irreducible, distinct representations, then

$$a_i = (\chi_{V_i}, \chi_V)$$

In particular, V is irreducible if and only if $(\chi_V, \chi_V) = 1$.

Corollary 3.3. Every irreducible representation appears in the regular representation $R = \mathbb{C}[G]$, and the multiplicity of it is equal to its dimension. In particular, $|\text{Irr}(G)| < \infty$.

Proof. Recall that $(e_g)_{g \in G}$ is a basis for R , and $ge_h = e_{gh}, \forall g, h \in G$. For the fixed point formula

$$\chi_R(g) = \begin{cases} 0, & g \neq e \\ |G|, & g = e \end{cases}$$

Then R is not irreducible unless G is trivial. Let V_i be any irreducible representation of G , then

$$(\chi_{V_i}, \chi_R) = \frac{1}{|G|} \chi_{V_i}(e) |G| = \dim V_i$$

So every irreducible representation appears in R and the multiplicity equals to its dimension. □

Remark 3.0.4. If $R = \bigoplus_i V_i^{\oplus a_i}$, $a_i = \dim V_i$, then

$$|G| = \dim R = \sum_i (\dim V_i)^2$$

Sometimes it's a good tool to determine what's dimension of irreducible representation we haven't found.

Remark 3.0.5. If $g \neq e$, then $0 = \chi_R(g) = \sum_i \dim V_i \chi_{V_i}(g)$. If we know all but one row of character table, we can calculate the remaining one using this remark.

Example 3.0.2 (Character table of S_4). We already have trivial representation, alternating representation and standard representation. Since $24 = 1 + 1 + 9 + \sum_i (\dim V_i)^2$, so there exist two irreducible¹ other representation $\tilde{V}, W$, such that $\dim \tilde{V} = 3, \dim W = 2$.

But how to construct them? Consider $\tilde{V} = U' \otimes V, \dim \tilde{V} = 3$, then

$$\chi_{\tilde{V}} = \chi_{U'} \chi_V = (3, -1, 0, 1, -1)$$

So we have

$$(\chi_{\tilde{V}}, \chi_{\tilde{V}}) = 1$$

So it is irreducible. And the remaining one can be calculated from Remark 2.16

	1	(12)	(123)	(1234)	(12)(34)
trivial U	1	1	1	1	1
alternating U'	1	-1	1	-1	1
standard V	3	1	0	-1	-1
$\tilde{V}$	3	-1	0	1	-1
W	2	0	-1	0	2
permutation P	4	2	1	0	0

Proposition 3.0.3. Let $\alpha : G \rightarrow \mathbb{C}$ be any function. Set $\varphi_{\alpha, V} = \sum_{g \in G} \alpha(g)g : V \rightarrow V$ for any representation V . Then $\varphi_{\alpha, V} \in \text{End}_G V$ for all V if and only if $\alpha \in \mathcal{C}_G$.

Proof. Condition for $\varphi_{\alpha, V}$ to be G -linear: For $h \in G$,

$$\begin{aligned} \varphi_{\alpha, V}(hv) &= \sum_g \alpha(g)g(hv) = \sum_g \alpha(h^{-1}gh)hgh^{-1}(hv) \\ &= h\left(\sum_g \alpha(hgh^{-1})gv\right) \\ &\stackrel{\alpha \text{ is class function}}{=} h\left(\sum_g \alpha(g)gv\right) = h\varphi_{\alpha, V}(v) \end{aligned}$$

Conversely, consider $\varphi_{\alpha, V}(hv) = h\varphi_{\alpha, V}(v)$ and take for V the regular representation R . For $x \in G$,

$$\varphi_{\alpha, R}(he_x) = \varphi_{\alpha, R}(e_{hx}) = \sum_g \alpha(g)e_{gx} = \sum_g \alpha(g)e_{ghx}$$

¹Why there is no other 1-dimensional representation? In fact, we will learn later that the number of irreducible representations is equal to the number of the conjugate classes. Or you can directly show that there are only two possible one dimension representations.

But we also have

$$h(\varphi_{\alpha,R}(e_x)) = h\left(\sum_g \alpha(g)ge_x\right) = \sum_g \alpha(g)hge_x = \sum_g \alpha(g)e_{hgx} = \sum_g \alpha(h^{-1}gh)e_{ghx}$$

Thus α is a class function by comparing the coefficient of two side. $\square$

Proposition 3.0.4. If $V = \bigoplus_i V_i^{\otimes a_i}$ is the isotypical decomposition, of a representation V . Then the projection $\pi_i : V \rightarrow V_i^{\otimes a_i}$ is given by

$$\pi_i = \frac{\dim V_i}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} g$$

Proof. Let W be fixed irreducible representation, V be any representation. Since $\overline{\chi_W} \in \mathcal{C}_G$, then

$$\psi_{\overline{\chi_W}, V} = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_W(g)} g \in \text{End}_G(V)$$

If V is irreducible, then Schur's lemma implies $\psi_{\overline{\chi_W}, V} = \lambda \text{id}$, where

$$\lambda = \frac{1}{\dim V} \text{tr}_V \psi_{\overline{\chi_W}, V} = \frac{1}{\dim V \cdot |G|} \sum_{g \in G} \overline{\chi_W(g)} \chi_V(g) = \begin{cases} \frac{1}{\dim V}, & V \cong W \\ 0, & \text{otherwise} \end{cases}$$

If V is arbitrary, then $\dim W \psi_{\overline{\chi_W}, V}$ is a projection onto W^a where a is the number of times W appears in V .

So, if $V = \bigoplus_i V_i^{\otimes a_i}$ is the isotypical decomposition, then

$$\pi_i = \frac{\dim V_i}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} g$$

is the projection onto $V_i^{\otimes a_i}$. $\square$

Proposition 3.0.5.

$$|\text{Irr}(G)| = |\text{Conj}(G)|$$

In other words, $\{\chi_{V_i} \mid V_i \in \text{Irr}(G)\}$ forms an orthogonal basis for $\mathcal{C}_G$.

Proof. Suppose $\alpha \in \mathcal{C}_G$, $(\alpha, \chi_V) = 0, \forall V \in \text{Irr}(G)$, we must show $\alpha = 0$.

For any representation V , consider $\varphi_{\alpha, V}$, Schur lemma implies $\varphi_{\alpha, V} = \lambda \text{id}_V$, let $n = \dim V$, this implies

$$\lambda = \frac{1}{n} \text{tr}(\varphi_{\alpha, V}) = \frac{1}{n} \sum_g \alpha(g) \chi_V(g) = \frac{|G|}{n} \overline{(\alpha, \chi_{V^\vee})} = 0$$

Thus $\varphi_{\alpha, V} = 0$, that is,

$$\sum_g \alpha(g)g = 0, \quad \text{for any representation } V \text{ of } G.$$

In particular, for $V = R$, the set $\{\rho(g) \in \text{End } R \mid g \in G\}$ consists of linearly independent elements, thus $\alpha(g) = 0, \forall g \in G$. $\square$

Corollary 3.4. If G is a finite group, the following are equivalent

1. G is abelian.

2. Every irreducible representation of G has dimension 1.

Proof. We have already proved (1) to (2), for (2) to (1):

$$|G| = \sum_{i=1}^{|\text{Conj}(G)|} (\dim V_i)^2 = |\text{Conj}(G)|$$

So $|K| = 1, \forall K \in \text{Conj}(G)$, that is, G is abelian. $\square$

Proposition 3.0.6 (Second orthogonality relation).

$$\sum_{i: V_i \in \text{Irr}(G)} \overline{\chi_{V_i}(g)} \chi_{V_i}(h) = \begin{cases} \frac{|G|}{|K_g|}, & K_g = K_h \\ 0, & \text{otherwise} \end{cases}$$

where K_g is the conjugacy class of g .

Proof. Let χ_V, χ_W be irreducible characters. First orthogonality relation implies

$$\delta_{V,W} = (\chi_V, \chi_W) = \frac{1}{|G|} = \sum_g \overline{\chi_V(g)} \chi_W(g) = \frac{1}{|G|} = \sum_{K \in \text{Conj}(G)} \overline{\chi_V(K)} \chi_W(K) |K|$$

Then

$$U = \left(\sqrt{\frac{|K|}{|G|}} \chi_V(K) \right)$$

is a unitary matrix. Orthogonality of the columns of U yields the claim $\square$

Example 3.0.3 (Monstrous Moonlight Conjecture). Let $G = \mathbb{M}$ be the monster group, i.e. the sporadic finite simple group with $|M| \sim 8 \cdot 10^{53}$. One can show that $|\text{Irr}(G)| = |\text{Conj}(G)| = 194$, a relatively small number.

To compare, $|\text{Irr} S_{15}| = 176, |\text{Irr} S_{16}| = 231$. Let $V_i \in \text{Irr}(G)$ be ordered by their dimension.

V	V_0	V_1	V_2	V_3
$\dim V$	1	196883	21296876	842609256

Complex analysis tells Eisenstein series

$$G_k(\tau) = \sum_{\substack{(m,n) \in \mathbb{Z}^2 \\ (m,n) \neq (0,0)}} \frac{1}{(m\tau + n)^k}$$

converges for $k \geq 3$ normally and defines a holomorphic function on $\mathbb{H}$. $G_k(\tau)$ admits a Fourier expansion

$$G_k(\tau) = \sum_{n=0}^{\infty} a_k(n) q^n, \quad q = e^{2\pi i \tau}$$

Consider

$$j(\tau) = \frac{172820G_4(\tau)^3}{20G_4(\tau)^3 + 49G_6(\tau)^2}$$

Then $j(\tau) - 744 = q^{-1} + 196884q + 21493690q^2 + 864299970q^3 + \dots$

Mckay 1978 wrote a letter to Thompson

$$196884 = 196883 + 1$$

Thompson: the next term work similarly.

Suggestion: there exists $V = \bigoplus_{i=0}^{\infty} V_i$ infinitely-dimensional graded representation of $\mathbb{M}$ such that

$$\sum_{n=0}^{\infty} \chi_{V_n} q^{n-1} = j(q) - 744$$

Moreover,

$$T_q(\tau) = \sum_{n=0}^{\infty} \chi_{V_n}(g) q^{n-1} = \text{other well-known functions in complex analysis}$$

Corway-Norton verified this in 1979 on a computer.

Borcherds proved this conjecture in 1992 by V the structure of a module over a vertex operator algebra.

Definition 3.0.4 (external tensor product representation). Let G, H be finite groups, V a representation of G , W a representation of H . Then we define the external tensor product representation $V \boxtimes W$ of $G \times H$ by

$$(g, h)(v, w) = gv \otimes hw, \quad \forall g \in G, h \in H, v \in V, w \in W.$$

and extension by linearity to $V \otimes W$. Similarly, we define a $G \times H$ action on $\text{Hom}(V, W)$ by

$$((g, h)\varphi)v = h\varphi(g^{-1}v), \quad g \in G, h \in H, v \in V, \varphi \in \text{Hom}(V, W).$$

and extension by linearity.

Remark 3.0.6. We have

$$\text{Hom}(V, W) \cong V^{\vee} \boxtimes W$$

as $G \times H$ representations.

Proposition 3.0.7. We have the following well-defined bijection:

$$\begin{aligned} \text{Irr}(G) \times \text{Irr}(H) &\rightarrow \text{Irr}(G \times H) \\ (V, W) &\rightarrow V \boxtimes W \end{aligned}$$

Proof. It suffices to look at characters. By properties of trace we have

$$\chi_{V \boxtimes W}((g, h)) = \chi_V(g) \chi_W(h)$$

Recall that

$$\dim \text{Hom}_G(V, W) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = (\chi_V, \chi_W)_G$$

Then

$$\begin{aligned}
(\chi_{V_1 \boxtimes W_1}, \chi_{V_2 \boxtimes W_2}) &= \frac{1}{|G \times H|} \sum_{g, h \in G \times H} \overline{\chi_{V_1}(g) \chi_{W_1}(g)} \chi_{V_2}(g) \chi_{W_2}(g) \\
&= \frac{1}{|G|} \sum_g \overline{\chi_{V_1}(g)} \chi_{V_2}(g) \frac{1}{|H|} \sum_{h \in H} \overline{\chi_{W_1}(g)} \chi_{W_2}(g) \\
&= (\chi_{V_1}, \chi_{V_2})_G (\chi_{W_1}, \chi_{W_2})_H
\end{aligned}$$

So $V \boxtimes W \in \text{Irr}(G \times H)$, if $V \in \text{Irr}(G), W \in \text{Irr}(H)$. By calculating the cardinality of both sides we get the desired result. $\square$

4. RESTRICTION AND INDUCED REPRESENTATION

4.1. restriction representation.

Definition 4.1.1 (restriction representation). Let $H < G$ be a subgroup, V a representation of G , we define $\text{Res } V = \text{Res}_H^G V : H \rightarrow \text{GL}(V)$ to be the restriction of V onto H , $\text{Res}_H^G V$ is a representation of H .

Remark 4.1.1. Restriction is transitive, i.e. for $K < H < G$, we have

$$\text{Res}_K^H \text{Res}_H^G = \text{Res}_K^G$$

Lemma 4.1.1. Let $H < G$, $W \in \text{Irr}(H)$, then there exists $V \in \text{Irr}(G)$ such that

$$(\text{Res}_H^G \chi_V, \chi_W)_H \neq 0$$

Proof. Consider the regular representation R , then

$$(\text{Res}_H^G \chi_R, \chi_W) = \frac{|G|}{|H|} \chi_W(e) \neq 0$$

But the left term also equals to $\sum_i \dim V_i (\text{Res}_H^G \chi_{V_i}, \chi_W)_H$, so there must be at least one V_i , such that

$$(\text{Res}_H^G \chi_{V_i}, \chi_W) \neq 0$$

$\square$

Lemma 4.1.2. Let $H < G$, $V \in \text{Irr}(G)$, $\text{Res}_H^G V = \bigoplus W_i^{\oplus a_i}$, $W_i \in \text{Irr}(H)$. Then $\sum a_i^2 \leq [G : H]$ with equality if and only if $\chi_V(\sigma) = 0, \forall \sigma \in G/H$.

Proof. We have

$$\frac{1}{|G|} \sum_{h \in H} |\chi_V(h)|^2 = (\text{Res}_H^G V, \text{Res}_H^G V) = \sum a_i^2$$

Since V is irreducible, we have

$$\begin{aligned}
 1 = (\chi_V, \chi_V)_G &= \frac{1}{|G|} \sum_{g \in G} |\chi_V(g)|^2 \\
 &= \frac{1}{|G|} \left(\sum_{h \in H} |\chi_V(h)|^2 + \sum_{\sigma \in G/H} |\chi_V(\sigma)|^2 \right) \\
 &= \frac{|H|}{|G|} \sum_i a_i^2 + \frac{1}{|G|} \sum_{\sigma \in G/H} |\chi_V(\sigma)|^2 \\
 &\geq \frac{|H|}{|G|} \sum_i a_i^2
 \end{aligned}$$

□

Proposition 4.1.1. Let V, W be representations of G . Then $V \cong W$ if and only if $\text{Res}_H^G V \cong \text{Res}_H^G W$, for all cyclic subgroup H of G .

Proof. One direction is obvious; Conversely, let $g \in G, H = \langle g \rangle$, then $\chi_V(g) = \chi_{\text{Res}_H^G V}(g)$, the claim follows from $V \cong W \iff \chi_V(g) = \chi_W(g), \forall g \in G$. □

4.2. Induced representation. Let $H < G$ be a subgroup, $\rho : G \rightarrow \text{GL}(V)$ be a representation, $W \subset V$ a H -invariant subspace. Then for any $g \in G$, the subspace $gW \subset V$ depends only on gH , since $ghW = g(hW) = gW$. Therefore, for $\sigma \in G/H$, we write σW for this subspace of V .

We say that V is induced by W if every element in V can be written uniquely as a sum of elements in such translates of W , that is

$$V = \bigoplus_{\sigma \in G/H} \sigma W$$

In this case, we write $V = \text{Ind } W = \text{Ind}_H^G W$.

Remark 4.2.1. Alternative formulations: for any $v \in V$, there exists a unique $v_\sigma \in \sigma W$, such that

$$v = \sum_{\sigma \in G/H} v_\sigma$$

or if $\{g_1, \dots, g_N\}$, $|N| = |G/H| = [G : H]$ is a complete system of representatives of G/H , then

$$V = \bigoplus_{i=1}^N g_i W$$

Remark 4.2.2. Clearly, if V is induced by W , then

$$\dim V = [G : H] \dim W$$

Example 4.2.1. Let R be the regular representation of G , and H is a subgroup of G . Then

$$W = \bigoplus_{h \in H} C e_h$$

is H -invariant. In fact, $W \cong R_H$ and clearly $R_G = \text{Ind}_H^G R_H$.

Example 4.2.2. Let $H < G$ and V the coset representation of G , i.e. V has basis $(e_\sigma)_{\sigma \in G/H}$ and $ge_\sigma = e_{g\sigma}$. Then

$$W = \mathbb{C}e_{eH}$$

is H -invariant, and is the trivial representation of H , then

$$V = \text{Ind}_H^G W$$

In particular, if $H = \{e\}$, then V is the permutation representation P of G , and $P = \text{Ind}_{\{e\}}^G \mathbb{C}$.

Example 4.2.3. If $V_i = \text{Ind}_H^G W_i, i = 1, 2$, then

$$V_1 \oplus V_2 = \text{Ind}_H^G (W_1 \oplus W_2)$$

Example 4.2.4. If $V = \text{Ind}_H^G W$, $W' \subset W$ is a H -invariant subspace, then

$$V' = \bigoplus_{\sigma \in G/H} \sigma W' \subset V$$

is G -invariant, and $V' = \text{Ind}_H^G W'$.

Proposition 4.2.1. Let $H < G$ be a subgroup, $\rho : G \rightarrow \text{GL}(V)$ is induced by $\psi : H \rightarrow \text{GL}(W)$, let $\rho' : G \rightarrow \text{GL}(V')$ be any representation, $\phi \in \text{Hom}_H(W, V')$, then there exists a unique $\Phi \in \text{Hom}_G(V, V')$, such that

$$\Phi|_W = \phi$$

Proof. For uniqueness: Let $\Phi \in \text{Hom}_G(V, V')$ with $\Phi|_W = \phi$, and let $w \in \rho(g)W, g \in G$, then

$$\Phi(w) = \Phi(\rho(g)\rho(g^{-1})w) = \rho'(g)\Phi(\rho(g)^{-1}w) = \rho'(g)\phi(\rho(g)^{-1}w)$$

This determines Φ on $\rho(g)W$ for all $g \in G$, hence on V .

For existence: we define

$$\Phi(w) = \rho'(g)\phi(\rho(g)^{-1}w)$$

if $w \in \rho(g)W$, this is independent of the choice of g , since

$$\begin{aligned} \rho'(gh)\phi(\rho(gh)^{-1}w) &= \rho'(g)\rho'(h)\phi(\rho(h)^{-1}\rho(g)^{-1}w) \\ &= \rho'(g)\phi(\rho(h)\rho(h)^{-1}\rho(g)^{-1}w) \\ &= \rho'(g)\phi(\rho(g)^{-1}w) \end{aligned}$$

□

Theorem 4.2.1. Let $H < G$ be a subgroup, and $\psi : H \rightarrow \text{GL}(W)$ be a representation. Then there exists a representation $\rho : G \rightarrow \text{GL}(V)$ induced by W , which is unique up to isomorphism.

Proof. For existence: By Example 3.10 we may assume $W \in \text{Irr}(H)$, W' is isomorphic to a subrepresentation of R_H , since any $W' \in \text{Irr}(H)$ appears in R_H . By Example 3.8 we have

$$R_G = \text{Ind}_H^G R_H$$

and by Example 3.11 with $V = R_G, W = R_H$, we get

$$V' = \text{Ind}_H^G W'$$

For uniqueness: Let $V = \text{Ind}_H^G W, V' = \text{Ind}_H^G W'$, then proposition 3.12 implies that there exists a unique $\Phi \in \text{Hom}_G(V, V')$ such that $\Phi|_W = \text{id}_W$, and $\Phi \circ \rho(g) = \rho'(g) \circ \Phi, \forall g \in G$. Then $\text{im } \Phi$ contains all $\rho'(g)W$, so $\text{im } \Phi = V'$.

By $\dim V = [G : H] \dim W = \dim V'$, we conclude Φ is an isomorphism. $\square$

Lemma 4.2.1. Let V be a representation of G , and $H < G$ be a subgroup. Then

$$V \otimes \text{Ind}_H^G W = \text{Ind}_H^G (\text{Res}_H^G V \otimes W)$$

Proof. Note that

$$\begin{aligned} V \otimes \text{Ind}_H^G W &= \bigoplus_{\sigma \in G/H} V \otimes \sigma W \\ &= \bigoplus_{\sigma \in G/H} \sigma(\text{Res}_H^G V) \otimes \sigma W = \text{Ind}_H^G (\text{Res}_H^G V \otimes W) \end{aligned}$$

$\square$

Corollary 4.1. We have

$$V \otimes P = \text{Ind}_H^G (\text{Res}_H^G V)$$

where P is permutation representation.

Proof. Take W as trivial representation, then this claim holds from Lemma 3.14. $\square$

Lemma 4.2.2. Ind is transitive.

Proof.

$$\begin{aligned} \text{Ind}_K^H \text{Ind}_H^G &= \text{Ind}_K^H \bigoplus_{\tau \in G/H} \tau V \\ &= \bigoplus_{\sigma \in H/K} \bigoplus_{\tau \in G/H} \sigma \tau V \\ &= \bigoplus_{\sigma' \in G/K} \sigma' V \\ &= \text{Ind}_K^G V \end{aligned}$$

$\square$

Remark 4.2.3. These results can also be obtained by looking at characters or using group algebra.

Theorem 4.2.2. Let $H < G$ be a subgroup, and $\rho : G \rightarrow \text{GL}(V), \psi : H \rightarrow \text{GL}(W)$ be two representations, such that $V = \text{Ind}_H^G W$. Then

$$\chi_V(g) = \sum_{\sigma \in G/H} \chi_W(g_\sigma^{-1} g g_\sigma) = \frac{1}{|H|} \sum_{\substack{x \in G, \\ x^{-1} g x \in H}} \chi_W(x^{-1} g x)$$

where g_σ is any representative of σ .

Proof. Let $V = \bigoplus_{\sigma \in G/H} \sigma W$, $\rho(g)$ permutes the σW among themselves, i.e. if $g_\sigma \in \sigma$ is a representative, we write $gg_\sigma = g_\tau h$ for some $\tau \in G/H, h \in H$.

$$g(g_\sigma W) = (g_\tau h)W = g_\tau(hW) = g_\tau W$$

Then we can calculate

$$\begin{aligned} \chi_V(g) &= \text{tr}_V(\rho(g)) = \sum_{\sigma \in G/H} \text{tr}_{\sigma W}(\rho(g)) \\ &= \sum_{\sigma \in G/H} \chi_W(g_\sigma^{-1} g g_\sigma) = \sum_{\tau \in G/H} \chi_W(h^{-1} g_\tau^{-1} g g_\tau h) \\ &= \frac{1}{|H|} \sum_{\tau \in G/H} \sum_{h \in H} \chi_W(h^{-1} g_\tau^{-1} g g_\tau h) = \frac{1}{|H|} \sum_{\substack{x \in G, \\ x^{-1} g x \in H}} \chi_W(x^{-1} g x) \end{aligned}$$

□

Theorem 4.2.3 (Frobenius reciprocity). Let $H < G$ be a subgroup, W be a representation of H , U be a representation of G . Assume that $V = \text{Ind}_H^G W$, then

$$\text{Hom}_H(W, \text{Res}_H^G U) \cong \text{Hom}_G(V, U)$$

i.e. for $\varphi \in \text{Hom}_H(W, \text{Res}_H^G U)$ extends uniquely to $\tilde{\varphi} \in \text{Hom}_G(V, U)$

Proof. We write $V = \bigoplus_{\sigma \in G/H} \sigma W$, define $\tilde{\varphi}$ on σW by the composition

$$\sigma W \xrightarrow{g_\sigma^{-1}} W \xrightarrow{\varphi} U \xrightarrow{g_\sigma} \sigma W$$

This is independent of the choice of g_σ since

$$g_\sigma h(\varphi(h^{-1} g_\sigma^{-1}(w))) = g_\sigma \varphi(h h^{-1} g_\sigma(w))$$

by $\varphi \in \text{Hom}_H(W, \text{Res}_H^G U)$

□

Corollary 4.2. Let $H < G$ be a subgroup, W be a representation of H , U be a representation of G . Then

$$(\chi_W, \text{Res}_H^G \chi_U)_H = (\text{Ind}_H^G \chi_W, \chi_U)_G$$

Proof. By linearity, we can assume W, U are irreducible representations. This claim follows from the Frobenius reciprocity and Schur's lemma

$$(\chi_V, \chi_U)_G = \dim \text{Hom}_G(V, U)$$

□

Example 4.2.5. Let $G = S_3, H = S_2$. In S_2 , the standard representation V_2 is isomorphic to the alternating representation U'_2 . We have seen that U_3, U'_3, V_3 are all irreducible representations of S_3 .

And we can write down their character tables as follows

	1	(12)		1	(12)	(123)
trivial U_2	1	1	,	trivial U_3	1	1
alternating U'_2	1	-1		alternating U'_3	1	-1
				standard V_3	2	-1

Note that

$$\text{Res } U_3 = U_2, \quad \text{Res } U'_3 = U'_2, \quad \text{Res } V_3 = U_2 \oplus U'_2$$

If we want to compute induced representation. Firstly note that we have seen

$$P \otimes U = \text{Ind}(\text{Res } U), \quad U \text{ is any representation of } G$$

For $U = U_3$, we have $P = U_3 \oplus V_3 = \text{Ind } U_2$. If we want to compute $\text{Ind } V_2$, it's a little bit complicated. By Frobenius reciprocity

$$\text{Hom}_{S_3}(\text{Ind } V_2, U_3) = \text{Hom}_{S_2}(V_2, \text{Res } U_3 = U_2) \stackrel{\text{Schur}}{=} 0$$

$$\text{Hom}_{S_3}(\text{Ind } V_2, U'_3) = \text{Hom}_{S_2}(V_2, \text{Res } U'_3 = U'_2) \stackrel{\text{Schur}}{=} \mathbb{C}$$

$$\text{Hom}_{S_3}(\text{Ind } V_2, V_3) = \text{Hom}_{S_2}(V_2, \text{Res } V_3 = U_2 \oplus U'_2) \stackrel{\text{Schur}}{=} \mathbb{C}$$

So

$$\text{Ind } V_2 = U'_3 \oplus V_3$$

Definition 4.2.1 (representation ring). Let G be a finite group, and $R_k(G)$ be the free abelian group generated by all isomorphism classes of representations of G over a field k , modulo the subgroup generated by elements of the form $V + W - (V \oplus W)$. $R(G)$ is called the representation ring of G , or the Grothendieck group of G , denoted by $K_0(G)$.

Definition 4.2.2 (virtual representation). Elements of $R(G)$ are called virtual representations.

Remark 4.2.4. The ring structure on $R(G)$ is the tensor product, defined on the generators of $R(G)$, and extended by linearity.

Remark 4.2.5. We have the following remarks:

1. A character defines a ring homomorphism from $R(G)$ to $\mathcal{C}_G$
2. χ is injective is equivalent to a representation is determined by its character, the image of χ are called virtual characters.
3. $\chi_{\mathbb{C}} : R(G) \otimes_{\mathbb{Z}} \mathbb{C} \rightarrow \mathcal{C}_G$ is an isomorphism.
4. The virtual characters form a lattice $\Lambda \cong \mathbb{Z}^c \subset \mathcal{C}_G$. The actual characters form a cone $\Lambda_0 \cong \mathbb{N}^0 \subset \Lambda$.
5. By 3. we can define an inner product on $R(G)$ by

$$(V, W) = \dim \text{Hom}_G(V, W)$$

Example 4.2.6. Let $G = C_n$, then $R(C_n) = \mathbb{Z}[x]/(x^n - 1)$, where X correspond to the representation of a primitive n -th root of unity.

Example 4.2.7. $R(S_3) \cong \mathbb{Z}[x, y]/(xy - y, x^2 - 1, y^2 - x - y - 1)$. We can identify x to the alternating representation U' , y to the standard representation V and 1 to the trivial representation.

Goal: Determine $R(S_n)$ for all n and determine all irreducible representations of S_n for all n .

Part 2. Symmetric functions

5. YOUNG TABLEAU

Definition 5.0.1 (Composition of n). A composition of n is an ordered sequence $(\alpha_1, \dots, \alpha_r)$ such that $\alpha_i \in \mathbb{Z}_{>0}$ and $\sum \alpha_i = n$; A weak composition of n is a (finite or infinite) ordered sequence $(\alpha_1, \dots)$ such that $\alpha_i \in \mathbb{Z}_{>0}$, $\sum \alpha_i = n$ and $|\{i \in \mathbb{Z}_{>0} \mid \alpha_i \neq 0\}| < \infty$.

Definition 5.0.2 (Partition). A partition is any weak composition $\lambda = (\lambda_1, \dots)$ such that $\lambda_i \geq \lambda_{i+1}$ for all i . The nonzero λ_i are called parts. The number of parts is the length of λ , denoted by $l(\lambda)$. $|\lambda| = \sum \lambda_i$ is the weight of λ . If $|\lambda| = n$, then we write $\lambda \vdash n$ and say λ is a partition of n .

Notation 5.0.1. The set consists of all partition of n is denoted by $\mathcal{P}_n$.

Notation 5.0.2 (Exponential notation). If j appears m_j times in λ , we write $\lambda = (1^{m_1} 2^{m_2} \dots)$

Lemma 5.0.1. We have the following correspondence

$$\text{Conj}(S_n) \longleftrightarrow \mathcal{P}_n$$

Proof. Recall that $w \in S_n$ factorizes uniquely as a product of disjoint cycles

$$w = (i_1 \dots i_{\alpha_1}) \dots (i_{n-\alpha_r+1} \dots i_n)$$

of order $\alpha_1, \dots, \alpha_r$. The order in which the cycles are listed is irrelevant.

If $\alpha_1 \geq \dots \geq \alpha_r$, then $\alpha = (\alpha_1, \dots, \alpha_r)$ is a partition of n , called the cycle type $\alpha(w)$ of w .

Let $v, w \in S_n$, if $v(i) = j$, then

$$w \circ v \circ w^{-1}(w(i)) = w(j)$$

so v and $w \circ v \circ w^{-1}$ have the same cycle type, i.e. $\alpha(v) = \alpha(w \circ v \circ w^{-1})$. So $\alpha(w)$ determines $w \in S_n$ up to conjugacy. $\square$

Theorem 5.0.1 ([Euler]). $p(n) = |\mathcal{P}_n|$, where

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$$

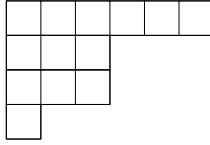
Exercise 5.0.1.

n	0	1	2	3	4	5	6	7	8	9	10
$p(n)$	1	1	2	3	5	7	11	15	22	30	42

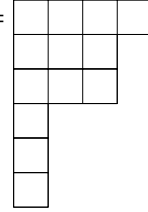
Definition 5.0.3 (Young subgroup). For $\lambda = (\lambda_1, \dots, \lambda_r) \in \mathcal{P}_n$. A Young subgroup is a subgroup of S_n given as

$$S_\lambda = S_{\{1, \dots, \lambda_1\}} \times S_{\{\lambda_1+1, \dots, \lambda_1+\lambda_2\}} \times \dots \times S_{\{n-\lambda_r+1, \dots, n\}}$$

Definition 5.0.4 (Young diagram). The Young diagram $D(\lambda)$ of $\lambda \in \mathcal{P}_n$ is $D(\lambda) = \{(i, j) \in \mathbb{Z}^2 \mid 1 \leq i \leq \lambda_j\}$. We draw a box for each point (i, j) .

Exercise 5.0.2. $D((6, 3, 3, 1)) =$ 

Definition 5.0.5 (Conjugate of a partition). The conjugate of $\lambda \in \mathcal{P}_n$ is the partition $\lambda' \in \mathcal{P}_n$ whose Young diagram $D(\lambda')$ is the transpose of $D(\lambda)$.

Exercise 5.0.3. $D((6, 3, 3, 1))' =$ 

Lemma 5.0.2. Let λ be a partition, and $m \geq \lambda_1, n \geq \lambda'_1$. The $m + n$ numbers $\lambda_i + n - i (1 \leq i \leq n)$, $n - 1 + j - \lambda'_j (1 \leq j \leq m)$ are a permutation of $\{0, 1, 2, 3, \dots, m + n - 1\}$

Proof. Clearly $D(\lambda) \subset D(m^n)$. Take a path corresponding to $D(\lambda)$ from the lower left corner to the upper right corner, number the segment of the path by $0, 1, \dots, m + n - 1$. The vertical segments are $\lambda_i + n - 1, 1 \leq i \leq n$. The horizontal segments (by transposition) are $(m + n - 1) - (\lambda'_j + m - j) = n - \lambda'_j + j - 1, 1 \leq j \leq m$. $\square$

Remark 5.0.1. The lemma is equivalent to the identity

$$f_{\lambda, n}(t) + t^{m+n-1} f_{\lambda', m}(t^{-1}) = \frac{1 - t^{m+n}}{1 - t}$$

Definition 5.0.6 (Operations on partitions). Let λ, μ be partitions. There are some operations:

1. $\lambda + \mu$ by $(\lambda + \mu)_i = \lambda_i + \mu_i$;
2. $\lambda \cup \mu$ is partition in which λ_i, μ_j are arranged decreasing in order;
3. $\lambda \mu$ is defined by $(\lambda \mu)_i = \lambda_i \mu_i$;
4. $\lambda \times \mu$ is the partition in which $\min\{\lambda_i, \mu_j\}$ are arranged in decreasing order.

Exercise 5.0.4. If we take $\lambda = (3, 2, 1)$ and $\mu = (2, 2)$, compute as follows to see what's going on

$$\begin{aligned} \lambda + \mu &= (5, 4, 1), & \lambda \mu &= (6, 4) \\ \lambda \cup \mu &= (3, 2, 2, 2, 1), & \lambda \times \mu &= (2, 2, 2, 2, 1, 1) \end{aligned}$$

Lemma 5.0.3. We have the following relation between above operations

$$\begin{aligned} (\lambda \cup \mu)' &= \lambda' + \mu' \\ (\lambda \times \mu)' &= \lambda' \mu' \end{aligned}$$

Proof. $D(\lambda \cup \mu)$ is obtained from the rows of $D(\lambda)$ and $D(\mu)$ and arranging in order of decreasing length, so we have

$$(\lambda \cup \mu)'_k = \lambda'_k + \mu'_k$$

And

$$(\lambda \times \mu)'_k = \{(i, j) \in \mathbb{Z}^2 \mid \lambda_i \geq k, \mu_j \geq k\} = \lambda'_k \mu'_k$$

□

Definition 5.0.7 (Containing ordering). Let λ, μ be two partitions, $\mu \leq \lambda$ with respect to containing ordering, denoted by $\mu \subseteq \lambda$ if and only if $\mu_i \leq \lambda_i, \forall i \geq 1$.

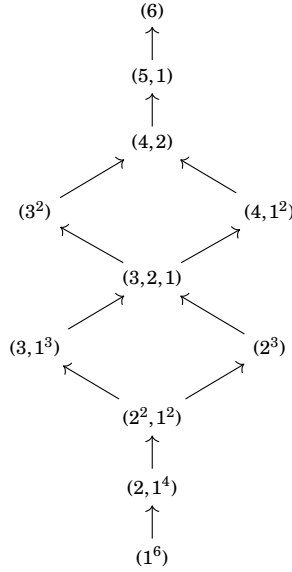
Definition 5.0.8 (Orderings). Let $\lambda, \mu \in \mathcal{P}_n$, then

1. Reverse lexicographic ordering L_n : $(\lambda, \mu) \in L_n$ if and only if for $\lambda = \mu$ or the first non-vanishing difference $\lambda_i - \mu_i$ is positive.
2. reverse lexicographic ordering L'_n : $(\lambda, \mu) \in L'_n$ if and only if $\lambda = \mu$ or the first non-vanishing difference $\lambda_i^* - \mu_i^*$ is negative, where $\lambda_i^* = \lambda_{n+1-i}$.
3. Natural/Dominance ordering N_n : $(\lambda, \mu) \in N_n$ if and only if $\lambda_1 + \dots + \lambda_i \geq \mu_1 + \dots + \mu_i$ for all $i \geq 1$. We write $\lambda \geq \mu$ instead of $(\lambda, \mu) \in N_n$.

Remark 5.0.2. Containing ordering and N_n are only partial orderings, but L_n and L'_n are total orderings.

Definition 5.0.9 (Cover & Hasse diagram). If $(A, \leq)$ is a poset, $b, c \in A$, we say that b is covered by c , written $b < c$, if $b < c$ and there is no $d \in A$ such that $b < d < c$; The Hasse diagram of A consists of vertices corresponding to element $a \in A$, and an arrow from the vertex b to vertex c if $b < c$.

Exercise 5.0.5. If we consider dominance ordering on $\mathcal{P}_6$ ²



Lemma 5.0.4. Let $\lambda, \mu \in \mathcal{P}_n$. Then $\lambda \geq \mu$ implies $(\lambda, \mu) \in L_n \cap L'_n$

Proof. Suppose that $\lambda \geq \mu$. Then either $\lambda_1 > \mu_1$, in which case $(\lambda, \mu) \in L_n$, or else $\lambda_1 = \mu_1$. In that case either $\lambda_2 > \mu_2$, in which case again $(\lambda, \mu) \in L_n$, or else $\lambda_2 = \mu_2$. Continuing in this way, we see that $(\lambda, \mu) \in L_n$.

²Here I really want to draw a Hasse diagram in the form of Young diagram, but there is no enough space for me to draw down all my ideas (smile).

Also, for each $i \geq 1$, we have

$$\begin{aligned}\lambda_{i+1} + \lambda_{i+2} + \dots &= n - (\lambda_1 + \dots + \lambda_i) \\ &\leq n - (\mu_1 + \dots + \mu_i) \\ &= \mu_{i+1} + \mu_{i+2} + \dots\end{aligned}$$

Hence the same reasoning as before shows that $(\lambda, \mu) \in L'_n$. $\square$

Lemma 5.0.5. Let $\lambda, \mu \in \mathcal{P}_n$, then $\lambda \geq \mu$ is equivalent to $\mu' \geq \lambda'$.

Proof. It suffices to show one direction. Suppose $\lambda' \not\geq \mu'$, then for some $i \geq 1$, we have

$$(*) \quad \begin{cases} \lambda'_1 + \dots + \lambda'_j \leq \mu'_1 + \dots + \mu'_j, & 1 \leq j \leq i-1 \\ \lambda'_1 + \dots + \lambda'_i > \mu'_1 + \dots + \mu'_i \end{cases}$$

which implies

$$\lambda'_i > \mu'_i$$

Let $l = \lambda'_i$ and $m = \mu'_i$. From $(*)$ it follows that

$$\lambda'_{i+1} + \lambda'_{i+2} + \dots < \mu'_{i+1} + \mu'_{i+2} + \dots$$

and denote this equation by $(**)$.

Now $\lambda'_{i+1} + \lambda'_{i+2} + \dots$ is equal to the number of nodes in the diagram of λ which lie to the right of the i -th column, and therefore

$$\lambda'_{i+1} + \lambda'_{i+2} + \dots = \sum_{j=1}^l (\lambda_j - i)$$

Likewise

$$\mu'_{i+1} + \mu'_{i+2} + \dots = \sum_{j=1}^m (\mu_j - i)$$

Hence from $(**)$ we have

$$\sum_{j=1}^m (\mu_j - i) > \sum_{j=1}^l (\lambda_j - i) \geq \sum_{j=1}^m (\lambda_j - i)$$

which implies

$$\mu_1 + \dots + \mu_m > \lambda_1 + \dots + \lambda_m$$

a contradiction. $\square$

Definition 5.0.10 (Young tableau). A Young tableau is a map $T(\lambda) : D(\lambda) \rightarrow \mathbb{N}$, defined by $(i, j) \mapsto T(\lambda)_{i,j} = k$. λ is called the shape of $T(\lambda)$.

Definition 5.0.11 (semistandard). For a Young tableau T . If $T_{i,j} \leq T_{i,j+1}$ and $T_{i,j} < T_{i+1,j}$ for all $(i, j) \in D(\lambda)$, then $T(\lambda)$ is called semistandard.

Definition 5.0.12 (weight). For a Young tableau T . Let $\alpha_k = |\{(i, j) \in D(\lambda) \mid T(\lambda)_{i,j} = k\}|$, then $\alpha = (\alpha_1, \dots)$ is called the weight of $T(\lambda)$.

Definition 5.0.13 (standard). For a Young tableau T , it's called standard, if its weight $\alpha = (1, 1, \dots, 1)$.

1	2	2	3
2	3	5	5
4	4	7	7
5	7		

1	3	7	12
2	5	10	14
4	8	11	16
6	9		

Lemma 5.0.8. Let $a \in P_n$. Then for each $w \in S_n$, we have $(a + \delta - w\delta)^+ \geq a$.

Proof. Since $\delta \in P_n$, then we have $\delta \geq w\delta$, hence

$$a + \delta - w\delta \geq a$$

Let $b = (a + \delta - w\delta)^+$. Then again by Lemma 4.28 we have

$$b \geq a + \delta - w\delta$$

Hence $b \geq a$. □

For each pair of integers i, j such that $1 \leq i < j \leq n$ define $R_{ij} : \mathbb{Z}^n \rightarrow \mathbb{Z}^n$ by

$$R_{ij}(a) = (a_1, \dots, a_i + 1, \dots, a_j - 1, \dots, a_n)$$

Any product $R = \prod_{i < j} R_{ij}^{r_{ij}}$ is called a raising operator. The order of the terms in the product is immaterial, since they commute with each other.

The following lemma explains why it is called raising:

Lemma 5.0.9. Let $a \in \mathbb{Z}^n$ and let R be a raising operator. Then

$$Ra \geq a$$

Proof. For we may assume that $R = R_{ij}$, in which case the result is obvious. □

However, the converse of the lemma still holds

Lemma 5.0.10. Let $a, b \in \mathbb{Z}^n$ be such that $a \leq b$ and $a_1 + \dots + a_n = b_1 + \dots + b_n$. Then there exists a raising operator R such that $b = Ra$.

Proof. We omit it here, since we won't use this result later. Readers may refer to [2] for more details. □

6. THE RING OF SYMMETRIC FUNCTIONS

The symmetric group S_n acts on the ring $\mathbb{Z}[x_1, \dots, x_n]$ of polynomials in n variables $x_1, \dots, x_n$ with integer coefficients by permuting the variables, that is

$$(wp)(x_1, \dots, x_n) = p(x_{w(1)}, \dots, x_{w(n)}), \quad w \in S_n, p \in \mathbb{Z}[x_1, \dots, x_n]$$

Definition 6.0.1 (Symmetric polynomial). $p \in \mathbb{Z}[x_1, \dots, x_n]$ is called symmetric if it is invariant under the action of S_n .

The symmetric polynomials form a subring

$$\Lambda_n = \mathbb{Z}[x_1, \dots, x_n]^{S_n} \subset \mathbb{Z}[x_1, \dots, x_n]$$

Note that Λ_n is a graded ring, i.e. $\Lambda_n = \bigoplus_{k \geq 0} \Lambda_n^k$, where $\Lambda_n^k = \{p \in \Lambda_n \mid \deg p = k\} \cup \{0\}$

There is a natural way to get a symmetric function with variables $x_1, \dots, x_n$: Given a monomial and use S_n to act on it, finally we do a summation. To be explicit, let's make the notation.

Notation 6.0.1. Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$. We set $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$.

Definition 6.0.2. Let λ be any partition of length $\leq n$. We define the polynomial

$$m_\lambda(x_1, \dots, x_n) = \sum_{\alpha} x^\alpha$$

where α runs over all distinct permutation of $\lambda = (\lambda_1, \dots, \lambda_n)$.

Exercise 6.0.1. Let $n = 3$ and $\lambda = (2, 1, 0)$ to see what's going on

$$m_{(2,1)} = x_1^2 x_2 + x_1^2 x_3 + x_1 x_2^2 + x_1 x_3^2 + x_2 x_3^2 + x_2^2 x_3$$

since we have all permutations of $(2, 1, 0)$ are listed as follows

$$(2, 1, 0), (2, 0, 1), (1, 2, 0), (1, 0, 2), (0, 1, 2), (0, 2, 1)$$

Remark 6.0.1. The $(m_\lambda)_{l(\lambda) \leq n}$ form a $\mathbb{Z}$ -basis of Λ_n . And $(m_\lambda)_{|\lambda|=k, l(\lambda) \leq n}$ form a $\mathbb{Z}$ -basis of Λ_n^k .

Definition 6.0.3 (Inverse system). Let $(I, \leq)$ be a directed set. Let $(A_i)_{i \in I}$ be a family of groups, rings, modules, indexed by I , and $(f_{ij})_{i, j \in I}$ be a family of morphisms with $f_{ij} : A_i \rightarrow A_j$, such that

1. $f_{ii} = \text{id}_{A_i}$;
2. $f_{ij} = f_{ij} \circ f_{jk}$ for all $i, j, k \in I$

The pair $(A_i, f_{ij})_{i, j \in I}$ is called an inverse system over I .

Definition 6.0.4 (Inverse limit). Let $(A_i, f_{ij})_{i, j \in I}$ be an inverse system. Let $x_i \in A_i, x_j \in A_j$. We define

$$x_i \sim x_j \iff \text{there exists } k \in I \text{ with } i \leq k, j \leq k \text{ and } f_{ki}(x_i) = f_{kj}(x_j)$$

We define the inverse limit of this inverse system by

$$\varprojlim_{i \in I} A_i = \prod_{i \in I} A_i / \sim$$

We can apply inverse limit to rings of symmetric functions. Let k be fixed, $m \geq n$, and consider

$$\mathbb{Z}[x_1, \dots, x_m] \rightarrow \mathbb{Z}[x_1, \dots, x_n]$$

which sends each of $x_{n+1}, \dots, x_m$ to zero and the other x_i to themselves. On restriction to Λ_m this gives a homomorphism as follows

$$\rho_{m,n} : \Lambda_m \rightarrow \Lambda_n$$

whose effect on the basis (m_λ) is easily described as follows

$$m_\lambda(x_1, \dots, x_m) \mapsto \begin{cases} m_\lambda(x_1, \dots, x_n), & l(\lambda) \leq n \\ 0, & \text{otherwise} \end{cases}$$

$\rho_{m,n}$ is a surjective ring homomorphism.

On restriction to Λ_m^k we have homomorphisms

$$\rho_{m,n}^k : \Lambda_m^k \rightarrow \Lambda_n^k$$

for all $k > 0$ and $m \geq n$, which are always surjective, and are bijective³ for $m \geq n \geq k$. So we have $(\Lambda_n^k, \rho_{m,n}^k)$ is an inverse system over $\mathbb{N}$. We define

$$\Lambda^k = \varprojlim_n \Lambda_n^k$$

Let us clarify the elements in Λ^k , as what we defined, an element of Λ^k is a sequence $f = (f_n)_{n \geq 0}$, where $f_n = f_n(x_1, \dots, x_n)$ is a homogenous symmetric polynomial of degree k in $x_1, \dots, x_n$, and $f_m(x_1, \dots, x_n, 0, \dots, 0) = f_n(x_1, \dots, x_n)$ whenever $m \geq n$. Since $\rho_{m,n}^k$ is an isomorphism for $m \geq n \geq k$, it follows that the projection

$$\rho_n^k : \Lambda^k \rightarrow \Lambda_n^k$$

which sends f to f_n is an isomorphism for all $n \geq k$, and hence that Λ^k has a $\mathbb{Z}$ -basis consisting of the monomial symmetric functions m_λ (for all partitions λ of k) defined by

$$\rho_n^k(m_\lambda) = m_\lambda(x_1, \dots, x_n)$$

for all $n \geq k$. Hence Λ^k is a free $\mathbb{Z}$ -module of rank $p(k)$, the number of partitions of k .

Exercise 6.0.2. The above discussion may be a little abstract, let's compute a concrete example to show what's going on

If we let $m = 3, n = 2$, and let $\lambda = (1, 1)$, then

$$m_{(1,1)}(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_1 + x_2x_3 + x_3x_1 + x_3x_2$$

So

$$\rho_{3,2}(m_{(1,1)}(x_1, x_2, x_3)) = m_{(1,1)}(x_1, x_2) = x_1x_2 + x_2x_1$$

and in this case, $l(\lambda) = 2 = n$. If we let $\lambda = (1, 1, 1)$, then

$$\rho_{3,2}(m_{(1,1,1)}(x_1, x_2, x_3)) = \rho_{3,2}(x_1x_2x_3) = 0$$

is quite natural.

Furthermore, if we let $k = n = 2, m = 3$, then obviously Λ_3^2 is spanned by

$$m_{(2,0)}(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$$

$$m_{(1,1)}(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_1 + x_2x_3 + x_3x_1 + x_3x_2$$

and Λ_2^2 is spanned by

$$m_{(2,0)}(x_1, x_2) = x_1^2 + x_2^2$$

$$m_{(1,1)}(x_1, x_2) = x_1x_2 + x_2x_1$$

So $\rho_{3,2}^2$ is clearly an isomorphism. Hope this example can help you to get a better understanding.

³Why?

Definition 6.0.5 (The ring of symmetric functions). We define

$$\Lambda = \bigoplus_{k \geq 0} \Lambda^k$$

Λ is the free $\mathbb{Z}$ -module generated by the m_λ for all partitions λ , and is called the ring of symmetric functions. The m_λ are called monomial symmetric functions.

Remark 6.0.2. We have the following remarks

1. For any commutative ring R in place of $\mathbb{Z}$, we can define a ring Λ_R satisfying $\Lambda_R \cong \Lambda \otimes_{\mathbb{Z}} R$.
2. We have surjective ring homomorphisms $\rho_n = \bigoplus_{k \geq 0} \rho_n^k : \Lambda \rightarrow \Lambda_n, n \geq 0$. ρ_n is an isomorphism in degrees $k \leq n$.

6.1. Elementary symmetric function. As we can see above, m_λ for any λ form a basis of the ring of symmetric functions. Now we will give several different basis of it, some of them are quite important to the representation theory of S_n .

First of them is elementary symmetric function

Definition 6.1.1 (Elementary symmetric function). Let $e_0 = 1$ and $e_r = \sum_{i_1 < \dots < i_r} x_{i_1} \dots x_{i_r} = m_{(1^r)}$ for some $r \geq 1$.

For each partition $\lambda = (\lambda_1, \lambda_2, \dots)$ define $e_\lambda = e_{\lambda_1} e_{\lambda_2} \dots$. Then e_λ is called elementary symmetric functions.

Remark 6.1.1. The generating function for the e_r is

$$E(t) = \sum_{r=0}^{\infty} e_r t^r = \prod_{i \geq 1} (1 + x_i t)$$

Remark 6.1.2. If the number of variables is finite, say n , then

$$\rho_n(e_r) = 0 \implies \sum_{r=0}^n e_r t^r = \prod_{i=1}^n (1 + x_i t) \in \Lambda_n[t]$$

Lemma 6.1.1. Let λ be a partition, λ' its conjugate. Then

$$e_{\lambda'} = m_\lambda + \sum_{\mu < \lambda} a_{\lambda\mu} m_\mu, \quad a_{\lambda\mu} \in \mathbb{Z}_{\geq 0}$$

Proof. When we multiply out the product $e_{\lambda'} = e_{\lambda'_1} e_{\lambda'_2} \dots$, we will obtain a sum of monomials, each of which is of the form

$$(x_{i_1} x_{i_2} \dots)(x_{j_1} x_{j_2} \dots) \dots = x^\alpha$$

where $i_1 < i_2 < \dots < i_{\lambda'_1}, j_1 < j_2 < \dots < j_{\lambda'_2}$, and so on.

Put the numbers $i_1, \dots, i_{\lambda'_1}$ into the first column of $D(\lambda)$ and similarly for the remaining numbers. The symbols $\leq r$ occur in the top r rows of $D(\lambda)$. Hence we have

$$\alpha_1 + \dots + \alpha_r \leq \lambda_1 + \dots + \lambda_r$$

for each $r \geq 1$, i.e. we have $\alpha \leq \lambda$. It follows Lemma 4.28 that

$$e_{\lambda'} = \sum_{\mu \leq \lambda} a_{\lambda\mu} m_\mu$$

with $a_{\lambda\mu} \geq 0$ for each $\mu \leq \lambda$, and the argument above also shows that the monomial x^λ occurs exactly once, so that $a_{\lambda\lambda} = 1$. $\square$

Proposition 6.1. We have

$$\Lambda \cong \mathbb{Z}[e_1, e_2, \dots]$$

and e_r are algebraically independent over $\mathbb{Z}$.

Proof. By above lemma, the e_r form a $\mathbb{Z}$ -basis since the m_λ do so. Then every $f \in \Lambda$ uniquely expressible as a polynomial in $e_r, r \geq 0$. $\square$

6.2. Complete symmetric function.

Definition 6.2.1 (complete symmetric function). Let $h_0 = 1$, and $h_r = \sum_{\mu \vdash r} m_\mu, r \geq 1$. For each partition $\lambda = (\lambda_1, \lambda_2, \dots)$, we define $h_\lambda = h_{\lambda_1} h_{\lambda_2} \dots$, called the complete symmetric functions.

Remark 6.2.1. Note that $e_1 = h_1$. And it will be convenient to define $h_r, e_r = 0$ to be zero for $r < 0$.

Lemma 6.2.1. The generating function of the h_r is

$$H(t) = \sum_{r \geq 0} h_r t^r = \prod_{i \geq 1} (1 - x_i t)^{-1}$$

Furthermore, we have

$$H(t)E(-t) = 1$$

Proof. To see the first, use the fact

$$\frac{1}{1 - x_i t} = \sum_k x_i^k t^k$$

and multiply these geometric series together.

Use the fact that the generating function of e_r is

$$E(t) = \sum_{r \geq 0} e_r t^r = \prod_{i \geq 1} (1 + x_i t)$$

together with what we have proven to see the second. $\square$

Remark 6.2.2. $H(t)E(-t) = 1$ is equivalent to

$$\sum_{r=0}^n (-1)^r e_r h_{n-r} = 0$$

for all $n \geq 1$.

Since e_r are algebraically independent, we may define a homomorphism of graded rings as follows

Definition 6.2.2.

$$\omega : \Lambda \rightarrow \Lambda$$

$$e_r \mapsto h_r$$

Lemma 6.2.2. ω is an involution.

Proof. The relations

$$\sum_{r=0}^n (-1)^r e_r h_{n-r} = 0, \quad \forall n \geq 1$$

are symmetric with respect to interchanging e_r and h_r . $\square$

Proposition 6.2. We have

$$\Lambda \cong \mathbb{Z}[h_1, h_2, \dots]$$

and h_r are algebraically independent over $\mathbb{Z}$.

Proof. Follows from that $\omega^2 = \text{Id}$, that is ω is an automorphism of Λ . $\square$

Remark 6.2.3. If the number of variables is finite, say n , then $\omega|_{\Lambda} = \text{id}|_{\Lambda}$, and $\Lambda_n \cong \mathbb{Z}[h_1, \dots, h_n]$ with h_r are algebraically independent over $\mathbb{Z}$, but $h_{r+1}, \dots$ are nonzero polynomials in $h_1, \dots, h_n$.

Remark 6.2.4. We could define $f_\lambda = \omega(m_\lambda)$ and would obtain another basis of Λ , but these play no role later on.

Remark 5.18 lead to a determinant identity which we shall make use of later. Let N be a positive integer and consider the matrices of $N+1$ rows and columns

$$H = (h_{i-j})_{0 \leq i, j \leq N}, \quad E = ((-1)^{i-j} e_{i-j})_{0 \leq i, j \leq N}$$

Then E, H are lower unitriangular, so we have $\det E = \det H = 1$. Moreover, Remark 5.18 shows that

$$\sum_{r=0}^N (-1)^r e_r h_{N-r} = 0$$

which implies that

$$EH = \text{Id}$$

It follows that each minor of H is equal to the complementary cofactor of E^T , the transpose of E .

Now let λ, μ be partitions of length $\leq p$ such that λ', μ' have length $\leq p$. $p+q = N+1$. And consider the minor of H with row indices $\lambda_i + p - i$ ($1 \leq i \leq p$) and columns indices $\mu_i + p - i$ ($1 \leq i \leq p$). By Lemma 4.13 the complementary cofactor of E^T has row indices $p - 1 + j - \lambda'_j$ ($1 \leq j \leq q$) and column indices $p - 1 + j - \mu'_j$ ($1 \leq j \leq p$). Hence we have

$$\det(h_{\lambda_i - \mu_j - i + j})_{1 \leq i, j \leq p} = (-1)^{|\lambda| + |\mu|} \det((-1)^{\lambda'_i - \mu'_j - i + j} e_{\lambda'_i - \mu'_j - i + j})_{1 \leq i, j \leq q}$$

The minus signs cancel out, and we have proven the following results:

Lemma 6.2.3. Let λ, μ be partitions of length $\leq p$ such that λ', μ' have length $\leq p$. $p+q = N+1$. Then

$$\det(h_{\lambda_i - \mu_j - i + j})_{0 \leq i, j \leq p} = \det(e_{\lambda'_i - \mu'_j - i + j})_{0 \leq i, j \leq q}$$

In particular, if $\mu = \emptyset$, then $\det(h_{\lambda_i - i + j}) = \det(e_{\lambda'_j - i + j})$.

6.3. Power sums.

Definition 6.3.1 (power sum). Let $p_r = \sum_i x_i^r = m_{(r)}$, $r \geq 1$, p_r is call the r -th power sum. For a partition $\lambda = (\lambda_1, \lambda_2, \dots)$, we define $p_\lambda = p_{\lambda_1} p_{\lambda_2} \dots$

Lemma 6.3.1. The generating function of p_r is

$$P(t) = \sum_{r \geq 1} p_r t^{r-1} = \frac{H(t)}{H'(t)}$$

Furthermore, we have the following properties

1. $P(-t) = \frac{E'(t)}{E(t)}$
2. $nh_n = \sum_{r=1}^n p_r h_{n-r}$
3. $ne_n = \sum_{r=1}^n (-1)^{r-1} p_r e_{n-r}$

Proof. We compute as follows

$$\begin{aligned} P(t) &= \sum_{i \geq 1} \sum_{r \geq 1} x_i^r t^{r-1} \\ &= \sum_{i \geq 1} \frac{x_i}{1 - x_i t} \\ &= \sum_{i \geq 1} \frac{d}{dt} \log\left(\frac{1}{1 - x_i t}\right) \\ &= \frac{d}{dt} \log \prod_{i \geq 1} (1 - x_i t)^{-1} \\ &= \frac{d}{dt} \log H(t) \\ &= \frac{H'(t)}{H(t)} \end{aligned}$$

Similarly we have $P(-t) = \frac{d}{dt} \log E(t)$.

From above we have

$$\begin{aligned} nh_n &= \sum_{r=1}^n p_r h_{n-r} \\ ne_n &= \sum_{r=1}^n (-1)^{r-1} p_r e_{n-r} \end{aligned}$$

for $n \geq 1$. □

Remark 6.3.1. The second and third equations enable us to express the h 's and the e 's in terms of the p 's, and vice versa. In fact, the third equations are due to Isaac Newton, and are known as Newton's formulas. And from the second formula, it is clear that $h_n \in \mathbb{Q}[p_1, \dots, p_n]$ and $p_n \in \mathbb{Z}[h_1, \dots, h_n]$, and hence

$$\mathbb{Q}[p_1, \dots, p_n] = \mathbb{Q}[h_1, \dots, h_n]$$

Since the h_r are algebraically independent over $\mathbb{Z}$, and hence also over $\mathbb{Q}$, it follows that:

Proposition 6.3. $\Lambda_{\mathbb{Q}} = \Lambda \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}[p_1, p_2, \dots]$ and the p_r are algebraically independent over $\mathbb{Q}$. The p_r form a $\mathbb{Q}$ -basis for $\Lambda_{\mathbb{Q}}$.

Definition 6.3.2. Let $\lambda = (1^{m_1} 2^{m_2} \dots)$ be a partition in exponential notation. We define

$$\varepsilon_{\lambda} = (-1)^{m_2 + m_4 + \dots} = (-1)^{|\lambda| - l(\lambda)}$$

$$z_{\lambda} = \prod_{j \geq 1} j^{m_j} m_j!$$

Remark 6.3.2. Let $w \in S_n$ with cycle type $\alpha(w) = (1^{m_1} 2^{m_2} \dots)$, then

$$\varepsilon_{\alpha(w)} = \begin{cases} 1, & w \text{ is even} \\ -1, & w \text{ is odd} \end{cases}$$

so we have $S_n \rightarrow \{\pm 1\}$ defined by $w \mapsto \varepsilon_{\alpha(w)}$ is the usual sign homomorphism.

Lemma 6.3.2. $\omega(p_{\lambda}) = \varepsilon_{\lambda} p_{\lambda}$

Proof. Since we have

$$\omega(E(t)) = H(t), \omega(H(t)) = E(t)$$

then we have

$$\omega(P(t)) = \omega\left(\frac{H'(t)}{H(t)}\right) = \frac{E'(t)}{E(t)} = P(-t)$$

then

$$\omega(p_n) = (-1)^{n-1} p_n, \quad \forall n \geq 1$$

then

$$\omega(p_{\lambda}) = (-1)^{\sum \lambda_i - \sum 1} p_{\lambda} = \varepsilon_{\lambda} p_{\lambda}$$

□

Lemma 6.3.3. We have

$$H(t) = \sum_{\lambda} \frac{1}{z_{\lambda}} p_{\lambda} t^{|\lambda|}, \quad h_n = \sum_{\lambda \vdash n} \frac{1}{z_{\lambda}} p_{\lambda}$$

$$E(t) = \sum_{\lambda} \frac{\varepsilon_{\lambda}}{z_{\lambda}} p_{\lambda} t^{|\lambda|}, \quad e_n = \sum_{\lambda \vdash n} \frac{\varepsilon_{\lambda}}{z_{\lambda}} p_{\lambda}$$

Proof. It suffices to prove the identity in the first row, since the one in the second row then follows by applying the involution ω and using the fact that p_k is an eigenvector of ω with respect to ε_{λ} .

We compute as follows,

$$\begin{aligned} H(z) &= \exp \sum_{r \geq 1} p_r t^r / r \\ &= \prod_{r \geq 1} \exp(p_r t^r / r) \\ &= \prod_{r \geq 1} \sum_{m_r=0}^{\infty} (p_r t^r)^{m_r} / r^{m_r} m_r! \\ &= \sum_{\lambda} z_{\lambda}^{-1} p_{\lambda} t^{|\lambda|} \end{aligned}$$

The first step follows from Lemma 5.26.

□

7. SCHUR FUNCTIONS

Lemma 7.0.1. Let $A_n = \{f \in \mathbb{Z}[x_1, \dots, x_n] \mid w(f) = \text{sgn}(w)f, \forall w \in S_n\}$, then A_n is a free module of rank 1 over Λ_n .

Proof. Let $f \in A_n$, then $x_i - x_j, i \neq j$ divides f , since $f|_{x_i=x_j} = 0$, so we have $\prod_{i < j} (x_i - x_j)$ divides f . Then

$$f = \prod_{i < j} (x_i - x_j)g, \quad g \in \Lambda_n$$

So A_n is generated by $\prod_{i < j} (x_i - x_j)$ over Λ_n , i.e. $A_n = \prod_{i < j} (x_i - x_j)\Lambda_n$ $\square$

Let $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$ be a monomial, and consider the polynomial a_α obtained by antisymmetrizing x^α , that is

$$a_\alpha = \sum_{w \in S_n} \text{sgn}(w)w(x^\alpha)$$

Clearly a_α is skew-symmetric, i.e. $a_\alpha \in A_n$. In particular, therefore a_α vanishes unless $\alpha_1, \dots, \alpha_n$ are all distinct. Hence we may as well assume that $\alpha_1 > \dots > \alpha_n \geq 0$. And we may write $\alpha = \lambda + \delta$, where λ is a partition⁴ with length $\leq n$ and $\delta = (n-1, n-2, \dots, 1, 0)$. Then

$$a_\alpha = a_{\lambda+\delta} = \sum_{w \in S_n} \text{sgn}(w)w(x^{\lambda+\delta})$$

which can be written as a determinant.

Lemma 7.0.2. Let λ be a partition $l(\lambda) \leq n$, then

1. $a_{\lambda+\delta} = \det(x_i^{\lambda_j+n-j})_{1 \leq i, j \leq n}$. In particular, $a_\delta = \det(x_i^{n-j})_{1 \leq i, j \leq n} = \prod (x_i - x_j)$ is the Vandermonde determinant.
2. $a_{\lambda+\delta}$ is divisible by a_δ .

Proof. 1. follows from the Leibniz formula for the determinant $\det A = \sum_{w \in S_n} \text{sgn}(w) \prod_{i=1}^n a_{i, w(i)}$.
2. follows from Lemma 6.1. $\square$

Definition 7.0.1. Let λ be a partition, $l(\lambda) \leq n$, and $\delta = (n-1, n-2, \dots, 0) \in \mathbb{Z}_{\geq 0}^n$. We define the Schur polynomial

$$s_\lambda = \frac{a_{\lambda+\delta}}{a_\delta} \in \Lambda_n$$

Notice that the definition of s_λ makes sense for any integer vector $\lambda \in \mathbb{Z}^n$ such that $\lambda + \delta$ has no negative parts. If $\lambda_i + n - i$ are not all distinct, then $s_\lambda = 0$. If they are all distinct, then we have $\lambda + \delta = w(\mu + \delta)$ for some $w \in S_n$ and some partition μ , and $s_\lambda = \text{sgn}(w)s_\mu$.

The polynomial $a_{\lambda+\delta}$ where λ runs through all partitions of length $\leq n$, form a basis of A_n . Multiplication by a_δ is an isomorphism of Λ_n onto A_n , since A_n is the free Λ_n -module generated by a_δ .

So we have proven

⁴ λ is indeed a partition. Take an example, $\alpha_1 + 1 - n \geq \alpha_2 + 2 - n$ holds, since $\alpha_1 > \alpha_2$ is equivalent to $\alpha_1 \geq \alpha_2 + 1$

Lemma 7.0.3. The Schur polynomial s_λ , where λ is a partition with $l(\lambda) \leq n$, form a $\mathbb{Z}$ -basis of Λ_n .

Proposition 7.1. The s_λ for all partitions λ form a $\mathbb{Z}$ -basis of Λ , called Schur functions. The s_λ for all partitions λ with $|\lambda| = k$ form a $\mathbb{Z}$ -basis of Λ^k .

Proof. From the definition it follows that

$$a_{\lambda+\delta+(k^n)} = \prod_{i=1}^n x_i^k a_{\lambda+\delta}, \quad s_{\lambda+(k^n)} = s_\lambda$$

□

Proposition 7.2.

$$\begin{aligned} s_\lambda &= \det(h_{\lambda_i-i+j})_{1 \leq i, j \leq n}, \quad n \leq l(\lambda) \\ s_\lambda &= \det(e_{\lambda'_i-i+j})_{1 \leq i, j \leq m}, \quad m \leq l(\lambda') \end{aligned}$$

Proof.

□

Corollary 7.1. We have the following properties

1. $\omega(s_\lambda) = s_{\lambda'}$
2. $s_{(n)} = h_n, s_{(1^n)} = e_n$

8. ORTHOGONALITY

Let $x = (x_1, x_2, x_3, \dots), y = (y_1, y_2, y_3, \dots)$ be finite or infinite sequences of variables. We denote the symmetric functions of the x 's by $s_\lambda(x), p_\lambda(x)$, etc. and the symmetric functions of the y 's by $s_\lambda(y), p_\lambda(y)$, etc.

Proposition 8.1. We give three series expansions for the product

$$\begin{aligned} \prod_{i,j} (1 - x_i y_j)^{-1} &= \sum_{\lambda} \frac{1}{z_\lambda} p_\lambda(x) p_\lambda(y) \\ &= \sum_{\lambda} h_\lambda(x) m_\lambda(y) \\ &= \sum_{\lambda} s_\lambda(x) s_\lambda(y) \end{aligned}$$

Proof. For the first one, Since we have

$$H(t) = \prod_i (1 - x_i t)^{-1} = \sum_{\lambda} z_\lambda^{-1} p_\lambda t^{|\lambda|}$$

Choose as variables $x_i y_j$, then

$$\begin{aligned} \prod_{i,j} (1 - x_i y_j t)^{-1} &= H(t) = \sum_{\lambda} \frac{1}{z_\lambda} p_\lambda(x_1 y_1, \dots, x_i y_j, \dots, x_n y_n) t^{|\lambda|} \\ &= \sum_{\lambda} \frac{1}{z_\lambda} p_\lambda(x) p_\lambda(y) t^{|\lambda|} \end{aligned}$$

and set $t = 1$ to get desired result.

For the second one,

$$\begin{aligned}
\prod_{i,j} (1 - x_i y_j t)^{-1} &= \prod_j H(y_j) \\
&= \prod_j \sum_{r=0}^{\infty} h_r(x) y_j^r \\
&= \sum_{\alpha} h_{\alpha}(x) y^{\alpha} \\
&= \sum_{\lambda} h_{\lambda}(x) m_{\lambda}(y)
\end{aligned}$$

where α runs through all sequences $(\alpha_1, \alpha_2, \dots)$ of non-negative integers such that $\sum \alpha_i < \infty$, and λ runs through all partitions.

For the third one is sometimes called Cauchy formula, we compute as

$$\begin{aligned}
a_{\delta}(x) a_{\delta}(y) \prod_{i,j=1}^n (1 - x_i y_j)^{-1} &= a_{\delta}(x) \sum_{w \in S_n} \text{sgn}(w) w(y^{\delta}) \sum_{\lambda} h_{\lambda}(x) m_{\lambda}(y) \\
&= a_{\delta}(x) \sum_{w \in S_n} \sum_{\lambda} \text{sgn}(w) y^{w\delta} h_{\lambda}(x) \sum_{\substack{\alpha \text{ is the} \\ \text{permutation of } \lambda}} y^{\alpha} \\
&= a_{\delta}(x) \sum_{w \in S_n, \alpha \in \mathbb{N}^n} \text{sgn}(w) h_{\alpha}(x) y^{\alpha + w\delta} \\
&= \sum_{w \in S_n, \beta \in \mathbb{N}^n} (a_{\delta}(x) \text{sgn}(w) h_{\beta - w\delta}(x)) y^{\beta} \\
&= \sum_{\beta \in \mathbb{N}^n} a_{\beta}(x) y^{\beta} \quad (\alpha_{\beta} = 0 \text{ if } \beta \neq w(\lambda + \delta), w \in S_n) \\
&= \sum_{w \in S_n} \sum_{\lambda} w(a_{\lambda + \delta}(x)) y^{w(\lambda + \delta)} \\
&= \sum_{\lambda} a_{\lambda + \delta}(x) \sum_{w \in S_n} \text{sgn}(w) w(y^{\lambda + \delta}) \\
&= \sum_{\lambda} a_{\lambda + \delta}(x) a_{\lambda + \delta}(y)
\end{aligned}$$

This proves in the case of n variables x_i and n variables y_i , now let $n \rightarrow \infty$ as usual to complete the proof. $\square$

Definition 8.0.1. We define a $\mathbb{Z}$ -valued bilinear form $\langle \cdot, \cdot \rangle : \Lambda \times \Lambda \rightarrow \mathbb{Z}$ by requiring

$$\langle h_{\lambda}, m_{\mu} \rangle = \delta_{\lambda\mu}$$

for all partitions λ, μ , where $\delta_{\lambda\mu}$ is the Kronecker delta.

Lemma 8.0.1. For each $n \geq 0$, let $(u_{\lambda}), (v_{\lambda})$ be $\mathbb{Q}$ -bases of $\Lambda_{\mathbb{Q}}^n$, indexed by the partition λ of n . Then the following condition are equivalent:

1. $\langle \mu_{\lambda}, v_{\mu} \rangle = \delta_{\lambda\mu}$ for all λ, μ .
2. $\sum_{\lambda} u_{\lambda}(x) v_{\lambda}(y) = \prod_{i,j} (1 - x_i y_j)^{-1}$.

Proof. Let

$$u_{\lambda} = \sum_{\rho} a_{\lambda\rho} h_{\rho}, \quad v_{\mu} = \sum_{\sigma} b_{\mu\sigma} m_{\sigma}$$

then

$$\langle u_\lambda, v_\mu \rangle = \sum_\rho a_{\lambda\rho} b_{\mu\rho}$$

so the first statement is equivalent to

$$\sum_\rho a_{\lambda\rho} b_{\mu\rho} = \delta_{\lambda\mu}$$

And note that the second statement is equivalent to

$$\sum_\lambda u_\lambda(x) v_\lambda(y) = \sum_\rho h_\rho(x) m_\rho(y)$$

so it is also equivalent to

$$\sum_\lambda a_{\lambda\rho} b_{\lambda\sigma} = \delta_{\rho\sigma}$$

This completes the proof. $\square$

So together with Proposition 7.1 with Lemma 7.3, it follows that

$$\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda$$

so that the p_λ form an orthogonal basis of $\Lambda_{\mathbb{Q}}$. Likewise we have

$$\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$$

so that s_λ form an orthonormal basis of Λ , and the s_λ such that $|\lambda| = n$ form an orthonormal basis of Λ^n .

Any other orthonormal basis of Λ^n must therefore be obtained from the basis (s_λ) by transformation by an orthonormal integer matrix. The only such matrices are signed permutation matrices, therefore the orthonormal relation s_λ satisfied characterizes the s_λ up to order and sign.

Lemma 8.0.2. $\omega : \Lambda \rightarrow \Lambda$ is an isometry for $\langle \cdot, \cdot \rangle$.

Proof. Since we have $\omega(p_\lambda) = \varepsilon_\lambda p_\lambda$, hence we

$$\langle \omega(p_\lambda), \omega(p_\mu) \rangle = \varepsilon_\lambda \varepsilon_\mu \langle p_\lambda, p_\mu \rangle = \varepsilon_\lambda \varepsilon_\mu z_\lambda \delta_{\lambda\mu} = \langle p_\lambda, p_\mu \rangle$$

since $(\varepsilon_\lambda)^2 = 1$. This completes the proof. $\square$

8.1. Transition matrices. Let λ, μ be partitions, we define

$$\begin{aligned} \{\lambda\}^j &= \{\mu \subset \lambda \mid |\mu| = |\lambda| - j, 0 \leq \lambda'_i - \mu'_i \leq 1, \forall i\} \\ \{\lambda\}_j &= \{\mu \subset \lambda \mid |\mu| = |\lambda| + j, \lambda'_i \leq \mu'_i \leq \lambda'_i + 1, \forall i\} \end{aligned}$$

Definition 8.1.1 (flag). A flag $\mu_\bullet$ is a sequence of partitions

$$\mu_n \subset \mu_{n-1} \subset \cdots \subset \mu_0 = \lambda$$

such that $\mu_i \in \{\mu_{i-1}\}^{a_i}$ for some $a_i \geq 0$, and all $1 \leq i \leq n$. The sequence $a = (a_1, \dots, a_n)$ is called the weight of μ_0 .

Definition 8.1.2 (complete). A flag is called complete if $n = |\lambda|$.

Exercise 8.1.1. Consider $\lambda = (6, 4, 4, 2)$, we can get a flag as follows by removing boxes.

1	2	2	3	3	5
2	3	5	5		
4	4	7	7		
5	7				
1	2	2			
2					

1	2	2	3	3	5
2	3	5	5		
4	4				
5					
1					

1	2	2	3	3	5
2	3	5	5		
4	4				
5					

1	2	2	3	3	
2	3				
4	4				

1	2	2	3	3	
2	3				

where we have

$$\mu_0 = (6, 4, 4, 2) \supset \mu_1 = (6, 4, 2, 1) \supset \mu_2 = (6, 4, 2, 1) \supset \mu_3 = (5, 2, 2) \supset$$

$$\mu_4 = (5, 2) \supset \mu_5 = (3, 1) \supset \mu_6 = (1) \supset \mu_7 = \emptyset$$

and

$$a_1 = 3, a_2 = 0, a_3 = 4, a_4 = 2, a_5 = 3, a_6 = 3, a_7 = 1$$

that is $a = (3, 0, 4, 2, 3, 3, 1)$

Lemma 8.1.1.

$$\{\text{semistandard Young tableau } T(\lambda) \longleftrightarrow \{\text{flag } \mu_\bullet \text{ such that } \mu_0 = \lambda\}$$

Proof. Let $n = |\lambda|$. Given $\mu_\bullet$ with $\mu_0 = \lambda$, define $T(\lambda)$ by filling all the a_i boxes of $\mu_i - \mu_{i+1}$ with $n - i$, $1 \leq i \leq n$. Then $u_i \in \{\mu_{i-1}\}^{a_i}$ implies all columns are strictly increasing and $a_i \geq 0$ implies all rows are increasing.

Given a semistandard Young tableau $T(\lambda)$ of weight $a = (a_1, \dots, a_n)$, remove a_i boxes whose entry is $n - i + 1$ to obtain μ_i and set $\mu_0 = \lambda$. Rows of $T(\lambda)$ are increasing implies $|\mu_i| - |\mu_{i-1}| = a_{i-1} \geq 0$ and columns of $T(\lambda)$ are strictly increasing implies at most one box in each column is removed, that is $0 \leq \mu'_{i-1} - \mu'_i \leq 1$. $\square$

Recall that we have

$$s_{(n)} = h_n, \quad s_{(1^n)} = e_n$$

Proposition 8.2 (Pier's formula). We have

1. $s_\lambda e_j = \sum_{\mu \in \{\lambda\}_j} s_\mu$
2. $s_\lambda h_j = \sum_{\mu' \in \{\lambda'\}_j} s_{\mu'}$

Proof. Let $\lambda = (\lambda_1, \dots, \lambda_n)$ with n sufficiently large by allowing some λ_i to be zero.

$$s_\lambda e_i a_\delta = a_{\lambda+\delta} e_i \in A_r$$

implies

$$a_{\lambda+\delta} = \sum_{\mu} B_{\lambda\mu} a_{\mu+\delta}$$

Let $l_i = \lambda_i + n - i$, then the only way to obtain a monomial $x_1^{m_1} \dots x_n^{m_n}$ with $m_1 > m_2 > \dots > m_n$ in $a_{\lambda+\delta} e_i$ is possibly by $x_1^{l_1} \dots x_n^{l_n} x_{j_1} \dots x_{j_n}$. This monomial

has strictly decreasing exponents if and only if the following is satisfied: Set

$$\mu_k = \begin{cases} \lambda_k, & k \notin \{j_1, \dots, j_i\} \\ \lambda_k + 1, & k \in \{j_1, \dots, j_i\} \end{cases}$$

Then $\mu_1 \geq \dots \geq \mu_n$, i.e. $\mu \in \{\lambda\}_i$. The coefficient of such a monomial is $B_{\lambda\mu} = 1$, so we have

$$a_{\lambda+\delta} e_i = \sum_{\mu \in \{\lambda\}_i} a_{\mu+\delta}$$

And the second equation follows from the first since $\omega(e_n) = h_n, \omega(s_\lambda) = s_{\lambda'}$. $\square$

Use the following, we can express s_λ with $x_n = 1$ in terms of s_μ in $n-1$ variables.

Lemma 8.1.2. $s_\lambda(x_1, \dots, x_{n-1}, 1) = \sum_{j=0}^{|\lambda|} \sum_{\mu \in \{\lambda\}_j} s_\mu(x_1, \dots, x_{n-1})$

Proof. By Cauchy formula

$$\begin{aligned} \sum_{\lambda} s_\lambda(x_1, \dots, x_{n-1}, 1) s_\lambda(y_1, \dots, y_n) &= \prod_{i=1}^{n-1} \prod_{j=1}^n (1 - x_i y_j)^{-1} \prod_{j=1}^n (1 - y_j)^{-1} \\ &= \sum_{\mu} s_\mu(x_1, \dots, x_{n-1}) s_\mu(y_1, \dots, y_n) \sum_{j=0}^{\infty} h_j(y_1, \dots, y_n) \\ &= \sum_{\mu} s_\mu(x_1, \dots, x_{n-1}) \sum_{j=0}^{\infty} \sum_{\lambda' \in \{\mu'\}_j} s_{\lambda'}(y_1, \dots, y_n) \end{aligned}$$

Comparing the coefficients of $s_\lambda(y_1, \dots, y_n)$, we have

$$\begin{aligned} s_\lambda(x_1, \dots, x_{n-1}, 1) &= \sum_{j=0}^{\infty} \sum_{\mu, \lambda' \in \{\mu'\}_j} s_\mu(x_1, \dots, x_{n-1}) \\ &= \sum_{j=0}^{|\lambda|} \sum_{\mu' \in \{\lambda\}_j} s_{\mu'}(x_1, \dots, x_{n-1}) \end{aligned}$$

since $\lambda' \in \{\mu'\}_j$ implies $j \leq |\lambda| = n$. $\square$

Lemma 8.1.3. We can write

$$s_\lambda(x_1, \dots, x_n) = \sum_{\substack{\mu \bullet = (\emptyset \subset \mu \subset \lambda) \\ a = |\lambda| - |\mu|}} x_n^a s_\mu(x_1, \dots, x_{n-1})$$

Proof. $s_\lambda(x_1, \dots, x_n)$ is homogenous of degree $|\lambda|$, then

$$\begin{aligned} s_\lambda(x_1, \dots, x_n) &= x_n^{|\lambda|} s_\lambda\left(\frac{x_1}{x_n}, \dots, \frac{x_{n-1}}{x_n}, 1\right) \\ &= x_n^{|\lambda|} \sum_{j=0}^{|\lambda|} \sum_{\mu \in \{\lambda\}_j} s_\mu\left(\frac{x_1}{x_n}, \dots, \frac{x_{n-1}}{x_n}\right) \\ &= \sum_{j=0}^{|\lambda|} \sum_{\mu \in \{\lambda\}_j} x_n^{|\lambda| - |\mu|} s_\mu(x_1, \dots, x_{n-1}) \end{aligned}$$

□

Theorem 8.1.1. We have

$$s_\lambda(x_1, \dots, x_n) = \sum_{\substack{T \text{ is semistandard} \\ \text{Young tableau of sharp } \lambda}} x^T$$

where

$$x^T = \prod_{i=1}^n x_i^{a_{n-i+1}}$$

and a is the weight of $T(\lambda)$.

Proof.

$$s_\lambda(x_1, \dots, x_n) = \sum x_n^{a_1} x_{n-1}^{a_2} \dots x_{n-i+1}^{a_i} s_\mu(x_1, \dots, x_{n-i})$$

where the summation runs over $\mu_\bullet = (\mu_i \subset \mu_{i-1} \subset \dots \subset \mu_0 = \lambda)$ such that $|\mu_i| - |\mu_{i-1}| = a_i$ and $0 \leq \mu'_i - \mu'_{i-1} \leq 1$. Then we have

$$\begin{aligned} s_\lambda(x_1, \dots, x_n) &= \sum_{\mu \text{ is a flag of } \lambda} \prod_{i=1}^n x_i^{a_{n-1+i}} \\ &= \sum x^T \end{aligned}$$

where T runs over all semistandard Young tableau as desired. □

Remark 8.1.1. In combinatorics this statement is taken as a definition, and all the properties of s_λ are derived from this. In particular, $s_\lambda \in \Lambda_n^k$ where $k = |\lambda|$.

Corollary 8.1. $s_\lambda = \sum_{\mu \leq \lambda} K_{\lambda\mu} m_\mu$, where $K_{\lambda\mu}$ is Kostka number.

Exercise 8.1.2. Let $n = 3$ and $\lambda = (3, 3, 1)$ to compute $s_\lambda(x_1, x_2, x_3)$ use above property. All we need to do is to find out all semistandard Young tableaux, and compute the weight of flags which correspond to them.

List as follows

1	1	1	$(3, 3, 1) \supset_1 (3, 3) \supset_3 (3) \supset_3 \emptyset$
2	2	2	
3			
1	1	1	$(3, 3, 1) \supset_2 (3, 2) \supset_2 (3) \supset_3 \emptyset$
2	2	3	
3			
1	1	1	$(3, 3, 1) \supset_3 (3, 1) \supset_1 (3) \supset_3 \emptyset$
2	3	3	
3			
1	1	2	$(3, 3, 1) \supset_2 (3, 2) \supset_3 (2) \supset_2 \emptyset$
2	2	3	
3			
1	1	2	$(3, 3, 1) \supset_3 (3, 1) \supset_2 (2) \supset_2 \emptyset$
2	3	3	
3			
1	2	2	$(3, 3, 1) \supset_3 (3, 1) \supset_3 (1) \supset_1 \emptyset$
2	3	3	
3			

so we have

$$s_{(3,3,1)} = x_1^3 x_2^3 x_3^3 + x_1^2 x_2^2 x_3^3 + x_1^3 x_2 x_3^3 + x_1^2 x_2^3 x_3 + x_1^3 x_2^2 x_3^2 + x_1^3 x_2^3 x_3$$

Now we have already know the relations between bases (s_λ) and (m_λ) , We also want to know

$$s_\lambda = \sum F_{\lambda\mu} p_\mu$$

Definition 8.1.3. We arrange partition with respect to the reverse lexicographic order L_n , i.e. (n) is first and (1^n) is last. A matrix $(M_{\lambda\mu})$ indexed by $\lambda, \mu \in \mathcal{P}_n$ is said to be **strictly upper triangle**, if $M_{\lambda\mu} = 0$ unless $\lambda \geq \mu$; And **strictly upper unitriangular** if also $M_{\lambda\lambda} = 1$ for all $\lambda \in \mathcal{P}_n$; Similarly for strictly lower unitriangular.

We set U_n be the set of all strictly upper unitriangular matrices and U'_n be the set of all strictly lower unitriangular matrices.

Lemma 8.1.4. U_n, U'_n are groups with respect to matrix multiplication.

Proof. Let $M, N \in U_n$, then we have

$$(MN)_{\lambda\mu} = \sum_\nu M_{\lambda\nu} N_{\nu\mu} = 0$$

unless there exists ν such that $\lambda \geq \nu \geq \mu$, i.e. unless $\lambda \geq \mu$. For the same reason we have

$$(MN)_{\lambda\lambda} = M_{\lambda\lambda} N_{\lambda\lambda} = 1$$

i.e. $MN \in U_n$.

Consider $\sum_{\mu} M_{\lambda\mu} x_{\mu} = y_{\lambda}$. If $\nu \leq \lambda$, these equations involve x_{μ} for $\mu \leq \nu$, hence $\mu \leq \lambda$. The same is true for the equivalent set of equations

$$\sum_{\mu} (M^{-1})_{\lambda\mu} y_{\mu} = x_{\lambda}$$

implies $(M^{-1})_{\lambda\mu} = 0$ unless $\mu \leq \lambda$. $\square$

Lemma 8.1.5. Let

$$J = \begin{cases} 1, & \mu = \lambda' \\ 0, & \text{otherwise} \end{cases}$$

Then $M \in U_n$ is equivalent to $JMJ \in U'_n$

Proof. If let $N = JMJ$, then we have $N_{\lambda\mu} = M_{\mu'\lambda'}$. Then by Lemma 4.23, we have $\lambda \geq \mu$ is equivalent to $\mu' \geq \lambda'$. This completes the proof. $\square$

Definition 8.1.4. Let $(u_{\lambda}), (v_{\lambda})$ be $\mathbb{Q}$ bases for Λ . We denote by $M(u, v)$ the matrix $(M_{\lambda\mu})$ of coefficients in the equations

$$u_{\lambda} = \sum_{\mu} M_{\lambda\mu} v_{\mu}$$

and $M(u, v)$ is called the transition matrix from (v_{λ}) to (u_{λ}) .

Lemma 8.1.6. Let $(u_{\lambda}), (v_{\lambda}), (w_{\lambda})$ be $\mathbb{Q}$ bases of Λ , and let $(u'_{\lambda}), (v'_{\lambda})$ be the dual bases of $(u_{\lambda}), (v_{\lambda})$ with respect to $\langle \cdot, \cdot \rangle$. Then

$$M(u, v)M(v, w) = M(v, w)$$

$$M(v, u) = M(u, v)^{-1}$$

$$M(v', u') = M(v, u)^T = M(u, v)^*$$

$$M(wv, wu) = M(u, v)$$

where T means transpose and $*$ means transpose of inverse.

Proposition 8.3. The matrix $(K_{\lambda\mu})$ is in U_n .

Proof. By Lemma 4.27, we have $K_{\lambda\mu} = 0$ unless $\lambda \geq \mu$. In particular, we have $K_{\lambda\lambda} = 1$. $\square$

Remark 8.1.2. In fact, all transition matrices between bases $e_{\lambda}, h_{\lambda}, m_{\lambda}, s_{\lambda}$ can be expressed in terms of J and K

Definition 8.1.5. Let L denote the transition matrix $M(p, m)$, i.e.

$$p_{\lambda} = \sum_{\mu} L_{\lambda\mu} m_{\mu}$$

Definition 8.1.6. Let λ be partition, $l(\lambda) = r$. Let $f : [1, r] \subset \mathbb{Z} \rightarrow \mathbb{Z}_{\geq 0}$. We define $f(\lambda)$ to be the vector whose i -th component is

$$f(\lambda)_i = \sum_{f(j)=i} \lambda_j, \quad i \geq 1$$

Proposition 8.4. $L_{\lambda\mu} = |\{f : \mathbb{Z} \rightarrow \mathbb{Z}_{\geq 0} \mid f(\lambda) = \mu\}|$

Proof. Note that

$$\begin{aligned}
 p\lambda &= p_{\lambda_1} p_{\lambda_2} \cdots \\
 &= \sum_{f: [1, l(\lambda)] \rightarrow \mathbb{Z}_{\geq 0}} x_{f(1)}^{\lambda_1} x_{f(2)}^{\lambda_2} \cdots \\
 &= \sum_f x^{f(\lambda)} \\
 &= \sum_{\mu} \sum_{f(\lambda)=\mu} \sum_{w \in S_n} x^{w(\mu)}
 \end{aligned}$$

and $\sum_{w \in S_n} x^{w(\mu)}$ is just m_{μ} . $\square$

Definition 8.1.7. Let λ, μ be partitions, λ is a refinement of μ if $\lambda = \bigcup_{i \geq 1} \lambda^{(i)}$ such that $\lambda^{(i)}$ is a partition of μ_j . We write $\lambda \leq_R \mu$.

Lemma 8.1.7. We have

1. $\lambda \leq_R \mu$ is equivalent to $\mu = f(\lambda)$ for some $f: [1, l(\lambda)] \rightarrow \mathbb{N}$.
2. $\leq_R$ is a partial order on $\mathcal{P}_n$.
3. $\lambda \leq_R \mu$ implies $\lambda \leq \mu$.

Proof. See problem set. $\square$

Corollary 8.2. We have

1. $L = (L_{\lambda\mu}) \in U'_n$
2. $M(p, s) = M(p, m)M(s, m)^{-1} = LK^{-1}$

9. REPRESENTATION OF S_n

Now finally we come back to our topic, representation theory, and use what we have learnt about symmetric functions to see what's the irreducible representation ring of S_n .

Recall we have a bilinear form on $C(G, \mathbb{C})$, defined by

$$(f, g)_G = \frac{1}{|G|} \sum_{x \in G} f(x)g(x^{-1})$$

We extend it to function $f: G \rightarrow A$, and A is any commutative $\mathbb{C}$ -algebra. We also extend restriction Res_H^G and induction Ind_H^G from $f: G \rightarrow \mathbb{C}$ to $f: G \rightarrow A$. Then Frobenius reciprocity still holds, i.e. For $H \leq G$, and $\chi: G \rightarrow A, \psi: H \rightarrow A$ are functions. If χ is a class function, then

$$(\text{Ind}_H^G \psi, \chi)_G = (\psi, \text{Res}_H^G \chi)_H$$

Lemma 9.0.1. Let $m, n \in \mathbb{N}$. We embed $S_m \times S_n$ into S_{m+n} by making S_m and S_n act on complementary subsets of $\{1, \dots, m+n\}$. Then:

1. All such subgroups are conjugate to each other
2. If $v \in S_n$ has cycle type $\alpha(v)$, $w \in S_n$ has cycle type $\alpha(w)$, then $v \times w \in S_{n+m}$ is well-defined up to conjugate in S_{m+n} with cycle type $\alpha(v \times w) = \alpha(v) \cup \alpha(w)$.
3. Let $\psi: S_n \rightarrow \Lambda, w \mapsto p_{\alpha(w)}$. Then in the setting of 2., $\psi(v \times w) = \psi(v)\psi(w)$.

Proof. Clear. $\square$

Definition 9.0.1. Let R^n denote the $\mathbb{Z}$ -module generated by $V \in \text{Irr}(S_n)$ modulo the relations $V + W - V \oplus W$. Set $R = \bigoplus_{n \geq 0} R^n$, where $S_0 = \{e\}$ and $R^0 = \mathbb{Z}$.

For $V \in R^m, W \in R^n$, let $V \boxtimes W$ be the corresponding representation of $S_m \times S_n$. Set

$$V \bullet W = \text{Ind}_{S_m \times S_n}^{S_{m+n}} (V \boxtimes W)$$

For $V = \bigoplus_{n \geq 0} V_n, W = \bigoplus_{n \geq 0} W_n$, where $V_n, W_n \in R^n$, we set

$$(V, W) = \sum_{n \geq 0} (V_n, W_n)_{S_n}$$

with

$$(V_n, W_n)_{S_n} = \dim \text{Hom}_{S_n}(V_n, W_n)$$

Proposition 9.1. For R , we have

1. $(R, \bullet)$ is a communicative graded ring.
2. $(\cdot, \cdot) : R \times R \rightarrow \mathbb{Z}$ is a well-defined scalar product on R .

Proof. Omit. □

Definition 9.0.2 (Frobenius characteristic). The Frobenius characteristic is the map

$$\begin{aligned} \text{ch} : R &\rightarrow \Lambda_{\mathbb{C}} = \Lambda \otimes \mathbb{C} \\ V &\mapsto \text{ch}(V) \end{aligned}$$

where $\text{ch}^n(V) = (\chi_V, \psi)_{S_n} = \frac{1}{n!} \sum_{w \in S_n} \chi_V(w) \psi(w^{-1})$ for $V \in R^n$.

Lemma 9.0.2. Let $V \in R^n$. Then

$$\text{ch}^n(V) = \sum_{|\lambda|=n} z_{\lambda}^{-1} \chi_V(K_{\lambda}) p_{\lambda}$$

where $\chi_V(K_{\lambda}) = \chi_V(w)$ for $w \in K_{\lambda} \in \text{Conj}(S_n)$.

Proof. Firstly, we have

$$\text{ch}^n(V) = \frac{1}{n!} \sum_{w \in S_n} \chi_V(w) p_{\alpha(w)}$$

since $\psi(w^{-1}) = p_{\alpha(w^{-1})} = p_{\alpha(w)}$. Note that $\chi_V(w) = \chi_V(w')$ if $\alpha(w) = \alpha(w') \in \text{Conj}(S_n)$ and $|K_{\lambda}| = n! z_{\lambda}^{-1}$, then

$$\text{ch}^n(V) = \frac{1}{n!} \sum_{\lambda \in \text{Conj}(S_n)} |K_{\lambda}| \chi_V(K_{\lambda}) p_{\lambda} = \sum_{|\lambda|=n} z_{\lambda}^{-1} \chi_V(K_{\lambda}) p_{\lambda}$$

as desired. □

Proposition 9.2. ch is an isometry, i.e. for $V, W \in R^n$, we have

$$\langle \text{ch}^n(V), \text{ch}^n(W) \rangle = (V, W)$$

Proof. Note that

$$\begin{aligned}
\langle \text{ch}^n(V), \text{ch}^n(W) \rangle &= \sum_{\lambda, \mu} z_\lambda^{-1} z_\mu^{-1} \chi_V(K_\lambda) \chi_W(K_\mu) \langle p_\lambda, p_\mu \rangle \\
&= \sum_{\lambda} z_\lambda^{-1} \chi_V(K_\lambda) \chi_W(K_\lambda) \\
&= \frac{1}{n!} \sum_{\lambda} |K_\lambda| \chi_V(K_\lambda) \chi_W(K_\lambda) \\
&= (\chi_V, \chi_W)_{S_n} \\
&= (V, W)_{R^n}
\end{aligned}$$

□

Proposition 9.3. ch is an isometric ring isomorphism $R \cong \Lambda_{\mathbb{C}}$.

Proof. It suffices to show ring isomorphism:

For $V \in R^m, W \in R^n$, we have

$$\begin{aligned}
\text{ch}(V \bullet W) &= \text{ch}(\text{Ind}_{S_m \times S_n}^{S_{m+n}} (V \boxtimes W)) \\
&= (\chi_{\text{Ind}_{S_m \times S_n}^{S_{m+n}} (V \boxtimes W)}, \psi)_{S_{m+n}} \\
&= (\text{Ind}_{S_m \times S_n}^{S_{m+n}} (\chi_V \boxtimes \chi_W), \psi)_{S_{m+n}} \\
&= (\chi_V \boxtimes \chi_W, \text{Res}_{S_m \times S_n}^{S_{m+n}} \psi)_{S_m \times S_n} \\
&= (\chi_V, \psi)_{S_m} (\chi_W, \psi)_{S_n} \\
&= \text{ch}(V) \text{ch}(W)
\end{aligned}$$

i.e. ch is a homomorphism.

Let $\eta = \chi_{U_n}$, where U_n is trivial representation of S_n . Then

$$\text{ch}(U_n) = \sum_{\lambda} z_\lambda^{-1} p_\lambda = h_\lambda$$

If $\lambda \vdash n$, let $\eta_\lambda = \eta_{\lambda_1} \eta_{\lambda_2}$, which implies η_λ is a character of S_n , and

$$H_\lambda = \text{Ind}_{S_{\lambda_1} \times \dots \times S_{\lambda_n}}^{S_n} (U_{\lambda_1} \boxtimes \dots \boxtimes U_{\lambda_n})$$

so we have $\text{ch}(H_\lambda) = h_\lambda$.

Recall that

$$s_\lambda = \det(h_{\lambda_i - i + j})_{i,j}$$

For each $\lambda \vdash n$. Let $V^\lambda \in R^n$ be the isomorphism class of a representation such that

$$\chi^\lambda = \chi_{V^\lambda} = \det(\eta_{\lambda_i - i + j})_{i,j}$$

Then $\text{ch}(V^\lambda) = s_\lambda$.

By the following computation

$$(\chi^\lambda, \chi^\mu) = \langle \text{ch}(V^\lambda), \text{ch}(V^\mu) \rangle = \langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$$

So $\pm \chi^\lambda$ is an irreducible character of S_n . Since we have $|\text{Conj}(S_n)| = p_n = |\text{Irr}(S_n)|$, then χ^λ are all characters of S_n , so $(V^\lambda)_{\lambda \vdash n}$ forms a basis of R^n , so we have $\text{ch}|_{R^n}$ is an isomorphism. This completes the proof. □

Theorem 9.0.1 (Frobenius). The irreducible characters of S_n are $\chi^\lambda, \lambda \vdash n$. Moreover, the dimension of V^λ is $K_{\lambda(1^n)}$, the number of standard Young tableau of shape λ .

Proof. It remains to show that χ^λ and not $-\chi^\lambda$ is an irreducible character. Need to show $\chi^\lambda(e) > 0$, where $e \in K_{(1^n)} \in \text{Conj}(S_n)$.

$$s_\lambda = \text{ch}(V^\lambda) = \sum_v z_v^{-1} \chi^\lambda(K_v) p_v$$

then

$$\langle s_\lambda, p_\mu \rangle = \sum_v z_v^{-1} \chi^\lambda(K_v) \langle p_v, p_\mu \rangle = \chi^\lambda(K_\mu)$$

since $\langle p_v, p_\mu \rangle = z_\mu \delta_{\mu v}$.

Then

$$\dim(V^\lambda) = \chi^\lambda(e) = \chi^\lambda(K_{(1^n)}) = \langle s_\lambda, p_{(1^n)} \rangle = K_{\lambda(1^n)}$$

□

Corollary 9.1. The transition matrix $M(p, s)$ is the character table of S_n .

Proof. Note that, from above proof we have

$$\chi^\lambda(K_\mu) = \langle s_\lambda, p_\mu \rangle$$

□

Exercise 9.0.1. Recall that we have computed $s_{(3,3,1)}(x_1, x_2, x_3)$ in Example 7.15. Use the same method, we can see

$$s_{(1^3)} = x_1 x_2 x_3$$

$$s_{(2,1)} = x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_1 x_3^2 + x_2^2 x_3 + x_2 x_3^2 + 2x_1 x_2 x_3$$

$$s_{(3)} = x_1^3 + x_2^3 + x_3^3 + x_1^2 x_2 + \cdots + x_2 x_3^2 + x_1 x_2 x_3$$

and we have

$$\begin{aligned} p_{(1^3)} &= p_1^3 = \left(\sum_{i=1}^3 x_i \right)^3 \\ &= x_1^3 + x_2^3 + x_3^3 + 3(x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_1 x_3^2 + x_2^2 x_3 + x_2 x_3^2 + x_1 x_2 x_3) \\ &= s_{(3)} + 2s_{(2,1)} + s_{(1^3)} \\ p_{(2,1)} &= p_2 p_1 = \left(\sum_{i=1}^3 x_i^2 \right) \left(\sum_{i=1}^3 x_i \right) \\ &= x_1^3 + x_2^3 + x_3^3 + x_1^2 x_2 + x_1 x_2^2 + x_1^2 x_3 + x_1 x_3^2 + x_2^2 x_3 + x_2 x_3^2 + x_1 x_2 x_3 \\ &= s_{(3)} + s_{(1^3)} \\ p_{(3)} &= \left(\sum_{i=1}^3 x_i^3 \right) = x_1^3 + x_2^3 + x_3^3 \\ &= s_{(3)} - s_{(2,1)} + s_{(1^3)} \end{aligned}$$

Hence we have

	$1 = (1^3)$	$(12) = (2, 1)$	$(123) = (3)$
$U = V^{(3)}$	1	1	1
$U' = V^{(1^3)}$	1	-1	1
$V = V^{(2,1)}$	2	0	-1

Definition 9.0.3. Let U'_n denote the sign representation of S_n . We define

$$\Omega : R \rightarrow R$$

$$V \mapsto V \otimes U'_n, \quad V \in R_n$$

Lemma 9.0.3. $\Omega^2 = \text{id}$.

Proof. Clearly we have

$$\chi_{U'_n \otimes U'_n}(g) = \chi_{U'_n}(g)\chi_{U'_n}(g) = 1, \quad \forall g \in S_n$$

□

Proposition 9.4. $\text{ch} \circ \Omega = \omega \circ \text{ch}$

Proof. Need to use the fact $\chi_{U'_n}(K_\mu) = \varepsilon_\mu = (-1)^{|\mu| - l(\mu)}$ and $\omega(P_\lambda) = \varepsilon_\lambda p_\lambda$.

Let V^λ be the representation such that $\chi_{V^\lambda} = \chi^\lambda = s_\lambda, |\lambda| = n$.

$$\begin{aligned} \text{ch}(\Omega(V^\lambda)) &= \text{ch}(V^\lambda \otimes U'_n) \\ &= \sum_{\mu} z_{\mu}^{-1} \chi^\lambda(K_{\mu}) \chi_{U'_n}(K_{\mu}) p_{\mu} \\ &= \sum_{\mu} z_{\mu}^{-1} \chi^\lambda(K_{\mu}) \omega(p_{\mu}) \\ &= \omega(\text{ch}(V^\lambda)) \end{aligned}$$

□

Definition 9.0.4 (hook length). Let λ be a partition, $D(\lambda)$ is its Young diagram. The hook length of λ at $x = (i, j) \in D(\lambda)$ is defined to be $h(x) = h(i, j) = \lambda_i - i + \lambda'_j - j + 1$. The hook length of λ is defined to be

$$h(\lambda) = \prod_{x \in D(\lambda)} h(x)$$

Corollary 9.2 (hook length formula).

$$\dim V^\lambda = \frac{n!}{h(\lambda)}$$

Proof. Compute directly

$$\begin{aligned} \dim V^\lambda &= K_{\lambda(1^n)} = \langle s_\lambda, p_{(1^n)} \rangle \\ &= \langle s_\lambda, (p_1)^n \rangle \\ &= \langle s_\lambda, (e_1)^n \rangle \\ &= \frac{n!}{h(\lambda)} \end{aligned}$$

□

Definition 9.0.5. Let λ be a partition of n and length r . Let T be a Young tableau of shape λ with range in $[1, n] \subset \mathbb{Z}$. We define an action of S_n on T by

$$(wT)_{i,j} = w(T_{i,j}), \quad w \in S_n$$

Definition 9.0.6 (row and column stabilizer). We define the row stabilizer

$$R_{T(\lambda)} = \{w \in S_n \mid w \text{ preserves each row of } T\} \subset S_n$$

and the column stabilizer

$$C_{T(\lambda)} = \{w \in S_n \mid w \text{ preserves each column of } T\} \subset S_n$$

Remark 9.0.1. For these stabilizers, we have following remarks.

1. Note that

$$R_{wT(\lambda)} = wR_{T(\lambda)}w^{-1}$$

$$C_{wT(\lambda)} = wC_{T(\lambda)}w^{-1}$$

so we always write $R_\lambda = R_{T(\lambda)}$ and $C_\lambda = C_{T(\lambda)}$.

2.

$$R_\lambda \cong S_{\lambda_1} \times \cdots \times S_{\lambda_r}$$

$$C_\lambda \cong S_{\lambda_1} \times \cdots \times S_{\lambda_c}$$

are Young subgroups.

3. $R_\lambda \cap C_{\lambda'} = \{e\}$.

4. Let $v \in C_\lambda, u \in R_\lambda, u' = vu v^{-1} \in R_{vT(\lambda)}$. Then $vuT(\lambda) = u'vT_\lambda$.

Remark 9.0.2. Let A be a ring, $x, y \in A$, we have Ax, Ay, Axy are A -modules, and $Axy \subset Ay$ is a submodule. Indeed, let $\varphi : A \rightarrow Ay$, defined by $a \mapsto ay$, is a module homomorphism. So we have $Axy = \varphi(Ax)$. Then the first isomorphism theorem implies

$$Axy = Ax / \ker \varphi$$

we will use this fact into what we have.

Definition 9.0.7. Let $A = \mathbb{C}[S_n]$ be group algebra. Consider

$$a_\lambda = \sum_{w \in R_\lambda} e_w \in A$$

$$b_\lambda = \sum_{w \in C_\lambda} \text{sgn}(w) e_w \in A$$

we define $c_\lambda = a_\lambda b_\lambda \in A$, and call it **Young symmetrizer**.

Remark 9.0.3. $a_\lambda, b_\lambda, c_\lambda$ depend implicitly on the tableau $T(\lambda)$. For example, we have

$$\begin{aligned} a_{wT(\lambda)} &= \sum_{w' \in R_{wT(\lambda)}} e_{w'} = \sum_{w' \in wR_{T(\lambda)}w^{-1}} e_{w'} \\ &= \sum_{w' \in R_{T(\lambda)}} e_{w^{-1}w'} e_w \\ &= w^{-1} \left(\sum_{w' \in R_{T(\lambda)}} e_{w'} \right) w \\ &= w^{-1} a_{T(\lambda)} w \end{aligned}$$

Remark 9.0.4. If $w \in S_n$ could be written as

$$w = u_1 v_1 = u_2 v_2, \quad u_1, u_2 \in R_\lambda, v_1, v_2 \in C_\lambda$$

then $u_2^{-1} u_1 = v_2 v_1^{-1} \in R_\lambda \cap C_\lambda = \{e\}$, so we have $u_1 = u_2, v_1 = v_2$. So it suffices to take the sum in c_λ over $w \in S_n$ which are of the form $w = uv, u \in R_\lambda, v \in C_\lambda$. In particular,

$$c_\lambda = e_{\text{id}} + \dots \neq 0$$

Lemma 9.0.4. Let U_n be the trivial representation of S_n , and U'_n be the sign representation of S_n . Let λ be a partition of n , $S_\lambda \subset S_n$ be the corresponding Young subgroup. Set

$$\begin{aligned} U_\lambda &= U_{\lambda_1} \boxtimes \cdots \boxtimes U_{\lambda_r}, & H_\lambda &= \text{Ind}_{S_\lambda}^{S_n} U_\lambda \\ U'_{\lambda'} &= U_{\lambda'_1} \boxtimes \cdots \boxtimes U'_{\lambda'_c}, & E_{\lambda'} &= \text{Ind}_{S'_{\lambda'}}^{S_n} U'_{\lambda'} \end{aligned}$$

Let $\eta_\lambda = \chi_{H_\lambda}$ and $\varepsilon_{\lambda'} = \chi_{E_{\lambda'}}$, χ^λ is the irreducible character corresponding to V^λ . Then

1.

$$\begin{aligned} H_\lambda &\cong \mathbb{C}[S_n] a_\lambda \\ E_{\lambda'} &\cong \mathbb{C}[S_n] b_\lambda \end{aligned}$$

2.

$$\begin{aligned} \eta_\lambda &= \chi^\lambda + \sum_{\mu > \lambda} K_{\lambda\mu} \chi^\mu \\ \varepsilon_{\lambda'} &= \chi^\lambda + \sum_{\mu < \lambda} K_{\lambda\mu} \chi^\mu \end{aligned}$$

Proof. See problem set. □

Finally, we can construct V^λ explicitly here.

Theorem 9.0.2. Let $\widehat{V}^\lambda = \mathbb{C}[S_n] c_\lambda$, where λ is a partition of n . Then $\widehat{V}^\lambda$ is an irreducible representations of S_n with character $\chi_{\widehat{V}^\lambda} = \chi^\lambda$. Every irreducible representation is of this form.

Proof. Let $A = \mathbb{C}[S_n]$. By the Remark 8.19 on algebra, $A c_\lambda = A a_\lambda b_\lambda$ is a submodule of $A a_\lambda \cong H_\lambda$ and is quotient of $A b_\lambda \cong E_{\lambda'}$. Lemma 8.23 implies that H_λ and $E_{\lambda'}$ have a unique common irreducible component, the irreducible representations V^λ of S_n , with character χ^λ . Thus we have $\widehat{V}^\lambda \cong V^\lambda$. □

Remark 9.0.5. $c_\lambda = c_{T(\lambda)}$ depends on the choice of $T(\lambda)$, since $c_{wT(\lambda)} = w c_{T(\lambda)} w^{-1}$, $\forall w \in S_n$, so we have

$$\widehat{V^{T(\lambda)}} \cong \widehat{V^{wT(\lambda)}}$$

Corollary 9.3 (Young's rule).

$$\begin{aligned} \text{Ind}_{S_\lambda}^{S_n} U_\lambda &= V^\lambda \oplus \bigoplus_{\mu \supset \lambda} (V^\mu)^{\oplus K_{\lambda\mu}} \\ \text{Ind}_{S_{\lambda'}}^{S_n} U_{\lambda'} &= V^\lambda \oplus \bigoplus_{\mu < \lambda} (V^\mu)^{\oplus K_{\lambda\mu}} \end{aligned}$$

Remark 9.0.6. If $\lambda = (1^n)$, then $\text{Ind}_{\{e\}}^{S_n} U_{(1^n)} = \mathbb{C}[S_n] = R$, where R is regular representation. But we have

$$R = \bigoplus_{\lambda} (V^{\lambda})^{\oplus \dim V^{\lambda}}$$

This shows again: $\dim V^{\lambda} = K_{\lambda(1^n)}$.

Remark 9.0.7. Let λ be a partition of n , μ be a partition of m , then

$$\begin{aligned} V^{\lambda} \bullet V^{\mu} &= \text{Ind}_{S_m \times S_n}^{S_{m+n}} V^{\lambda} \boxtimes V^{\mu} \\ &= \bigoplus_{\gamma} N_{\lambda\mu}^{\gamma} V^{\gamma} \end{aligned}$$

where V^{γ} is an irreducible representation of S_{m+n} , and the sum runs over all partitions γ of $m+n$. $N_{\lambda\mu}^{\gamma}$ can be determined combinatorially using the Littlewood-Richardson rule.

Exercise 9.0.2. Let $G = S_3$. There are three partitions of 3, that is, $(3), (2, 1), (1^3)$.

For $\lambda = (3)$, that is, the Young tableau is just one row, so every element of S_3 lie in row stabilizer, so we have

$$V^{(3)} = \mathbb{C} \sum_{w \in S_3} e_w = U, \quad \text{trivial representation.}$$

For $\lambda = (1^3)$, the Young tableau is just one column, so every element lie in column stabilizer, so we have

$$V^{(1^3)} = \mathbb{C} \sum_{w \in S_3} \text{sgn}(w) e_w = U', \quad \text{alternating representation.}$$

For $\lambda = (2, 1)$, things are a little complicated. Since we have $R_{(2,1)} \cong S_2 \times S_1$. We can take Young tableau as follows for an example

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

then we have

$$\begin{aligned} a_{(2,1)} &= e_{\text{id}} + e_{(12)} \\ b_{(2,1)} &= e_{\text{id}} - e_{(13)} \\ c_{(2,1)} &= (e_{\text{id}} + e_{(12)})(e_{\text{id}} - e_{(13)}) \\ &= e_{\text{id}} + e_{(12)} - e_{(13)} - e_{(123)} \end{aligned}$$

so

$$V^{(2,1)} = \mathbb{C}[S_n] c_{(2,1)}$$

By simply computation, we have

$$\begin{aligned} v_1 &= c_{(2,1)} = e_{(12)} c_{(1,2)} \\ v_2 &= e_{(13)} c_{(2,1)} = e_{(13)} + e_{(123)} - e_{\text{id}} - e_{(23)} \\ e_{(23)} c_{(2,1)} &= e_{(23)} + e_{(123)} - e_{(132)} - e_{(13)} = -v_1 - v_2 \end{aligned}$$

So we have

$$V^{(2,1)} = \mathbb{C} c_{(2,1)} \oplus \mathbb{C} e_{(13)} c_{(2,1)}$$

that is standard representation.

Proposition 9.5. Let λ be a partition of n , U'_n be the alternating representation of S_n . Then $V^{\lambda'} \cong V^\lambda \otimes U'_n$.

Proof.

$$\begin{aligned} (\text{ch} \circ \Omega)(V^\lambda) &= \text{ch}(V^\lambda \otimes U'_n) \\ (\omega \circ \text{ch})(V^\lambda) &= \omega(s_\lambda) = s_{\lambda'} = \text{ch}(V^{\lambda'}) \end{aligned}$$

□

Proposition 9.6. For any λ , $c_\lambda c_\lambda = d_\lambda c_\lambda$, where $d_\lambda = h(\lambda)$.

Proof. Let $A = \mathbb{C}[S_n]$, $\varphi_\lambda : A \rightarrow A$, defined by $v \mapsto v c_\lambda$, then

$$\varphi_\lambda(V^\lambda) = V^\lambda c_\lambda = A c_\lambda^2 \subset A c_\lambda = V^\lambda$$

Since V^λ is irreducible, Schur's lemma tells us that

$$\varphi_\lambda|_{V^\lambda} = \alpha_\lambda \text{id}_{V^\lambda}$$

then

$$c_\lambda^2 = \varphi_\lambda(c_\lambda) = \alpha_\lambda c_\lambda$$

then

$$\varphi_\lambda^2(v) = v c_\lambda^2 = \alpha_\lambda v c_\lambda = \alpha_\lambda \varphi_\lambda(v)$$

implies that eigenvalues of φ_λ are zero and α_λ and the multiplicity of α_λ is $\dim V^\lambda$. So

$$\text{tr } \varphi_\lambda = \alpha \dim V^\lambda = \alpha_\lambda \frac{n!}{h(\lambda)}$$

□

Lemma 9.0.5. Let E be a finite dimensional vector space over $\mathbb{C}$, S_n acts on $E^{\otimes n}$ by permuting the factors. View a_λ, b_λ as a representation of $\mathbb{C}[S_n]$

$$\mathbb{C}[S_n] \rightarrow \text{End}(E^{\otimes n})$$

Then

1. $\text{im}(a_\lambda) = \bigotimes_{i=1}^r \text{Sym}^{\lambda_i} E \subset E^{\otimes n}$
2. $\text{im}(b_\lambda) = \bigotimes_{i=1}^c \bigwedge^{\lambda'_i} E \subset E^{\otimes n}$

Proof. Clear.

□

Remark 9.0.8. In particular, we have

$$\begin{aligned} c_{(n)} &= a_{(n)} = \sum_{w \in S_n} e_w \\ c_{(1^n)} &= b_{(1^n)} = \sum_{w \in S_n} \text{sgn}(w) e_w \end{aligned}$$

then

$$\begin{aligned} \text{im } c_{(n)} &= \text{Sym}^n E \subset E^{\otimes n} \\ \text{im } c_{(1^n)} &= \bigwedge^n E \subset E^{\otimes n} \end{aligned}$$

Part 3. Representation theory of Lie groups and Lie algebras

10. LIE GROUPS

10.1. Basic definitions about Lie groups.

Definition 10.1.1 (Lie group). A Lie group is a group G that is also a smooth manifold in which the multiplication $\mu : G \times G \rightarrow G$ and inversion $\iota : G \rightarrow G$ are differentiable maps.

Definition 10.1.2 (morphism of Lie groups). A morphism of Lie groups is a map $f : G \rightarrow H$ between Lie groups G, H that is also a group homomorphism and differentiable.

Definition 10.1.3 ((closed) Lie subgroup). A (closed) Lie subgroup $H \subset G$ is a subset H of G that is a subgroup and a closed submanifold.

Definition 10.1.4 (immersed Lie group). An immersed Lie group is the image of a Lie group H under an injective morphism to G .

Definition 10.1.5 (complex Lie group). A complex Lie group is a group G that is also a complex manifold in which multiplication and inversion are holomorphic maps.

Definition 10.1.6 (morphism of complex Lie groups). A morphism of complex Lie groups is a map $f : G \rightarrow H$ between complex Lie groups G, H that is also a group homomorphism and a holomorphic map.

Example 10.1.1. $(\mathbb{R}^n, +)$ is a Lie group.

Example 10.1.2 (general linear group). $\mathrm{GL}(n, \mathbb{R})$ is an open subset of $\mathrm{Mat}(n, \mathbb{R}) \cong \mathbb{R}^{n^2}$. The manifold structure is induced from $\mathbb{R}^{n^2}$, so multiplication is differentiable. And Cramer's rule implies the inversion is differentiable. In fact, $\mathrm{GL}(n, \mathbb{R})$ is an algebraic group. Consider

$$U = \{(A_{ij}, t) \in \mathbb{R}^{n^2+1} \mid \det(A_{ij})t - 1 = 0, \text{ a polynomial in } A_{ij} \text{ and } t\}$$

Let

$$\phi : \mathrm{GL}(n, \mathbb{R}) \rightarrow U$$

$$(a_{ij}) \mapsto (a_{ij}, \det(a_{ij})^{-1})$$

This is a bijection, making $\mathrm{GL}(n, \mathbb{R})$ as a zero set of a polynomial in $n^2 + 1$ variables. Furthermore, you can show that this polynomial is irreducible.

Example 10.1.3 (special linear group). Consider

$$\mathrm{SL}(n, \mathbb{R}) = \{A \in \mathrm{GL}(n, \mathbb{R}) \mid \det A = 1\} = \ker(\det : \mathrm{GL}(n, \mathbb{R}) \rightarrow \mathrm{GL}(1, \mathbb{R}))$$

is also a Lie group.

Our goal is to study the representation theory of a Lie group G . We will reduce this problem in several steps

1. Reduce to G is connected.
2. Reduce to G is simply connected.

3. Reduce to the tangent space of G , that is, its Lie algebra. In this case, representation theory of G equals to the one of its Lie algebra.
4. Reduce to complex semisimple Lie algebra.
5. Reduce to $SU(2)$.

10.2. Review of geometry. This section is a mixture of a review of concepts and notations of differential geometry and motivational arguments for reduction process. We omit the proofs of theorem we mentioned in this section, you can find them in almost every standard textbook for differential manifold and algebraic topology.

10.2.1. Differentiable manifold.

Definition 10.2.1 (smooth&diffeomorphism). Let M, N be differentiable manifolds, a map $f : M \rightarrow N$ is called smooth or differentiable, if it is continuous and for all $p \in M$, there exists a chart (φ, U) for p and a chart (ψ, V) of $f(p)$ such that $\psi \circ f \circ \varphi^{-1}$ is smooth; f is called a diffeomorphism if it is bijective and f, f^{-1} are smooth.

Remark 10.2.1. If we replace differentiable by complex and smooth by holomorphic, we define a holomorphic map $f : M \rightarrow N$ between complex manifolds; f is called biholomorphic if it is bijective and f, f^{-1} is holomorphic.

Since a manifold is a topological space satisfying additional properties such as Hausdorff and separation axiom, the notions of topological space apply to manifolds.

Definition 10.2.2 (connectedness). A topological space is disconnected, if $X = X_1 \sqcup X_2$ with $X_1, X_2 \neq \emptyset$, otherwise it is connected. The maximal connected subsets of X are called connected components of X .

Remark 10.2.2. For connectedness, we have the following remarks

1. X is connected if and only if the only subsets of X that are both open and closed are X and $\emptyset$.
2. A manifold is connected if and only if it is path connected.
3. The connected components of a manifold are still manifolds.

Proposition 10.2.1. Let X, Y be topological spaces. If $f : X \rightarrow Y$ is continuous and X is connected, then $f(X)$ is connected.

Definition 10.2.3 (compactness). A topological space X is called compact if each of its open covering admits a finite subcover.

Remark 10.2.3. If X is a subset of $\mathbb{R}^n$, then the Heine-Borel theorem says that X is compact if and only if X is closed and bounded.

Example 10.2.1. $GL(n, \mathbb{R})$ is an open submanifold of $\mathbb{R}^{n^2}$ and a closed submanifold of $\mathbb{R}^{n^2+1}$, and one chart gives an atlas. $GL(n, \mathbb{R})$ has two connected components.

$$GL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) \mid \det A > 0\} \sqcup \{A \in GL(n, \mathbb{R}) \mid \det A < 0\}$$

Similarly we can define $\mathrm{GL}(n, \mathbb{C})$. However, it is connected, and $\mathrm{GL}(n, \mathbb{C}) \subset \mathrm{GL}(2n, \mathbb{R})$. To be more explicit, if $A = A_1 + iA_2$, then

$$A \mapsto \begin{pmatrix} A_1 & A_2 \\ -A_2 & A_1 \end{pmatrix} \in \mathrm{GL}(2n, \mathbb{R})$$

Example 10.2.2. $\mathrm{SL}(n, \mathbb{R}) = \{A \in \mathrm{GL}(n, \mathbb{R}) \mid \det A = 1\}$ is a manifold with dimension $n^2 - 1$. Take $n = 2$ for an example, then

$$G = \mathrm{SL}(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid ad - bc = 1 \right\}$$

that is, G is the zero locus of $p(a, b, c, d) = ad - bc - 1$, and $dp \neq 0$ on the locus $p = 0$. The implicit function theorem implies we can solve one variable in terms of other three. Near the identity⁵, we have

$$d = \frac{1}{a}(1 + bc), \quad a = \frac{1}{d}(1 + bc)$$

So we have $\psi_1 : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightarrow (a, b, c)$ is a local homomorphism, since we have its inverse

$$(a, b, c) \mapsto \begin{pmatrix} a & b \\ c & \frac{1}{a}(1 + bc) \end{pmatrix}$$

Similarly we can define a local homomorphism $\psi_2 : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightarrow (b, c, d)$. Furthermore,

$$(a, b, c) \xrightarrow{\psi_1} \begin{pmatrix} a & b \\ c & \frac{1}{a}(1 + bc) \end{pmatrix} \xrightarrow{\psi_2^{-1}} (b, c, \frac{1}{a}(1 + bc))$$

is smooth, so these two charts are compatible. Arguing in this way for any matrix in G , we get a differentiable atlas.

Using such atlas, we can check the multiplication and inversion are smooth.

Take inversion for an example. If we use ψ_i to denote $\begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \mapsto (a_i, b_i, c_i), i = 1, 2$. Then

$$\psi_2 \circ \iota \circ \psi_1^{-1} : (a_1, b_1, c_1) \mapsto \left(\frac{1}{a_1}(1 + b_1 c_1), -b_1, -c_1 \right)$$

is smooth.

Example 10.2.3. Let $\{e_1, \dots, e_n\}$ be a basis of $\mathbb{R}^n$, $V_i = \mathbb{R}\langle e_1, \dots, e_i \rangle$ and consider the flag $0 = V_0 \subset V_1 \subset \dots \subset V_n = \mathbb{R}^n$.

$$\begin{aligned} B_n &= \{A \in \mathrm{GL}(n, \mathbb{R}) \mid A \text{ preserves } V_\bullet\} \\ &= \{A \in \mathrm{GL}(n, \mathbb{R}) \mid A \text{ is upper triangular}\} \end{aligned}$$

And we can define

$$\begin{aligned} N_n &= \{A \in \mathrm{GL}(n, \mathbb{R}) \mid A \text{ preserves } V_\bullet, A|_{V_{i+1}/V_i} = \mathrm{id}\} \\ &= \{A \in \mathrm{GL}(n, \mathbb{R}) \mid A \text{ is upper triangular, and } A_{ii} = 1\} \end{aligned}$$

⁵That is, $a \neq 0, d \neq 0$.

Example 10.2.4. Let V be a real vector space with dimension n . $Q \in (V^\vee)^{\otimes 2}$ is symmetric, positive definite.

$$\mathrm{SO}(n, \mathbb{R}) = \{A \in \mathrm{GL}(n, \mathbb{R}) \mid Q(Av, Aw) = Q(v, w), v, w \in V\}$$

If we choose Q is skew-symmetric, non-degenerate and n is even, then

$$\mathrm{Sp}(n, \mathbb{R}) = \{A \in \mathrm{GL}(n, \mathbb{R}) \mid Q(Av, Aw) = Q(v, w), v, w \in V\}$$

Example 10.2.5. $\mathbb{R}^n/\mathbb{Z}^n = (S^1)^n$ is a Lie group.

Example 10.2.6. Any finite group is a Lie group of dimension 0, with respect to discrete topology.

Remark 10.2.4. A closed subgroup of $\mathrm{GL}(n, \mathbb{C})$ or $\mathrm{GL}(n, \mathbb{R})$ is often called a closed linear group or linear Lie group or matrix Lie group. Most examples are matrix Lie groups as they are defined by polynomial equations. An example of a subgroup of $\mathrm{GL}(n, \mathbb{C})$ which is not closed is $\mathrm{GL}(n, \mathbb{Q})$. Another example is irrational line on the torus. Take $a \in \mathbb{R} \setminus \mathbb{Q}$, and consider

$$G = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{ait} \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

Then G is a subgroup of $\mathrm{GL}(2, \mathbb{C})$, but not closed.

Our first reduction process allow us to consider only connected Lie groups, and it mainly rely on the following proposition.

Proposition 10.2.2. Let G be a real or complex Lie group, use G^o to denote the connected component of the identity. Then G^o is a normal subgroups of G and is a Lie group itself. The quotient group G/G^o is discrete.

10.2.2. *Homotopy theory.*

Definition 10.2.4 (path). Let M be a manifold, $p, q \in M$. A path from p to q in M is a continuous map $\gamma : I = [0, 1] \rightarrow M$ such that $\gamma(0) = p, \gamma(1) = q$.

Notation 10.1. Let $\mathcal{P}(p, q)$ be the set of all such paths.

Definition 10.2.5 (loop). A loop is an element of $\mathcal{P}(p, p)$.

Definition 10.2.6 (fixed-point homotopy). Let $\gamma, \tilde{\gamma} \in \mathcal{P}(p, q)$, a fixed-endpoint homotopy from γ to $\tilde{\gamma}$ is a continuous map $H : I \times I \rightarrow M$ such that

$$\begin{aligned} H(t, 0) &= \gamma(t), & H(0, s) &= p \\ H(t, 1) &= \tilde{\gamma}(t), & H(1, s) &= q \end{aligned}$$

for all $t, s \in I$. If such a homotopy exists, γ and $\tilde{\gamma}$ are fixed-endpoint homotopic, written $\gamma \simeq \tilde{\gamma}$.

Definition 10.2.7 (null homotopy). A loop γ is called null homotopy, if it is homotopic to the constant loop.

Lemma 10.1. Fixed-endpoint homotopy is an equivalence relation on $\mathcal{P}(p, q)$.

Definition 10.2.8 (concatenation). Let $\gamma, \tilde{\gamma} \in \mathcal{P}(p, q), p, q \in M$, and define

$$\gamma * \tilde{\gamma} = \begin{cases} \gamma(2t), & 0 \leq t \leq \frac{1}{2} \\ \tilde{\gamma}(2t-1), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

$\gamma * \tilde{\gamma}$ is called the concatenation of γ and $\tilde{\gamma}$.

Definition 10.2.9 (reverse path). The reverse path γ^{-1} is defined by $\gamma^{-1}(t) := \gamma(1-t)$.

Proposition 10.2.3 (fundamental group). Let $p \in M$ and $\pi_1(M, p)$ is the homotopy classes of $\mathcal{P}(p, p)$. Then it is a group with respect to concatenation, called fundamental group.

Proof. Standard conclusion in homotopy theory. $\square$

Proposition 10.2.4. Let M be connected, then $\pi_1(M, p)$ are all isomorphic to each other for all $p \in M$.

Proof. For any two points p, q , consider $\gamma \in \mathcal{P}(p, q)$ and the map

$$[\tilde{\gamma}] \rightarrow [\gamma * \tilde{\gamma} * \gamma^{-1}]$$

$\square$

Notation 10.2. So if M is connected, the base point of fundamental group doesn't matter, so we can write $\pi_1(M)$ in this case.

Definition 10.2.10 (simply connected). Let M be connected, if $\pi_1(M)$ is trivial, then M is called simply connected.

Example 10.2.7. $\mathbb{R}^n$ is simply connected, since any $\gamma \in \mathcal{P}(0, 0)$ is homotopic to constant loop e_0 under $H(s, t) = s\gamma(t)$.

Example 10.2.8. S^1 is not simply connected, we will see later $\pi_1(S^1) = \mathbb{Z}$.

Proposition 10.2.5. Let M, N be connected manifolds. Then

$$\pi_1(M \times N) \cong \pi_1(M) \times \pi_1(N)$$

Proposition 10.2.6. Let $\phi : M \rightarrow N$ be a continuous map. Then there exists a group homomorphism

$$\begin{aligned} \phi_{\#} : \pi_1(M, p) &\rightarrow \pi_1(N, \phi(p)) \\ [\gamma] &\mapsto [\phi \circ \gamma] \end{aligned}$$

Proposition 10.2.7. Let M be a manifold, $p, q \in M, \gamma \in \mathcal{P}(p, q)$. Then there exists a piecewise smooth path $\tilde{\gamma} \in \mathcal{P}(p, q)$ homotopic to γ .

Definition 10.2.11 (covering map). Let M, N be manifolds. A smooth, surjective map $\pi : M \rightarrow N$ is a covering map, if for all $p \in N$, there exists a neighborhood $U(p)$ such that $U(p)$ is evenly covered, i.e. π maps each connected components of $\pi^{-1}(U(p))$ diffeomorphically onto $U(p)$, such a component is called a sheet.

Example 10.2.9. $\pi : \mathbb{R} \rightarrow S^1$, defined by $t \mapsto e^{it}$ is a covering map. But its restriction to any interval $[a, b]$ is not.

Example 10.2.10. A map from S^1 to S^1 defined by $z \mapsto z^n$ is a covering map for $n \in \mathbb{Z}_{>0}$.

Lemma 10.2 (multiplicity). Let $\pi : M \rightarrow N$ be a covering map, N is connected. Then $|\pi^{-1}(p)| \in \mathbb{N} \cup \{\infty\}$ is constant for all $p \in N$. This number is called the multiplicity of π .

Example 10.2.11. The multiplicity of map $z \mapsto z^n$ is n , and the multiplicity of $t \mapsto e^{it}$ is ∞ .

Definition 10.2.12 (lift). Let $\pi : M \rightarrow N, \phi : P \rightarrow N$ be smooth maps of manifolds. A lift of ϕ through π is a smooth map $\tilde{\phi} : P \rightarrow M$ such that $\pi \circ \tilde{\phi} = \phi$.

$$\begin{array}{ccc} P & \xrightarrow{\tilde{\phi}} & M \\ & \searrow \phi & \downarrow \pi \\ & & N \end{array}$$

Lemma 10.3 (path lifting property). Let $\pi : M \rightarrow N$ be a covering map, $\gamma : I \rightarrow N$ be a smooth curve. Then there exists a lift $\tilde{\gamma} : I \rightarrow M$ of γ through π .

Corollary 10.2.1. Let $\pi : M \rightarrow N$ be a covering map, γ_1, γ_2 be fixed-endpoint homotopic paths in N . For the lifts $\tilde{\gamma}_1, \tilde{\gamma}_2$ of γ_1, γ_2 through π such that $\tilde{\gamma}_1(0) = \tilde{\gamma}_2(0)$, we have $\tilde{\gamma}_1, \tilde{\gamma}_2$ are still fixed-endpoint homotopic.

Corollary 10.2.2. $\pi_{\#} : \pi_1(M) \rightarrow \pi_1(N)$ is injective.

Proof. It suffices to show, if two loops γ_1, γ_2 are homotopic in $\pi(N)$, then their lifts in M must be homotopic. That's what Proposition 9.51 says. $\square$

Proposition 10.2.8. Let $\pi : M \rightarrow N$ be a covering map, $\phi : P \rightarrow N$ be a smooth map. Let $p_0 \in P, q_0 \in M$ such that $\pi(q_0) = \phi(p_0)$. Then

1. If P is connected, then there exists at most one lift $\tilde{\phi}$ of ϕ through π , such that $\tilde{\phi}(p_0) = q_0$.
2. If P is simply connected, such a lift exists.

Manifold properties attributed to a covering refer to the covering manifold M . For example, a simply connected covering $\pi : M \rightarrow N$ is one for which M is simply connected.

Theorem 10.2.1 (uniqueness of universal covering). Any connected manifold has a simply connected covering. Any two simply connected covering are diffeomorphic.

Definition 10.2.13 (universal covering). Let M be a connected manifold. Any simply connected covering is called universal covering of M , denoted by $\tilde{M}$.

Corollary 10.2.3. Let N be connected, H be a subgroup of $\pi_1(N)$. Then there is a connected covering $\pi : M \rightarrow N$ such that $\pi_{\#}(\pi_1(M)) = H$.

Corollary 10.2.4. Every covering $\pi : M \rightarrow N$ of a simply connected manifold is trivial.

Example 10.2.12. $\mathbb{R} \rightarrow S^1$ is the universal covering of S^1 .

Example 10.2.13. We will see later, $SU(2) \rightarrow SO(3)$ is a two to one covering. Furthermore, $SU(2)$ is simply connected, thus this covering is also a universal covering.

Theorem 10.2.2. Let G be a connected real or complex Lie group. Then its universal covering $\tilde{G}$ has a unique structure of Lie group such that the covering map π is a morphism of Lie groups. In this case, $\ker \pi \cong \pi_1(G)$ as a group and $\ker \pi$ is discrete subgroup of $Z(\tilde{G})$.

This is reduction process two.

Remark 10.2.5. If M is a connected manifold and $\tilde{M}$ is its universal covering, then there exists an isomorphism of groups

$$\begin{aligned} \{f \in \text{Aut}(\tilde{M}) \mid \pi \circ f = \pi\} &\cong \pi_1(M) \\ f &\mapsto [\pi \circ \gamma] \end{aligned}$$

where $\gamma \in \mathcal{P}(\tilde{p}, f(\tilde{p}))$, $\tilde{p} \in \tilde{M}$. In fact, this group is the group of Deck transformations.

Example 10.2.14. The covering map $\phi : \mathbb{R} \rightarrow S^1, t \mapsto e^{it}$ is the universal covering map of S^1 , we have $\ker \phi = 2\pi\mathbb{Z}$. Any continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\phi \circ f = \phi$ must satisfy $f(t) = t + 2\pi n(t)$, since

$$e^{if(t)} = e^{it}$$

What's more, $n(t)$ is a constant function, since f is continuous. Then

$$\begin{aligned} \pi_1(S^1) &\cong \{f \in \text{Aut}(\mathbb{R}, +) \mid \phi \circ f = \phi\} \\ &= \{f_n \in \text{Aut}(\mathbb{R}, +) \mid f_n(t) = t + 2\pi n, n \in \mathbb{Z}\} \end{aligned}$$

So we have a clear isomorphism $\ker \phi \cong \pi_1(S^1)$.

11. LIE ALGEBRA

Now let G be connected and simply connected, we want to reduce the case to its Lie algebra. Firstly, recall some basic definitions about tangent space of smooth manifolds.

11.1. Tangent space.

Definition 11.1.1 (curves which are tangential at a point). Let M be a manifold, $p \in M$, and (ψ, V) is a chart at p . Two smooth curves $\gamma_i : I \rightarrow M, i = 1, 2$ with $\gamma_i(0) = p$ are called tangential at with respect to ψ , if

$$(\psi \circ \gamma_1)'(0) = (\psi \circ \gamma_2)'(0)$$

Remark 11.1.1. Clearly, this definition is independent of the choice of ψ . Furthermore, tangential at a point gives an equivalence relation for curves starting at this point. Use this equivalent relation, we can define what is a tangent space.

Definition 11.1.2 (tangent space). Let M be a manifold, $p \in M$. The tangent space of M at p is defined by

$$T_p M := \{\gamma \mid \gamma : I \rightarrow M, \gamma(0) = p\} / \sim$$

where $\sim$ is the tangential equivalence relation, we use $[\gamma]_p$ to denote a representative element.

Definition 11.1.3 (tangent map). Let M, N be manifolds, $f : M \rightarrow N$ be a smooth map. We call $T_p f : T_p M \rightarrow T_p N, [\gamma]_p \mapsto [f \circ \gamma]_{f(p)}$ the tangent map of f at p .

Proposition 11.1.1 (chain rule). Let M, N, P be manifolds, $f : M \rightarrow N, g : N \rightarrow P$ be smooth maps, take $p \in M$, then

$$T_p(g \circ f) = T_{f(p)}g \circ T_p f$$

Moreover, since $T_p(\text{id}_M) = \text{id}_{T_p M}$, then for any diffeomorphism $f : M \rightarrow N$, $T_p f$ is bijective and $(T_p f)^{-1} = T_{f(p)}f^{-1}$.

Lemma 11.1. Let $U \subset \mathbb{R}^n$ be open, $p \in U$. Then $\iota : T_p U \rightarrow \mathbb{R}^n, [\gamma]_p \mapsto \gamma'(0)$ is bijective, so that $T_p U$ can be identified with $\mathbb{R}^n$. Furthermore, for any smooth map $f : U \rightarrow V$, $V \subset \mathbb{R}^n$ is an open subset, $\iota \circ T_p f = Df(p) \circ \iota$, where $Df(p)$ is the Jacobi matrix of f at point p .

Proof. It's almost trivial that $T_p U \cong \mathbb{R}^n$. Since U is already an open subset in $\mathbb{R}^n$, then it is a chart of itself. If two curves γ_1, γ_2 such that $\gamma_1'(0) = \gamma_2'(0)$, then clearly they are same element in $T_p U$ since it's exactly the equivalent relation we killed. For any $v = (v_1, \dots, v_n) \in \mathbb{R}^n$, clearly $\gamma(t) = p + tv$ is the curve such that $\gamma(0) = p, \gamma'(0) = v$.

Now let's see what is $T_p f$. For $[\gamma]_p \in T_p U$, we take an representative element $\gamma(t) = p + tv$. Then by definition

$$\begin{aligned} \iota \circ T_p f([\gamma]_p) &= \iota([f \circ \gamma]_{f(p)}) \\ &= (f \circ \gamma)'(0) \\ &= Df(p)\gamma'(0) \\ &= Df(p)v \\ &= Df(p) \circ \iota([\gamma]_p) \end{aligned}$$

□

Remark 11.1.2. In other words, we can draw the following communicative diagram:

$$\begin{array}{ccc} T_p U & \xrightarrow{T_p f} & T_p V \\ \downarrow \iota & & \downarrow \iota \\ \mathbb{R}^n & \xrightarrow{Df(p)} & \mathbb{R}^n \end{array}$$

With above isomorphism ι , we always regard $v \in \mathbb{R}^n$ and $[\gamma]_p \in T_p U$ where $\gamma(t) = p + tv$ the same thing.

Proposition 11.1.2. Let M be a manifold, $p \in M$, (ψ, V) is a chart at p . Then the vector space structure of $T_p M$ is induced by the bijection $T_p \psi : T_p M \rightarrow T_{\psi(p)} \psi(V) \cong \mathbb{R}^n$.

Remark 11.1.3. Any chart ψ allows us to choose a particular basis for $T_p M$. Let (ψ, V) be a chart of M centered at p , that is $\psi = (x^1, \dots, x^n) : V \rightarrow \mathbb{R}^n$ is a diffeomorphism such that $\psi(p) = (0, \dots, 0)$. Let $(e_1, \dots, e_n)$ be the standard basis of $\mathbb{R}^n$. Then

$$\begin{aligned} \left. \frac{\partial}{\partial x^i} \right|_p &:= (T_p \psi)^{-1}(e_i) \\ &= (T_p \psi)^{-1}([\gamma]_0), \quad \gamma(t) = te_i \\ &= [\psi^{-1} \circ \gamma]_p \end{aligned}$$

Then $\{\left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p\}$ is a basis of $T_p M$.

Remark 11.1.4 (directional derivative). Note that for any $v = [\gamma]_p \in T_p M$ and $f \in C^\infty(M)$. Then we define the directional derivative $\partial_v : C^\infty(M) \rightarrow \mathbb{R}$ by

$$\partial_v(f) := T_p f(v) = T_p f([\gamma]_p) = [f \circ \gamma]_{f(p)} = (f \circ \gamma)'(0)$$

Furthermore, ∂_v satisfies the Leibniz rule. Indeed,

$$\begin{aligned} \partial_v(fg) &= (fg \circ \gamma)'(0) \\ &= (f \circ \gamma)'(0)g(p) + f(p)(g \circ \gamma)'(0) \\ &= \partial_v(f)g + f \partial_v(g) \end{aligned}$$

Furthermore, it's crucial to note that $\partial_v(f)$ only depends on the local property of f at p .

Now let's describe tangent vector in another point of view, that's regard a tangent vector as a derivation on germs of differential functions. First we define an equivalent relation $\sim$ on the algebra of smooth functions $C^\infty(M)$ to describe the local property at p .

For any $f, g \in C^\infty(M)$, we say $f \sim g$ if there exists a neighborhood U of p such that f agrees with g on U . Then

Definition 11.1.4 (germ). The germ at p is the equivalent class $C^\infty(M)/\sim$, where $\sim$ is the equivalent relation we mentioned above.

Definition 11.1.5 (derivation on germs). Let M be a manifold. A map $\partial : C^\infty(M) \rightarrow \mathbb{R}$ is called a derivation at p if for all $f, g \in C^\infty(M)/\sim$, where $f \sim g$

means there exists a neighborhood U of p such that f agrees with g in U , we have

1. $\partial(f + \alpha g) = \partial f + \alpha \partial g, \quad \forall \alpha \in \mathbb{R}$
2. $\partial(fg) = \partial f g + f \partial g$

Notation 11.1. We denote the set of all derivation at p on M by $\text{Der}_p(C^\infty(M), \mathbb{R})$.

Remark 11.1.5. So as we have seen in Remark 10.10, ∂_v is a derivation on germ $C^\infty(M)/\sim$. Here comes the definition of derivations.

Theorem 11.1.1. The map

$$\begin{aligned} \Phi : T_p M &\rightarrow \text{Der}_p(C^\infty(M), \mathbb{R}) \\ v &\mapsto \partial_v \end{aligned}$$

is a linear isomorphism.

11.2. First and second principles of Lie group. Now let's focus on the case of Lie groups. Lie group is a very special manifold with quite nice symmetry. Here is a very important diffeomorphism on Lie groups.

Definition 11.2.1 (left/right translation). Let G be a Lie group, $g \in G$. The left translation by g is defined as $L_g : G \rightarrow G, h \mapsto gh$. Analogously, the right translation by g is $R_g : G \rightarrow G, h \mapsto hg$.

Lemma 11.2. Let G be a Lie group, $g \in G$. Then L_g is an automorphism of Lie group. Furthermore,

$$\begin{aligned} L : G &\rightarrow \text{Aut}(G) \\ g &\mapsto L_g \end{aligned}$$

is a group homomorphism.

Proof. We have $L_g(h) = \mu(g, h)$, so $L_g = \mu(g, -)$ is differentiable. And $(L_g)^{-1} = L_{g^{-1}}$. So L_g is a diffeomorphism. Furthermore,

$$L_g \circ L_h = L_{gh}, L_e = \text{id}_G$$

So L is a group homomorphism. $\square$

Lemma 11.3. Let G be a connected Lie group. Let $U \subset G$ be any neighborhood of the identity e . Then U generates G .

Proof. We may assume $U = U^{-1}$, otherwise we replace U by $U \cap U^{-1}$. Let $U^k = \{g_1 \dots g_k \mid g_i \in U\}$, $S = \bigcup_{k \geq 0} U^k$. We claim that $S \neq \emptyset$, S is both open and closed, then $S = G$ by the connectedness of G .

Note that $U^2 = \bigcup_{g \in U} L_g U$, and L_g is a diffeomorphism. So we have U^2 is open, since U is. By induction we have U^k is open. Thus S is open. Also note that

$$G = \bigcup_{g \in G} gS = \bigcup_{g \in S} gS \cup \bigcup_{g \in G \setminus S} gS$$

But $\bigcup_{g \in S} gS = S$, so $G \setminus S$ is open. Thus S is closed. $\square$

What information can you see from above lemma? This statement implies that any morphism of Lie groups $\rho : G \rightarrow H$ will be determined by what it does on any open set containing the identity. In other word, ρ is determined by its germ at $e \in G$. In fact, here is the first principle of Lie groups, we will prove it later.

Theorem 11.2.1 (First principle of Lie groups). Let G, H be Lie groups, G is connected. A group homomorphism $\rho : G \rightarrow H$ is uniquely determined by its differential $T_e \rho : T_e G \rightarrow T_e H$ at the identity.

From above theorem we get an inclusion of sets

$$\text{Hom}_{gp}(G, H) \subset \text{Hom}_{vect}(T_e G, T_e H)$$

But we want an intrinsic criterion which can tell us when a linear map $T_e G \rightarrow T_e H$ comes from a group homomorphism ρ .

We look closer at $\text{Hom}_{gp}(G, H)$. If $\rho : G \rightarrow H$ is a group homomorphism, then

$$\rho(L_{g_1} g_2) = L_{\rho(g_1)} \rho(g_2)$$

In other words, the following diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\rho} & H \\ \downarrow L_g & & \downarrow L_{\rho(g)} \\ G & \xrightarrow{\rho} & H \end{array}$$

But L_g has no fixed point, hence tangent spaces at different points are mapped to each other.

If we choose $\Psi_g = R_{g^{-1}} \circ L_g$, things will be better. Then $\rho : G \rightarrow H$ is a group homomorphism if the following diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\rho} & H \\ \downarrow \Psi_g & & \downarrow \Psi_{\rho(g)} \\ G & \xrightarrow{\rho} & H \end{array}$$

Take differential of Ψ_g at e , we have

$$\text{Ad}(g) : T_e \Psi_g : T_e G \rightarrow T_e G, \quad \forall g \in G$$

We get a map $\text{Ad} : G \rightarrow \text{GL}(T_e G)$, called the adjoint representation of G on $T_e G$.

Then for a group homomorphism ρ , we have that its differential $T_e \rho$ must satisfy the following communicative diagram

$$\begin{array}{ccc} T_e G & \xrightarrow{T_e \rho} & T_e H \\ \downarrow \text{Ad}(g) & & \downarrow \text{Ad}(\rho(g)) \\ T_e G & \xrightarrow{T_e \rho} & T_e H \end{array}$$

This is equivalent to

$$T_e \rho(\text{Ad}(g)X) = \text{Ad}(\rho(g))(T_e \rho(X)), \quad \forall X \in T_e G$$

However, this is still not intrinsic, since this condition still depends on the map $\rho(g)$. Let's take differential of Ad . Note that for any $\phi \in \text{GL}(T_e G)$, we have

$$T_\phi \text{GL}(T_e G) \cong \text{End}(T_e G)$$

Then we have

$$\begin{aligned} \text{ad} &:= T_e \text{Ad} : T_e G \rightarrow \text{End}(T_e G) \\ X &\mapsto (Y \mapsto \text{ad}_X Y) \end{aligned}$$

In other words, we have a bilinear map which we call it a Lie bracket

$$\begin{aligned} [\cdot, \cdot] &: T_e G \times T_e G \rightarrow T_e G \\ (X, Y) &\mapsto \text{ad}_X Y \end{aligned}$$

As desired, the map ad involves only the tangent space $T_e G$ and have nothing with ρ itself. This gives us our final characterization as the following communicative diagram

$$\begin{array}{ccc} T_e G & \xrightarrow{T_e \rho} & T_e H \\ \downarrow \text{ad}_X & & \downarrow \text{ad}_X \circ T_e \rho \\ T_e G & \xrightarrow{T_e \rho} & T_e H \end{array}$$

Equivalently, we have

$$T_e \rho(\text{ad}_X Y) = \text{ad}_{T_e \rho(X)}(T_e \rho(Y)), \quad \forall X, Y \in T_e G$$

In other words,

$$T_e \rho([X, Y]) = [T_e \rho(X), T_e \rho(Y)], \quad \forall X, Y \in T_e G$$

So we have seen that, if ρ arises as the differential of some group homomorphism, it must preserve the Lie bracket. However, it's the only requirement it needs to satisfy. This is the second principle of Lie groups.

Theorem 11.2.2 (Second principle of Lie group). Let G, H be Lie groups, G is connected and simply connected. A linear map $f : T_e G \rightarrow T_e H$ is the differential of group homomorphism from G to H if and only if

$$[f(X), f(Y)] = f([X, Y]), \quad \forall X, Y \in T_e G$$

Let's compute a concrete example to get a feeling of Ad and ad .

Example 11.2.1. Let $G = \text{GL}(n, \mathbb{R})$. Since G is an open set in $\mathbb{R}^{n^2}$, thus its tangent space at identity $\mathfrak{g}$ can be viewed as $\text{Mat}(n, \mathbb{R})$. Then for any $g \in G$, let's compute $\text{Ad}(g)$ as follows: Take $X \in \mathfrak{g}$

$$\begin{aligned} \text{Ad}(g)(X) &= (\Psi_g)_*(X) \\ &= \left. \frac{d}{dt} \right|_{t=0} g e^{tX} g^{-1} \\ &= gXg^{-1} \end{aligned}$$

Now let's take $X, Y \in \mathfrak{g}$, then

$$\begin{aligned}
 [X, Y] &= \text{ad}_X(Y) \\
 &= (\text{Ad})_*(X)(Y) \\
 &= \left. \frac{d}{dt} \right|_{t=0} (\text{Ad}(e^{tX})(Y)) \\
 &= \left. \frac{d}{dt} \right|_{t=0} (e^{tX} Y e^{-tX}) \\
 &= (X e^{tX} Y e^{-tX} - e^{tX} Y X e^{-tX}) \Big|_{t=0} \\
 &= XY - YX
 \end{aligned}$$

In this case, we can see clearly Lie bracket has the following properties

$$\begin{cases} [Y, X] = -[X, Y] \\ [[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0 \end{cases}$$

And that's what we use in the general definition.

11.3. Lie algebra.

Definition 11.3.1 (Lie algebra). A Lie algebra $\mathfrak{g}$ is a vector space with a skew-symmetric bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying the Jacobi identity

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0, \quad \forall X, Y, Z \in \mathfrak{g}$$

Notation 11.2. If $\mathfrak{a}, \mathfrak{b}$ are subsets of a Lie algebra $\mathfrak{g}$, then we write

$$[\mathfrak{a}, \mathfrak{b}] := \{[X, Y] \mid X \in \mathfrak{a}, Y \in \mathfrak{b}\}$$

Definition 11.3.2 (morphism of Lie algebras). Let $\mathfrak{g}, \mathfrak{h}$ be two Lie algebras, then $\rho : \mathfrak{g} \rightarrow \mathfrak{h}$ is called a morphism of Lie algebras if

$$\rho([X, Y]) = [\rho(X), \rho(Y)], \quad \forall X, Y \in \mathfrak{g}$$

Thus, in a summary we have:

1. The tangent space of a Lie group G is naturally endowed with a Lie algebra structure;
2. If G and H are Lie groups with G is connected and simply connected, then morphisms between Lie groups are in one to one correspondence with morphisms of their Lie algebras, by associating to $\rho : G \rightarrow H$ its differential $T_e \rho : \mathfrak{g} \rightarrow \mathfrak{h}$.

Recall that a representation of Lie group G is a morphism $\rho : G \rightarrow \text{GL}(V)$. So for a connected and simply connected Lie group G , its representation is in one to one correspondence to Lie algebra morphism

$$\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V) := \text{End}(V)$$

Here comes the definition of representation of Lie algebras.

Definition 11.3.3 (representation of Lie algebras). A representation of a Lie algebra $\mathfrak{g}$ on a finite-dimensional vector space V is a morphism of Lie algebras $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V) := \text{End}(V)$.

Example 11.3.1 (abelian Lie algebra). Let V be a vector space, define $[v, w] = 0, \forall v, w \in V$. Then $(V, [\ , \])$ is an abelian Lie algebra.

Example 11.3.2. Let A be an associative algebra, define $[X, Y] = XY - YX, \forall X, Y \in A$. Then $(A, [\ , \])$ is a Lie algebra.

Example 11.3.3. $\mathfrak{sl}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid \text{tr}(X) = 0\}$ is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$.

Example 11.3.4. $\mathfrak{so}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{R}) \mid X + X^T = 0\}$ is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$.

Example 11.3.5. Let $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. Then $\mathfrak{sp}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(2n, \mathbb{R}) \mid JX + X^T J = 0\}$ is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$.

Example 11.3.6. Similarly, we have $\mathfrak{sl}(n, \mathbb{C}), \mathfrak{so}(n, \mathbb{C}), \mathfrak{sp}(n, \mathbb{C})$.

Example 11.3.7. $\mathfrak{u}(n) = \{X \in \mathfrak{gl}(n, \mathbb{C}) \mid X + \overline{X}^T = 0\}, \mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n, \mathbb{C})$.

Exercise 11.1. Verify that the defining conditions are preserved under $[X, Y]$ and under $X \mapsto gXg^{-1}, \forall g \in G$.

11.4. Exponential map.

11.4.1. Vector field.

Definition 11.4.1 (vector field - first definition). Let M be a smooth manifold. A vector field v on M is a functions that assigns to each $p \in M$ a tangent vector $v_p \in T_p M$.

Remark 11.4.1. We have already know that for any tangent vector v_p at p , we can give a real number $v_p(f)$, called the directional derivative at p . So if v is a vector field on M and $f \in C^\infty(M)$, then $v(f)$ denotes the function $p \mapsto v(f)(p) := v_p(f)$.

Definition 11.4.2 (smooth vector field). A vector field v is called smooth, if $v(f)$ is smooth for all $f \in C^\infty(M)$.

Notation 11.3. We use $\mathfrak{X}(M)$ to denote the set of all smooth vector fields on M .

Since we already know the fact that $v_p(f)$ satisfies the Leibniz rule, it follows that $v(f)$ also satisfies the Leibniz rule. Here comes the second definition

Definition 11.4.3 (vector field - second definition). A vector field on M is a linear map

$$D : C^\infty(M) \rightarrow C^\infty(M)$$

such that

$$D(fg) = D(f)g + fD(g), \quad \forall f, g \in C^\infty(M)$$

Remark 11.4.2. Theorem 10.15 implies that the two definitions for vector field is the same. So we can see a vector field as a derivation on the algebra $C^\infty(M)$ of smooth functions. Use such point of view, we can easily define the Lie bracket of two vector fields.

Proposition 11.4.1 (Lie bracket of vector field). Let v, w be two vector fields, then the commutator

$$[v, w] : vw - wv : C^\infty(M) \rightarrow C^\infty(M)$$

is again a vector field.

Proof. It suffices to check the commutator is a derivation. For any $f, g \in C^\infty(M)$, compute directly as follows

$$\begin{aligned} [v, w](fg) &= v(w(fg)) - w(v(fg)) \\ &= v(w(f)g + fw(g)) - w(v(f)g + fv(g)) \\ &= vw(f)g + w(f)v(g) + v(f)w(g) + fvw(g) \\ &\quad - wv(f)g - v(f)w(g) - w(f)v(g) - fwv(g) \\ &= (vw(f) - wv(f))g + f(vw(g) - wv(g)) \\ &= [v, w](f)g + f[v, w](g) \end{aligned}$$

This completes the proof. $\square$

Theorem 11.4.1. $(\mathfrak{X}(M), [\cdot, \cdot])$ is a Lie algebra.

Proof. It suffices to check Jacobi identity, we omit it. $\square$

Remark 11.4.3. Let $f : M \rightarrow N$ be a differentiable map between smooth manifolds. Recall that we can pushforward a tangent vector in $T_p M$ for any $p \in M$. However, we can not pushforward a vector field in general. For example, if f is not surjective, then values for $q \in N \setminus f(M)$ is undetermined and if f is not injective, then there may be several distinct vectors in $T_{f(p)}N$.

Definition 11.4.4 (f -related). Let M, N be smooth manifold. $f : M \rightarrow N$ be a smooth map. For $v \in \mathfrak{X}(M)$, if there exists $w \in \mathfrak{X}(N)$ such that $(T_p f)(v_p) = w_{f(p)}$, $\forall p \in M$. Then v, w are called f -related.

Notation 11.4. If two vector fields v, w are f -related, we write as $v \sim_f w$

Lemma 11.4. Let M, N be smooth manifolds, $f : M \rightarrow N$ be a smooth map. For $v \in \mathfrak{X}(M), w \in \mathfrak{X}(N)$. Then

$$v \sim_f w \iff v(\phi \circ f) = w(\phi) \circ f, \quad \forall \phi \in C^\infty(N)$$

Proposition 11.4.2 (pushforward of vector fields). Let M, N be smooth manifolds, $f : M \rightarrow N$ be a diffeomorphism. Then for all $v \in \mathfrak{X}(M)$ there exists a unique $w \in \mathfrak{X}(N)$ such that $v \sim_f w$. This vector field is called the push-forward of v , and denoted by $f_* v$.

Corollary 11.4.1. Let M, N be smooth manifolds, $f : M \rightarrow N$ be a diffeomorphism and $v \in \mathfrak{X}(M)$. Then

$$f_* v(\phi) = v(\phi \circ f), \quad \forall \phi \in C^\infty(M)$$

Lemma 11.5. Let M, N be smooth manifolds. $f : M \rightarrow N$ be smooth map. For $v_1, v_2 \in \mathfrak{X}(M)$ and $w_1, w_2 \in \mathfrak{X}(N)$ such that $v_i \sim_f w_i, i = 1, 2$. Then

$$[v_1, v_2] \sim_f [w_1, w_2]$$

Corollary 11.4.2. Let M, N be smooth manifolds, f be a diffeomorphism and $v_1, v_2 \in \mathfrak{X}(M)$. Then

$$f_*[v_1, v_2] = [f_*v_1, f_*v_2]$$

Recall that left translation $L_g : G \rightarrow G$ is a diffeomorphism, and the tangent map at identity $T_e L_g : T_e G \rightarrow T_g G$ is an isomorphism of vector spaces.

Definition 11.4.5 (left-invariant vector field). Let G be a Lie group, and v is a vector field on G . v is called left-invariant if $(L_g)_*v = v, \forall g \in G$. In other words,

$$T_h L_g(v_h) = v_{Lg(h)} = v_{gh}, \quad \forall g, h \in G$$

Lemma 11.6. For left-invariant vector field, we have

1. Any left-invariant vector field is smooth.
2. $\mathfrak{X}_L(G)$ is a Lie subalgebra of $\mathfrak{X}(G)$.

Proof. Let $v \in \mathfrak{X}_L(G)$, we need to show that for all $\phi \in C^\infty(G)$, $v(\phi) \in C^\infty(M)$. Let $\gamma : I \rightarrow G$ be a smooth curve such that $\gamma(0) = e, \gamma'(0) = v_e \in T_e G$. Then

$$\begin{aligned} v(\phi)(g) &= v_g(\phi) \\ &= T_e L_g(v_e)(\phi) \\ &= v_e(\phi \circ L_g) \\ &= \gamma'(0)(\phi \circ L_g) \\ &= \left. \frac{d}{dt} \right|_{t=0} (\phi \circ L_g \circ \gamma)(t) \end{aligned}$$

If we define

$$\begin{aligned} \psi : I \times G &\rightarrow \mathbb{R} \\ (t, g) &\mapsto \phi(g\gamma(t)) \end{aligned}$$

then from above computation we can see

$$v(\phi)(g) = \frac{\partial \psi}{\partial t}(0, g)$$

ψ is a composition of smooth maps; hence it is smooth, and so is $v(\phi)(g)$.

For the second. Clearly $(L_g)_*(\alpha v + \beta w) = \alpha(L_g)_*v + \beta(L_g)_*w = \alpha v + \beta w$. And the corollary says that

$$(L_g)_*([v, w]) = [(L_g)_*v, (L_g)_*w] = [v, w]$$

That is $[v, w] \in \mathfrak{X}_L(G)$. Thus $\mathfrak{X}_L(G)$ is a Lie subalgebra. □

Lemma 11.7. Let G be a Lie group, $X \in T_e G$. Define a vector field v_X by $g \mapsto v_{X,g} := T_e L_g X \in T_g G$. Then $v_X \in \mathfrak{X}_L(G)$.

Proof. Clearly

$$\begin{aligned} T_h L_g(v_{X,h}) &= T_h L_g(T_e L_h X) \\ &= T_e((L_g \circ L_h)X) \\ &= T_e(L_{gh}X) \\ &= v_{X,gh}, \quad \forall g, h \in G \end{aligned}$$

□

Theorem 11.4.2. Let G be a Lie group. Let $\varepsilon : \mathfrak{X}_L(G) \rightarrow T_e G$ defined by $v \mapsto v_e$. Then the map $T_e G \rightarrow \mathfrak{X}_L(G), X \mapsto v_X$ is a linear isomorphism with inverse ε .

Proof. Linearity. For any $g \in G$ we have $v_{\alpha X + \beta Y, g} = T_e L_g(\alpha X + \beta Y) = \alpha T_e L_g X + \beta T_e L_g Y = \alpha v_{X, g} + \beta v_{Y, g}$; If $v_{X, g} = T_e L_g X = 0$, since L_g is a diffeomorphism, then $T_e L_g$ is an isomorphism so we must have $X = 0$, this is injectivity; And by Lemma 10.52, it's surjective.

Finally let's check the inverse of $X \mapsto v_X$ is ε . Let $X \in T_e G$. Then

$$\varepsilon(v_X) = v_{X, e} = T_e L_e X = \text{id}_{T_e G} X = X$$

And conversely let $v \in \mathfrak{X}_L(G)$, then

$$v_g = T_e L_g v_e = v_{\varepsilon(v), g}$$

as desired. □

This theorem induces a Lie algebra structure on $T_e G$, since $\mathfrak{X}_L(G)$ is a Lie algebra.

Definition 11.4.6 (Lie algebra). Let G be a Lie group. The Lie algebra $\mathfrak{g} = \text{Lie}(G)$ of G is defined as $\mathfrak{g} = \mathfrak{X}_L(G) \cong T_e G$. For $X, Y \in T_e G$, we define Lie bracket as

$$[X, Y] = \varepsilon([v_X, v_Y])$$

Proposition 11.4.3. The composition of the natural maps

$$\text{Lie}(\text{GL}(n, \mathbb{R})) \rightarrow T_{I_n} \text{GL}(n, \mathbb{R}) \rightarrow \mathfrak{gl}(n, \mathbb{R})$$

gives a Lie algebra isomorphism

$$\text{Lie}(\text{GL}(n, \mathbb{R})) \cong \mathfrak{gl}(n, \mathbb{R})$$

Proof. The Theorem 10.53 gives a vector space isomorphism $\text{Lie}(\text{GL}(n, \mathbb{R})) \cong T_{I_n} \text{GL}(n, \mathbb{R})$. Since $\text{GL}(n, \mathbb{R}) \subset \mathfrak{gl}(n, \mathbb{R}) = \mathbb{R}^{n^2}$ as an open subset, then

$$T_{I_n} \text{GL}(n, \mathbb{R}) \xrightarrow{\cong} \mathfrak{gl}(n, \mathbb{R})$$

as vector spaces. More explicitly, for $A \in \text{GL}(n, \mathbb{R})$, we use $A_j^i, i, j = 1, 2, \dots, n$ as global coordinates on $\text{GL}(n, \mathbb{R}) \subset \mathfrak{gl}(n, \mathbb{R})$. So we can make the following identification

$$T_{I_n} \text{GL}(n, \mathbb{R}) \ni \sum_{i,j=1}^n X_j^i \frac{\partial}{\partial A_j^i} \Big|_{I_n} \longleftrightarrow (X_j^i) \in \mathfrak{gl}(n, \mathbb{R})$$

Let $\mathfrak{g} = \text{Lie}(\text{GL}(n, \mathbb{R}))$, $X \in \mathfrak{gl}(n, \mathbb{R})$, $A \in \text{GL}(n, \mathbb{R})$. Then the left-invariant vector field which corresponds to A is

$$\begin{aligned} v_{X,A} &= T_{I_n} L_A X \\ &= T_{I_n} L_A \left(\sum_{i,j=1}^n X_j^i \frac{\partial}{\partial A_j^i} \right) \Big|_{I_n} \\ &= \sum_{i,j,k=1}^n A_j^i X_k^j \frac{\partial}{\partial A_i^k} \Big|_A \end{aligned}$$

where L_A is the restriction of $X \mapsto AX$ to $\text{GL}(n, \mathbb{R})$. Now let's compute the Lie bracket for $X, Y \in \mathfrak{gl}(n, \mathbb{R})$ as follows:

$$\begin{aligned} [v_X, v_Y] &= \sum_{i,j,k=1}^n \sum_{p,q,r=1}^n [A_j^i X_k^j \frac{\partial}{\partial A_i^k}, A_q^p Y_r^q \frac{\partial}{\partial A_p^r}] \\ &= \sum_{i,j,k=1}^n \sum_{p,q,r=1}^n (A_j^i X_k^j \frac{\partial}{\partial A_i^k} (A_q^p Y_r^q) \frac{\partial}{\partial A_p^r} - A_q^p Y_r^q \frac{\partial}{\partial A_p^r} (A_j^i X_k^j) \frac{\partial}{\partial A_i^k}) \\ &= \sum_{i,j,k,r=1}^n A_j^i (X_k^j Y_r^k - Y_k^j X_r^k) \frac{\partial}{\partial A_r^i} \end{aligned}$$

Thus we have

$$[v_X, v_Y] \Big|_{I_n} = [A, B]_r^i \frac{\partial}{\partial A_i^r} \Big|_{I_n} = v_{[A,B]} \Big|_{I_n}$$

Since a left-invariant vector field is determined by its value at identity, then

$$[v_X, v_Y] = v_{[X,Y]}$$

□

We have already defined how to push forward a vector field using diffeomorphism. Recall Remark 10.42, what's the obstruction if we want to use a morphism which is not injective or surjective? But left-invariant vector field is totally determined by its value at identity, so above bad things won't happen.

Thus for any morphism of Lie groups, we can use it to push-forward left-invariant vector fields, or elements in Lie algebras.

Definition 11.4.7. Let G, H be Lie groups with Lie algebras $\mathfrak{g}, \mathfrak{h}$, $\rho : G \rightarrow H$ a morphism of Lie groups. For $X \in \mathfrak{g}$, we define

$$\rho_*(X) = T_e \rho(v_{X,e}) \in \mathfrak{h}$$

Theorem 11.4.3. Let G, H be Lie groups with Lie algebras $\mathfrak{g}, \mathfrak{h}$, $\rho : G \rightarrow H$ a morphism of Lie groups. Then

1. $\rho_* X \sim_\rho X$ for all $X \in \mathfrak{g}$.
2. $\rho_* : \mathfrak{g} \rightarrow \mathfrak{h}$ is a morphism of Lie algebras.

Proof. Let $X \in \mathfrak{g}$ and $Y = \rho_* X \in \mathfrak{h}$, that is

$$v_{Y,e} = T_e \rho(v_{X,e})$$

Since ρ is a group homomorphism, that is $\rho(gh) = \rho(g)\rho(h), \forall g, h \in G$. Then

$$\rho(L_g h) = L_{\rho(g)} \rho(h) \implies \rho \circ L_g = L_{\rho(g)} \circ \rho$$

So we have

$$T\rho \circ TL_g = TL_{\rho(g)} \circ T\rho$$

Then

$$\begin{aligned} (T_g \rho)v_{X,g} &= T_g \rho(T_e L_g v_{X,e}) \\ &= T_e L_{\rho(g)}(T_e \rho(v_{X,e})) \\ &= T_e L_{\rho(g)}(v_{Y,e}) \\ &= v_{Y,\rho(g)} \end{aligned}$$

Thus $v_X \sim_\rho v_Y$. For the second. From above we have

$$[v_{X_1}, v_{X_2}] \sim_\rho [v_{Y_1}, v_{Y_2}]$$

where $Y_i = \rho_* X_i, i = 1, 2$. In particular, we have

$$T_e \rho([v_{X_1}, v_{X_2}]_e) = [v_{Y_1}, v_{Y_2}]_e$$

But $\rho_*([v_{X_1}, v_{X_2}])$ is the unique left-invariant vector field such that

$$\rho_*([v_{X_1}, v_{X_2}]_e) = T_e \rho([v_{X_1}, v_{X_2}]_e)$$

So

$$\rho_*([X_1, X_2]) = [\rho_* X_1, \rho_* X_2] = [Y_1, Y_2]$$

□

Corollary 11.4.3. Let V be a finite dimensional vector space over $\mathbb{R}$, G is a Lie group and $\rho : G \rightarrow \text{GL}(V)$ is a representation. Then

$$\rho_* : \mathfrak{g} \rightarrow \text{Lie}(\text{GL}(V))$$

is a representation of Lie algebras.

Corollary 11.4.4. Let G be an abelian group, then $\mathfrak{g}$ is also abelian.

Proof. If G is abelian, then inversion $\iota : G \rightarrow G, g \mapsto g^{-1}$ is a morphism. Indeed, clearly ι is smooth and it's a group homomorphism since

$$\iota(gh) = (gh)^{-1} = h^{-1}g^{-1} = g^{-1}h^{-1} = \iota(g)\iota(h)$$

Then $\iota_* : \mathfrak{g} \rightarrow \mathfrak{g}$ is a morphism of Lie algebras. Let's compute ι_* explicitly. For $X \in \mathfrak{g}$,

$$\begin{aligned} \iota_*(X) &= T_e \iota(X) \\ &= \left. \frac{d}{dt} \right|_{t=0} \iota(\gamma(t)), \quad \gamma(0) = e, \gamma'(0) = X \\ &= \left. \frac{d}{dt} \right|_{t=0} \gamma(t)^{-1} \end{aligned}$$

So we need to compute the derivative of $\gamma(t)^{-1}$ at $t = 0$. Note that

$$\gamma(t)\gamma(t)^{-1} = e$$

So take derivative and take $t = 0$ we have

$$\left. \frac{d\gamma(t)}{dt} \right|_{t=0} \gamma(0)^{-1} + \gamma(0) \left. \frac{d\gamma(t)^{-1}}{dt} \right|_{t=0} = 0 \implies X + \left. \frac{d}{dt} \right|_{t=0} \gamma(t)^{-1} = 0$$

Then we have

$$\left. \frac{d}{dt} \right|_{t=0} \gamma(t)^{-1} = -X$$

In other words, $\iota_* = -\text{id}_{\mathfrak{g}}$. So

$$-[X, Y] = \iota_*[X, Y] = [\iota_*X, \iota_*Y] = [-X, -Y] = [X, Y] \quad \forall X, Y \in \mathfrak{g}$$

Thus $[X, Y] = 0, \forall X, Y \in \mathfrak{g}$. □

Proposition 11.4.4. We have the following properties:

1. $(\text{id}_G)_* : \mathfrak{g} \rightarrow \mathfrak{g}$ is the identity.
2. If $\rho : G \rightarrow H, \sigma : H \rightarrow K$ are morphisms of Lie groups. Then $(\sigma \circ \rho)_* = \sigma_* \circ \rho_*$.
3. If $G \cong H$, then $\mathfrak{g} \cong \mathfrak{h}$.

Proof. The first and second hold since

$$\begin{aligned} T_e \text{id}_G &= \text{id}_{T_e G} \\ T_e(\sigma \circ \rho) &= T_e \sigma \circ T_e \rho \end{aligned}$$

Then the third holds from above, since

$$\rho_* \circ (\rho^{-1})_* = (\rho \circ \rho^{-1})_* = \text{id} = (\rho^{-1})_* \circ \rho_*$$

□

Proposition 11.4.5. Let $H \leq G$ be a Lie subgroup, $i : H \rightarrow G$ the inclusion map. Then there exists a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$, canonically isomorphic to $\text{Lie}(H)$, given by

$$\mathfrak{h} = i_* \text{Lie}(H)$$

Proof. Since $i : H \rightarrow G$ is a morphism of Lie groups, i_* is a morphism of Lie algebras. Thus $i_* \text{Lie}(H)$ is a Lie subalgebra of $\text{Lie}(G)$. □

11.4.2. One parameter subgroups.

Definition 11.4.8 (integral curve). Let M be a smooth manifold. A curve $\gamma : I \rightarrow M$ is called an integral curve of a vector field $v \in \mathfrak{X}(M)$ if $\gamma'(t) = v_{\gamma(t)}, \forall t \in I$.

Remark 11.4.4. In local coordinates $(x^1, \dots, x^n)$ of $U \subset M$, this condition yields a system of first order ordinary differential equations

$$\frac{d(x^i \circ \gamma)}{dt} = F^i(x^1 \circ \gamma, \dots, x^n \circ \gamma)$$

where F^i is the coordinate expression of vx^i . The fundamental theorem for existence and uniqueness of solutions of such systems yields the existence and uniqueness of integral curves. That's following proposition.

Proposition 11.4.6. Let M be a smooth manifold, $v \in \mathfrak{X}(M)$. For any $p \in M$, there exists an open interval I around 0 and a unique integral curve $\gamma : I \rightarrow M$ of v such that $\gamma(0) = p$.

Definition 11.4.9 (maximal integral curve). Let M be a smooth manifold. An integral curve $\gamma : I \rightarrow M$ is called maximal if it can not be extended to any larger open interval.

Definition 11.4.10 (complete). Let M be a smooth manifold, $v \in \mathfrak{X}(M)$ is called complete if each of its maximal integral curves is defined on $\mathbb{R}$.

Lemma 11.8. Let M be a smooth manifold, $v \in \mathfrak{X}(M)$. $\gamma : I \rightarrow M$ is an integral curve of v , then for any $b \in \mathbb{R}$, $\tilde{\gamma} : \tilde{I} \rightarrow M, t \mapsto \gamma(b + t)$ is also an integral curve of v , where $\tilde{I} = \{t \in \mathbb{R} \mid t + b \in I\}$

Proof. Clear. □

Lemma 11.9. Let M, N be manifolds, $f : M \rightarrow N$ a smooth map and $v \in \mathfrak{X}(M), w \in \mathfrak{X}(N)$. Then $v \sim_f w$ is equivalent to for all integral curve γ of v the curve $f \circ \gamma$ is the integral curve of w .

Corollary 11.4.5. Let G, H be two Lie groups, $\rho : G \rightarrow H$ a morphism of Lie groups, then for any $v \in \mathfrak{X}_L(G)$, we have

$$\gamma_{\rho_* v} = \rho \circ \gamma_v$$

Proof. By the properties of ρ_* , we know that $\rho_* v \sim_\rho v$, so $\rho \circ \gamma_v$ is an integral curve of $\rho_* v$. But both $\gamma_{\rho_* v}$ and $\rho \circ \gamma_v$ are integral curves of $\rho_* v$, and by uniqueness of integral curves, they must coincide. □

Definition 11.4.11 (one parameter subgroup). A one parameter subgroup in a Lie group G is a morphism of Lie groups $\gamma : (\mathbb{R}, +) \rightarrow G$.

Lemma 11.10. Let G be a Lie group, $v \in \mathfrak{X}_L(G)$ and $\gamma : I \rightarrow M$ is an integral curve of v . Then I can be extended to $\mathbb{R}$.

Proof. $v \in \mathfrak{X}_L(G)$ is equivalent to $v \sim_{L_g} v$ for all $g \in G$. Let γ be the unique integral curve for v such that $\gamma(0) = e$, defined on $(-\varepsilon, \varepsilon)$. Then $\gamma_g := L_g \gamma$ is an integral curve for v such that $\gamma_g(0) = g$. Indeed,

$$\gamma'_g(t) = T_{\gamma(t)} L_g(\gamma'(t)) = T_{\gamma(t)} L_g(v_{\gamma(t)}) = v_{L_g \gamma(t)} = v_{\gamma_g(t)}$$

In particular, for $t_0 \in (-\varepsilon, \varepsilon)$, the curve $t \mapsto \gamma(t_0)\gamma(t)$ is an integral curve for v starting at $\gamma(t_0)$. By uniqueness, this curve coincides with $\gamma(t_0 + t)$ for all $t \in (-\varepsilon, \varepsilon) \cap (-\varepsilon - t_0, \varepsilon - t_0)$. Define

$$\tilde{\gamma}(t) = \begin{cases} \gamma(t), & t \in (-\varepsilon, \varepsilon) \\ \gamma(t_0)\gamma(t), & t \in (-\varepsilon - t_0, \varepsilon - t_0) \end{cases}$$

Repeat above operations to get our desired extension. □

Remark 11.4.5. In other words, above lemma says that every left-invariant vector field on a Lie group is complete.

Theorem 11.4.4. Let G be a Lie group. Then there is a one to one correspondence

$$\{\text{one parameter subgroups of } G\} \iff \{\text{maximal integral curves } \gamma \text{ of } v, v \in \mathfrak{X}_L(G), \gamma(0) = e\}$$

Proof. Let $\gamma : \mathbb{R} \rightarrow G$ be a one parameter subgroup. View $\frac{d}{dt}$ as a left invariant vector field on $\mathbb{R}$, let $v = \gamma_*(\frac{d}{dt}) \in \mathfrak{X}_L(G)$. It suffices to show γ is a integral curve of v . In other words, we need to check $\gamma'(t_0) = v_{\gamma(t_0)}$. Indeed,

$$\gamma'(t_0) = T_{t_0}\gamma\left(\frac{d}{dt}\Big|_{t=t_0}\right) = v_{\gamma(t_0)}$$

On the other direction, let $v \in \mathfrak{X}_L(G)$, and γ is the corresponding maximal integral curves such that $\gamma(0) = e$. By Lemma 10.63, we know that γ is defined on $\mathbb{R}$. Now it's suffices to show $\gamma(s+t) = \gamma(s)\gamma(t)$, $\forall s, t \in \mathbb{R}$.

Note that v is left-invariant, so L_g will maps integral curves of v to integral curves of v . Then

$$t \mapsto L_{\gamma(s)}(\gamma(t))$$

is an integral curve for v starting at $\gamma(s)$. And Lemma 10.60 tells us that $t \mapsto \gamma(s+t)$ is also an integral curve for v starting at $\gamma(s)$. Thus by the uniqueness of integral curves we have $\gamma(s)\gamma(t) = \gamma(s+t)$. This completes the proof. $\square$

Definition 11.4.12 (exponential map). Let G be a Lie group with Lie algebra $\mathfrak{g}$. The exponential map for G is the map $\exp : \mathfrak{g} \rightarrow G$, sending X to $\gamma_{v_X}(1)$, where $\gamma_{v_X}(t)$ is the one parameter subgroup determined by $v_X \in \mathfrak{X}_L(G)$, i.e. $\gamma'_{v_X}(0) = X$.

The following proposition shows the power of exponential map, that's we can use exponential map to characterize the one parameter subgroup generated by some $X \in \mathfrak{g}$.

Proposition 11.4.7. Let G be a Lie group. For any $X \in \mathfrak{g}$, $\gamma(t) = \exp(tX)$ is the one parameter subgroup for G generated by X .

Proof. Let γ be the one parameter subgroup generated by X , that is, the integral curve $\gamma = \gamma_{v_X}$ with $\gamma(0) = e$. We need to show $\gamma(t) = \exp(tX)$ for all $t \in \mathbb{R}$.

Let $s \in \mathbb{R}$ be fixed. Consider $\tilde{\gamma}(t) = \gamma(st)$ we have $\tilde{\gamma}'(t) = s\gamma'(st) = sv_{X, \gamma(st)} = sv_{X, \tilde{\gamma}(t)}$. Thus $\tilde{\gamma}$ is an integral curve for sv_X starting at $\tilde{\gamma}(0) = \gamma(0) = e$. By definition of exponential map, we have

$$\gamma(s) = \tilde{\gamma}(1) = \exp(sX), \quad \forall s \in \mathbb{R}$$

as desired. $\square$

Corollary 11.4.6. Let G be Lie group with Lie algebra $\mathfrak{g}$, $X \in \mathfrak{g}$ and $v_X \in \mathfrak{X}_L(G)$, $\phi \in C^\infty(G)$. Then

$$(v_X \phi)(\exp(tX)) = \frac{d}{dt}(\phi(\exp(tX)))$$

Proof. Let $\gamma(t) = \exp(tX)$ be integral curve for v_X with $\gamma(0) = e$, that is, $\gamma'(t) = v_{X, \gamma(t)} = v_{X, \exp(tX)}$. Thus

$$(v_X \phi)(\exp(tX)) = v_{X, \exp(tX)} \phi = \gamma'(t) \phi = \frac{d}{dt}(\phi \circ \gamma)(t) = \frac{d}{dt} \phi(\exp(tX))$$

□

Definition 11.4.13 (flow). Let M be a smooth manifold, $v \in \mathfrak{X}(M)$ complete. Then $\Phi : M \times \mathbb{R} \rightarrow M$, given by $\Phi(p, t) = \gamma_p(t)$, where γ_p is the maximal integral curve for v with $\gamma_p(0) = p$, is called the flow of v .

Remark 11.4.6. For p fixed, $t \mapsto \Phi(p, t)$ is just the integral curve γ_p . For t fixed, $p \mapsto \Phi(p, t)$ defines a map $\Phi_t : M \rightarrow M$ which lets every point $p \in M$ flow along the vector field for the time t .

Lemma 11.11. Let Φ be the flow of a complete vector field $v \in \mathfrak{X}(M)$. For $t \in \mathbb{R}$, let $\Phi_t : M \rightarrow M$ be the corresponding map. Then

1. $\Phi_0 = \text{id}_M$;
2. $\Phi_s \circ \Phi_t = \Phi_{s+t}$;
3. For $t \in \mathbb{R}$, Φ_t is a diffeomorphism with $(\Phi_t)^{-1} = \Phi_{-t}$.

Proof. Clear. □

Theorem 11.4.5. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Then

1. $\exp : \mathfrak{g} \rightarrow G$ is smooth;
2. $\forall X \in \mathfrak{g}, s, t \in \mathbb{R}, \exp((s+t)X) = \exp(sX)\exp(tX)$;
3. $\forall X \in \mathfrak{g}, (\exp(X))^{-1} = \exp(-X)$;
4. $\forall X \in \mathfrak{g}, n \in \mathbb{Z}, (\exp X)^n = \exp(nX)$;
5. $T_0 \exp : T_0 \mathfrak{g} \rightarrow T_e G$ is the identity map under the canonical identifications $T_0 \mathfrak{g} \cong \mathfrak{g}$ and $T_e G \cong \mathfrak{g}$;
6. $\exp$ is a local diffeomorphism;
7. Let H be a Lie group, $h \in \mathfrak{h}, \rho : G \rightarrow H$ a morphism of Lie groups. Then the following diagram commutes

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\rho_*} & \mathfrak{h} \\ \downarrow \exp & & \downarrow \exp \\ G & \xrightarrow{\rho} & H \end{array}$$

8. The flow of $v \in \mathfrak{X}_L(G)$ is given by $\Phi_t(X) = R_{\exp(tX)}$.

Proof. For smoothness, take $X \in \mathfrak{g}$ and let Φ_X be the flow of v_X . We need to show $\Phi_X(e, 1)$ depends smoothly on X , since by definition we have

$$\Phi_X(e, 1) = \gamma_{v_X}(1) = \exp(X), \quad \gamma_{v_X}(0) = e, \gamma'_{v_X}(0) = X$$

Define a vector field Ξ on $G \times \mathfrak{g}$ by

$$\Xi_{(g, X)} = (v_{X, g}, 0) \in T_g G \oplus T_X \mathfrak{g} \cong T_{(g, X)}(G \times \mathfrak{g})$$

Let x^i be global coordinates on $\mathfrak{g}$, with respect to a basis X_i of $\mathfrak{g}$, ω^i a local coordinates on G , $\phi \in C^\infty(G \times \mathfrak{g})$. Then locally we can write

$$\Xi(\phi) = \sum x^i v_{X_i}(\phi)$$

where v_{X_i} differentiates ϕ only in the w^i directions. Ξ is smooth if and only if $\Xi(\phi)$ is smooth for all ϕ . Thus Ξ is smooth. The flow of Ξ is given by

$$\Theta_t((g, X)) = (\Phi_t(t, g), X)$$

hence Θ is smooth. But $\exp X = \pi_G(\Theta_1(e, X))$, where $\pi_G : G \times \mathfrak{g} \rightarrow G$ is the projection. So $\exp$ is smooth.

(2) and (3) follow from the Proposition 10.68 that $\gamma(t) = \exp(tX)$ is the one-parameter subgroup generated by X . (4) follows from (2) by induction on $n > 0$ and from (3) for $n < 0$.

Now let's see (5). Let $X \in \mathfrak{g}, \gamma : \mathbb{R} \rightarrow \mathfrak{g}, t \mapsto tX$. Then

$$\begin{aligned} T_0 \exp X &= T_0 \exp(\gamma'(0)) \\ &= (\exp \circ \gamma)'(0) \\ &= \left. \frac{d}{dt} \right|_{t=0} \exp(tX) \\ &= X \end{aligned}$$

So we have $T \exp : \mathfrak{g} \rightarrow \mathfrak{g}$ is the identity map. Immediately we have (6) from (5) and inverse function theorem.

For (7). It suffices to show $\exp(t\rho_* X) = \rho(\exp(tX)), \forall t \in \mathbb{R}$ and take $t = 1$ to get desired result. By Proposition 10.68, $\exp(t\rho_* X)$ is the one parameter subgroup generated by $\rho_* X$. Let $\gamma(t) = \rho(\exp(tX))$. It suffices to show γ is a morphism of Lie groups satisfying

$$\gamma'(0) = \rho_* X$$

Note that γ is the composition of the morphisms of Lie groups ρ and $t \mapsto \exp(tX)$. We have

$$\begin{aligned} \gamma'(0) &= \left. \frac{d}{dt} \right|_{t=0} \rho(\exp tX) \\ &= T_0 \rho \left(\left. \frac{d}{dt} \right|_{t=0} \exp tX \right) \\ &= T_0 \rho(X) \\ &= \rho_* X \end{aligned}$$

For (8). Any $g \in G, t \mapsto L_g \exp(tX)$ is an integral curve for v_X starting at g . Hence, it equals to $\Phi_{X,t}(g)$, where $\Phi(X)$ is the flow of X . Then

$$\begin{aligned} R_{\exp(tX)}(g) &= g \exp(tX) \\ &= L_g \exp(tX) \\ &= \Phi_{X,t}(g) \end{aligned}$$

□

Corollary 11.4.7 (First principle). Let G, H be Lie groups, with Lie algebras $\mathfrak{g}, \mathfrak{h}$. If G is connected, $\rho : G \rightarrow H$ is a morphism of Lie groups. Then ρ is determined by ρ_* .

Proof. By (5) of Theorem 10.73, $T_0 \exp = \text{id}_g$. So im exp contains a neighborhood U_e of $e \in G$. Since G is connected, U_e generates all of G . Then the claim follows from (7) of Theorem 10.73. $\square$

Let's compute explicitly in the case of linear Lie group to see what does exponential map look like. In fact, it's just the exponential function we met in analysis.

Example 11.4.1. $G = \text{GL}(n, \mathbb{R})$. For any $X \in \mathfrak{gl}(n, \mathbb{R})$, we define

$$\exp(X) := \sum_{k=0}^{\infty} \frac{1}{k!} X^k$$

This is an infinitely summation, we need to consider the convergence. In fact, we can show that this series do converges if we give a suitable norm and $\exp(X) \in \text{GL}(n, \mathbb{R})$

Consider the norm $\|X\| = (\sum_{i,j} (X_j^i)^2)^{\frac{1}{2}}$, the Cauchy inequality implies that $\|XY\| \leq \|X\| \|Y\|$. So by induction, we have $\|X^k\| \leq \|X\|^k$. Hence the series converges uniformly on any bounded subset of $\mathfrak{gl}(n, \mathbb{R})$, by comparison to $\sum \frac{1}{k!} x^k = e^x$.

To $X \in \mathfrak{gl}(n, \mathbb{R})$ corresponding to $v_X = \sum_{i,j} X_j^i \frac{\partial}{\partial A_j^i}$. The one parameter subgroup generated by X is an integral curve γ of v_X satisfying $\gamma'(t) = v_{X, \gamma(t)}$, $\gamma(0) = I_n$. In other words, if we use matrix notation, we have the following first order ODEs

$$\gamma'(t) = \gamma(t)X$$

We claim that $\gamma(t) = \exp(tX)$ is a solution to this equation. Indeed,

$$\begin{aligned} \gamma'(t) &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (tX)^k \right)' \\ &= \sum_{k=1}^{\infty} \frac{k}{k!} t^{k-1} X^k \\ &= \left(\sum_{k=1}^{\infty} \frac{1}{(k-1)!} t^{k-1} X^{k-1} \right) X \\ &= \gamma(t)X \end{aligned}$$

Termwise differentiation is justified since the differentiated series also converges uniformly on bounded subsets. By the smoothness of solutions to ODEs, γ is smooth.

For invertibility, let $\sigma(t) = \gamma(t)\gamma(-t)$. Consider

$$\begin{aligned} \sigma'(t) &= \gamma'(t)\gamma(-t) - \gamma(t)\gamma'(-t) \\ &= \gamma(t)X\gamma(-t) - \gamma(t)X\gamma(-t) \\ &= 0 \end{aligned}$$

So $\sigma(t)$ is constant, that is $\sigma(t) = \sigma(0) = I_n$. So we have $\gamma(-t) = \gamma^{-1}(t)$ as desired.

Proposition 11.4.8. Let G be a Lie group with Lie algebra $\mathfrak{g}$, $X \in \mathfrak{g}, \phi \in C^\infty(G)$. Then

$$(v_X^n \phi)(g \exp tX) = \frac{d^n}{dt^n} (\phi(g \exp tX))$$

for all $g \in G$. If $\|\cdot\|$ denotes a norm on $\mathfrak{g}$ and X is restricted to a bounded subset in $\mathfrak{g}$. Then

$$\phi(\exp X) = \sum_{k=0}^n \frac{1}{k!} (v_X^k \phi)(e) + R_n$$

with $|R_n(X)| \leq C \|X\|^{n+1}$.

Proof. The first statement for $g = e$ follows from applying $v_X(\phi)(\exp tX) = \frac{d}{dt}(\phi(\exp tX))$ iteratively. Replace $\phi(h)$ by $\phi_g(h) = (\phi \circ L_g)(h)$ and use left invariance of v_X yields the statement for general $g \in G$.

For the half part, expand $t \mapsto \exp(tX)$ in a Taylor series about $t = 0$ and evaluate at $t = 1$.

$$\begin{aligned} \phi(\exp X) &= \sum_{k=0}^n \frac{1}{k!} \left(\frac{d}{dt} \right)^k \phi(\exp tX) \Big|_{t=0} + \frac{1}{n!} \int_0^1 (1-s)^n \left(\frac{d}{ds} \right)^{n+1} \phi(\exp sX) ds \\ &= \sum_{k=0}^n \frac{1}{k!} (v_X^k \phi)(e) + \underbrace{\frac{1}{n!} \int_0^1 (1-s)^n (v_X^{n+1} \phi)(\exp sX) ds}_{R_n} \end{aligned}$$

Now it suffices to estimate the remainder term R_n . Write $X = \sum \lambda_j X_j$ in some basis and expand v_X^{n+1} . Since X lies in a compact set, then $\exp sX$ also lies in a compact set. So

$$\int_0^1 (1-s)^n (v_X^{n+1} \phi)(\exp sX) ds = \|\lambda\|^{n+1} \int_0^1 (1-s)^n \dots$$

Thus $R_n(X) \leq C \|X\|^{n+1}$ as desired. $\square$

Corollary 11.4.8. Let G be a Lie group with Lie algebra $\mathfrak{g}, X \in \mathfrak{g}, \phi \in C^\infty(G)$. Then

$$(v_X \phi)(g) = \frac{d}{dt} \Big|_{t=0} \phi(g \exp tX)$$

Lemma 11.12. Let G be a Lie group with Lie algebra $\mathfrak{g}$. For $X, Y \in \mathfrak{g}, t \in \mathbb{R}$. We have

1. $\exp(tX) \exp(tY) = \exp(t(X+Y) + \frac{t^2}{2}[X, Y] + O(t^3));$
2. $\exp(tX) \exp(tY) \exp(tX)^{-1} = \exp(tY + t^2[X, Y] + O(t^3));$
3. $\lim_{n \rightarrow \infty} (\exp(\frac{t}{n}X) \exp(\frac{t}{n}Y))^n = \exp(t(X+Y)).$

Proof. Since $\exp$ is a diffeomorphism on some neighborhood of $0 \in \mathfrak{g}$, there is $\varepsilon > 0$ such that

$$Z : (-\varepsilon, \varepsilon) \rightarrow \mathfrak{g}$$

$$t \mapsto \exp^{-1}(\exp tX \exp tY)$$

is smooth, $Z(0) = 0$ and $\exp(Z(t)) = \exp tX \exp tY$. Expand $Z(t)$ as follows

$$Z(t) = tZ_1 + t^2Z_2 + O(t^3), \quad Z_1, Z_2 \in \mathfrak{g}$$

Let $\phi \in C^\infty(G)$. Then by the Proposition 10.83, we can expand $\phi(\exp(Z(t)))$ as follows

$$\begin{aligned}\phi(\exp(Z(t))) &= \sum_{k=0}^2 \frac{1}{k!} (tv_{Z_1} + t^2v_{Z_2} + O(t^3))^k \phi(e) + O(t^3) \\ &= \phi(e) + t(v_{Z_1}\phi)(e) + t^2\left(\frac{1}{2}v_{Z_1}^2\phi + v_{Z_2}\phi\right)(e) + O(t^3)\end{aligned}$$

We can do the same thing for $\phi(\exp tX \exp sY)$ for $s, t \in \mathbb{R}$

$$\begin{aligned}\phi(\exp tX \exp sY) &= \sum_{k=0}^2 \frac{1}{k!} s^k v_Y^k \phi(\exp tX) + O_t(s^3) \\ &= \sum_{k=0}^2 \sum_{l=0}^2 \frac{1}{k!} \frac{1}{l!} s^k t^l v_X^l v_Y^k \phi(e) + O_t(s^3) + O(t^3)\end{aligned}$$

Set $t = s$, then

$$\phi(\exp tX \exp tY) = \phi(e) + t(v_X + v_Y)\phi(e) + t^2\left(\frac{1}{2}v_X^2 + v_X v_Y + \frac{1}{2}v_Y^2\right)\phi(e) + O(t^3)$$

Replace ϕ by $\phi \circ L_g$ and use the left-invariance of $v_X, v_Y, v_{Z_1}, v_{Z_2}$, then we have $\phi(\exp Z(t)) = \phi(\exp tX \exp tY)$. By comparing coefficient, we have

$$\begin{cases} v_{Z_1} = v_X + v_Y \\ \frac{1}{2}v_{Z_1}^2 + v_{Z_2} = \frac{1}{2}v_X^2 + v_X v_Y + \frac{1}{2}v_Y^2 \end{cases}$$

which implies

$$\begin{cases} Z_1 = X + Y \\ Z_2 = \frac{1}{2}[X, Y] \end{cases}$$

For second,

$$\begin{aligned}\exp(tX) \exp(tY) \exp(tX)^{-1} &= \exp(t(X + Y) + \frac{t^2}{2}[X, Y] + O(t^3)) \exp(-tX) \\ &= \exp(t(X + Y - X) + \frac{t^2}{2}[X + Y, -X] + \frac{t^2}{2}[X, Y] + O(t^3)) \\ &= \exp(tY + t^2[X, Y] + O(t^3))\end{aligned}$$

For third,

$$\left(\exp\left(\frac{t}{n}X\right) \exp\left(\frac{t}{n}Y\right)\right)^n = \exp\left(t(X + Y) + \frac{t^2}{n}[X, Y] + O\left(\frac{t^3}{n^2}\right)\right)$$

Fix t and let $n \rightarrow \infty$ to get desired result. $\square$

Definition 11.4.14 (adjoint representation). Let G be a Lie group with Lie algebra $\mathfrak{g}$. For $g \in G$, let $c_g = L_g \circ R_{g^{-1}} \in \text{Aut}(G)$. We define the adjoint representation of G on $\mathfrak{g}$ by

$$\begin{aligned}\text{Ad} : G &\rightarrow \text{GL}(\mathfrak{g}) \\ g &\mapsto \text{Ad}(g) := T_e c_g\end{aligned}$$

Proposition 11.4.9. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Then

1. Ad is a morphism of Lie groups;
2. The differential of Ad is ad
3. $\text{Ad}(\exp X) = \exp(\text{ad}_X)$, $\forall X \in \mathfrak{g}$.

Proof. By (7) of Theorem 10.80, we have $\exp \circ T_e c_g = c_g \circ \exp$, that is

$$\exp(\text{Ad}(g)X) = g \exp(X) g^{-1}, \quad \forall X \in \mathfrak{g}$$

And (6) of Theorem 10.80 says that $\exp$ is a diffeomorphism in a local neighborhood of $0 \in \mathfrak{g}$, hence it has a smooth inverse. Thus $g \mapsto \text{Ad}(g)X$ is a smooth map from a neighborhood of $e \in G$ into $\text{GL}(\mathfrak{g})$. Obviously, $\text{Ad}(g_1 g_2) = \text{Ad } g_1 \circ \text{Ad } g_2$, since

$$T_e c_{g_1 g_2} = T_e(c_{g_1} \circ c_{g_2}) = T_e c_{g_1} \circ T_e c_{g_2}$$

Thus $g \mapsto \text{Ad}(g)$ is smooth everywhere.

For second, let's take $X, Y \in \mathfrak{g}$ and compute directly as follows

$$\begin{aligned} \exp(\text{Ad}(\exp tX)tY) &= \exp tX \exp tY (\exp tX)^{-1} \\ &= \exp(tY + t^2[X, Y] + O(t^3)) \end{aligned}$$

Thus

$$\text{Ad}(\exp tX)tY = tY + t^2[X, Y] + O(t^3) \implies \text{Ad}(\exp tX)Y = Y + t[X, Y] + O(t^2)$$

So for any $X, Y \in \mathfrak{g}$, we have

$$\begin{aligned} T_e \text{Ad}(X)(Y) &= \left. \frac{d}{dt} \right|_{t=0} \text{Ad}(\exp tX)Y \\ &= \left. \frac{d}{dt} \right|_{t=0} (Y + t[X, Y] + O(t^2)) \\ &= [X, Y] \\ &= \text{ad}_X Y \end{aligned}$$

as desired.

The third holds also from (7) of Theorem 10.80. □

Definition 11.4.15. Let V be a finite dimensional vector space, $A \in \text{End } V$, we define

$$f(A) = \frac{1 - \exp(-A)}{A} = \int_0^1 \exp(-sA) ds = \sum_{k=0}^{\infty} \frac{1}{(k+1)!} (-A)^k$$

We also define g to be the convergent power series expansions of $\frac{z \log z}{z-1}$ in the disk $|z-1| < r$, that is

$$g(1+u) = \frac{(1+u) \log(1+u)}{u} = 1 + \frac{u}{2} - \frac{u^2}{6} + \dots$$

We define $g(A)$ by this series for A such that $\|A - \text{id}\| < 1$.

Remark 11.4.7. $\exp(\log A) = A$ for $\|A - \text{id}\| < 1$ and $\log(\exp A) = A$ for $\|A\| < 2$.

Thus

$$f(A)g(\exp A) = \text{id}, \quad \text{for } \|A\| < 2$$

Theorem 11.4.6. Let G be a Lie group with Lie algebra $\mathfrak{g}$, $X \in \mathfrak{g}$. Then linear map $T_X \exp : \mathfrak{g} \rightarrow T_{\exp X} G$ is

$$\begin{aligned} T_X \exp &= T_e R_{\exp X} \circ f(-\text{ad}_X) \\ &= T_e L_{\exp X} \circ f(\text{ad}_X) \end{aligned}$$

where $f(A) = \frac{1 - \exp(-A)}{A} = \int_0^1 \exp(-sA) ds$.

Proof. For the first equality: From differential geometry we have the following fact: Let $v^\varepsilon \in \mathfrak{X}(M)$ be a smooth vector field on a manifold M depending smoothly on a parameter ε , and Φ_t^ε is its flow. Then the map $\varepsilon \mapsto \Phi_t^\varepsilon$ is smooth and

$$\frac{\partial}{\partial \varepsilon} \Phi_t^\varepsilon(p) = \int_0^t T_{\Phi_s^\varepsilon(p)}(\Phi_{t-s}^\varepsilon) \frac{\partial v^\varepsilon}{\partial \varepsilon}(\Phi_t^\varepsilon(p)) ds \in T_{\Phi_t^\varepsilon(p)}(M)$$

We apply this to $v_{X+\varepsilon Y} \in \mathfrak{X}_L(G)$, $X, Y \in T_e G$. Let Φ^ε denote the flow of $v_{X+\varepsilon Y}$. By (8) of Theorem 10.80, we have $\Phi_t^\varepsilon = R_{\exp(t(X+\varepsilon Y))}$. So

$$\begin{aligned} \frac{\partial}{\partial \varepsilon} \Phi_t^\varepsilon(e) &= \frac{\partial}{\partial \varepsilon} R_{\exp(X+\varepsilon Y)}(e) \\ &= \frac{\partial}{\partial \varepsilon} \exp(X + \varepsilon Y) \\ &= \int_0^1 T_{\exp s(X+\varepsilon Y)}(R_{\exp(1-s)(X+\varepsilon Y)}) \frac{\partial}{\partial \varepsilon} v_{X+\varepsilon Y}(R_{\exp s(X+\varepsilon Y)}(e)) ds \end{aligned}$$

Now we prove the second equality:

$$\begin{aligned} (T_e R_{\exp X})^{-1} \circ T_e L_{\exp X} \circ \int_0^1 \exp(-s \text{ad}_X) ds &= T_{\exp X} R_{(\exp X)^{-1}} \circ T_e L_{\exp X} \circ \int_0^1 \exp(-s \text{ad}_X) ds \\ &= T_e(\Psi_{\exp X}) \circ \int_0^1 \exp(-s \text{ad}_X) ds \\ &= \text{Ad}(\exp X) \circ \int_0^1 \exp(-s \text{ad}_X) ds \\ &= \exp(\text{ad}_X) \circ \int_0^1 \exp(-s \text{ad}_X) ds \\ &= \int_0^1 \exp((1-s) \text{ad}_X) ds \\ &= \int_0^1 \exp(u \text{ad}_X) du \\ &= f(\text{ad}_X) \end{aligned}$$

This completes the proof. $\square$

Now it's time to show the second principle, Recall that the second principle says: Let G, H be Lie groups, G is connected and simply connected. A linear map $T_e G \rightarrow T_e H$ is the differential of a morphism of Lie groups if and only if it preserves the Lie bracket.

So given a morphism of Lie algebras $\psi : \mathfrak{g} \rightarrow \mathfrak{h}$, we want to recover a $\rho : G \rightarrow H$ such that $T_e \rho = \psi$. The tool we use is the exponential map.

Let $U_e \subset G$ be a neighborhood of $e \in G$ such that $\log(g) = \exp^{-1}(g)$ exists for some $g \in G$.

We define

$$\rho(g) = \exp(\psi(\log(g))), \quad \forall g \in U_e \subset G$$

If we define in such a way, then we have

$$\rho(\exp(X)) = \exp(\psi(X)), \quad \forall X \in U_0 \subset \mathfrak{g}$$

We also need to show ρ is a group homomorphism. Suppose $g = \exp(X), h = \exp(Y)$ for $X, Y \in V \subset U_0 \subset \mathfrak{g}$ such that $\exp(X), \exp(Y), \exp(X)\exp(Y)$ are all in $U_e \subset G$.

$$\begin{aligned} \rho(gh) &= \rho(\exp(X)\exp(Y)) \\ &= \rho(\exp Z) \\ &= \exp(\psi(Z)) \end{aligned}$$

where $Z = \log(\exp(X)\exp(Y))$. We have seen last time

$$\exp(tX)\exp(tY) = \exp(t(X+Y) + \frac{t^2}{2}[X, Y] + O(t^3))$$

Assume that $Z = X+Y+F([X, Y])$, i.e. F depends on X, Y only through $[X, Y]$. Since ψ is a morphism of Lie algebras, then

$$\begin{aligned} \psi(Z) &= \psi(\log(\exp X \exp Y)) \\ &= \psi(X + Y + F([X, Y])) \\ &= \psi(X) + \psi(Y) + F([\psi(X), \psi(Y)]) \\ &= \log(\exp(\psi(X))\exp(\psi(Y))) \end{aligned}$$

Applying $\exp$ we have

$$\begin{aligned} \rho(gh) &= \exp(\psi(Z)) \\ &= \exp(\log(\exp(\psi(X))\exp(\psi(Y)))) \\ &= \exp(\psi(X))\exp(\psi(Y)) \\ &= \rho(g)\rho(h) \end{aligned}$$

So what is left is to show F do have the property we need. In fact, it's called Baker-Campbell-Hausdorff formula. And all the questions can be answered by looking at the differential of $\exp$.

Lemma 11.13. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Then $\exp : \mathfrak{g} \rightarrow G$ is a local diffeomorphism near $X \in \mathfrak{g}$ if and only if $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ has no eigenvalues of the form $2\pi i k$, where $k \in \mathbb{Z} \setminus \{0\}$.

Theorem 11.4.7. Let $\mathfrak{g}$ be a finite-dimensional Lie group over $\mathbb{R}$. Let S be the set of all singular points of $\exp$, and $V = \mathfrak{g} \setminus S$. V is an open neighborhood of 0 in $\mathfrak{g}$, and

$$f(\text{ad}_X) = \frac{1 - \exp(-\text{ad}_X)}{\text{ad}_X}$$

is invertible for $X \in V$. Thus $X \mapsto f(\text{ad}_X)^{-1}$ is an analytic map $V \rightarrow \text{End } \mathfrak{g}$. Let $t \mapsto Z(t)$ be a solution to the ODE

$$\begin{cases} \frac{d}{dt}Z(t) = f(\text{ad}_{Z(t)})^{-1}(X) \\ Z(0) = Y \end{cases}$$

Let $W = \{(X, Y) \in \mathfrak{g} \times V \mid Z(t) \text{ is defined for all } t \in [0, 1]\}$. Set $\mu(X, Y) = Z(1)$ for $(X, Y) \in W$. Then W is an open neighborhood of $(0, 0) \in \mathfrak{g} \times V$ and $\mu : W \rightarrow \mathfrak{g}$ is analytic. If $\mathfrak{g} = \text{Lie}(G)$ for a Lie group G with exponential map $\exp$, then

$$\exp(X)\exp(Y) = \exp(\mu(X, Y)), \quad (X, Y) \in W$$

Proof. Integral curves of smooth vector fields are solutions to an ODE, hence depends smoothly on initial values and on parameters. The map $(X, Z) \mapsto \frac{\text{ad}_Z}{1 - \exp(-\text{ad}_Z)}(X)$ is a smooth map $\mathfrak{g} \times V \rightarrow \mathfrak{g}$. So W is open and μ is smooth.

$$\begin{aligned} \frac{d}{dt} \exp(Z(t)) &= T_{Z(t)} \exp\left(\frac{dZ(t)}{dt}\right) \\ &= T_e L_{\exp Z(t)} \circ \frac{1 - \exp(-\text{ad}_{Z(t)})}{\text{ad}_{Z(t)}}\left(\frac{dZ(t)}{dt}\right) \\ &= T_e L_{\exp Z(t)}(X) \\ &= v_{X, \exp(Z(t))} \end{aligned}$$

Thus $t \mapsto \exp(Z(t))$ is an integral curve for the left-invariant vector field v_X starting at $\exp(Z(0)) = \exp(Y)$. By (8) of Theorem 10.80, we have

$$\exp(Z(t)) = R_{\exp tX} \exp Y = \exp tX \exp Y$$

for t for which $Z(t)$ is defined. Thus $\exp(Z(1)) = \exp X \exp Y$. $\square$

Remark 11.4.8. Let V, V_0 be neighborhoods of 0 in $\mathfrak{g}$, U a neighborhood of e in G such that $\exp|_V : V \rightarrow U$ is a diffeomorphism and for all $X, Y, Z \in V_0$ such that

$$(X, -Y) \in W, \quad (\mu(X, -Y), Z) \in W, \mu(\mu(X, -Y), Z) \in U$$

Their existence follows from the invertibility of $T_X \exp$. For each $g \in G$ we write $U_0^g = L_g \exp(V_0)$, $\psi^g(W) = \log(g^{-1}h)$, $h \in U_0^g$. Then

$$\{\psi^g : U_0^g \rightarrow V_0 \mid g \in G\}$$

from an atlas for G such that the group operations induced by μ are smooth.

Corollary 11.4.9 (Baker-Campbell-Hausdorff formula). Let G be a Lie group with Lie algebra $\mathfrak{g}$, V a connected open neighborhood of 0 in $\mathfrak{g}$, U an open neighborhood of e in G such that $\exp|_V$ is an isomorphism. Let $\log : U \rightarrow V$ such that

$$\log(\exp X) = X, \quad \forall X \in V \subset \mathfrak{g}$$

$$\exp(\log h) = h, \quad \forall h \in U \subset G$$

Let $V' \subset V$ be connected such that $\|\text{ad}_X\| \leq \frac{1}{2} \log 2$ for all $X \in V'$. Then for all $X, Y \in V'$

$$\log(\exp X \exp Y) = Y + \int_0^1 g(\exp(t \text{ad}_X) \exp(\text{ad}_Y)) dt$$

Proof. Recall

$$g(A) = \frac{(1+A)\log(1+A)}{A} = 1 + \frac{A}{2} - \frac{A^2}{6} + \dots$$

Define $t \rightarrow Z(t)$ by $\exp(Z(t)) = \exp tX \exp Y$. We have $Z(0) = Y, Z'(0) = X$. We want to prove

$$\frac{d}{dt}Z(t) = g(\exp(t \operatorname{ad}_X) \exp(\operatorname{ad}_Y))(X)$$

We know from the proof of the theorem that

$$\frac{d}{dt}Z(t) = f(\operatorname{ad}_{Z(t)})^{-1}(X)$$

We have:

$$\begin{aligned} \exp(\operatorname{ad}_{Z(t)}) &= \operatorname{Ad}(\exp(Z(t))) \\ &= \operatorname{Ad}(\exp tX \exp Y) \\ &= \operatorname{Ad}(\exp tX) \operatorname{Ad}(\exp Y) \\ &= \exp(\operatorname{ad}_{tX}) \exp(\operatorname{ad}_Y) \\ &= \exp(t \operatorname{ad}_X) \exp(\operatorname{ad}_Y) \end{aligned}$$

From Remark 10.89, choose $A = \operatorname{ad}_{Z(t)}$ and $\|X\| < \frac{1}{2} \log 2$ for all $X \in V'$, we have

$$f^{-1}(\operatorname{ad}_{Z(t)})^{-1} = g(\exp(\operatorname{ad}_{Z(t)})) = g(\exp t \operatorname{ad}_X \exp \operatorname{ad}_Y)$$

□

Remark 11.4.9. Working with Taylor series expansion for $Z(t)$ one finds

$$Z(1) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \dots$$

Theorem 11.4.8. Let G be a Lie group with Lie algebra $\mathfrak{g}$, for any Lie subalgebra $\mathfrak{h}$ of $\mathfrak{g}$, there exists a unique immersed connected Lie group H of G such that $\operatorname{Lie}(H) = \mathfrak{h}$. As a subset of G , H is equal to the subgroup of G generated by $\exp(\mathfrak{h})$.

Remark 11.4.10. This subgroup is not necessarily a closed subgroup of G . Let $G = \operatorname{GL}(2, \mathbb{C}), \mathfrak{g} = \mathfrak{gl}(2, \mathbb{C}), a \in \mathbb{Q}$, consider

$$\mathfrak{h} = \left\{ \begin{pmatrix} it & 0 \\ 0 & ita \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

Then

$$H = \exp(\mathfrak{h}) = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{ita} \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

We have $\dim H = 1$, but

$$\overline{H} = \exp(\mathfrak{h}) = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{i\varphi} \end{pmatrix} \mid \theta, \varphi \in \mathbb{R} \right\}$$

We have $\dim \overline{H} = 2$.

Example 11.4.2. Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and ad a morphism of Lie algebras. Then $\text{ad } \mathfrak{g}$ is a Lie subalgebra of $\mathfrak{gl}(\mathfrak{g})$. Let $\text{Ad } \mathfrak{g}$ be the unique connected subgroup of $\text{GL}(\mathfrak{g})$ generated by $\exp(\text{ad}_X), X \in \mathfrak{g}$ with $\text{Lie}(\text{Ad } \mathfrak{g}) = \text{ad } \mathfrak{g}$.

Definition 11.4.16 (adjoint group). $\text{Ad } \mathfrak{g}$ is called the adjoint group of $\mathfrak{g}$.

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Since $\exp(\text{ad}_X) = \text{Ad}_{\exp X}$, then $\text{Ad}(\mathfrak{g})$ is also the image of $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ if G is connected. In this situation, $\text{Ad } \mathfrak{g}$ is called the adjoint form of G .

Definition 11.4.17 (isogony). Let G, H be Lie groups. A morphism of Lie groups $\rho : G \rightarrow H$ is called isogony if ρ is a covering map.

Remark 11.4.11. Among all the Lie groups that are isogenous to each other, there are two distinguished ones:

1. $\tilde{G}$: The universal covering of G which is simply connected;
2. If $\rho : G \rightarrow H$ is an isogony, then $Z(G)$ is discrete if and only if $Z(H)$ is discrete. In that case, we have $G/Z(G) \cong H/Z(H)$. In particular, if $Z(\tilde{G})$ is discrete, then $\tilde{G}/Z(\tilde{G})$ coincides with $\text{Ad } \mathfrak{g}$, the adjoint form of G .

Isogenous Lie groups with discrete center have isomorphic Lie algebras.

Proposition 11.4.10 (Second principle). Let G, H be Lie groups with G connected and simply connected, and $\mathfrak{g}, \mathfrak{h}$ are their Lie algebras. A linear map $\psi : \mathfrak{g} \rightarrow \mathfrak{h}$ is the differential of a morphism of Lie groups $\rho : G \rightarrow H$ if and only if ψ is a morphism of Lie algebras.

Proof. Consider the product $G \times H$, its Lie algebra is $\mathfrak{g} \oplus \mathfrak{h}$. Let $\kappa \subset \mathfrak{g} \oplus \mathfrak{h}$ be the graph of ψ . Then ψ is a morphism of Lie algebras if and only if κ is a Lie subalgebra of $\mathfrak{g} \oplus \mathfrak{h}$. Indeed,

$$[X + \psi(X), Y + \psi(Y)] = [X, Y] + [\psi(X), \psi(Y)] = \psi([X, Y])$$

By the theorem on Lie subalgebra and Lie subgroups, we know there exists an immersed Lie subgroup $K \subset G \times H$ such that $T_e K \cong \kappa$. Let $\pi : K \rightarrow G$ be the projection onto the first factor. $T_e \pi : T_e K \rightarrow T_e G$ is an isomorphism. So $\pi : K \rightarrow G$ is an isogony. But G is simply connected, then π is an isomorphism. Let $\rho : K \cong G \rightarrow H$ be the projection to the second factor, then $T_e \rho = \psi$. $\square$

Example 11.4.3. Let $G = \text{SU}(2), H = \text{SO}(3, \mathbb{R})$. Let $\rho : G \rightarrow H$ be the covering homomorphism.

Remark 11.4.12 (Ado's theorem). Every finite dimensional Lie algebra over $\mathbb{R}$ has a finite-dimensional faithful representation. In other words, it's a subalgebra of $\mathfrak{gl}(V)$ for some finite dimensional vector space V .

Remark 11.4.13 (Lie's third theorem). Every finite dimensional Lie algebra over $\mathbb{R}$ is the Lie algebra of a connected Lie subgroup of $\text{GL}(n, \mathbb{C})$ for some n .

We end this section by the tensor product of representations of Lie algebras. Recall that for two representations of Lie groups $\rho_1 : G \rightarrow \text{GL}(V), \rho_2 : G \rightarrow \text{GL}(W)$. We have

$$(\rho_1 \otimes \rho_2)(g) := \rho_1(g) \otimes \rho_2(g)$$

Take $[\gamma]_e \in T_e G$ with $\gamma'(0) = X$. We know that X acts on $v \in V$ by

$$\begin{aligned} X(v) &= \left. \frac{d}{dt} \right|_{t=0} (\rho(\gamma(t))v) \\ &= T_{\gamma(t)}\rho \circ \left. \frac{d}{dt} \right|_{t=0} \rho(\gamma(t)) \end{aligned}$$

So we can define how does X acts on $v \otimes w$ for $v \in V, w \in W$.

$$\begin{aligned} X(v \otimes w) &= \left. \frac{d}{dt} \right|_{t=0} (\rho_1(\gamma(t)) \otimes \rho_2(\gamma(t)))(v \otimes w) \\ &= \left. \frac{d}{dt} \right|_{t=0} \rho_1(\gamma(t))(v) \otimes \rho_2(\gamma(t))(w) + \rho_1(\gamma(t))(v) \otimes \left. \frac{d}{dt} \right|_{t=0} \rho_2(\gamma(t))(w) \\ &= X(v) \otimes \text{id}_W(w) + \text{id}_V(v) \otimes X(w) \end{aligned}$$

That's why we define tensor product of representations of Lie algebras as follows:

Definition 11.4.18 (tensor product of representations of Lie algebras). Let $\rho_1 : \mathfrak{g} \rightarrow \mathfrak{gl}(V), \rho_2 : \mathfrak{g} \rightarrow \mathfrak{gl}(W)$ be representation of a Lie algebra $\mathfrak{g}$. Then we define

$$\begin{aligned} \rho_1 \otimes \rho_2 : \mathfrak{g} &\rightarrow \mathfrak{gl}(V \otimes W) \\ X &\mapsto (v \otimes w \mapsto X(v) \otimes \text{id}_W(w) + \text{id}_V(v) \otimes X(w)) \end{aligned}$$

12. ROUGH CLASSIFICATION OF LIE ALGEBRAS

12.1. Solvable and nilpotent Lie algebras.

Definition 12.1.1 (lower center series). Let $\mathfrak{g}$ be a Lie algebras, we define the lower center series $\mathfrak{g}_i$ by

$$\mathfrak{g}_0 = \mathfrak{g}, \mathfrak{g}_1 = [\mathfrak{g}_0, \mathfrak{g}], \dots, \mathfrak{g}_{j+1} = [\mathfrak{g}_j, \mathfrak{g}]$$

Definition 12.1.2 (derived series). Let $\mathfrak{g}$ be a Lie algebras, we define the derived series $\mathfrak{g}^i$ by

$$\mathfrak{g}^0 = \mathfrak{g}, \mathfrak{g}^1 = [\mathfrak{g}^0, \mathfrak{g}^0], \dots, \mathfrak{g}^{j+1} = [\mathfrak{g}^j, \mathfrak{g}^j]$$

Definition 12.1.3 (nilpotent). $\mathfrak{g}$ is called nilpotent, if $\mathfrak{g}^k = 0$ for some $k \geq 0$.

Definition 12.1.4 (solvable). $\mathfrak{g}$ is called solvable, if $\mathfrak{g}^k = 0$ for some $k \geq 0$.

Definition 12.1.5 (semisimple). $\mathfrak{g}$ is semisimple, if $\mathfrak{g}$ has no nonzero solvable ideals.

Definition 12.1.6 (simple). $\mathfrak{g}$ is simple, if $\dim \mathfrak{g} > 1$ and $\mathfrak{g}$ has no nonzero ideals.

Lemma 12.1. Each $\mathfrak{g}^i$ and each $\mathfrak{g}_i$ is an ideal in $\mathfrak{g}$. Moreover $\mathfrak{g}^i \subset \mathfrak{g}_i$ for all i .

Lemma 12.2. The following are equivalent:

1. $\mathfrak{g}$ is solvable;
2. $\mathfrak{g}$ has a sequence of Lie subalgebras $\mathfrak{g} = \mathfrak{h}_0 \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_k = 0$ such that $\mathfrak{h}_{i+1}$ is an ideal in $\mathfrak{h}_i$ and $\mathfrak{h}_i/\mathfrak{h}_{i+1}$ is abelian;

Lemma 12.3. The following are equivalent:

1. $\mathfrak{g}$ is nilpotent;
2. $\mathfrak{g}$ has a sequence of ideals $\mathfrak{g} = \mathfrak{h}_0 \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_k = 0$ such that $\mathfrak{h}_i/\mathfrak{h}_{i+1} \in Z(\mathfrak{g}/\mathfrak{h}_{i+1})$;
3. $\text{ad}_{X_1} \circ \cdots \circ \text{ad}_{X_k} Y = 0$ for some $k \in \mathbb{N}$ and all $X_1, X_2, \dots, X_k, Y \in \mathfrak{g}$.

Proposition 12.1.1. Any subalgebra or quotient algebra of a solvable(nilpotent) Lie algebra is solvable(nilpotent).

Proof. If $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$, then by induction $\mathfrak{h}^k \subset \mathfrak{g}^k$. Hence $\mathfrak{g}$ is solvable implies $\mathfrak{h}$ is solvable; If $\pi : \mathfrak{g} \rightarrow \mathfrak{h}$ is a surjective morphism of Lie algebra, then

$$\pi(\mathfrak{g}^k) = \pi([\mathfrak{g}^{k-1}, \mathfrak{g}^{k-1}]) = [\pi(\mathfrak{g}^{k-1}), \pi(\mathfrak{g}^{k-1})] = [\mathfrak{h}^{k-1}, \mathfrak{h}^{k-1}] = \mathfrak{h}^k$$

Hence $\mathfrak{g}$ is solvable implies $\mathfrak{h}$ is solvable. For the nilpotent, the argument is analogous. $\square$

Proposition 12.1.2. If $\mathfrak{a}$ is a solvable ideal in $\mathfrak{g}$ and if $\mathfrak{g}/\mathfrak{a}$ is solvable, then $\mathfrak{g}$ is solvable.

Proof. Let $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ be the projection morphism. Suppose $(\mathfrak{g}/\mathfrak{a})^k = 0$, since $\mathfrak{g}/\mathfrak{a}$ is solvable. By induction, $\pi(\mathfrak{g}^j) = (\mathfrak{g}/\mathfrak{a})^j$ for all j , so we have $\pi(\mathfrak{g})^k = 0$. In other words, $\mathfrak{g}^k \subset \mathfrak{a}$. Together with the fact that $\mathfrak{a}$ is solvable, we obtain the desired result. $\square$

Remark 12.1.1. The analogous statement for nilpotent Lie algebra is false. Let $\mathfrak{n}(n, \mathbb{R})$ be the Lie algebra of strictly upper triangular matrices. $\mathfrak{n}(n, \mathbb{R}) \subset \mathfrak{b}(n, \mathbb{R})$ is a nilpotent subalgebra. The quotient $\mathfrak{b}(n, \mathbb{R})/\mathfrak{n}(n, \mathbb{R})$ is the diagonal matrices, and it's nilpotent since it's abelian, but $\mathfrak{b}(n, \mathbb{R})$ is not nilpotent.

Proposition 12.1.3. Let $\mathfrak{g}$ is a finite-dimensional Lie algebra. There is a unique solvable ideal of $\mathfrak{g}$ containing all solvable ideals.

Proof. Since $\mathfrak{g}$ is finite-dimensional, it suffices to show that the sum of two solvable ideals is solvable. Let $\mathfrak{a}, \mathfrak{b}$ be solvable ideals, $\mathfrak{h} = \mathfrak{a} + \mathfrak{b}$ is an ideal and

$$\mathfrak{h}/\mathfrak{a} \cong \mathfrak{b}/\mathfrak{a} \cap \mathfrak{b}$$

But since $\mathfrak{b}$ is solvable, then $\mathfrak{b}/\mathfrak{a} \cap \mathfrak{b}$ are solvable, so $\mathfrak{h}/\mathfrak{a}$ is solvable, but $\mathfrak{a}$ is solvable, so $\mathfrak{h}$ is solvable. $\square$

Definition 12.1.7 (radical). The unique maximal solvable ideal of a finite-dimensional Lie algebra $\mathfrak{g}$ is called the radical of $\mathfrak{g}$, denoted by $\text{rad}(\mathfrak{g})$.

Proposition 12.1.4. Some properties about simple and semisimple:

1. $\mathfrak{g}$ is simple then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$;
2. $\mathfrak{g}$ is simple implies $\mathfrak{g}$ is semisimple;

3. $\mathfrak{g}$ is semisimple implies $Z(\mathfrak{g}) = 0$.

Proof. (1) Let $\mathfrak{g}$ be simple, then $[\mathfrak{g}, \mathfrak{g}]$ is an ideal, hence either 0 or $\mathfrak{g}$. since $\mathfrak{g}$ is not abelian, then it is $\mathfrak{g}$.

(2) $\text{rad}(\mathfrak{g})$ is an ideal, hence either 0 or $\mathfrak{g}$. If $\text{rad}(\mathfrak{g}) = \mathfrak{g}$, then $\mathfrak{g}$ is solvable and $[\mathfrak{g}, \mathfrak{g}] \subsetneq \mathfrak{g}$, a contradiction. So $\text{rad}(\mathfrak{g}) = 0$, which means $\mathfrak{g}$ is semisimple.

(3) Since $Z(\mathfrak{g})$ is an abelian ideal, $\mathfrak{g}$ is semisimple implies that $Z(\mathfrak{g})$ is zero. $\square$

Proposition 12.1.5. Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. Then $\mathfrak{g}/\text{rad}(\mathfrak{g})$ is semisimple.

Proof. Let $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/\text{rad}(\mathfrak{g})$ be the projection. Let $\mathfrak{h} \subset \mathfrak{g}/\text{rad}(\mathfrak{g})$ be a solvable ideal, we need to show $\mathfrak{h}$ is trivial.

Consider $\mathfrak{a} = \pi^{-1}(\mathfrak{h}) \subset \mathfrak{g}$, so we have $\pi(\mathfrak{a})$ is solvable. Note that $\ker \pi|_{\mathfrak{a}} \subset \text{rad}(\mathfrak{g})$, since $\ker \pi = \text{rad}(\mathfrak{g})$. So we have $\ker \pi|_{\mathfrak{a}}$ is solvable, since any subalgebra of a solvable algebra is solvable. Together with $\mathfrak{h}$ is solvable and Proposition 11.11, we have $\mathfrak{a}$ is solvable, then $\mathfrak{a} \subset \text{rad}(\mathfrak{g})$, that is $\mathfrak{h} = 0$. $\square$

Remark 12.1.2. This proposition means that any finite-dimensional Lie algebra $\mathfrak{g}$ fits into a short exact sequence

$$0 \rightarrow \underbrace{\text{rad}(\mathfrak{g})}_{\text{solvable}} \rightarrow \mathfrak{g} \rightarrow \underbrace{\mathfrak{g}/\text{rad}(\mathfrak{g})}_{\text{semisimple}} \rightarrow 0$$

In fact, one can show that this sequence always splits. The Levi decomposition of $\mathfrak{g}$. Let $\mathfrak{g}$ be a Lie algebra with radical $\mathfrak{r}$. Then there exists a semisimple Lie subalgebra $\mathfrak{s}$ such that

$$\mathfrak{g} = \mathfrak{r} + \mathfrak{s}$$

Proposition 12.1.6. $\mathfrak{g}$ is a finite-dimensional Lie algebra, $\mathfrak{g}$ is semisimple if and only if $\mathfrak{g}$ has no abelian ideals.

Proof. If $\mathfrak{g}$ is semisimple, then $\text{rad}(\mathfrak{g}) = 0$. Suppose $\mathfrak{a} \neq 0$ is an abelian ideal of $\mathfrak{g}$, then $\mathfrak{a}$ is a solvable ideal, thus $\mathfrak{a} \subset \text{rad}(\mathfrak{g}) = 0$.

For the other direction, if $\mathfrak{g}$ is not semisimple, so $\mathfrak{r} = \text{rad}(\mathfrak{g})$ is non-zero. Let k be the smallest integer such that $\mathfrak{r}^k = 0$ and $\mathfrak{r}^{k-1} \neq 0$, so $\mathfrak{r}^{k-1}$ is an abelian ideal. $\square$

Definition 12.1.8 (invariant subspace). Let $\mathfrak{g}$ be a Lie algebra, V a finite-dimensional vector space, $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ a representation. A subspace $W \subset V$ such that $\rho(g)W \subset W$ is called invariant.

Definition 12.1.9 (irreducible). A representation V of $\mathfrak{g}$ is called irreducible if the only invariant subspace are 0 and V , also called simple representation.

Definition 12.1.10 (complete reducible). A representation V of $\mathfrak{g}$ is called complete reducible if every invariant subspace of V has a complementary invariant subspace, also called semisimple.

Proposition 12.1.7. $\mathfrak{g}$ is a finite-dimensional Lie algebra, $\mathfrak{g}$ is nilpotent if and only if $\text{ad } \mathfrak{g}$ is nilpotent.

12.2. Engel and Lie's themsub.

Theorem 12.2.1 (Engel's theorem). Let $V \neq 0$ be a finite-dimensional vector space, $\mathfrak{g}$ a Lie algebra of nilpotent endomorphisms of V . Then there exists $0 \neq v \in V$ such that $Xv = 0$ for all $X \in \mathfrak{g}$.

Proof. Induction on dimension of $\mathfrak{g}$: If $\dim \mathfrak{g} = 1$, then $X \in \mathfrak{g}$ is nilpotent implies $X = 0$; Suppose the claim holds for dimensions $< \dim \mathfrak{g}$ and $\dim \mathfrak{g} > 1$.

We construct a nilpotent ideal $\mathfrak{h} \subset \mathfrak{g}$ of codimension 1 as follows: Let $\mathfrak{h}$ be a proper Lie subalgebra of maximal dimension in $\mathfrak{g}$. We need to show $\mathfrak{h}$ is an ideal and $\text{codim}_{\mathfrak{g}} \mathfrak{h} = 1$. Since $\text{ad} \mathfrak{h}$ leaves $\mathfrak{h}$ invariant, then we have a representation

$$\begin{aligned} \rho : \mathfrak{h} &\rightarrow \mathfrak{gl}(\mathfrak{g}/\mathfrak{h}) \\ X &\mapsto (Y + \mathfrak{h} \mapsto [X, Y] + \mathfrak{h}) \end{aligned}$$

We show $\rho(X)$ is nilpotent for all $X \in \mathfrak{h}$ as follows: Let $Y \in \mathfrak{g}$, then

$$\begin{aligned} \text{ad}_X^{2m} Y &= \text{ad}_X^{2m-1}(XY - YX) \\ &= \text{ad}_X^{2m-2}(X^2Y - 2XYX + YX^2) \\ &= \sum_{j=0}^{2m} c_j X^j Y X^{2m-j} \end{aligned}$$

But $X^m = 0$ for some $m \in \mathbb{N}$, so we will have $\text{ad}_X^{2m} = 0$, that is ad_X is nilpotent, so ad_X is nilpotent on $\mathfrak{g}/\mathfrak{h}$.

We know that the dimension of $\mathfrak{h}$ is strictly less than the dimension of $\mathfrak{g}$ so the induction hypothesis implies that there exists $Y + \mathfrak{h} \neq \mathfrak{h}$ in $\mathfrak{g}/\mathfrak{h}$ such that

$$\rho(X)(Y + \mathfrak{h}) = \mathfrak{h}, \quad \forall X \in \mathfrak{h}$$

In other words, $[X, Y] \in \mathfrak{h}, \forall X \in \mathfrak{h}$. Let $\mathfrak{s} = \mathfrak{h} + \mathbb{C}Y$, $\mathfrak{s}$ is a subalgebra of $\mathfrak{g}$ properly containing $\mathfrak{h}$ so $\mathfrak{s} = \mathfrak{g}$ by maximality of $\mathfrak{h}$. Thus the codimension of $\mathfrak{h}$ is 1. Moreover, $\mathfrak{h}$ is an ideal.

Let $W = \{v \in V \mid Xv = 0, \forall X \in \mathfrak{h}\}$, induction hypothesis implies $W \neq 0$, let $v \in W, X \in \mathfrak{h}, Y \in \mathfrak{g}$, then

$$XYv = [X, Y]v + YXv = 0$$

Therefore we see that $Y(W) \subseteq W$. By assumption, Y is nilpotent, so 0 is its only eigenvalue. Y has an eigenvector $w \in W$ such that $Yw = 0$. But $\mathfrak{h}w = 0$, so $\mathfrak{g}(w) = 0$, since $\mathfrak{g} = \mathfrak{h} + \mathbb{C}Y$. This completes the proof. $\square$

Corollary 12.2.1. Let $\mathfrak{g}$ be a nilpotent Lie algebra, $V \neq 0$ be a finite-dimensional vector space, $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ a representation. Then there exists a sequence of subspaces $V = V_0 \supset V_1 \supset \cdots \supset V_m = 0$ such that $\rho(X)V_i \subset V_{i+1}, \forall X \in \mathfrak{g}$. Hence V has a basis in terms of which the matrix representation of each $X \in \mathfrak{g}$ is strictly upper triangular.

Proof. Standard. $\square$

Corollary 12.2.2. $\mathfrak{g}$ is a Lie algebra, if $\text{ad}_X \in \mathfrak{gl}(\mathfrak{g})$ is nilpotent for all $X \in \mathfrak{g}$, then $\mathfrak{g}$ is nilpotent.

Proof. Engel's theorem implies that $\text{ad } g$ is nilpotent, by Proposition 11.22, we have g is nilpotent. $\square$

Lemma 12.4. Let $\mathfrak{h}$ be an ideal in a Lie algebra g , $\rho : g \rightarrow \mathfrak{gl}(V)$ a representation, $\lambda : \mathfrak{h} \rightarrow \mathbb{C}$ a linear function on $\mathfrak{h}$. Set

$$W = \{v \in V \mid X(v) = \lambda(X)v, \forall X \in \mathfrak{h}\}$$

Then $YW \subset W$, for all $Y \in g$.

Proof. Let $0 \neq w \in W, X \in \mathfrak{h}, Y \in g$, then $[X, Y] \in \mathfrak{h}$, since $\mathfrak{h}$ is an ideal. Use X to act on Yw , we have

$$\begin{aligned} XYw &= [X, Y]w + YXw \\ &= \lambda([X, Y])w + \lambda(X)Yw \end{aligned}$$

So $Yw \in W$ if and only if $\lambda([X, Y]) = 0$ for all $X \in \mathfrak{h}$.

Let $U = \langle Y^n w, n \in \mathbb{N} \rangle_{\mathbb{C}}$, clearly $YU \subset U$. We show that $XU \subset U$ for all $X \in \mathfrak{h}$. Indeed, we first show

$$XY^n w \equiv \lambda(X)Y^n w \pmod{\langle w, Yw, \dots, Y^{n-1}w \rangle_{\mathbb{C}}}$$

This is true for $n = 0$ since $w \in W$. Assume it's true for n . Then

$$\begin{aligned} XY^{n+1} &= XY^ny \\ &= [X, Y]Y^n w + YXY^n w \\ &\equiv \lambda([X, Y])Y^n w + YXY^n w \pmod{\langle w, Yw, \dots, Y^{n-1}w \rangle_{\mathbb{C}}} \\ &\equiv \lambda([X, Y])Y^n w + \lambda(X)Y^{n+1}w \pmod{\langle w, Yw, \dots, Y^{n-1}w, Yw, \dots, Y^n w \rangle_{\mathbb{C}}} \\ &\equiv \lambda(X)Y^{n+1}w \pmod{\langle w, Yw, \dots, Y^n w \rangle_{\mathbb{C}}} \end{aligned}$$

This completes the induction, so we have $XU \subset U$. In fact, if we use $\{w, Yw, \dots, Y^n w\}$ as a basis for U , we have the matrix representation for $\rho(X)$ as a upper triangular matrix, with all of diagonal elements are $\lambda(X)$. So trace of X is $\dim U \lambda(X)$, then

$$\lambda([X, Y])\dim U = \text{tr}([X, Y]) = 0$$

So $\lambda([X, Y]) = 0$. $\square$

Theorem 12.2.2 (Lie's theorem). Let g be a solvable Lie algebra, $0 \neq V$ a finite-dimensional vector space, $\rho : g \rightarrow \mathfrak{gl}(V)$ a representation. Then there exists a simultaneous eigenvector $0 \neq v \in V$ for all $\rho(X), X \in g$.

Proof. Induction on $\dim g$. If $\dim g = 1$, then $\rho(X)$ consists of multiples of a single $X \in g$. If $\dim g > 1$, construct an ideal $\mathfrak{h} \subset g$ of codimension 1 as follows: Since g is solvable, then $g^1 \neq g$, then $\mathfrak{a} = g/g^1$ is a non-zero abelian Lie algebra. Choose a subspace $\mathfrak{h} \subset g$ with codimension 1, and $[g, g] \subseteq \mathfrak{h}$. Then

$$[\mathfrak{h}, g] \subset [g, g] = \mathfrak{h}$$

Thus $\mathfrak{h}$ is an ideal and solvable. By induction, we can assume that there exists $v_0 \in V$ such that $Xv_0 = \lambda(X)v_0$ for all $X \in \mathfrak{h}$ for a linear function $\lambda : \mathfrak{h} \rightarrow \mathbb{C}$.

Let $W = \{v \in V \mid Xv = \lambda(X)v, \forall X \in \mathfrak{h}\} \neq 0$. Fix $Y \in g \setminus \mathfrak{h}$ with $g = \mathfrak{h} + CY$. Now it suffices to show any $w \in W$ is an eigenvector of Y . By Lemma 11.26, we

have $YW \subset W$ for all $Y \in \mathfrak{g}$. So $v_0 \in V$ is an eigenvector for all $Y \in \mathfrak{g}$. This completes the proof. $\square$

Corollary 12.2.3. Let $\mathfrak{g}$ be a solvable Lie algebra, $0 \neq V$ a finite-dimensional vector space, $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ a representation. Then there exists a sequence of subspaces

$$V = V_0 \supset V_1 \supset \cdots \supset V_m = 0$$

such that each V_i is an invariant subspace and $\dim V_i/V_{i+1} = 1$. Hence V has a basis in terms of which the matrix representation of each $X \in \mathfrak{g}$ is upper triangular.

Corollary 12.2.4. Any irreducible representation ρ of a solvable Lie algebra $\mathfrak{g}$ is of dimension 1. Moreover, $\rho([\mathfrak{g}, \mathfrak{g}]) = 0$.

Proof. By Lie's theorem, there exists $0 \neq v \in V$ such that for all $X \in \mathfrak{g}$, $\rho(X)v = \mathbb{C}v$, i.e. $\mathbb{C}v$ is an invariant subspace. So we know that any irreducible representation of a solvable Lie algebra is 1-dimensional. For any $X, Y \in \mathfrak{g}$ and $v \in V$,

$$\begin{aligned} \rho([X, Y])v &= \rho(X)\rho(Y)v - \rho(Y)\rho(X)v \\ &= \lambda(X)\lambda(Y)v - \lambda(Y)\lambda(X)v \\ &= 0 \end{aligned}$$

This completes the proof. $\square$

Proposition 12.2.1. Let $\mathfrak{g}$ be a complex Lie algebra, $\mathfrak{g}_{ss} = \mathfrak{g}/\text{rad}(\mathfrak{g})$. Every irreducible representation of $\mathfrak{g}$ is of the form $V = V_0 \otimes L$ where V_0 is an irreducible representation of $\mathfrak{g}_{ss}$ and $\dim L = 1$.

Proof. We apply Lie's theorem to $\text{rad}(\mathfrak{g})$, that is, there exists a linear function $\lambda : \mathfrak{g} \rightarrow \mathbb{C}$ such that

$$W = \{v \in V \mid Xv = \lambda(X)v, \forall X \in \text{rad}(\mathfrak{g})\} \neq 0$$

By Lemma 11.26, we have $YW \subset W, \forall Y \in \mathfrak{g}$. But V is irreducible, so we have $W = V$. Thus $\text{tr}(X) = \dim V \lambda(X), \forall X \in \text{rad}(\mathfrak{g})$. So we have

$$\lambda|_{\text{rad}(\mathfrak{g}) \cap [\mathfrak{g}, \mathfrak{g}]} = 0$$

since $\text{tr}([\mathfrak{g}, \mathfrak{g}]) = 0$. Let $\tilde{\lambda} : \mathfrak{g} \rightarrow \mathbb{C}$ be a linear function such that

$$\begin{cases} \tilde{\lambda}|_{\text{rad}(\mathfrak{g})} = \lambda \\ \tilde{\lambda}|_{[\mathfrak{g}, \mathfrak{g}]} = 0 \end{cases}$$

Thus $\tilde{\lambda}$ determines a representation $\mathfrak{g} \rightarrow \mathfrak{gl}(L)$ with $\dim L = 1$. In other words, $Y(z) = \tilde{\lambda}(Y)z, \forall Y \in \mathfrak{g}$.

$X \in \text{rad}(\mathfrak{g})$ acts on $L^\vee = \text{Hom}(L, \mathbb{C})$ by

$$(Xf)(z) = -f(Xz) = -\lambda(X)f(z), \quad f \in \text{Hom}(L, \mathbb{C})$$

The action of $\mathfrak{g}$ on $V \otimes L^\vee$ is then

$$Y(v \otimes f) = Yv \otimes f + v \otimes Yf, \quad v \in V, f \in L^\vee$$

So

$$X(v \otimes f) = \lambda(X)v \otimes f - \lambda(X)v \otimes f = 0, \quad \forall X \in \text{rad}(\mathfrak{g})$$

Thus $V \otimes L^\vee$ is a trivial representation on $\text{rad}(\mathfrak{g})$, it comes from a representation of $\mathfrak{g}/\text{rad}(\mathfrak{g})$. $\square$

Theorem 12.2.3. Let V be a representation of a semisimple Lie algebra $\mathfrak{g}$, $W \subset V$ an invariant subspace. Then there exists a complementary invariant subspace W' .

Proof. Fulton-Harris, Appendix C. $\square$

12.3. Jordan–Chevalley decomposition. Recall from linear algebra: Let V be a vector space over an algebraically closed field, $X \in \text{End}(V)$. Then X has a Jordan normal form. From the Jordan normal form, we can directly see that there is a decomposition of $X = X_s + X_n$, such that X_s is diagonalizable, and X_n is nilpotent. But is this decomposition unique? or it has any other properties? That's what Jordan–Chevalley decomposition tells us.

Proposition 12.3.1 (Jordan–Chevalley decomposition). Let V be a finite-dimensional vector space, $X \in \text{End}(V)$, then

1. There exists unique $X_s, X_n \in \text{End}(V)$ such that $X = X_s + X_n$, where X_s is semisimple, X_n is nilpotent and $[X_s, X_n] = 0$;
2. There exists $p, q \in \mathbb{C}[T], p(0) = q(0) = 0$ such that $X_s = p(X), X_n = q(X)$. In particular, X_s, X_n commute with any endomorphisms commuting with X ;
3. If $A \subset B \subset V$ are subspaces, $X(B) \subseteq A$, then $X_s(B) \subseteq A, X_n(B) \subseteq A$.

Definition 12.3.1 (Jordan–Chevalley decomposition). Given a finite-dimensional vector space V and $X \in \text{End}(V)$. The decomposition $X = X_s + X_n$ is called the Jordan–Chevalley decomposition of X .

Remark 12.3.1. Let $\mathfrak{g}$ be an arbitrary Lie algebra, $X \in \mathfrak{g}$, how can we define the Jordan–Chevalley decomposition of X ? Choose $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ a representation and consider the Jordan–Chevalley decomposition of $\rho(X)$ may be a good way.

But $\rho(X)_s, \rho(X)_n$ sometimes do not lie in $\rho(\mathfrak{g})$. For example, let $\mathfrak{g} = \mathbb{C}$, and consider the following representations

$$\begin{aligned} \rho_1 : t &\mapsto (t) \\ \rho_2 : t &\mapsto \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \\ \rho_3 : t &\mapsto \begin{pmatrix} t & t \\ 0 & 0 \end{pmatrix} \end{aligned}$$

For the former two representations, we have

$$\begin{cases} \rho_1(X)_s = \rho_1(X) \\ \rho_1(X)_n = 0 \end{cases} \quad \begin{cases} \rho_2(X)_s = 0 \\ \rho_2(X)_n = \rho_2(X) \end{cases}$$

But for the third representation, $\rho_3(X)_s, \rho_3(X)_n \notin \rho(\mathfrak{g})$.

Theorem 12.3.1. Let $\mathfrak{g}$ be a semisimple Lie algebra. For any $X \in \mathfrak{g}$, there exists $X_s, X_n \in \mathfrak{g}$ such that for all representation $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, we have

$$\begin{aligned}\rho(X)_s &= \rho(X_s) \\ \rho(X)_n &= \rho(X_n)\end{aligned}$$

Proof. Fulton-Harris, Appendix C. □

13. EXAMPLES OF LIE ALGEBRAS IN SMALL DIMENSIONS

13.1. Dimension one. Firstly, let's consider the complex Lie algebra: Any complex Lie algebra of dimension 1 is abelian, that is $\mathbb{C}$ with $[X, X] = 0, \forall X \in \mathbb{C}$. The simply connected Lie group G with $\text{Lie}(G) = \mathbb{C}$ is $(\mathbb{C}, +)$.

All the other connected Lie groups with $\text{Lie}(G) = \mathbb{C}$ are of the form $\mathbb{C}/\Lambda$ for a discrete subgroup Λ . Now the rank of Λ matters.

If rank of Λ is 1, then $\mathbb{C}/\Lambda \sim \mathbb{R} \times \mathbb{R}/\Lambda \sim \mathbb{R} \times S^1 \sim \mathbb{C}^*$.

If rank of Λ is 2, then $\mathbb{C}/\Lambda$ is a complex torus of dimension 1. We have other equivalent descriptions, such as a Riemann surface with genus 1, elliptic curves over $\mathbb{C}$ or more explicitly

$$\{(x, y) \in \mathbb{C}^2 \mid y^3 = 4x^3 + ax + b, a, b \in \mathbb{C}, 4a^3 - 27b^2 \neq 0\} \cup \{\infty\}$$

In course of Riemann surface, you will learn that $\Lambda = 1 \cdot \mathbb{Z} + \tau \mathbb{Z}, \tau \in \mathbb{H} = \{z \in \mathbb{C} \mid \text{Im } z > 0\}$. And $\mathbb{C}/\Lambda_\tau \cong \mathbb{C}/\Lambda_{\tau'}$ if and only if there exists $A \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z})$

such that $\tau' = \frac{a\tau + b}{c\tau + d}$. So the set of isomorphism classes is $\mathbb{H}/\text{SL}(2, \mathbb{Z})$. Or in a more fancy word, the moduli space of complex tori is $\mathbb{H}/\text{SL}(2, \mathbb{Z})$.

Over real numbers, $(\mathbb{R}, [\cdot, \cdot] = 0)$ is the only Lie algebra of dimension 1, and $(\mathbb{R}, +)$ is the corresponding simply connected Lie group. $(\mathbb{R}/\mathbb{Z}, +) \cong S^1$ is the only other connected real Lie groups.

13.2. Dimension two. $\mathfrak{g} = \mathbb{C}^2$. Now the abelian cases come out, since $[\cdot, \cdot]$ is not always trivial.

13.2.1. Abelian case. Assume $\mathfrak{g}$ is abelian. If G is simply connected, then $G = \mathbb{C}^2$. If G is connected, then $\mathbb{C}^2/\Lambda$, where $\Lambda \subset \mathbb{C}^2$ is a discrete subgroup of rank 1, 2, 3, 4.

If rank of Λ is 1, then $\Lambda = \mathbb{Z}e_1 \subset \mathbb{C}e_1 + \mathbb{C}e_2$, so $G \cong \mathbb{C}^* \times \mathbb{C}$.

If rank of Λ is 2, there are two possibilities: either Λ lies in a one-dimensional complex subspace of $\mathbb{C}^2$ or not. If it does not, then basis of Λ will also be a basis of $\mathbb{C}^2$, thus in this case we have $G = \mathbb{C}^* \times \mathbb{C}^*$. If Λ do lies in a complex subspace of dimension 1. Then $\Lambda = (\mathbb{Z} + \mathbb{Z}\tau)e_1, \tau \in \mathbb{C} \setminus \mathbb{R}$. So $G = \mathbb{C}/\Lambda \times \mathbb{C} \cong E \times E$, where E is a elliptic curve.

If rank of Λ is 3. If any rank 2 sublattice $\Lambda' \subset \Lambda$ lies a complex subspace of dimension 1, then

$$\Lambda = (\mathbb{Z} + \tau \mathbb{Z})e_1 + \mathbb{Z}e_2$$

In this case we have $G = E \times \mathbb{C}^*$. If no such sublattice exists, then

$$\Lambda = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}(ae_1 + \beta e_2), \quad \beta \in \mathbb{C} \setminus \mathbb{R}$$

Consider

$$\begin{array}{ccc} \mathbb{C}^2 & \longrightarrow & \mathbb{C}^2/\mathbb{C}e_1 = \mathbb{C} \\ \downarrow & & \downarrow \\ G = \mathbb{C}^2/\mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}(\alpha e_1 + \beta e_2) & \longrightarrow & \mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}\beta) \end{array}$$

So G is a bundle over $E = \mathbb{C}/\mathbb{Z} + \mathbb{Z}\beta$, with fiber $\mathbb{C}^*$. Exchanging e_1 and e_2 yields a non isomorphic elliptic curves E' as the basis. As algebraic varieties, the two bundles are not isomorphic, but the corresponding groups are isomorphic as complex Lie groups.

If rank of Λ is 4, we only know how to describe those complex tori of dimension 2 that can be embedded into complex projective space. These complex tori are known as abelian varieties. However, the set of those only forms a dense subset in the set of complex tori.

13.2.2. *Real cases.* Over $\mathbb{R}$, there are only three cases, that is

$$G = \mathbb{R} \times \mathbb{R}$$

$$G = \mathbb{R} \times S^1$$

$$G = S^1 \times S^1$$

13.2.3. *Non abelian cases.* Assume $\mathfrak{g}$ is non abelian. View $[\cdot, \cdot] : \wedge^2 \mathfrak{g} \rightarrow \mathfrak{g}$, linear, and $[\cdot, \cdot] \neq 0$, then $\dim \text{im}[\cdot, \cdot] = 1$. Choose a basis $\{X, Y\}$ for $\mathfrak{g}$ such that $\text{im}([\cdot, \cdot]) = \mathbb{C}X$, then

$$[X, Y] = \alpha X, \alpha \neq 0$$

If we replace Y by $\frac{1}{\alpha}Y$, then we have $[X, Y] = X$. This determines $\mathfrak{g}$ uniquely. There exists unique non abelian Lie algebra with dimension 2 over $\mathbb{R}$ or $\mathbb{C}$.

In order to find out the Lie groups with such Lie algebra. Recall $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is faithful if $\mathfrak{g}$ is semisimple. In this case

$$\begin{cases} \text{ad}_X X = 0 \\ \text{ad}_X Y = X \end{cases} \quad \begin{cases} \text{ad}_Y X = -X \\ \text{ad}_Y Y = 0 \end{cases}$$

that is

$$\text{ad}_X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \text{ad}_Y = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$$

So we have

$$\text{ad } \mathfrak{g} = \left\{ \begin{pmatrix} * & * \\ 0 & 0 \end{pmatrix} \right\} \subset \mathfrak{gl}(2, \mathbb{C})$$

Consider

$$\begin{aligned} G_0 = \exp(\text{ad } \mathfrak{g}) &= \left\{ \begin{pmatrix} e^t & s \\ 0 & 1 \end{pmatrix} \mid s, t \in \mathbb{C} \right\} \subset \text{GL}(2, \mathbb{C}) \\ &\cong \mathbb{C} \times \mathbb{C}^*, \quad \text{as complex manifold} \end{aligned}$$

To take its universal cover, we write a general number of G_0 as

$$\begin{pmatrix} e^t & s \\ 0 & 1 \end{pmatrix}$$

And the product of two such matrices is given by

$$\begin{pmatrix} e^t & s \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} e^{t'} & s' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{t+t'} & s + e^t s' \\ 0 & 1 \end{pmatrix}$$

So we can realize the universal cover G of G_0 as $G = (\mathbb{C}^2, *)$ with $(t, s) * (t', s') = (t + t', s + e^t s')$. The center is

$$Z(G) = \{(2\pi i n, 0) \mid n \in \mathbb{Z}\} \cong \mathbb{Z}$$

So that the connected Lie group with Lie algebra $\mathfrak{g}$ forms a partially order tower

$$G \rightarrow \cdots \rightarrow G_n = G/n\mathbb{Z} \rightarrow \cdots \rightarrow G_0 = G/Z(G)$$

13.2.4. *Real cases.* In the real cases,

$$\exp(\text{ad } \mathfrak{g}) = \left\{ \begin{pmatrix} e^t & s \\ 0 & 1 \end{pmatrix} \mid s, t \in \mathbb{R} \right\} \cong \mathbb{R} \times \mathbb{R}_{>0}$$

is already simply connected, so is the unique connected Lie group with this Lie algebra.

13.3. **Dimension three.** Consider $[\cdot, \cdot]: \wedge^2 \mathfrak{g} \rightarrow \mathfrak{g}$, linear. Recall from linear algebra if $\dim V = 3$, then $\wedge^2 V$ is isomorphic to the dual space of V . Thus $\wedge^2 \mathfrak{g}$ is isomorphic the set of all linear functions from $\mathfrak{g}$ to $\mathbb{C}$.

We need to consider the rank of mapping $[\cdot, \cdot]$, which is equal to $\dim \mathfrak{g}^1$, the only possibilities are 0, 1, 2, 3.

If $\dim \mathfrak{g}^1 = 0$, thus $\mathfrak{g}$ is abelian, and the discussion for abelian $\mathfrak{g}$ of dimension 2 generalizes.

13.3.1. *Rank one.* If $\dim \mathfrak{g}^1 = 1$, let $X \in \mathfrak{g}$ such that $\mathfrak{g}^1$ is generated by X , extend this to a basis of $\{Y, Z, X\}$ of $\mathfrak{g}$. So there exists $\alpha, \beta, \gamma \in \mathbb{C}$, and not all 0, such that

$$\begin{aligned} [X, Y] &= \alpha X \\ [X, Z] &= \beta X \\ [Y, Z] &= \gamma X \end{aligned}$$

If $\alpha \neq 0$, replace Y by $\frac{1}{\alpha}Y$ and Z by $\alpha Z - \beta Y + \gamma X$, so we have

$$\begin{aligned} [X, Y] &= X \\ [X, Z] &= \alpha[X, Z] - \beta[X, Y] = 0 \\ [Y, Z] &= \frac{1}{\alpha}\alpha[Y, Z] + \frac{1}{\alpha}\gamma[X, Y] \\ &= \gamma X - \frac{1}{\alpha}\gamma\alpha X = 0 \end{aligned}$$

If $\beta \neq 0$, exchange Y and Z , yields the same Lie bracket as for $\alpha \neq 0$. So if either α or β is not 0, we have

$$\mathfrak{g} = \underbrace{(\mathbb{C}Z, 0)}_{\text{abelian}} \oplus \underbrace{(\mathbb{C}X \oplus \mathbb{C}Y, [\cdot, \cdot])}_{\text{unique nonabelian Lie algebra of dimension 2}}$$

In this case, $\mathfrak{g}$ is just the product of one-dimensional abelian Lie algebra with the non-abelian two-dimensional Lie algebra in the preceding discussion, so we may ignore this case.

If we assume both α and β is 0, replace Z by γZ we have

$$[X, Y] = 0, [X, Z] = 0, [Y, Z] = X$$

To find Lie groups with Lie algebra $\mathfrak{g}$, we need a faithful representation of $\mathfrak{g}$, but here adjoint representation $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is not faithful, since $X = \ker \text{ad}$.

So if we want to find a faithful representation $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ such that $\rho([Y, Z]) = \rho(X)$. A good way is to find a pair of endomorphisms Y and Z on some vector space that do not commute, but that do commute with their commutator $X = [Y, Z]$. After directly computing we have

$$[X, Y] = [[Y, Z], Y] = 2YZY - ZY^2 - Y^2Z = 0$$

$$[X, Z] = [[Y, Z], Z] = YZ^2 - 2ZY Z + Z^2Y = 0$$

So the easiest way is to choose Y, Z such that $Y^2Z, YZY, Z^2Y, \dots$ in above equations are zero. Let

$$\rho(X) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \rho(Y) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \rho(Z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

So we have a faithful representation, that is $\mathfrak{g} = \mathfrak{n}(3, \mathbb{C})$. Thus

$$G = \exp(\mathfrak{n}(3, \mathbb{C})) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in \mathbb{C} \right\}$$

which is simply connected, and called Heisenberg group.

$$Z(G) \cong \left\{ \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid b \in \mathbb{C} \right\} \cong \mathbb{C}$$

The same analysis holds for $\mathbb{R}$, G is the group of unipotent real 3×3 matrices, simply connected. Only one other connected Lie group is $H = G/\mathbb{Z}$.

H is a Lie group which is not a linear or matrix Lie group, i.e. it is not isomorphic to a subgroup of $\text{GL}(n, \mathbb{R})$, equivalently it does not admit any faithful finite dimensional representation.

Indeed, assume that there exists vector space V , $\dim V < \infty$ such that $\rho : H \rightarrow \text{GL}(V)$ is irreducible. Note that $Z(H) = Z(G)/H = \mathbb{R}/\mathbb{Z} \cong S^1$. Thus $\rho(S^1)$ must be diagonalizable. Let $\rho_* : \mathfrak{n}(3, \mathbb{R}) \rightarrow \mathfrak{gl}(V)$. Then $\rho_*(X)$ must be diagonalizable. Let $v \in V$ be an eigenvector for $\rho_*(X)$ with eigenvalue λ .

$$\begin{aligned} XYv &= [X, Y]v + YXv \\ &= \lambda Yv \end{aligned}$$

Similarly $XZv = \lambda Zv$. So Yv, Zv are eigenvectors of $\rho_*(X)$ with eigenvalue λ . Y, Z generates $\mathfrak{g}$, V irreducible. So

$$\rho_*(X)v = \lambda \text{id}_V$$

But $\text{tr } \rho_*(X) = \text{tr } \rho_*([Y, Z]) = 0$. So $\lambda = 0$, that is, ρ_* is not faithful.

However, H do has an important infinitely-dimensional representation. Let $V_1 = C^\infty(\mathbb{R}, \mathbb{C})$, $V_2 = \{s \mapsto e^{-\pi s^2} P(s) \mid P \in \mathbb{R}[x]\} \subset V_1$ and $\hbar \in \mathbb{R}_{\geq 0}$.

Let

$$\begin{aligned}\rho(Y)f &= -i\hbar s f \\ \rho(Z)f &= \frac{df}{ds} \\ \rho(X)f &= i\hbar\end{aligned}$$

Let $L = 2X + i\hbar Y$, so

$$\begin{aligned}L(e^{-\pi s^2} P(s)) &= -2i\hbar \pi s e^{-\pi s^2} P(s) + 2\pi \hbar s e^{-\pi s^2} P(s)_e^{-\pi s^2} P'(s) \\ &= e^{-\pi s^2} P'(s) \\ &= e^{-\pi s^2} Q(s), \quad \deg Q = \deg P - 1\end{aligned}$$

Let $U \neq 0$ be an invariant subspace of V_2 , $f = e^{-\pi s^2} P(s) \in U$, $P \neq 0$, $\deg P = n$. So $L^n f = e^{-\pi s^2} \in U$, since U is invariant.

Consider the exponential map $\exp : \mathfrak{h} \rightarrow H$,

$$e^{tY} f(s) = (\cos(\hbar ts) - \sin(\hbar ts))f(s)$$

momentum.

$$e^{tZ} f(s) = f(t + s)$$

translation.

$$e^{tX} = e^{i\hbar t} f(s)$$

multiplication by a phase. This is representation of the Heisenberg uncertainty relation

$$[x, p] = -i\hbar$$

13.3.2. *Rank two.* If $\dim \mathfrak{g}^1 = 2$, let $\mathfrak{g}^1 = \langle Y, Z \rangle_{\mathbb{C}}$. So

$$[Y, Z] = \alpha Y + \beta Z$$

$\text{ad}_Y Y = 0$, $\text{ad}_Y Z = \alpha Y + \beta Z$, so $\text{tr ad}_Y = \beta = 0$. Similarly we have $\alpha = -\text{tr ad}_Z = 0$. Thus $\mathfrak{g}^1$ is abelian. $\text{ad}_X : \mathfrak{g}^1 \rightarrow \mathfrak{g}^1$ is an isomorphism for all $X \notin \mathfrak{g}^1$. There are two possibilities: ad_X is diagonalizable or not.

If ad_X is diagonalizable. Let Y, Z be an eigenbasis of $\mathfrak{g}^1$ for ad_X . Thus

$$\begin{aligned}[X, Y] &= \lambda_1 Y \\ [X, Z] &= \lambda_2 Z \\ [Y, Z] &= 0\end{aligned}$$

with $\lambda_1, \lambda_2 \neq 0$. So we can rewrite these relations as

$$\begin{aligned}[X, Y] &= Y \\ [X, Z] &= \alpha Z \\ [Y, Z] &= 0\end{aligned}$$

with $\alpha = \frac{\lambda_1}{\lambda_2} \neq 0$.

Exercise 13.1. Show that two Lie algebras $\mathfrak{g}_\alpha, \mathfrak{g}_{\alpha'}$ are isomorphic if and only if

$$\alpha = \alpha' \text{ or } \alpha = \frac{1}{\alpha'}$$

This is the first time we see a continuously varying family of nonisomorphic complex Lie algebras.

Proof. Clearly, if $\alpha = \alpha'$ or $\alpha = \frac{1}{\alpha'}$, $\mathfrak{g}_\alpha \cong \mathfrak{g}_{\alpha'}$. Conversely, if $\mathfrak{g}_\alpha \cong \mathfrak{g}_{\alpha'}$, there is a Lie algebra isomorphism φ , that is a vector space isomorphism such that

$$\varphi([X, Y]) = [\varphi(X), \varphi(Y)], \quad \forall X, Y \in \mathfrak{g}$$

In other words, we have $\varphi \circ \text{ad}_X = \text{ad}_{\varphi(X)} \circ \varphi$. So operators ad_X and $\text{ad}_{\varphi(X)}$ are similar, thus they have the same characteristic polynomials, traces and determinants.

Since we have the matrix representation of ad_X as

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \alpha \end{pmatrix}$$

Assume $\varphi(X) = aX + bY + cZ$, then we can also compute the matrix representation of $\text{ad}_{\varphi(X)}$

$$\begin{pmatrix} 0 & -b & -c\alpha' \\ 0 & a & 0 \\ 0 & 0 & a\alpha' \end{pmatrix}$$

After computing the characteristic polynomials, we have

$$\lambda^2 - (\alpha + 1)\lambda + \alpha = \lambda^2 - a(1 + \alpha')\lambda + a^2\alpha'$$

Thus by comparing coefficients and cancelling a we have

$$\frac{\alpha'}{(1 + \alpha')^2} = \frac{\alpha}{(1 + \alpha)^2}$$

The only possibilities are $\alpha = \alpha'$ or $\alpha = \frac{1}{\alpha'}$. □

From above computation we also see that the adjoint representation of $\mathfrak{g}$ is faithful. Let's rewrite in a more beautiful way: A general element $aX - bY - cZ$ of Lie algebras with respect to the basis $\{Y, Z, X\}$ by the matrix

$$\begin{pmatrix} a & 0 & b \\ 0 & aa & ac \\ 0 & 0 & 0 \end{pmatrix}$$

Exponentiating, we find that a group with Lie algebra $\mathfrak{g}$ is

$$G = \left\{ \begin{pmatrix} e^t & 0 & u \\ 0 & e^{at} & v \\ 0 & 0 & 1 \end{pmatrix} \mid t, u, v \in \mathbb{C} \right\} \subset \mathrm{GL}(3, \mathbb{C})$$

If ad_X is not diagonalizable. It's natural to choose a basis $\{Y, Z\}$ of $\mathfrak{g}^1$ with respect to which ad_X is in Jordan form. Replacing X by a multiple, we may assume both its eigenvalues are 1 so that we have Lie algebra

$$\begin{aligned} [X, Y] &= Y \\ [X, Z] &= Y + Z \\ [Y, Z] &= 0 \end{aligned}$$

With respect to the basis $\{Y, Z, X\}$ for $\mathfrak{g}$, then the adjoint action of general element $aX - bY - cZ$ of Lie algebra is represented by the matrix

$$\begin{pmatrix} a & a & b+c \\ 0 & a & c \\ 0 & 0 & 0 \end{pmatrix}$$

Exponentiating we have

$$G = \left\{ \begin{pmatrix} e^t & te^t & u \\ 0 & e^t & v \\ 0 & 0 & 1 \end{pmatrix} \mid t, u, v \in \mathbb{C} \right\}$$

Over $\mathbb{R}$, there is a third possibility: ad_X acts on $\mathfrak{g}^1$ with distinct complex eigenvalues $\lambda, \bar{\lambda}$.

$$\mathrm{ad}_X(Y + iZ) = \lambda(Y + iZ), \quad \lambda = \alpha + i\beta$$

So

$$[Y, X] = \frac{\alpha}{\beta} Y$$

13.3.3. *Rank three.* $\dim \mathfrak{g}^1 = 3$. Claim: there exists $H \in \mathfrak{g}$ such that $\mathrm{ad}_H : \mathfrak{g} \rightarrow \mathfrak{g}$ has an eigenvector with eigenvalue $\lambda \neq 0$.

Indeed, note that for any $X \in \mathfrak{g}, X \neq 0$, we have rank of ad_X is 2 since $[X, Y], [X, Z]$ must be linearly independent, otherwise $\dim \mathfrak{g}^1 < 3$. So $\ker \mathrm{ad}_X = \mathbb{C}X$. Now for any $X \in \mathfrak{g}$, either ad_X has non zero eigenvalue or ad_X is nilpotent. If it's nilpotent, choose $Y \in \mathfrak{g}^1$ such that Y is not in the kernel of ad_X but is in the kernel of ad_X^2 . In other words,

$$[X, Y] = \alpha Y, \quad \alpha \in \mathbb{C}^*$$

So of course X is a eigenvector for ad_Y with non-zero eigenvalue.

So choose H and $X \in \mathfrak{g}$ such that X is an eigenvector with non-zero eigenvalue for ad_H , and write $[H, X] = \alpha X$. $H \in \mathfrak{g}^1$, ad_H is a commutator in $\mathfrak{gl}(\mathfrak{g})$, so $\mathrm{tr} \mathrm{ad}_H = 0$. Thus ad_H must have a third eigenvector Y with eigenvalue $-\alpha$.

So in order to describe the structure of $\mathfrak{g}$, it remains to find the commutator of X and Y . Jacobi identity implies that

$$\begin{aligned} [H, [X, Y]] &= -[X, [Y, H]] - [Y, [H, X]] \\ &= -[X, \alpha Y] - [Y, \alpha X] \\ &= 0 \end{aligned}$$

Thus $[X, Y] = \beta H$. By multiply X or Y by a scalar to make $\beta = 1$. Thus, there is only one possible complex Lie algebra of this type. We can realize it or give a faithful representation as follows

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with

$$\begin{aligned} [H, X] &= 2X \\ [H, Y] &= -2Y \\ [X, Y] &= H \end{aligned}$$

this is exactly the Lie algebra of special linear group $\mathrm{SL}(2, \mathbb{C})$, and we denote it by $\mathfrak{sl}(2, \mathbb{C})$, which is the smallest simple Lie algebra.

14. REPRESENTATIONS OF $\mathfrak{sl}(n, \mathbb{C})$

14.1. Representations of $\mathfrak{sl}(2, \mathbb{C})$. Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V)$ be a representation. From problem set we know $\mathfrak{sl}(2, \mathbb{C})$ is semisimple, then Jordan–Chevalley decomposition says that $\rho(H)$ is diagonalizable. Set

$$V = \bigoplus_{\alpha} V_{\alpha}, \quad V_{\alpha} = \{v \in V \mid \rho(H)v = \alpha v\}$$

Notation 14.1. For the convenience of symbols, we use H to denote $\rho(H)$.

Lemma 14.1. Let $v \in V$ be an eigenvector of H with eigenvalue α . If Xv, Yv are nonzero, then

$$\begin{aligned} H(Xv) &= (\alpha + 2)Xv \\ H(Yv) &= (\alpha - 2)Yv \end{aligned}$$

Proof. Directly compute as follows

$$\begin{aligned} H(Xv) &= [H, X]v + XHv \\ &= 2Xv + \alpha Xv \end{aligned}$$

□

Remark 14.1.1. Above lemma implies that

$$\begin{aligned} X : V_{\alpha} &\rightarrow V_{\alpha+2} \\ Y : V_{\alpha} &\rightarrow V_{\alpha-2} \end{aligned}$$

If V is irreducible, then for any eigenvalue α , $W = \bigoplus_{m \in \mathbb{Z}} V_{\alpha+2m}$ is an invariant subspace. Hence $W = V$. We have the following picture of the action of $\mathfrak{sl}(2, \mathbb{C})$ on the vector space V :

$$\begin{array}{ccccccc} & & X & & X & & X \\ & \nearrow & & \nearrow & & \nearrow & \\ \cdots & \xleftarrow{Y} & V_{n-4} & \xleftarrow{Y} & V_{n-2} & \xleftarrow{Y} & V_n \\ & \nwarrow & \nwarrow & \nwarrow & \nwarrow & \nwarrow & \\ & (H) & (H) & (H) & (H) & (H) & \end{array}$$

Since the representation is finite dimensional, there exists $k \geq 0$ such that $V_\beta \neq 0$ for all $\beta \in \underbrace{\{\alpha, \alpha+2, \dots, \alpha+2k\}}_{\text{unbroken string of complex numbers}}$

We use n to denote the last element in this sequence, until now we only know n is a complex numbers, but later we will see it's an integer. Since $V_{n+2} = 0$, then there exists $v \in V_n, v \neq 0$ such that $Xv = 0$. Let's see what will happen if we use Y to act on it.

Lemma 14.2. Given $v \in V_n, v \neq 0$ such that $Xv = 0$, then $V = \langle v, Yv, Y^2v, \dots \rangle_{\mathbb{C}}$.

Proof. It suffices to show that $W = \langle v, Yv, Y^2v, \dots \rangle_{\mathbb{C}}$ is an invariant subspace since V is irreducible. $YW \subset W$ is clear. $HW \subset W$, since

$$HY^m v = (n - 2m)Y^m v, \quad \forall m \in \mathbb{Z}_{\geq 0}$$

For X ,

$$\begin{aligned} XYv &= [X, Y]v + YXv \\ &= Hv \\ &= nv \end{aligned}$$

We claim $X(Y^m v) = m(n - m + 1)Y^{m-1}v$. Assume this is true for all value $\leq m$. Then

$$\begin{aligned} X(Y^{m+1}v) &= XY^m Yv \\ &= [X, Y]Y^m v + YXY^m v \\ &= HY^m v + m(n - m + 1)Y^m v \\ &= [(n - 2m) + m(n - m + 1)]Y^m v \\ &= (m + 1)(n - m)Y^m v \end{aligned}$$

As desired. $\square$

Corollary 14.1.1. $\dim V_\alpha = 1$ for all α such that $V_\alpha \neq 0$. Moreover, V is completely determined by integer n .

Lemma 14.3. n is an integer.

Proof. Since $\dim V < \infty$, then there exists k such that $Y^k v = 0$. If m is the smallest power of Y such that $Y^m v = 0$. Then

$$0 = XY^m v = m(n - m + 1)Y^{m-1}v \neq 0$$

This means $n - m + 1 = 0$, therefore n must be an integer. $\square$

Theorem 14.1.1. The representations $V^{(n)} = \bigoplus_{\alpha=-n}^n V_\alpha, \alpha \equiv n \pmod{2}, n \in \mathbb{N}$ of dimension $n + 1$ form complete set of irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$ up to isomorphism. In particular, the eigenvalues of H are $-n, -n + 2, \dots, n - 2, n$.

Proof. Only the existence remains to be checked. Let $n \in \mathbb{N}$, V a vector space of dimension $n + 1$ with basis $\{v_0, \dots, v_n\}$. Define

$$\begin{aligned} H v_m &= (n - 2m) v_m \\ X v_m &= m(n - m + 1) v_{m-1}, \quad X v_0 = 0 \\ Y v_m &= v_{m+1}, \quad Y v_n = 0 \end{aligned}$$

Clearly:

$$\begin{aligned} [H, X] v_m &= 2X v_m \\ [H, Y] v_m &= -2X v_m \\ [X, Y] v_m &= H v_m \end{aligned}$$

□

Corollary 14.1.2. Any representation V of $\mathfrak{sl}(2, \mathbb{C})$ such that the eigenvalues of H all have same parity occur with multiplicity 1, is irreducible.

Corollary 14.1.3. The numbers of irreducible summands in a representation V of $\mathfrak{sl}(2, \mathbb{C})$ is equal to the sum of multiplicities of eigenvalue 0 and 1 of H .

In fact, we can rewrite these irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$ in terms of the standard representation of $\mathfrak{sl}(2, \mathbb{C})$.

Example 14.1.1. $V^{(0)}$ is the trivial representation.

Example 14.1.2. $V^{(1)}$ is the standard representation.

Proof. Let x, y be a basis of $\mathbb{C}^2$, and consider the standard representation ρ of $\mathfrak{sl}(2, \mathbb{C})$, that is, $\rho : H \rightarrow (v \mapsto H v)$. Thus we have

$$\begin{aligned} H x &= x \\ H y &= -y \end{aligned}$$

implies $V = \mathbb{C}x \oplus \mathbb{C}y = V_{-1} \oplus V_1 = V^{(1)}$. This completes the proof. □

Theorem 14.1.2. $V^{(n)} \cong \text{Sym}^n V^{(1)}$.

Proof. Note that if $\{x, y\}$ be a basis of V , then $\text{Sym}^n V$ has basis $\{x^n, x^{n-1}y, \dots, xy^{n-1}, y^n\}$. Directly compute as follows:

$$\begin{aligned} H(x^{n-k}y^k) &= (n-k)H(x)x^{n-k-1}y^k + kH(y)x^{n-k}y^{k-1} \\ &= (n-2k)x^{n-k}y^k \end{aligned}$$

So eigenvalues of H on $\text{Sym}^n V$ are exactly $-n, -n+2, \dots, n-2, n$. Thus $\text{Sym}^n V$ is irreducible with dimension $n + 1$, then

$$\text{Sym}^n V \cong V^{(n)}$$

□

Corollary 14.1.4. $V^{(2)}$ is the adjoint representation.

14.2. Representations of $\mathfrak{sl}(3, \mathbb{C})$. Recall

$$\mathfrak{sl}(3, \mathbb{C}) = \{A \in \mathfrak{gl}(3, \mathbb{C}) \mid \text{tr}(A) = 0\}$$

Clearly we have $\dim \mathfrak{sl}(3, \mathbb{C}) = 8$, and we can write the basis explicitly using elementary matrices E_{ij} , that is

$$\begin{cases} E_{ij}, & i \neq j, 1 \leq i, j \leq 3 \\ E_{11} - E_{22} \\ E_{11} - E_{33} \end{cases}$$

Furthermore, we have

$$[E_{ij}, E_{i'j'}] = \delta_{ij'} E_{i'j} - \delta_{i'j} E_{ij'}$$

which determines the Lie algebra structure of $\mathfrak{sl}(3, \mathbb{C})$.

The distinguished $H \in \mathfrak{sl}(2, \mathbb{C})$ whose eigenvalue yields a decomposition $V = \bigoplus_{\alpha} V_{\alpha}$ is replaced by subspace $\mathfrak{h} \subset \mathfrak{sl}(3, \mathbb{C})$, where

$$\begin{aligned} \mathfrak{h} &= \{A \in \mathfrak{sl}(3, \mathbb{C}) \mid A_{ij} = 0, i \neq j\} \\ &= \left\{ \begin{pmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{pmatrix} \mid a_1, a_2, a_3 \in \mathbb{C}, a_1 + a_2 + a_3 = 0 \right\} \end{aligned}$$

From problem set we know $\mathfrak{h}$ is center of $\mathfrak{sl}(3, \mathbb{C})$. Commuting diagonalizable matrices are simultaneously diagonalizable. Then

$$V = \bigoplus_{\alpha} V_{\alpha}, \quad v \in V_{\alpha} \text{ is an eigenvector for every element } H \in \mathfrak{h}$$

We need to generalize the notion of eigenvector and eigenspace from a single $H \in \mathfrak{h}$ to all $H \in \mathfrak{h}$. If $v \in V$ is a simultaneous eigenvector for all $H \in \mathfrak{h}$, we have $Hv = \alpha(H)v, \forall H \in \mathfrak{h}$, where $\alpha(H)$ denotes the eigenvalue for action of $H \in \mathfrak{h}$. α depends linearly on H and therefore defines an element in $\mathfrak{g}^{\vee}$.

Definition 14.2.1 (weight). Let V be a finite dimensional representation of $\mathfrak{sl}(3, \mathbb{C}), \alpha \in \mathfrak{h}^{\vee}$.

$$V_{\alpha} = \{v \in V \mid Hv = \alpha(H)v, \forall H \in \mathfrak{h}\}$$

If $V_{\alpha} \neq 0$, then α is called a weight and V_{α} is called weight space, and

$$V = \bigoplus_{\alpha} V_{\alpha}$$

is called weight space decomposition.

Definition 14.2.2 (root). If $V = \mathfrak{g} = \mathfrak{sl}(3, \mathbb{C})$ is the adjoint representation, then α is called a root, $\mathfrak{g}_{\alpha}$ is called a root space and

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha} \mathfrak{g}_{\alpha}$$

is called root space decomposition. The set of all roots is called root system, denoted by Φ .

Proposition 14.2.1. Let V be a finite dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$. Then there exists a weight space decomposition $V = \bigoplus_{\alpha} V_{\alpha}$, where α runs over a finite subset of weights in $\mathfrak{h}^{\vee}$.

Having decided what the analogous for $\mathfrak{sl}(3, \mathbb{C})$ of $H \in \mathfrak{sl}(2, \mathbb{C})$, let's see what will play the role of X and Y . The relations $[H, X] = 2X, [H, Y] = -2Y$ in $\mathfrak{sl}(2, \mathbb{C})$ should be understood $\text{ad}_H X = 2X, \text{ad}_H Y = -2Y$, that is, X, Y are eigenvectors for the adjoint action of H on $\mathfrak{sl}(2, \mathbb{C})$.

The correct way to interpret these is to look for eigenvectors of the adjoint action of $\mathfrak{h}$ on $\mathfrak{sl}(3, \mathbb{C})$, that is to consider the root space decomposition of $\mathfrak{g}$

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha} \mathfrak{g}_{\alpha}$$

$\mathfrak{h}$ acts on $\mathfrak{g}_{\alpha}$ by scalar multiplication, i.e.

$$\text{ad}_H X = [H, X] = \alpha(H)X, \quad \forall H \in \mathfrak{h}, X \in \mathfrak{g}_{\alpha}$$

If $H = \sum_{k=1}^3 a_k E_{kk} \in \mathfrak{h}$, then

$$\begin{aligned} [H, E_{ij}] &= \sum a_k (\delta_{ik} E_{kj} - \delta_{jk} E_{ki}) \\ &= (a_i - a_j) E_{ij} \end{aligned}$$

Thus $\mathbb{C}E_{ij}, 1 \leq i \neq j \leq 3$ are simultaneously eigenspaces of $\mathfrak{h}$.

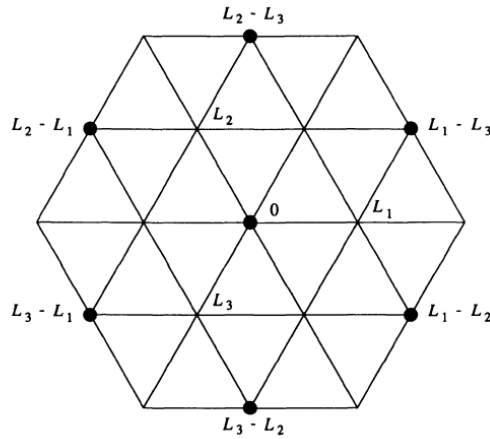
Since $\mathfrak{h} = \left\{ \begin{pmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{pmatrix} \mid a_1, a_2, a_3 \in \mathbb{C}, a_1 + a_2 + a_3 = 0 \right\}$, we can write $\mathfrak{h}^{\vee}$ explicitly as

$$\mathfrak{h}^{\vee} = \mathbb{C}L_1 \oplus \mathbb{C}L_2 \oplus \mathbb{C}L_3 / \langle L_1 + L_2 + L_3 = 0 \rangle$$

with $L_i \begin{pmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{pmatrix} = a_i$. So roots α of $\mathfrak{sl}(3, \mathbb{C})$ are $\alpha(H)_{ij} = a_i - a_j$. In other words,

$$\Phi = \{L_i - L_j \mid 1 \leq i \neq j \leq 3\}$$

We can draw a picture as follows



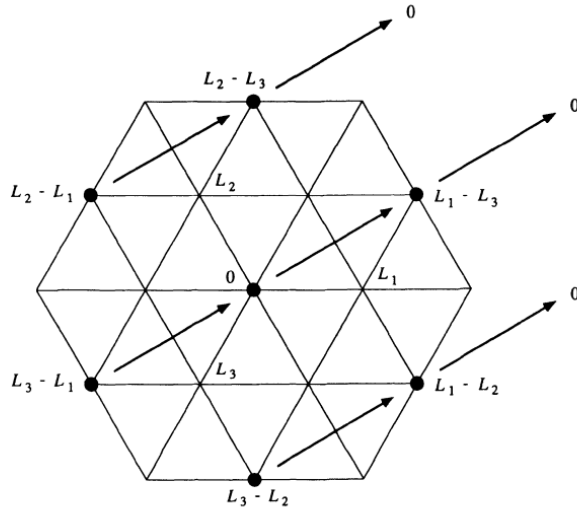
We can read off from this picture almost everything. For example: if $X \in \mathfrak{g}_\alpha, Y \in \mathfrak{g}_\beta$, what is $\text{ad}_X Y$? One way is to compute directly as follows:

$$\begin{aligned} [H, \text{ad}_X Y] &= [H, [X, Y]] \\ &= [X, [H, Y]] + [[H, X], Y] \\ &= [X, \beta(H)Y] + [\alpha(H)X, Y] \\ &= (\alpha(H) + \beta(H))[X, Y] \\ &= (\alpha(H) + \beta(H))\text{ad}_X Y \end{aligned}$$

So $\text{ad}_X Y$ is an eigenvector for H with eigenvalue $\alpha + \beta$. In other words,

$$\text{ad}_{\mathfrak{g}_\alpha} : \mathfrak{g}_\beta \rightarrow \mathfrak{g}_{\alpha+\beta}$$

So we can see directly from the picture. In the following picture, $\mathfrak{g}_\alpha$ acts by translation in the direction $\alpha = L_1 - L_3$.



More explicitly, it carries $\mathfrak{g}_{L_2-L_1}$ to $\mathfrak{g}_{L_2-L_3}$; $\mathfrak{g}_{L_3-L_1} \rightarrow \mathfrak{h}$; $\mathfrak{h}$ to $\mathfrak{g}_{L_1-L_3}$; $\mathfrak{g}_{L_3-L_2}$ to $\mathfrak{g}_{L_1-L_2}$ and kills $\mathfrak{g}_{L_2-L_3}$, $\mathfrak{g}_{L_1-L_3}$ and $\mathfrak{g}_{L_1-L_2}$.

The same structure holds for any representation V of $\mathfrak{sl}(3, \mathbb{C})$. The action of $\mathfrak{h}$ on V yields the weight space decomposition $V = \bigoplus_{\alpha} V_{\alpha}$. The action of $\mathfrak{g}_{\alpha}$ on V_{β} is determined by the commutation relations: Let $v \in V_{\beta}, X \in \mathfrak{g}_{\alpha}$.

$$\begin{aligned} HXv &= [H, X]v + XHv \\ &= \alpha(H)Xv + \beta Xv \\ &= (\alpha(H) + \beta(H))Xv \end{aligned}$$

We see from this that $X(v)$ is again an eigenvector for the action for the action of $\mathfrak{h}$, with eigenvalue $\alpha + \beta$. In other words:

$$\mathfrak{g}_{\alpha} : V_{\alpha} \rightarrow V_{\alpha+\beta}$$

We can again represent V_α by dots in a plane diagram such that $\mathfrak{g}_\alpha$ acts by translation.

So we have the following important observation: Let V be an irreducible representation, $\alpha, \alpha' \in \mathfrak{h}^\vee$ be two weights, then

$$\alpha - \alpha' = \sum n_{ij}(L_i - L_j) \in \mathfrak{h}^\vee, \quad n_{ij} \in \mathbb{Z}$$

Definition 14.2.3 (root lattice). The lattice $\Lambda_{\mathbb{R}} \subset \mathfrak{h}^\vee$ generated by the roots Φ is called the root lattice.

Definition 14.2.4 (simple root). A subset of roots $\Delta \subset \Phi$ forming a lattice basis is called the set of simple roots.

We have a decomposition

$$\Phi = \Phi^+ \cup \Phi^-$$

with $\Phi^+ = \Phi \cap \bigoplus_{\alpha \in \Delta} \mathbb{Z}_{\geq 0}$, $\Phi^- = -\Phi^+$. The roots in Φ^+ are called positive roots and roots in Φ^- are called negative roots.

Example 14.2.1.

$$\Delta = \{L_1 - L_2, L_2 - L_3\}$$

$$\Phi^+ = \{L_1 - L_2, L_2 - L_3, L_1 - L_3\}$$

Consider the representation with $V = \mathbb{C}^3$, $\rho : \mathfrak{sl}(3, \mathbb{C}) \rightarrow \mathfrak{gl}(\mathbb{C}^3)$ defined by $X \mapsto (v \mapsto Xv)$, called the standard representation of $\mathfrak{sl}(3, \mathbb{C})$.

The standard basis vectors e_1, e_2, e_3 of $\mathbb{C}^3$ are weight vectors with weight L_1, L_2, L_3 , then

$$\mathbb{C}^3 = (\mathbb{C}^3)_{L_1} \oplus (\mathbb{C}^3)_{L_2} \oplus (\mathbb{C}^3)_{L_3}$$

Part 4. Solutions to problem sets

15. PROBLEM SET 1: BASIC AND FAITHFUL REPRESENTATION

Exercise 15.0.1 (morphisms of representations). Let G be a finite group, V, W be G -modules.

1. Let $\varphi : V \rightarrow W$ a morphism of G -modules. Show that $\ker \varphi$ is a G -submodule of V , $\operatorname{im} \varphi, \operatorname{coker} \varphi$ are G submodules of W .
2. Let $\rho : G \rightarrow \operatorname{GL}(V)$ be a representation, $\rho^\vee : G \rightarrow \operatorname{GL}(V^\vee)$ the dual representation defined as $\rho^\vee(g)(\alpha) = \rho(g^{-1})^T \alpha$ for $g \in G, \alpha \in V^\vee$. Show that

$$\langle \rho^\vee(g)(\alpha), \rho(g)(v) \rangle = \langle \alpha, v \rangle \quad \text{for all } g \in G, v \in V, \alpha \in V^\vee$$

3. Let $\operatorname{Hom}_G(V, W)$ be the vector space of morphisms of G -modules V, W . Let $\operatorname{Hom}(V, W)^G$ be the subspace $\operatorname{Hom}(V, W)^G = \{\varphi \in \operatorname{Hom}(V, W) \mid g \circ \varphi = \varphi\}$ of $\operatorname{Hom}(V, W)$. Show that $\operatorname{Hom}_G(V, W) \cong \operatorname{Hom}(V, W)^G$.

Proof. For (1). Routines.

For (2). Check by definition

$$\begin{aligned} \langle \rho^\vee(g)(\alpha), \rho(g)(v) \rangle &= \langle \rho(g^{-1})^T \alpha, \rho(g)v \rangle \\ &= \langle \alpha, \rho(g^{-1})\rho(g)v \rangle \\ &= \langle \alpha, v \rangle \end{aligned}$$

For (3). It's important to keep in mind if $\rho_V : G \rightarrow \text{GL}(V), \rho_W : G \rightarrow \text{GL}(W)$ are two representations of G . how does G acts on $\text{Hom}(V, W)$ to make it as a representation of G . In fact, G acts on $\varphi \in \text{Hom}(V, W)$ as follows

$$g \circ \varphi(v) = \rho_W(g)\varphi(\rho_V^{-1}(g)v)$$

So the condition $g \circ \varphi = \varphi$ is equivalent to

$$\rho_W(g)\varphi = \varphi\rho_V(g)$$

and that's the definition of morphisms of G -modules. $\square$

Exercise 15.0.2 (Calculus of representations). Let G be a finite group, V, W be G -modules. Show that the following are isomorphisms of G -modules:

1. $\text{Hom}(V, W) \cong W \otimes V^\vee$
2. $\wedge^k(V \oplus W) \cong \bigoplus_{a+b=k} \wedge^a V \otimes \wedge^b W$
3. $A^k(V^\vee) \cong (A^k V)^\vee$
4. $A^k(V^\vee) \simeq A^{n-k} V \otimes \det V^\vee, n = \dim V$ where $\det V := \wedge^n V$.

Proof. Routines. $\square$

Exercise 15.0.3 (Faithful representations). Let G be a finite group, $\rho : G \rightarrow \text{GL}(V)$ be a representation. Its kernel is defined to be $\ker \rho = \{g \in G \mid \rho(g) = \text{id}_V\}$. A representation is called faithful if it is injective.

1. Show that $\ker \rho$ is a normal subgroup of G and find a condition on $\ker \rho$ equivalent to the representation being faithful
2. Let $H \subset G$ be a subgroup. Show that for the coset representation, $\ker \rho = \bigcap_i g_i H g_i^{-1}$, where $\{g_i\}$ is a complete set of representatives of G/H .
3. For each of the following representations, under what conditions are they faithful: trivial, regular, coset, alternating for S_n , defining for S_n , dimension 1 for C_n , where C_n is the cyclic group of order n ?

Proof. For (1). If $x \in \ker \rho$, for any $g \in G$, we need to show gxg^{-1} still lies in $\ker \rho$. Check by definition we have

$$\begin{aligned} \rho(gxg^{-1}) &= \rho(g)\rho(x)\rho(g)^{-1} \\ &= \rho(g)\rho(g)^{-1} \\ &= \text{id}_V \end{aligned}$$

This shows $\ker \rho$ is normal. Clearly ρ is faithful if and only if $\ker \rho$ is trivial.

For (2). If $g \in G$ lies in kernel of coset representation, then for any g_i , we have

$$g(g_i H) = g_i H$$

where g_i runs over the complete set of representatives of G/H . Thus for each i we have

$$g \in g_i H g_i^{-1}$$

So we have $g \in \bigcap_i g_i H g_i^{-1}$, that is $\ker \rho \subseteq \bigcap_i g_i H g_i^{-1}$. Conversely it's trivial.

For (3). Trivial representation is faithful if and only if G is trivial; Regular representation is always faithful; From (2), coset representation is faithful if and only if $\bigcap_i g_i H g_i^{-1}$ is trivial. $\square$

Exercise 15.0.4 (Lifting representations). Let G be a finite group, $\rho : G \rightarrow \text{GL}(V)$ be a representation. Set $N = \ker \rho$ and define a map $\psi : G/N \rightarrow \text{GL}(V)$ by setting $\psi(gN) = \rho(g)$ for $gN \in G/N$.

1. Show that ψ is a well-defined faithful representation of the factor group G/N .
2. Show that ψ is irreducible if and only if ρ is.
3. If ρ is the coset representation for a normal subgroup H of G , what is the corresponding representation ψ ?

Now we turn this around. Let N be any normal subgroup of G and let $\psi : G/N \rightarrow \text{GL}(V)$ be a representation. Define a map $\rho : G \rightarrow \text{GL}(V)$ by setting $\rho(g) = \psi(gN)$ for $g \in G$. Show that the following holds:

4. ρ is a representation. We say that the representation ψ has been lifted to a representation ρ .
5. If ψ is faithful, then $\ker \rho = N$.
6. ρ is irreducible if and only if ψ is.

Proof. For (1). Firstly show ψ is well-defined, that is if $g_1N = g_2N$, then we need to show $\rho(g_1) = \rho(g_2)$. Clearly $g_1 = g_2n, n \in N$, then

$$\rho(g_1) = \rho(g_2)\rho(n) = \rho(g_2)$$

as desired. It's faithful since if $\psi(gN) = \rho(g) = \text{id}_V$, then $g \in N$, that is $gN = N$. So ψ is a well-defined faithful representation.

For (2). It suffices to show $W \subset V$ is ρ invariant if and only if ψ invariant. Clearly if W is ρ invariant then it's ψ invariant; Conversely, for any $g \in G$, write it as $g'n$ for some $n \in N$, then

$$\begin{aligned} \rho(g)W &= \rho(g'n)W \\ &= \rho(g')W \\ &= \psi(g'N)W \subset W \end{aligned}$$

As desired.

For (3). It's the restriction of ρ to G/H , obviously.

For (4). By definition we need to check for any $g_1, g_2 \in G$, we have $\rho(g_1g_2) = \rho(g_1)\rho(g_2)$, that is to show

$$\psi(g_1g_2N) = \psi(g_1N)\psi(g_2N)$$

It suffices to show $g_1g_2N = g_1Ng_2N$, but that's clear.

For (5). $g \in G$ is in $\ker \rho$ if and only if $\psi(gN) = \text{id}_V$. Since ψ is faithful, this is equivalent to $gN = N$, and that's equivalent to $g \in N$. So we have $\ker \rho = N$.

For (6). The same as (2). □

16. PROBLEM SET 2: CHARACTERS OF FINITE GROUPS

Exercise 16.0.1. Decompositions of tensor and alternating powers

1. Let V be a faithful representation of G . Show that any irreducible representation of G is contained in $V^{\otimes n}$ for some $n \in \mathbb{N}$.
2. Let V be the standard representation of S_n . Show that $\wedge^k V$ is irreducible for every $0 \leq k \leq n-1$. Hint: Recall that the permutation representation $P = R_{\{1, \dots, n\}}$ decomposes as $P = U \oplus V$ with U the trivial representation, and consider $\wedge^k P$. Use Problem 14.2 to argue that it suffices to show that $(\chi_{\wedge^k P}, \chi_{\wedge^k P}) = 2$. Compute $(\chi_{\wedge^k P}, \chi_{\wedge^k P})$.

Proof. For (1). Let W be any irreducible representation of G , the trivial representation, I will show instead W appears in some representation of $(V \oplus \underline{1})^{\otimes N}$. This is equivalent since

$$(V \oplus \underline{1})^{\otimes N} = \bigoplus_{k=0}^N \binom{N}{k} V^{\otimes k}$$

It suffices to show

$$\dim \text{Hom}_G(W, (V \oplus \underline{1})^{\otimes N}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} (\psi(g) + 1)^N \geq 1$$

for sufficiently large n , where χ is the character of W and ψ is the one of V .

We have $|\psi(g)| \leq \dim V$ for all $g \in G$. And since V is faithful, $\psi(g)$ is $\dim V$ if and only if $g = e$. So $|\psi(g) + 1| \leq \dim V + 1$ with equality precisely for $g = e$. So the right hand side is a finite sum of exponentials, and the term $(\dim W)(\dim V + 1)^N$ has a larger base than any of others.

For (2). Let's prove $\langle \chi_{\wedge^k V}, \chi_{\wedge^k V} \rangle = 1$ if and only if $\langle \chi_{\wedge^k P}, \chi_{\wedge^k P} \rangle = 2$. Since $\wedge^k U = 0, k \geq 2$, then

$$\wedge^k P = \bigoplus_{a+b=k} \wedge^a U \otimes \wedge^b V = \wedge^k V \oplus \wedge^{k-1} V$$

So

$$\langle \chi_{\wedge^k P}, \chi_{\wedge^k P} \rangle = \langle \chi_{\wedge^k V}, \chi_{\wedge^k V} \rangle + \langle \chi_{\wedge^{k-1} V}, \chi_{\wedge^{k-1} V} \rangle$$

It suffices to show $\langle \chi_{\wedge^k P}, \chi_{\wedge^k P} \rangle = 2$, which will force the two on the right hand side to be 1. Now let's try to compute $\langle \chi_{\wedge^k P}, \chi_{\wedge^k P} \rangle$. □

Exercise 16.0.2. Character table of S_5 .

1. Show that the character table of S_5 takes the following form

	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
U	1	1	1	1	1	1	1
U'	1	-1	1	-1	1	1	-1
V	4	2	1	0	-1	0	-1
V'	4	-2	1	0	-1	0	1
$\wedge^2 V$	6	0	0	0	1	-2	0
W	5	1	-1	-1	0	1	1
W'	5	-1	-1	1	0	1	-1

Hints: Determine the cardinality of the conjugacy classes. Verify the characters of the trivial representation U , the alternating representation U' , the standard representation $V = R/U$. Show that $V' = V \otimes U'$ is irreducible. Use Problem 15.1 for $\wedge^2 V$. Determine the character for $\text{Sym}^2 V$. Without using any knowledge of the character table show that $\text{Sym}^2 V$ is the sum of three distinct irreducible representations of S_5 . Using the character table show that $\text{Sym}^2 V \cong U \oplus V \oplus W$. Complete the table.

2. Find the decomposition into irreducible representations of $\wedge^2 W, \text{Sym}^2 W$ and $V \otimes W$. Remark: We will see later in the course a completely general way for determining the character table for any S_n .

Proof. For (1). Firstly, the number of conjugacy classes of S_5 equals to the number of partitions of 5. And for 5, clearly we have following partitions:

$$5 = \begin{cases} 5 \\ 4+1 \\ 3+2 \\ 3+1+1 \\ 2+2+1 \\ 2+1+1+1 \\ 1+1+1+1+1 \end{cases}$$

Thus there are seven conjugacy classes of S_5 , and we can write them explicitly as

$$1, (12), (123), (1234), (12345), (12)(34), (12)(345)$$

It's easy to compute trivial representation U and alternating representation U' . Now let's consider standard representation, that is $V = R/U$, where R is regular representation. By fixed point formula, we can easily determine the character of R as follows

	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
R	5	3	2	1	0	1	0

Thus we have the character of V as follows:

	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
V	4	2	1	0	-1	0	-1

It's easy to check V is irreducible. Furthermore, $V' = U \otimes U'$ is also irreducible. (2) of Problem 15.1 implies that $\wedge^2 V$ is also irreducible. By formula $\chi_{\wedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$, we have the character of $\wedge^2 V$ as follows

	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
$\wedge^2 V$	6	0	0	0	1	-2	0

Since there are only two irreducible representations W, W' left to determine, we have

$$(\dim W)^2 + (\dim W')^2 = 50 \implies \dim W = \dim W' = 5$$

Clearly $W' = W \otimes U'$, since if W is irreducible, then $W \otimes U'$ is also irreducible.

It's easy to compute character of $\text{Sym}^2 V$, since $\chi_{\text{Sym}^2 V} + \chi_{\wedge^2 V} = (\chi_V)^2$

	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
$\text{Sym}^2 V$	10	4	1	0	0	2	1

If we directly compute $\langle \chi_{\text{Sym}^2 V}, \chi_{\text{Sym}^2 V} \rangle$

$$\langle \chi_{\text{Sym}^2 V}, \chi_{\text{Sym}^2 V} \rangle = 3$$

Thus $\text{Sym}^2 V$ is the sum of three distinct irreducible representation, we must have $\text{Sym}^2 V \cong U \oplus V \oplus W$, and we can determine W using this isomorphism.

For (2). Firstly we compute characters as follows

	1	(12)	(123)	(1234)	(12345)	(12)(34)	(12)(345)
$\wedge^2 W$	10	-2	1	0	0	-2	1
$\text{Sym}^2 W$	15	3	0	1	0	3	0
$V \otimes W$	20	2	-1	0	0	0	-1

So we have

$$\begin{aligned} \wedge^2 W &\cong \wedge^2 V \oplus V' \\ \text{Sym}^2 W &\cong U \oplus V \oplus W \oplus W' \\ V \otimes W &\cong V \oplus \wedge^2 V \oplus W \oplus W' \end{aligned}$$

□

Exercise 16.0.3. Character table of A_5 .

1. Let C be a conjugacy class in S_5 of cycle type $\lambda = (\lambda_1, \lambda_2, \dots)$. Show that when viewed as subset of A_5 :

$$C = \begin{cases} C_1 \cup C_2 & \text{all } \lambda_i \text{ are odd and distinct} \\ C & \text{otherwise} \end{cases}$$

where C_1, C_2 are distinct conjugacy classes of A_5 . Determine the conjugacy classes for A_5 and their cardinality.

2. Let $U, U', V, V', W, W', \wedge^2 V$ be the irreducible representations of S_5 of Problem 15.2. Show that

$$\begin{aligned} \text{Res}_{A_5}^{S_5} U &\cong \text{Res}_{A_5}^{S_5} U' \\ \text{Res}_{A_5}^{S_5} V &\cong \text{Res}_{A_5}^{S_5} V' \\ \text{Res}_{A_5}^{S_5} W &\cong \text{Res}_{A_5}^{S_5} W' \\ \text{Res}_{A_5}^{S_5} \wedge^2 V &\cong Y \oplus Z \end{aligned}$$

for some irreducible representations Y, Z of A_5 .

3. Show that the character table of A_5 takes the following form

	1	(123)	(12)(34)	(12345)	(21345)
$\text{Res}_{A_5}^{S_5} U$	1	1	1	1	1
$\text{Res}_{A_5}^{S_5} V$	4	1	0	-1	-1
$\text{Res}_{A_5}^{S_5} W$	5	-1	1	0	0
Y	3	0	-1	$\frac{1}{2}(1 + \sqrt{5})$	$\frac{1}{2}(1 - \sqrt{5})$
Z	3	0	-1	$\frac{1}{2}(1 - \sqrt{5})$	$\frac{1}{2}(1 + \sqrt{5})$

4. Find the decomposition of the permutation representation of A_5 corresponding to the vertices, the faces, and the edges of the icosahedron.

Proof. For (1). For any $\alpha \in A_n$, we use $\text{Stab}_{S_n}(\alpha)$ to denote the stabilizer of α with respect to the conjugate action of S_n , and use $\text{orb}_{S_n}(\alpha)$ to denote orbits of this action, that is all conjugacy classes of S_n . On the other hand, we use $\text{orb}_{A_n}(\alpha)$ to denote the orbits of α with respect to the conjugation action of A_n . By basic facts of group action, we have the following isomorphisms

$$\text{orb}_{S_n}(\alpha) = S_n / \text{Stab}_{S_n}(\alpha)$$

$$\text{orb}_{A_n}(\alpha) = A_n / \text{Stab}_{A_n}(\alpha) = A_n / \text{Stab}_{S_n}(\alpha) \cap A_n$$

So we want what's the difference between $\text{orb}_{S_n}(\alpha)$ and $\text{orb}_{A_n}(\alpha)$. There are two cases:

1. If $\text{Stab}_{S_n}(\alpha) \subseteq A_n$, in this case $\text{Stab}_{S_n}(\alpha) \cap A_n = \text{Stab}_{S_n}(\alpha)$, and since A_n has just half as many elements as S_n , we have

$$\text{orb}_{A_n}(\alpha) \cong A_n / \text{Stab}_{S_n}(\alpha)$$

has half as many elements as $\text{orb}_{S_n}(\alpha)$.

2. If $\text{Stab}_{S_n}(\alpha) \not\subseteq A_n$. In this case $\text{orb}_{A_n}(\alpha) \cong \text{orb}_{S_n}(\alpha)$. Indeed, by assumption there is some odd element τ in $\text{Stab}_{S_n}(\alpha)$. Then if σ is any odd elements of S_n , then

$$\sigma(\alpha) = \sigma(\tau\alpha) = \sigma\tau(\alpha) \in \text{orb}_{A_n}(\alpha)$$

If σ is even, then $\sigma\alpha$ is already in $\text{orb}_{A_n}(\alpha)$. So this shows they're equal.

So let's see what will happen if we let $n = 5$. For all even permutation, there are only follows possible types: 3-cycles, 5-cycles, products of 2-cycles and 2-cycles and identity element.

1. For 3-cycles, take $\alpha = (123)$. There are $|\text{orb}_{S_5}(\alpha)| = 20$, and (45) is an odd element in $\text{Stab}_{S_5}(\alpha)$, then $|\text{orb}_{A_5}(\alpha)| = |\text{orb}_{S_5}(\alpha)| = 20$;
2. For 5-cycles, take $\alpha = (12345)$, and $\text{Stab}_{S_5}(\alpha) = \langle \alpha \rangle \subset A_5$, thus this conjugacy class splits into two pieces, each of size 12. For example, (21345) is not A_5 -conjugate to (12345);
3. For products of 2-cycles and 2-cycles, take $\alpha = (12)(34)$, and $|\text{orb}_{S_5}(\alpha)| = 15$. Note that (12) is an odd element in $\text{Stab}_{S_5}(\alpha)$, thus $|\text{orb}_{A_5}(\alpha)| = 15$
4. Identity element is fixed by conjugation.

Thus A_n has the following conjugacy classes:

1. (123), with 20 elements;

2. (12345), with 12 elements;
3. (21345), with 12 elements;
4. (12)(34), with 15 elements;
5. (1), with 1 element.

Note that $60 = 20 + 12 + 12 + 15 + 1$, as it should⁶. This completes the proof of (1).

For (2). It's easy to see first three isomorphism by computing characters of these representations. For forth, we still determine the number of irreducible representations in $\text{Res}_{A_5}^{S_5} \wedge^2 V$ by computing its character as follows:

	1	(123)	(12)(34)	(12345)	(21345)
$\text{Res}_{A_5}^{S_5} \wedge^2 V$	6	0	-2	1	1

Thus

$$\langle \chi_{\text{Res}_{A_5}^{S_5} \wedge^2 V}, \chi_{\text{Res}_{A_5}^{S_5} \wedge^2 V} \rangle = \frac{1}{|A_5|} (6^2 \times 1 + 0^2 \times 20 + (-2)^2 \times 15 + 1^2 \times 12 + 1^2 \times 12) = \frac{120}{60} = 2$$

Thus $\text{Res}_{A_5}^{S_5} \wedge^2 V = Y \oplus Z$, where Y, Z are some irreducible representations of A_5 .

For (4). It suffices to compute the characters of Y, Z . Use the fact $\chi_{\text{Res}_{A_5}^{S_5} \wedge^2 V} = \chi_Y + \chi_Z$ and the orthonormal relations of character table to conclude. $\square$

Exercise 16.0.4 (The dihedral group D_{2n}). Let D_{2n} be the group of symmetries (rotations and reflections) of a regular n -gon. This group is called a dihedral group.

1. Show that $D_{2n} \cong \langle s, r \mid s^2 = r^n = (sr)^2 = e \rangle$.
2. Conclude that every element of D_{2n} is uniquely expressible as $s^i r^j$ where $0 \leq i \leq 1$ and $0 \leq j \leq n-1$.
3. Find the conjugacy classes of D_{2n} .
4. Find all the irreducible representations of D_{2n} . Hint: Use the fact that the cyclic group C_n is a normal subgroup of D_{2n} .

Proof. Since the cases of $n = 1, 2$ are quite trivial, here we assume $n \geq 3$.

For (1). A regular polygon with n sides has $2n$ different symmetries: n rotational symmetries and n reflection symmetries. If we use s to represent reflection about a given symmetric axis and r to represent rotate $\frac{2\pi}{n}$. Then clearly

$$D_{2n} \cong \langle a, b \mid s^2 = r^n = (sr)^2 = e \rangle$$

For (2). It suffices to show for any $m \in \mathbb{Z}$, we have $r^m s = sr^{-m}$. We prove this by induction on m . For $m = 1$, we need to show $rs = s^{-1}$. From definition

⁶This can be used to show A_5 is simple, since any normal subgroup is invariant under conjugation, and hence must be a union of conjugacy classes. Furthermore, it contains identity element. So the number of elements in a normal subgroup is a sum of the some of the above numbers, including 1, and it divides 60, but the only possible number of this form is 1 and 60. This completes the proof.

we have

$$\begin{aligned} srsr = e &\iff rsr = s \\ &\iff rs = sr^{-1} \end{aligned}$$

Let $m > 1$ and suppose $r^m s = sr^{-m}$, then

$$\begin{aligned} r^{m+1}s &= (r^m r)s \\ &= r^m(rs) \\ &= (r^m s)r^{-1} \\ &= (sr^{-m})r^{-1} \\ &= sr^{-(m+1)} \end{aligned}$$

For (3). It's a long to find all conjugacy classes of D_{2n} , let's state it as a theorem.

Theorem 16.0.1 (conjugacy classes of D_{2n}). Let $D_{2n}, n \geq 3$ be dihedral group, then

1. If $n = 2k$ for $k \geq 2$, then D_{2n} has $k + 3$ conjugacy classes;
2. If $n = 2k + 1$ for $k \geq 1$, then D_{2n} has $k + 2$ conjugacy classes.

Before prove of the theorem, let's state a lemma which will be used.

Lemma 16.0.1. If $x \in D_{2n}$ such that $x \neq r^m$ for any $0 \leq m \leq n$, then $rx = xr^{-1}$.

Proof. If $x \in D_{2n}$ such that $x \neq r^m$ for any $0 \leq m \leq n$, then $x = sr^m$ for some $0 \leq m \leq n$. Thus

$$\begin{aligned} rx &= r(sr^m) \\ &= (rs)r^m \\ &= (sr^{-1})r^m \\ &= s(r^{-1}r^m) \\ &= s(r^m r^{-1}) \\ &= (sr^m)r^{-1} \\ &= xr^{-1} \end{aligned}$$

□

Let's show the first part of Theorem 15.4, that is conjugacy classes of D_{2n} when $n = 2k, k \geq 2$ are listed as follows

$$\begin{aligned}
 [1] &= \{1\} \\
 [r] &= \{r, r^{-1}\}, \\
 [r^2] &= \{r^2, r^2\} \\
 &\vdots \\
 [r^{k-1}] &= \{r^{k-1}, r^{-(k+1)}\} \\
 [r^k] &= \{r^k\} \\
 [sr] &= \{sr^{2b-1} : b = 1, \dots, k\} \\
 [sr^2] &= \{sr^{2b} : b = 1, \dots, k\}
 \end{aligned}$$

Firstly, if $x \in Z(D_{2n})$, then $[x] = \{x\}$. Thus we see that $[1] = \{1\}$ and $[r^k] = \{r^k\}$ since $Z(D_{2n}) = \{1, r^k\}$.

Now let's check $[r^m] = \{r^m, r^{-m}\}$ for all $1 \leq m \leq k-1$. Clearly $r^m \in [r^m]$, and for r^{-m} . Note that Lemma 15.5 implies that $rx = xr^{-1}$, that is $r^{-1} = x^{-1}rx$. Then

$$\begin{aligned}
 r^{-m} &= (x^{-1}rx)^m \\
 &= x^{-1}r^mx
 \end{aligned}$$

Thus $r^{-m} \in [r^m]$; Conversely, let $g \in [r^m]$, then $g = x^{-1}r^mx$ for some $x \in D_{2n}$. Let's consider what does x look like:

1. If $x = r^t, 0 \leq t \leq n-1$, then

$$\begin{aligned}
 g &= (r^t)^{-1}r^mr^t \\
 &= r^{-t}r^mr^t \\
 &= r^m \in \{r^m, r^{-m}\}
 \end{aligned}$$

2. If $x = sr^t, 0 \leq t \leq n-1$, then

$$\begin{aligned}
 g &= (sr^t)^{-1}r^m(sr^t) \\
 &= r^{-t}sr^msr^t \\
 &= r^{-t}ssr^{-m}r^t \\
 &= r^{-t}r^{-m}r^t \\
 &= r^{-m} \in \{r^m, r^{-m}\}
 \end{aligned}$$

Thus we have $[r^m] = \{r^m, r^{-m}\}$.

Now let's check $[sr] = \{sr^{2b-1} : b = 1, \dots, k\}$. Firstly we check $sr^{2b-1}, 1 \leq b \leq k$ do lies in $[sr]$. By induction on b . If $b = 1$, clearly $sr \in [sr]$. Let $1 < b < k$ and suppose $sr^{2b-1} \in [sr]$, that is $sr^{2b-1} = x^{-1}sr^mx$ for some $x \in D_{2n}$, we need to show $sr^{2b+1} \in [sr]$. By Lemma 15.5 we have

$$r(sr^{2b+1}) = (sr^{2b+1})r^{-1}$$

Thus

$$\begin{aligned}
 sr^{2b+1} &= r^{-1}(sr^{2b+1})r^{-1} \\
 &= r^{-1}sr^{2b} \\
 &= r^{-1}sr^{2b-1}r \\
 &= r^{-1}x^{-1}srxxr \\
 &= (xr)^{-1}sr(xr) \in [sr]
 \end{aligned}$$

Conversely, let $g \in [sr]$ and write $g = x^{-1}srx$ for some $x \in D_{2n}$. Similarly we consider what does x look like:

1. If $x = r^t, 0 \leq t \leq n-1$, then

$$\begin{aligned}
 g &= r^{-t}sr^t \\
 &= r^{-t}st^{t+1} \\
 &= sr^t r^{t+1} \\
 &= sr^{2t+1} \\
 &\equiv sr^{2b-1} \pmod{2k}, \quad \text{for some } 1 \leq b \leq k
 \end{aligned}$$

2. If $x = sr^t, 0 \leq t \leq n-1$, then

$$\begin{aligned}
 g &= (sr^t)^{-1}sr(sr^t) \\
 &= r^{-t}ssr^t \\
 &= r^{-t}sr^{-1}r^t \\
 &= sr^t r^{-1}r^t \\
 &= sr^{2t-1} \\
 &\equiv sr^{2b-1} \pmod{2k}, \quad \text{for some } 1 \leq b \leq k
 \end{aligned}$$

Thus we have proven $[sr] = \{sr^{2b-1} : b = 1, \dots, k\}$. Finally let's check $[sr^2] = \{sr^{2b} : b = 1, \dots, k\}$, we use the same method in the case of $[sr]$. This completes the proof of conjugacy classes of D_{2n} when n is even.

For $n = 2k + 1$, it can be proved similarly and we omit the proof there, but list these conjugacy classes as follows:

$$\begin{aligned}
 [1] &= \{1\} \\
 [r] &= \{r, r^{-1}\}, \\
 [r^2] &= \{r^2, r^{-2}\} \\
 &\vdots \\
 [r^{k-1}] &= \{r^{k-1}, r^{-(k+1)}\} \\
 [r^k] &= \{r^k, r^{-k}\} \\
 [s] &= \{sr^b : b = 1, \dots, n\}
 \end{aligned}$$

For (4). Firstly let's try to determine the possible dimensions of irreducible representation of D_{2n} . Whether n is even or odd, there are two irreducible representations of D_{2n} , that is trivial one and alternating one:

$$\begin{array}{ll} \phi_1 : D_{2n} \rightarrow \mathbb{C} & \phi_2 : D_{2n} \rightarrow \mathbb{C} \\ r \mapsto 1 & r \mapsto 1 \\ s \mapsto 1 & s \mapsto -1 \end{array}$$

If $n = 2k$ is even, then there are still $k + 1$ irreducible representations left, and only possible dimensions are $2, 4, k$. In this case there is only one possibility, that is

$$4k = 1 + 1 + 1 + 1 + (k - 1) \times 4$$

If $n = 2k + 1$ is odd, then there are still k irreducible representations left, and only possible dimensions are $2, 4, 2k + 1$. In this case there is only one possibility, that is

$$4k + 2 = 1 + 1 + k \times 4$$

So as we can see, the case of $n = 2k + 1$ is easier, we just need to find k two-dimensional irreducible representations. By considering its geometric explanations, we have the following k representations $\rho_m, 1 \leq m \leq k$

$$\begin{array}{l} \rho_m : D_{2n} \rightarrow \text{GL}_2(\mathbb{C}) \\ r \mapsto \begin{pmatrix} e^{\frac{2m\pi i}{n}} & 0 \\ 0 & e^{-\frac{2m\pi i}{n}} \end{pmatrix} \\ s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{array}$$

After directly checking the character, we can prove all these ρ_m are irreducible, thus we already have all irreducible representations of D_{2n} , when $n = 2k + 1$ is odd.

For the case $n = 2k$, we still can consider above representations ρ_m , but when $m = k$, we have

$$\begin{array}{l} \rho_m : D_{2n} \rightarrow \text{GL}_2(\mathbb{C}) \\ r \mapsto \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \\ s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{array}$$

which is not irreducible. So until now we only get $k + 1$ irreducible representations of D_{2n} when $n = 2k$ is even. In fact, there are two other one-dimensional irreducible representations of D_{2n} , since we can prove the cyclic subgroup $C_n = \langle r^2 \rangle$ is the commutator of D_{2n} , and thus the number of one dimension irreducible representations are $|D_{2n}/C_n|$. The other two ones are listed as follows

$$\begin{array}{ll} \phi_3 : D_{2n} \rightarrow \mathbb{C} & \phi_4 : D_{2n} \rightarrow \mathbb{C} \\ r \mapsto -1 & r \mapsto -1 \\ s \mapsto 1 & s \mapsto -1 \end{array}$$



17. PROBLEM SET 3: INDUCED REPRESENTATIONS

Exercise 17.0.1 (Induction in terms of class functions). Let $H \leq G$ be a subgroup of a finite group G with embedding $\iota : H \rightarrow G$. We define linear maps for class functions by

$$\text{Ind}_H^G : \mathcal{C}_H \rightarrow \mathcal{C}_G, \beta \mapsto \left(\text{Ind}_H^G \beta \right)(g) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \beta(x^{-1}gx)$$

and

$$\text{Res}_H^G : \mathcal{C}_G \rightarrow \mathcal{C}_H, \alpha \mapsto \left(\text{Res}_H^G \alpha \right)(h) = \alpha(\iota(h))$$

Let $(\cdot, \cdot)_G$ denote the standard bilinear form on $\mathcal{C}_G$. Verify that

1. $\text{Ind}_H^G \beta \in \mathcal{C}_G$ for all $\beta \in \mathcal{C}_H$.
2. $(\text{Ind}_H^G \beta, \alpha)_G = (\beta, \text{Res}_H^G \alpha)_H$ for all $\alpha \in \mathcal{C}_G, \beta \in \mathcal{C}_H$.
3. $\text{Ind}_H^G (\beta \cdot \text{Res}_H^G \alpha) = (\text{Ind}_H^G \beta) \cdot \alpha$ for all $\alpha \in \mathcal{C}_G, \beta \in \mathcal{C}_H$.
4. $\text{Ind}_H^G \circ \text{Ind}_K^H = \text{Ind}_K^G$ for $K \leq H \leq G$.

Proof. For (1). We need to show for any $g, h \in G$, we have

$$\text{Ind}_H^G \beta(h^{-1}gh) = \text{Ind}_H^G \beta(g)$$

By definition

$$\begin{aligned} \text{Ind}_H^G \beta(h^{-1}gh) &= \frac{1}{|H|} \sum_{\substack{x' \in G \\ x'^{-1}h^{-1}ghx' \in H}} \beta(x'^{-1}h^{-1}ghx') \\ &\stackrel{y=hx'}{=} \frac{1}{|H|} \sum_{\substack{y \in G \\ y^{-1}gy \in H}} \beta(y^{-1}gy) \\ &= \text{Ind}_H^G \beta(g) \end{aligned}$$

For (2).

□

18. PROBLEM SET 4: SYMMETRIC GROUPS I

Exercise 18.0.1 (Induction and restriction from product subgroups). Let S_2 act on $\{1, 2\}$, S_3 act on $\{3, 4, 5\}$ and embed $H = S_2 \times S_3$ into $G = S_5$ acting on $\{1, 2, 3, 4, 5\}$

1. Let $U_2, V_2 \in \text{Irr}(S_2), U_3, U'_3, V_3 \in \text{Irr}(S_3)$. Compute the character table of H .
2. Compute $\text{Res}_H^G V = \bigoplus_{j: W_j \in \text{Irr}(H)} W_j^{\oplus b_j}$ for each $V \in \text{Irr}(G)$.
3. Compute $\text{Ind}_H^G W = \bigoplus_{i: V_i \in \text{Irr}(G)} V_i^{\oplus a_i}$ for each $W \in \text{Irr}(H)$.

Proof. For (1). Firstly, we list the conjugacy class of H as follows

$$(1), (12), (34), (12)(34), (345), (12)(345)$$

And by Proposition 2.26 all irreducible representations are as follows

$$U_2 \boxtimes U_3, V_2 \boxtimes U_3, U_2 \boxtimes U'_3, V_2 \boxtimes U'_3, U_2 \boxtimes V_3, V_2 \boxtimes V_3$$

We can use formula $\chi_{V \boxtimes W} = \chi_V \chi_W$ to compute the character table of H as follows

	1	(12)	(34)	(12)(34)	(345)	(12)(345)
$U_2 \boxtimes U_3$	1	1	1	1	1	1
$V_2 \boxtimes U_3$	1	-1	1	-1	1	-1
$U_2 \boxtimes U'_3$	2	2	0	0	-1	-1
$V_2 \boxtimes U'_3$	2	-2	0	0	-1	1
$U_2 \boxtimes V_3$	1	1	-1	-1	1	1
$V_2 \boxtimes V_3$	1	-1	-1	1	1	-1

For (2). We compute the characters of these restriction as follows

	1	(12)	(34)	(12)(34)	(345)	(12)(345)
$\text{Res } U$	1	1	1	1	1	1
$\text{Res } U'$	1	-1	-1	1	1	-1
$\text{Res } V$	4	2	2	0	1	-1
$\text{Res } V'$	4	-2	-2	0	1	1
$\text{Res } \bigwedge^2 V$	6	0	0	-2	0	0
$\text{Res } W$	5	1	1	1	-1	1
$\text{Res } W'$	5	-1	-1	1	-1	-1

From the character, we can directly have the following decomposition

$$\text{Res } U = U_2 \boxtimes U_3$$

$$\text{Res } U' = V_2 \boxtimes V_3$$

$$\text{Res } V = (U_2 \boxtimes U_3) \oplus (V_2 \boxtimes U_3) \oplus (U_2 \boxtimes U'_3)$$

$$\text{Res } V' = (V_2 \boxtimes U'_3) \oplus (U_2 \boxtimes V_3) \oplus (V_2 \boxtimes V_3)$$

$$\text{Res } \bigwedge^2 V = (V_2 \boxtimes U_3) \oplus (U_2 \boxtimes U'_3) \oplus (V_2 \boxtimes U'_3) \oplus (U_2 \boxtimes V_3)$$

$$\text{Res } W = (U_2 \boxtimes U_3) \oplus (U_2 \boxtimes U'_3) \oplus (V_2 \boxtimes U'_3)$$

$$\text{Res } W' = (U_2 \boxtimes U'_3) \oplus (V_2 \boxtimes U'_3) \oplus (V_2 \boxtimes V_3)$$

For (3). Use Frobenius reciprocity

$$\langle \chi_{\text{Ind}_H^G W}, \chi_V \rangle_G = \langle \chi_W, \chi_{\text{Res}_H^G V} \rangle_H$$

to compute a_i , since we already know $\chi_{\text{Res}_H^G V}$. For example, take $W = U_2 \boxtimes U_3$.

Then

$$\begin{aligned} \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_U \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G U} \rangle_H = \frac{1}{12}(1 + 1 + 1 \times 3 + 1 \times 3 + 1 \times 2 + 1 \times 2) = 1 \\ \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_{U'} \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G U'} \rangle_H = \frac{1}{12}(1 - 1 - 1 \times 3 + 1 \times 3 + 1 \times 2 - 1 \times 2) = 0 \\ \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_V \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G V} \rangle_H = \frac{1}{12}(4 + 2 + 2 \times 3 + 1 \times 2 - 1 \times 2) = 1 \\ \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_{V'} \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G V'} \rangle_H = \frac{1}{12}(4 - 2 - 2 \times 3 + 1 \times 2 + 1 \times 2) = 0 \\ \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_{\wedge^2 V} \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G \wedge^2 V} \rangle_H = \frac{1}{12}(6 - 2 \times 3) = 0 \\ \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_W \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G W} \rangle_H = \frac{1}{12}(5 + 1 + 1 \times 3 + 1 \times 3 - 1 \times 2 + 1 \times 2) = 1 \\ \langle \chi_{\text{Ind}_H^G U_2 \boxtimes U_3}, \chi_{W'} \rangle_G &= \langle \chi_{U_2 \boxtimes U_3}, \chi_{\text{Res}_H^G W'} \rangle_H = \frac{1}{12}(5 - 1 - 1 \times 3 + 1 \times 3 - 1 \times 2 - 1 \times 2) = 0 \end{aligned}$$

Thus we have

$$\text{Ind}_H^G U_2 \boxtimes U_3 \cong U \oplus V \oplus W$$

Similarly we will have

$$\begin{aligned} \text{Ind}_H^G V_2 \boxtimes U_3 &\cong V \oplus \bigwedge^2 V \\ \text{Ind}_H^G U_2 \boxtimes U'_3 &\cong V \oplus \bigwedge^2 V \oplus W \oplus W' \\ \text{Ind}_H^G V_2 \boxtimes U'_3 &\cong V' \oplus \bigwedge^2 V \oplus W \oplus W' \\ \text{Ind}_H^G U_2 \boxtimes U_3 &\cong V' \oplus \bigwedge^2 V \\ \text{Ind}_H^G V_2 \boxtimes V_3 &\cong U' \oplus V' \oplus W' \end{aligned}$$

□

Exercise 18.0.2 (Induction and product groups). Let G_1, G_2 be finite groups, and $G_1 \times G_2$ the direct product.

1. Show that $\mathbb{C}[G_1 \times G_2] \cong \mathbb{C}[G_1] \otimes \mathbb{C}[G_2]$ as $\mathbb{C}$ -algebras.
2. Let H_i be subgroups of $G_i, i = 1, 2$, and let $\rho_i : H_i \rightarrow \text{GL}(V_i)$ be representations, $i = 1, 2$. Denote by $V_1 \otimes V_2$ the exterior tensor product. Show that

$$\text{Ind}_{H_1 \times H_2}^{G_1 \times G_2} (V_1 \otimes V_2) \cong \text{Ind}_{H_1}^{G_1} V_1 \otimes \text{Ind}_{H_2}^{G_2} V_2.$$

Hint: Use the universal property of the tensor product for the direct product $\mathbb{C}[G_1 \times G_2] \times (V_1 \otimes V_2)$ and verify that the involved maps are module homomorphisms.

Proof. For (1). Consider the following isomorphism

$$\begin{aligned}\mathbb{C}[G_1 \times G_2] &\cong \mathbb{C}[G_1] \otimes \mathbb{C}[G_2] \\ (g_1, g_2) &\mapsto g_1 \otimes g_2\end{aligned}$$

For (2). The following isomorphisms complete the proof

$$\begin{aligned}\mathbb{C}[G_1 \times G_2] \otimes_{\mathbb{C}[H_1 \times H_2]} V_1 \otimes_{\mathbb{C}} V_2 &\cong (\mathbb{C}[G_1] \otimes_{\mathbb{C}} \mathbb{C}[G_2]) \otimes_{\mathbb{C}[H_1 \times H_2]} V_1 \otimes_{\mathbb{C}} V_2 \\ &\cong (\mathbb{C}[G_1] \otimes_{\mathbb{C}[H_1]} V_1) \otimes_{\mathbb{C}} (\mathbb{C}[G_2] \otimes_{\mathbb{C}[H_2]} V_2) \\ &= \text{Ind}_{H_1}^{G_1} V_1 \otimes \text{Ind}_{H_2}^{G_2} V_2\end{aligned}$$

□

Exercise 18.0.3 (A ring structure coming from product subgroups). Let R^n denote the $\mathbb{Z}$ -module generated by the irreducible representations of S_n , and set $R = \bigoplus_{n \geq 0} R^n$ with $S_0 = \{e\}, R_0 = \mathbb{Z}$. For $V \in R^m, W \in R^n$, let $V \otimes W$ be the corresponding representation of $S_m \times S_n$. Set

$$V \bullet W = \text{Ind}_{S_m \times S_n}^{S_{m+n}} (V \otimes W)$$

For $V = \sum_{n \geq 0} V_n, W = \sum_{n \geq 0} W_n \in R$ with $V_n, W_n \in R^n$ set

$$(V, W) = \sum_{n \geq 0} (V_n, W_n)_{S_n}$$

where $(V_n, W_n)_{S_n} = \dim \text{Hom}_{S_n}(V_n, W_n)$. Show that

1. $V \bullet W \in R^{m+n}$ is well-defined
2. $(R, \bullet)$ is a commutative, associative, graded ring with unit.
3. $(\cdot, \cdot) : R \times R \rightarrow R$ is a well defined scalar product.

Proof.

□

19. PROBLEM SET 5: SYMMETRIC FUNCTIONS

Exercise 19.0.1. Character of a special representation of S_n .

1. Let $H \leq G$ be a subgroup, let $K \in \text{Conj}(G)$. Assume that $K \cap H = K_1 \sqcup \cdots \sqcup K_r$ for $K_j \in \text{Conj}(H)$, $j = 1, \dots, r$. Let W be a representation of H . Show that

$$(\text{Ind}_H^G \chi_W)(K) = \frac{|G|}{|H|} \sum_{i=1}^r \frac{|K_i|}{|K|} \chi_W(K_i)$$

2. Let $\lambda \vdash n$ be a partition of n . Let $S_\lambda = S_{\lambda_1} \times S_{\lambda_2} \times \cdots \times S_{\lambda_r}$ be a Young subgroup of S_n . Let $a_\lambda = \sum_{g \in S_\lambda} e_g \in \mathbb{C}[S_n]$ and set $U_\lambda = \mathbb{C}[S_n]a_\lambda$. Show that $U_\lambda = \text{Ind}_{S_\lambda}^{S_n} U$ where U is the trivial representation of S_λ .
3. Suppose that λ is of the form $\lambda = (1^{m_1}, 2^{m_2}, \dots, r^{m_r})$, and let $g \in S_n$ have cycle type λ . Let $Z_g = \{h \in G \mid h^{-1}gh = g\}$ be the centralizer of $g \in G$. Show that $|Z_g|$ depends only on λ and that $z_\lambda := |Z_g| = \prod_{j=1}^r j^{m_j} m_j!$.
4. Let $K_\lambda = K_g$ be the conjugacy class of $g \in S_n$. Use Problem (c) to derive a formula for $|K_\lambda|$.
5. Let μ be a partition of n . Compute $\chi_{U_\lambda}(K_\mu)$ and show that $\chi_{U_\lambda}(K_\mu) = [p_\mu]_\lambda$, the coefficient of $x^\lambda = x_1^{\lambda_1} x_2^{\lambda_2} \cdots$ in p_μ . Here, p_μ is defined as the symmetric polynomial $p_\mu = p_{\mu_1} p_{\mu_2} \cdots$ with $p_r(x_1, \dots, x_n) = \sum_{i=1}^n x_i^r$.
6. Compute $\chi_{U_\lambda}(K_\mu)$ for $n = 5$, $\lambda = (3, 2)$, $\mu \vdash 5$, and compare with the result of Problem 14(c).

Proof. For (1). For induced character we have the following formula:

$$\text{Ind}_H^G \chi_W(g) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \chi_W(x^{-1}gx)$$

But if $g \in K$, then $g^{-1}xg$ in the summation lies in $K \cap H$ in fact. $\square$

20. PROBLEM SET 8: LIE GROUPS: TOPOLOGY

Exercise 20.0.1 (Component group of a Lie group). Let G be a Lie group, G^0 its connected component of the identity e . Show that G^0 is a normal subgroup of G , and a Lie group itself. Any connected component of G is a coset of G^0 . Hint: The image of a continuous map of a connected topological space is connected.

Remark 20.1. The (discrete) quotient group G/G^0 is called the group of components of G , denoted by $\pi_0(G)$.

Proof. It suffices to show that for all $g \in G, x \in G^0$, we have $gxg^{-1} \in G^0$. Consider the map $\Psi_g : x \mapsto gxg^{-1}$, it's continuous since the multiplication and inversion are continuous, so is their composition. Furthermore, $\text{im } \Psi_g$ contains e since $e \mapsto geg^{-1} = e$. So $\text{im } \Psi_g \subseteq G^0$, by the definition of G^0 , so we have proved $gxg^{-1} \in G^0$ for any $g \in G, x \in G^0$. In particular, G^0 is a Lie group itself. $\square$

Exercise 20.0.2. The center of a connected Lie group:

1. Let G be a connected Lie group, $N \leq G$ a discrete normal subgroup. Show that $N \subset Z(G)$, the center of G . Hint: Consider the map $G \rightarrow N, g \mapsto ghg^{-1}h^{-1}$ where $h \in N$ is fixed element. A topological space is connected if and only if every discrete-valued map is constant.
2. Let G be a connected Lie group with discrete center $Z(G)$. Show that $G/Z(G)$ has trivial center. Hint: Reduce to problem (1).

Proof. For (1), we need to show for any $h \in N, g \in G$, we have $gh = hg$, in other words, $ghg^{-1}h^{-1} = e$. Fix $h \in N$ and consider the map

$$\begin{aligned} \varphi_h : G &\rightarrow N \\ g &\mapsto ghg^{-1}h^{-1} \end{aligned}$$

It suffices to show φ_h is a constant map valued e . However, N is discrete, thus φ_h is always constant, and we can value it for arbitrary g . So we take $g = e$ to complete the proof of (1).

For (2), if $h + Z(G)$ lies in the center of $G/Z(G)$, that is for any $g_1, g_2 \in G$, we have

$$hg_1 - g_2h \in Z(G)$$

and what we need to show is $h \in Z(G)$. If we want to reduce to the case of (1), it suffices to show that the set of h is a discrete normal subgroup of G . To be explicit, let

$$N = \{h \in G \mid h + Z(G) \text{ lies in the center of } G/Z(G)\}$$

Clearly N is a normal subgroup. If N is not discrete, there exists an accumulation point h in N , thus for any $g_1, g_2 \in G$, $hg_1 - g_2h$ will be an accumulation point of $Z(G)$. A contradiction to the fact that $Z(G)$ is discrete. $\square$

Exercise 20.0.3. Coverings of Lie groups

1. Let M, N be connected manifolds, $\pi : \widetilde{M} \rightarrow M, \psi : \widetilde{N} \rightarrow N$ be their universal coverings. Let $f : M \rightarrow N$ be a continuous map. Choose $m \in M, n \in N, \tilde{m} \in \widetilde{M}, \tilde{n} \in \widetilde{N}$ such that $\pi(\tilde{m}) = m, \psi(\tilde{n}) = n$ and $f(m) = n$. Show that there exists a unique continuous map $\tilde{f} : \widetilde{M} \rightarrow \widetilde{N}$ such that $\psi \circ \tilde{f} = f \circ \pi$ and $\tilde{f}(\tilde{m}) = \tilde{n}$. Hint: Consider $\phi = f \circ \pi$ and use the results from the lecture.
2. Let H be a group, G be a connected Lie group, and $\pi : H \rightarrow G$ be a covering map. Show that H has a unique structure as a Lie group such that f is a morphism of Lie groups. Hint: Argue first that it suffices to show this statement for the universal covering $H = \tilde{G}$. Use (a) and the results from the lecture.
3. In the situation of (2), show that $\ker \pi$ is a discrete subgroup of $Z(H)$. Hint: Use Problem 30 (1).

Proof. For (1),

□

Exercise 20.0.4 ($SU(2)$ is a double covering of $SO(3)$). Let $\mathbb{H}$ denote the algebra of quaternions. Show that the following holds:

1. S^n is simply connected for $n > 1$. Hint: Show that for any $p \in S^n$ any map $f : S^1 \rightarrow S^n$ is homotopic to a map $g : S^1 \rightarrow S^n \setminus \{p\}$.
2. $SU(2)$ is diffeomorphic to S^3 .
3. $SU(2) \cong Sp(2) \cong \{q \in \mathbb{H} \mid q\bar{q} = 1\}$.
4. For any $q \in \mathbb{H}$ such that $q\bar{q} = 1$, the map $f_q : \mathbb{R}^3 \rightarrow \mathbb{R}^3, v \rightarrow qv\bar{q}$ is an isometry of the Euclidean space $\mathbb{R}^3$. Hint: Use that the subspace $\{q \in \mathbb{H} \mid q = -\bar{q}\} \subset \mathbb{H}$ of imaginary quaternions is isomorphic to $\mathbb{R}^3$.
5. The induced map $SU(2) \rightarrow SO(3), q \rightarrow f_q$ is a covering map of $SO(3)$ of multiplicity 2. Conclude that $SU(2)$ is the universal covering of $SO(3)$.

Proof. For (1), we need to show that for any map $f : S^1 \rightarrow S^n$ is null homotopy. Since $S^n \setminus \{p\}$ is homomorph to $\mathbb{R}^n$, and any loop in $\mathbb{R}^n$ is null homotopy since $\mathbb{R}^n$ is a convex set. So it suffices to show for any $p \in S^n$ and any $f : S^1 \rightarrow S^n$, f is homotopic to a map $g : S^1 \rightarrow S^n \setminus \{p\}$. If $p \notin \text{im } f$, it's clear. If $p \in \text{im } f$, we can disturb f slightly in order to avoid p , and this is clearly a homotopy.

For (2), by definition we have

$$SU(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}$$

Since $\mathbb{R}^4 \cong \mathbb{C}^2$, we may think S^3 as

$$S^3 = \{(a, b) \in \mathbb{C}^2 : |a|^2 + |b|^2 = 1\}$$

We can define a map as follows

$$\begin{aligned} F : S^3 &\rightarrow SU(2) \\ (a, b) &\mapsto \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \end{aligned}$$

Clearly F is a bijection, it remains to show it's smooth. This is also clear since we can regard S^3 and $SU(2)$ as a submanifold of $\mathbb{R}^8$, and in this view F is a smooth function since its components are linear functions of the coordinates. So is its restriction on S^3 and $SU(2)$.

For (3), take any

$$\begin{pmatrix} x & -\bar{y} \\ y & \bar{x} \end{pmatrix} \in SU(2)$$

we can write $x = a + ib, y = c + id$, consider the following map

$$G : SU(2) \rightarrow \mathbb{H}$$

$$\begin{pmatrix} x & -\bar{y} \\ y & \bar{x} \end{pmatrix} \mapsto a + bi + cj + dk$$

Since $|x|^2 + |y|^2 = 1$, then

$$(a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2 = 1$$

So G is an isomorphism between $SU(2)$ and $\{q \in \mathbb{H} \mid q\bar{q} = 1\}$. Furthermore, it preserves the Lie group structures of them, thus it's a Lie group isomorphism.

For (4). Firstly regard $\mathbb{R}^3$ as the imaginary quaternions, that is $v = (x, y, z) \cong xi + yj + zk$. For any $q = a + bi + cj + dk \in \mathbb{H}$, directly compute as follows

$$\begin{aligned} qv\bar{q} &= (a + bi + cj + dk)(xi + yj + zk)(a - bi - cj - dk) \\ &= (a + bi + cj + dk) \end{aligned}$$

Thus it's an isometry, since $a^2 + b^2 + c^2 + d^2 = 1$.

For (5), Since $SU(2)$ is homomorphic to S^3 , thus it's simply connected, and it's a covering space of $SO(3)$, thus it's the universal covering space of $SO(3)$. $\square$

21. PROBLEM SET 9: LIE GROUPS: EXAMPLES

Exercise 21.0.1 (Orthogonal and unitary groups). Let $O(n) = \{A \in GL(n, \mathbb{R}) \mid A^T A = \mathbf{1}\}$ and $U(n) = \{A \in GL(n, \mathbb{C}) \mid \bar{A}^T A = \mathbf{1}\}$. Show that the following holds:

1. $O(n)$ and $U(n)$ are compact. Hint: What properties do these groups have as subsets of $\text{End}(\mathbb{R}^n)$ and $\text{End}(\mathbb{C}^n)$, respectively.
2. $O(n)$ and $U(n)$ are Lie groups. Hint: In the first case, apply the inverse function theorem to $f : \text{End}(\mathbb{R}^n) \rightarrow \text{Sym}^2 \mathbb{R}^n, f : A \mapsto A^T A$.
3. $U(n) \subset SO(2n)$ and $GL(n, \mathbb{C}) \subset GL^+(2n, \mathbb{R}) = \{A \in GL(2n, \mathbb{R}) \mid \det A > 0\}$ as Lie subgroups.
4. $GL(n, \mathbb{C}) \cap SO(2n) = U(n)$ and $GL(n, \mathbb{R}) \cap U(n) = O(n)$.

Proof. For (1), it suffices to show $O(n)$ and $U(n)$ are bounded and closed. Let's see $O(n)$ first: Clearly $O(n)$ is a closed subset of $\mathbb{R}^{n^2}$, since it's the zero set of some polynomials defined by $A^T A = \mathbf{1}$. And it's bounded since all its components $|t_{ij}| \leq 1$. Indeed, for each i ,

$$a_{1i}^2 + \cdots + a_{ni}^2 = (A^T A)_{ii} = 1$$

The story for $U(n)$ is almost the same: $U(n)$ is also a closed subset of $\mathbb{R}^{2n^2}$ for the same reason. And it's bounded since for each i we have

$$|a_{1i}|^2 + \cdots + |a_{ni}|^2 = (\bar{A}^T A)_{ii} = 1$$

For (2), it suffices to use regular value theorem to show $O(n)$ and $U(n)$ are smooth manifolds, the multiplication and inversion is automatically smooth since they're just operations of polynomials. Let's see $O(n)$ first: Since $O(n)$ is the preimage of $\mathbf{1}$ of $f : A \mapsto A^T A$. So it suffices to show $\mathbf{1}$ is the regular value of f . Directly computing we have

$$\left. \frac{d}{dt} \right|_{t=0} f(A + tX) = \left. \frac{d}{dt} \right|_{t=0} (A + tX)^T (A + tX) = A^T X + X^T A$$

So the differential of f at $\mathbf{1}$ is a linear map from $\text{End}(\mathbb{R}^n) \rightarrow \text{Sym}^2 \mathbb{R}^n$, defined by $X \mapsto A^T X + X^T A$. As $A \in f^{-1}(\mathbf{1})$, any symmetric matrix B is $A^T X + X^T A$ when $X = \frac{1}{2}AB$. Indeed,

$$\begin{aligned} A^T X + X^T A &= A^T \left(\frac{1}{2} AB \right) + \frac{1}{2} B^T A^T A \\ &= \frac{1}{2} B + \frac{1}{2} B^T \\ &= B \end{aligned}$$

This shows the differential is surjective and hence $\mathbf{1}$ is a regular value of f . The story for $U(n)$ is totally same, just use $\bar{A}^T$ to replace A^T .

For the second part of (3), take $Z \in GL(n, \mathbb{C})$, we can define a $\mathbb{R}$ linear transformation $\mathbb{R}Z$ as follows: For each entry Z_{jk} , we write it as

$$Z_{jk} = X_{jk} + iY_{jk}$$

and replace it by the 2×2 matrix

$$\begin{pmatrix} X_{jk} & -Y_{jk} \\ Y_{jk} & X_{jk} \end{pmatrix}$$

This map is denoted by $Z \mapsto \mathbb{R}Z$. Clearly it's injective, and we have

$$\det \mathbb{R}Z = |\det Z|^2$$

It follows immediately that the map $Z \mapsto \mathbb{R}Z$ from $\mathrm{GL}(n, \mathbb{C}) \rightarrow \mathrm{GL}(2n, \mathbb{R})$.

So we have an inclusion $\mathrm{GL}(n, \mathbb{C}) \subset \mathrm{GL}^+(2n, \mathbb{R})$. Let's see more explicitly. The matrix $X \in \mathrm{GL}(2n, \mathbb{R})$ belongs to $\mathrm{GL}(n, \mathbb{C})$ if and only if it is built out of 2×2 matrices of the form

$$\begin{pmatrix} x & -y \\ y & x \end{pmatrix}$$

Let

$$iI \mapsto J = \begin{pmatrix} 0 & -1 & & \\ 1 & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \\ & & & \ddots \end{pmatrix}$$

Since any scalar multiple of the identity commutes with all matrices, we have

$$X \in \mathrm{GL}(n, \mathbb{C}) \implies (iI)X = X(iI)$$

Applying the operator $\mathbb{R}$, we have

$$X \in \mathrm{GL}(2n, \mathbb{R}) \implies JX = XJ$$

Conversely, if $JX = XJ$, then it is readily to verified X is of the required form, thus

$$\mathrm{GL}(n, \mathbb{C}) = \{X \in \mathrm{GL}(2n, \mathbb{R}) \mid JX = XJ\}$$

For the first part of (3), note that there is an important property of $Z \mapsto \mathbb{R}Z$, that is

$$\mathbb{R}(\overline{Z}^T) = (\mathbb{R}Z)^T$$

So if $Z \in \mathrm{U}(n)$, then $\overline{Z}^T Z = \mathbf{1}$, that is $(\mathbb{R}Z)^T(\mathbb{R}Z) = \mathbf{1}$, and the module of determinant of unitary matrix is 1, so is $\mathbb{R}Z$, that is $\mathrm{U}(n) \subset \mathrm{SO}(2n)$.

For (4),

□

Exercise 21.0.2. More on unitary and orthogonal groups:

1. Show that $\mathrm{O}(2n+1) \cong \mathrm{SO}(2n+1) \times \mathbb{Z}/2\mathbb{Z}$ as Lie groups.
2. Show that $\mathrm{O}(2n) \cong \mathrm{SO}(2n) \times \mathbb{Z}/2\mathbb{Z}$ as manifolds. Describe the multiplication $\mathrm{SO}(2n) \times \mathbb{Z}/2\mathbb{Z}$ inherits from $\mathrm{O}(2n)$.
3. Show that $\mathrm{U}(n) \cong \mathrm{SU}(n) \times S^1$ as manifolds.
4. Show that there is a surjective group homomorphism $S^1 \times \mathrm{SU}(n) \rightarrow \mathrm{U}(n)$, $(\zeta, A) \mapsto \zeta \cdot A$ whose kernel is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Exercise 21.0.3 (The relation between unitary, orthogonal and symplectic groups). Let V be a complex vector space with underlying real vector space $V_{\mathbb{R}}$. Let $H : V \times V \rightarrow \mathbb{C}$ be a Hermitian form, $g = \operatorname{Re} H$ and $\omega = \operatorname{Im} H$. Show that

1. $g : V_{\mathbb{R}} \times V_{\mathbb{R}} \rightarrow \mathbb{R}$ is symmetric, $\omega : V_{\mathbb{R}} \times V_{\mathbb{R}} \rightarrow \mathbb{R}$ is skew-symmetric.
2. $g(iv, iw) = g(v, w)$, $\omega(iv, iw) = \omega(v, w)$ and $\omega(v, w) = -g(iv, w)$.
3. Any symmetric form $g : V_{\mathbb{R}} \times V_{\mathbb{R}} \rightarrow \mathbb{R}$ such that $g(iv, iw) = g(v, w)$ is the real part of a unique Hermitian form $H : V \times V \rightarrow \mathbb{C}$.
4. If V has a basis such that $H(v, w) = \bar{v}^T w$, then $g(v, w) = v^T w$ and ω corresponds to the matrix $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. Deduce that $U(n) = O(2n) \cap \operatorname{Sp}(2n, \mathbb{R})$.

Proof. For (1), since H is Hermitian, then for $v, w \in V$, we have

$$\operatorname{Re} H(v, w) + i \operatorname{Im} H(v, w) = H(v, w) = \overline{H(w, v)} = \operatorname{Re} H(w, v) - i \operatorname{Im} H(w, v)$$

By comparing the real part and imaginary part, we have

$$\begin{cases} g(v, w) = g(w, v) \\ \omega(v, w) = -\omega(w, v) \end{cases}$$

For (2), we still need to use the fact that H is Hermitian. Directly compute as follows

$$\begin{aligned} g(v, w) + i\omega(v, w) &= -(i)^2 H(v, w) \\ &= H(iv, iw) \\ &= g(iv, iw) + i\omega(iv, iw) \end{aligned}$$

So we have

$$\begin{cases} g(iv, iw) = g(v, w) \\ \omega(iv, iw) = \omega(v, w) \end{cases}$$

And consider

$$\begin{aligned} -ig(v, w) + \omega(v, w) &= -iH(v, w) \\ &= H(v, iw) \\ &= g(v, iw) + i\omega(v, iw) \end{aligned}$$

So we have

$$\begin{cases} g(v, w) = -\omega(v, iw) \\ \omega(v, w) = g(v, iw) = -g(iv, w) \end{cases}$$

For (3). Note that we already have a relation between g and ω in (2), so we may define

$$\omega(v, w) = -g(iv, w)$$

and define

$$H(v, w) = g(v, w) + i\omega(v, w)$$

We directly check $H(v, w), v, w \in V$ is Hermitian as follows

$$\begin{aligned}
 \overline{H(v, w)} &= g(v, w) - i\omega(v, w) \\
 &= g(w, v) + ig(iv, w) \\
 &= g(w, v) - ig(v, iw) \\
 &= g(w, v) - ig(iw, v) \\
 &= g(w, v) + i\omega(w, v) \\
 &= H(w, v) \\
 H(iv, w) &= g(iv, w) + i\omega(iv, w) \\
 &= -\omega(v, w) + ig(v, w) \\
 &= i(g(v, w) + i\omega(v, w)) \\
 &= iH(v, w)
 \end{aligned}$$

For (4). If we already choose a basis such that $H(v, w) = \bar{v}^T w$ for all $v, w \in V$.
Then □

22. PROBLEM SET 10: LIE ALGEBRAS I

Exercise 22.0.1. The Lie algebra of derivations

1. Let A be an algebra over a field k . Show that $\text{Der}_k(A)$ is a Lie algebra with $[D, E] = D \circ E - E \circ D, D, E \in \text{Der}_k(A)$.
2. Let $\mathfrak{g}$ be a Lie algebra. Show that $\text{ad} : \mathfrak{g} \rightarrow \text{Der}_k(\mathfrak{g}) \subseteq \text{End}_k(\mathfrak{g}), X \mapsto \text{ad}_X = [X, -]$ is a morphism of Lie algebras.

Proof. The proof for (1) is the same as proof of Proposition 10.40.

For (2). ad_X is a derivation since there exists Jacobi identity.

$$\begin{aligned} \text{ad}_X([Y, Z]) &= [X, [Y, Z]] \\ &= -[Y, [Z, X]] - [Z, [X, Y]] \\ &= [Y, \text{ad}_X(Z)] + [\text{ad}_X(Y), Z] \end{aligned}$$

In fact, Jacobi identity is equivalent to ad_X is a derivation of $\mathfrak{g}$. Furthermore, we need to check $\text{ad}_{[X, Y]} = \text{ad}_X \circ \text{ad}_Y$, this still holds thanks to Jacobi identity. $\square$

Exercise 22.0.2 (Lie algebras of classical Lie groups). Verify that the Lie algebras $\mathfrak{so}(n), \mathfrak{u}(n), \mathfrak{sl}(n, \mathbb{R}), \mathfrak{sl}(n, \mathbb{C})$ and $\mathfrak{sp}(n)$ are closed under the Lie product of matrices and are invariant under conjugation by elements of the corresponding groups $\text{SO}(n), \text{U}(n), \text{SL}(n, \mathbb{R}), \text{SL}(n, \mathbb{C})$ and $\text{Sp}(n)$, respectively.

Proof. $\square$

Exercise 22.0.3 (Kernel and quotient of Lie algebra morphisms). Let $\mathfrak{g}$ be a Lie algebra.

1. Let $\mathfrak{h} \subset \mathfrak{g}$ be an ideal. Show that the quotient space $\mathfrak{g}/\mathfrak{h}$ has a unique structure of Lie algebra such that the canonical projection is a morphism of Lie algebras.
2. Let $\mathfrak{h} \subset \mathfrak{g}$ a subspace. Show that $\mathfrak{h}$ is an ideal if and only if it is the kernel of a morphism of Lie algebras.

Proof. For (1), clearly the Lie structure of $\mathfrak{g}/\mathfrak{h}$ is defined as follows: For any $g_1 + \mathfrak{h}, g_2 + \mathfrak{h} \in \mathfrak{g}/\mathfrak{h}$, we define

$$[g_1 + \mathfrak{h}, g_2 + \mathfrak{h}] := [g_1, g_2] + \mathfrak{h}$$

For (2). If $\mathfrak{h}$ is a kernel of some Lie algebra morphisms, it's clearly an ideal; Conversely, if $\mathfrak{h}$ is an ideal, then it's the kernel of canonical projection we construct in (1). $\square$

Exercise 22.0.4 (Second isomorphism theorem). Let $\mathfrak{g}$ be a Lie algebra.

1. Let $\mathfrak{a}, \mathfrak{b}$ be ideals in $\mathfrak{g}$. Show that $\mathfrak{a} + \mathfrak{b}, \mathfrak{a} \cap \mathfrak{b}$ are ideals in $\mathfrak{g}$.
2. Let $\mathfrak{a}, \mathfrak{b}$ be ideals in $\mathfrak{g}$ such that $\mathfrak{a} + \mathfrak{b} = \mathfrak{g}$. Show that $\mathfrak{g}/\mathfrak{a} = \mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$ is an isomorphism of Lie algebras.

Proof. Routines. $\square$

23. PROBLEM SET 11: THE EXPONENTIAL MAP

Exercise 23.0.1 (Abelian Lie groups and Lie algebras). Let G be a connected Lie group and $\mathfrak{g}$ its Lie algebra.

1. For any $X, Y \in \mathfrak{g}$ show that $[X, Y] = 0$ if and only if $\exp(tX)\exp(sY) = \exp(sY)\exp(tX)$ for all $s, t \in \mathbb{R}$.
2. Show that G is abelian if and only if $\mathfrak{g}$ is abelian
3. Give a counterexample to (2) if G is not connected
4. Assume that G is not necessarily connected. Show that $\exp(X + Y) = \exp(X)\exp(Y)$ for all $X, Y \in \mathfrak{g}$ if and only if the identity component of G is abelian.

Proof. For (1). It's clear if $[X, Y] = 0$ then

$$\exp(tX)\exp(sY) = \exp(sY)\exp(tX), \quad \forall s, t \in \mathbb{R}$$

Conversely, take differential of t and s respectively we have

$$X \exp(tX)Y(\exp sY) = Y \exp(sY)X \exp(tX)$$

and set $t = s = 0$ to get

$$XY = YX$$

that is $[X, Y] = 0$.

For (2). Corollary 10.59 says that if G is abelian, then its Lie algebra $\mathfrak{g}$ is also abelian. Here we don't use the assumption of connectedness of G . If $\mathfrak{g}$ is abelian, then by (1) we have

$$\exp(tX)\exp(sY) = \exp(sY)\exp(tX), \quad \forall X, Y \in \mathfrak{g}$$

However, $\exp$ is a local diffeomorphism near $0 \in \mathfrak{g}$, and if G is connected, then any neighborhood of $e \in G$ can generate the whole group. So the converse is still true.

For (3). Let H be any non abelian finite group, G an abelian group. Then $G \times H$ is a non abelian Lie group with abelian Lie algebra.

For (4). □

Exercise 23.0.2. The adjoint form of a Lie group

1. Let G, H be connected Lie groups and $\pi : H \rightarrow G$ is a covering map. Show that $Z(G)$ is discrete if and only if $Z(H)$ is discrete, and in this case $G/Z(G) \cong H/Z(H)$.
2. Let G be a connected Lie group, $\tilde{G}$ its universal covering, and $\mathfrak{g}$ its Lie algebra. Show that $\text{im Ad} = \tilde{G}/Z(\tilde{G})$. The Lie group $\tilde{G}/Z(\tilde{G})$ is called the adjoint form of G .
3. Let $H = \exp(Z(\mathfrak{g}))$, where $Z(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0, \forall Y \in \mathfrak{g}\}$. Show that H is the connected component of the identity in $Z(G)$.

Proof. For (1). Firstly note that $\pi(Z(H)) \subset Z(G)$, since for any $x \in Z(H)$, then $xyx^{-1}y^{-1} = e_H, \forall y \in H$, then

$$\pi(x)\pi(y)\pi(x)^{-1}\pi(y)^{-1} = e_G$$

And since π is a covering map, then it's surjective, then any element in G can be written as some $\pi(y)$, which implies $\pi(x) \in Z(G)$, that is $\pi(Z(H)) \subset Z(G)$; Conversely, $\pi^{-1}(Z(G))$ is a discrete normal subgroup in H , since fiber of covering map is discrete, then by the second problem in Problem set 8, we have $\pi^{-1}(Z(G)) \subset Z(H)$, thus $\pi(Z(H)) = Z(G)$. So $Z(G)$ is discrete if and only if $Z(H)$ is. Furthermore, let $\pi : H \rightarrow G$ descend on $H/Z(H)$, we have

$$\tilde{\pi} : H/Z(H) \rightarrow G/Z(G)$$

is an isomorphism, since it's injective and surjective.

For (2). Together (1) and first isomorphism theorem, it suffices to show $\ker \text{Ad} = Z(G)$. Suppose $x \in \ker \text{Ad}$, for any $y \in G$, the connectedness of G implies that we can write y as $e^{Y_1} \dots e^{Y_n}$, $Y_i \in \mathfrak{g}$. Then⁷

$$\begin{aligned} xyx^{-1} &= xe^{Y_1} \dots e^{Y_n} x^{-1} \\ &= xe^{Y_1} x^{-1} \dots xe^{Y_n} x^{-1} \\ &= e^{xY_1 x^{-1}} \dots e^{xY_n x^{-1}} \\ &= e^{Y_1} \dots e^{Y_n} \\ &= y \end{aligned}$$

proving $x \in Z(G)$; Conversely, suppose $y \in Z(G)$, then we can define a group homomorphism

$$\begin{aligned} \psi_y : G &\rightarrow G \\ x &\mapsto yxy^{-1} \end{aligned}$$

Let ϕ_y be the induced map on Lie algebras, then

$$\begin{aligned} e^{t\phi_y(X)} &= e^{\phi_y(tX)} \\ &= \psi_y(e^{tX}) \\ &= ye^{tX}y^{-1} \\ &= e^{t(yXy^{-1})} \end{aligned}$$

Take differential at $t = 0$, then we have

$$\phi_y(X) = yXy^{-1}$$

then ϕ_y is exactly Ad_y . However, $ye^{tX}y^{-1} = e^{tX}$, since $y \in Z(G)$. It follows $\phi_y(X) = \text{Ad}_y(X) = X$ for all $X \in \mathfrak{g}$. This shows $y \in \ker \text{Ad}$. This completes the proof of (2).

For (3). □

⁷For convenience of computing and WLOG, we regard G as a linear Lie group, so we have $xe^{Y_i}x^{-1} = e^{xY_ix^{-1}} = e^{\text{Ad}_x Y_i}$

24. PROBLEM SET 12: THE EXPONENTIAL MAP II

Exercise 24.0.1 (The Heisenberg group). Let $H = N(3, \mathbb{R})$ be the Heisenberg group, i.e. the group of upper triangular unipotent 3×3 matrices and $\mathfrak{h} = \text{Lie}(H) = \mathfrak{n}(3, \mathbb{R})$ with basis $\{X, Y, Z\}$ such that $[Y, Z] = X, [X, Y] = [X, Z] = 0$

1. Show that $\exp : \mathfrak{h} \rightarrow H$ is injective and surjective.
2. Let $Y, Z \in \text{Mat}(n \times n, \mathbb{C})$ such that $[Y, [Y, Z]] = [Z, [Y, Z]] = 0$. Show that $e^Y e^Z = e^{Y+Z+\frac{1}{2}[Y, Z]}$. Hint: Show that $t \mapsto e^{tY} e^{tZ} e^{-\frac{t^2}{2}[Y, Z]}$ and $t \mapsto e^{t(Y+Z)}$ satisfy the same differential equation.
3. Let $G \subset \text{GL}(n, \mathbb{R})$ be a Lie group with Lie algebra $\text{Lie}(G) = \mathfrak{g}$. Let $\phi : \mathfrak{h} \rightarrow \mathfrak{g}$ be a morphism of Lie algebras. Show that there exists a unique morphism of Lie groups $\Phi : H \rightarrow G$ such that $\Phi(\exp X) = \exp(\phi(X))$ for all $X \in \mathfrak{h}$.
4. Let $V = \langle Y, Z \rangle_{\mathbb{R}}$ be a vector subspace of $\mathfrak{h}$ and $K = \exp(V)$. Show that $K = H$ and conclude that $\text{Lie}(K) \neq V$.

Proof. For (1). For any Heisenberg group H

$$H = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

we have

$$\exp(aY + bZ + cX) = H$$

Furthermore, $\exp(aY + bZ + cX) = 0$ if and only if $a = b = c = 0$. So $\exp$ is both injective and surjective.

For (2). From hint, it suffices to show if we denote $\gamma(t) = e^{tY} e^{tZ} e^{-\frac{t^2}{2}[Y, Z]}$, then

$$\gamma'(t) = (Y + Z)\gamma(t)$$

Directly compute we have

$$\gamma'(t) = Y e^{tY} e^{tZ} e^{-\frac{t^2}{2}[Y, Z]} + e^{tY} Z e^{tZ} e^{-\frac{t^2}{2}[Y, Z]} - e^{tY} e^{tZ} [Y, Z] t e^{-\frac{t^2}{2}[Y, Z]}$$

Note that we have $[Y, [Y, Z]] = [Z, [Y, Z]] = 0$, thus we have

$$e^{tY} e^{tZ} [Y, Z] t e^{-\frac{t^2}{2}[Y, Z]} = t[Y, Z] e^{tY} e^{tZ} e^{-\frac{t^2}{2}[Y, Z]}$$

So it suffices to show

$$[e^{tY}, Z] e^{tZ} e^{-\frac{t^2}{2}[Y, Z]} = t[Y, Z] e^{tY} e^{tZ} e^{-\frac{t^2}{2}[Y, Z]}$$

And this is equivalent to show

$$[e^{tY}, Z] = t[Y, Z] e^{tY}$$

It's quite easy to check these two functions satisfying the same differential equation.

For (3). Since $\exp : \mathfrak{h} \rightarrow H$ is an isomorphism, it's clear to define $\Phi(h), h \in H$ as

$$\Phi(h) := \exp(\phi(\exp^{-1}(h)))$$

and this is unique.

For (4). Note that $(Y+Z)^2 = H$, thus $\exp(V) = H$. As a consequence $\text{Lie}(K) = \text{Lie}(H) = \mathfrak{h} \neq V$. $\square$

Exercise 24.0.2. Surjectivity of the exponential map

1. Show that there does not exist a matrix $X \in \mathfrak{sl}(2, \mathbb{C})$ with $\exp(X) = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \in \text{SL}(2, \mathbb{C})$.
2. Give an example of matrices X and Y in $\mathfrak{sl}(2, \mathbb{C})$ such that $[X, Y] = 2\pi i Y$ but such that there does not exist any Z in $\mathfrak{sl}(2, \mathbb{C})$ with $\exp X \exp Y = \exp Z$.
3. Let $X, Y \in \mathfrak{gl}(n, \mathbb{C})$ such that $[X, Y] = \alpha Y$ for some $\alpha \in \mathbb{C}$. Suppose further that there is no nonzero integer n such that $\alpha = 2\pi i n$. Show that $\exp X \exp Y = \exp(X + \frac{\alpha}{1-e^{-\alpha}} Y)$.

Proof. For (1). Note that $\exp(P^{-1}AP) = P^{-1}\exp(A)P$, so WLOG we may assume $X \in \mathfrak{sl}(2, \mathbb{C})$ is in its Jordan form. But for $X \in \mathfrak{sl}(2, \mathbb{C})$, its Jordan form J are in following forms

$$\begin{pmatrix} -\lambda & 0 \\ 0 & \lambda \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

So any matrix in the image of exponential map must have Jordan forms which are listed as follows

$$\begin{pmatrix} e^{-\lambda} & 0 \\ 0 & e^{\lambda} \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

But $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \in \text{SL}(2, \mathbb{C})$ is already its Jordan form and it is not in any form listed above. Thus it doesn't lie in the image of exponential map.

For (2). Consider

$$X = \begin{pmatrix} i\pi & 0 \\ 0 & -i\pi \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$$

It's easy to check $[X, Y] = 2\pi i Y$, and we have

$$\exp X \exp Y = \begin{pmatrix} e^{i\pi} & 0 \\ 0 & e^{-i\pi} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$$

From (1) we know that $\exp X \exp Y$ does not lie in the image of exponential.

For (3). Let

$$A(t) = e^X e^{tY} = e^{Z(t)}, \quad B(t) = \exp\{X + \frac{\alpha}{1-e^{-\alpha}} Y\} = e^{C(t)}$$

It suffices to show $Z(t)$ and $C(t)$ satisfy the same differential equation and initial conditions, then $A(t) = B(t)$ and let $t = 1$ to conclude.

$$\begin{aligned} \frac{d}{dt} e^{Z(t)} &= e^{Z(t)} \left(\frac{I - e^{-\text{ad}_{Z(t)}}}{\text{ad}_{Z(t)}} \left(\frac{dZ(t)}{dt} \right) \right) \\ \frac{d}{dt} e^X e^{tY} &= e^X Y e^{tY} = e^{Z(t)} Y \end{aligned}$$

Thus

$$\frac{dZ(t)}{dt} = \left(\frac{I - e^{-\text{ad}_{Z(t)}}}{\text{ad}_{Z(t)}} \right)^{-1} (Y)$$

For $C(t)$, we have

$$\begin{aligned}\frac{dC(t)}{dt} &= \frac{\alpha}{1 - e^{-\alpha}} Y \\ &= \frac{\text{ad}_{C(t)}}{I - e^{-\text{ad}_{C(t)}}}(Y) \\ &= \left(\frac{I - e^{-\text{ad}_{C(t)}}}{\text{ad}_{C(t)}} \right)^{-1}(Y)\end{aligned}$$

This completes the proof. $\square$

Exercise 24.0.3 (The relation between Ad and ad for $\text{GL}(n, \mathbb{C})$). Let $X, Y \in \mathfrak{gl}(n, \mathbb{C})$, $\text{ad}_X^n = \underbrace{\text{ad}_X \circ \dots \circ \text{ad}_X}_n$. Show that:

1.

$$\text{ad}_X^n(Y) = \sum_{m=0}^n \binom{n}{m} X^m Y (-X)^{n-m}$$

2.

$$\exp(\text{ad}_X Y) = \text{Ad}_{\exp X} Y = \exp(X) Y \exp(-X)$$

Proof. For (1). We prove it by induction. If $n = 1$, then

$$\text{ad}_X Y = -YX + XY$$

If this relation holds for $n \leq k$. Then consider $n = k + 1$ we have

$$\begin{aligned}\text{ad}_X^{k+1}(Y) &= \text{ad}_X(\text{ad}_X^k(Y)) \\ &= \text{ad}_X\left(\sum_{m=0}^k \binom{k}{m} X^m Y (-X)^{k-m}\right) \\ &= \sum_{m=0}^k \binom{k}{m} X^m \text{ad}_X(Y) (-X)^{k-m} \\ &= \sum_{m=0}^k \binom{k}{m} X^m (XY - YX) (-X)^{k-m} \\ &= \sum_{m=0}^k \binom{k}{m} X^{m+1} Y (-X)^{k-m} + \sum_{m=0}^k \binom{k}{m} X^m Y (-X)^{k-m+1} \\ &= \sum_{m'=m+1}^{k+1} \binom{k}{m'-1} X^{m'} Y (-X)^{k-m'+1} + \sum_{m=0}^k \binom{k}{m} X^m Y (-X)^{k-m+1} \\ &= X^{k+1} Y + Y (-X)^{k+1} + \sum_{m=1}^k \left(\binom{k}{m-1} + \binom{k}{m} \right) X^m Y (-X)^{k-m+1} \\ &= \sum_{m=0}^{k+1} \binom{k+1}{m} X^m Y (-X)^{k-m+1}\end{aligned}$$

In the last equality we use the identity

$$\binom{k}{m-1} + \binom{k}{m} = \binom{k}{m+1}$$

For (2). Expand $\exp(\operatorname{ad}_X(Y))$ in power series, we have

$$\begin{aligned}\exp(\operatorname{ad}_X Y) &= \sum_{n=0}^{\infty} \frac{1}{n!} \operatorname{ad}_X^n(Y) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \binom{n}{m} X^m Y (-X)^{n-m}\end{aligned}$$

And expand $\exp(X)Y \exp(-X)$ in power series, we have

$$\begin{aligned}\exp(X)Y \exp(-X) &= \left(\sum_{n=0}^{\infty} \frac{1}{n!} X^n \right) Y \left(\sum_{m=0}^{\infty} \frac{1}{m!} (-X)^m \right) \\ &= \sum_{n=0}^{\infty} \sum_{m=0}^n \frac{1}{m!(n-m)!} X^m Y (-X)^{n-m} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^n \binom{n}{m} X^m Y (-X)^{n-m} \\ &= \exp(\operatorname{ad}_X Y)\end{aligned}$$

This completes the proof. $\square$

Exercise 24.0.4 (Lower central and derived series). Let $\mathfrak{g}$ be a finite dimensional Lie algebra. We define recursively for $j \geq 0$

$$\begin{aligned}\mathfrak{g}^0 &= \mathfrak{g}, & \mathfrak{g}^1 &= [\mathfrak{g}, \mathfrak{g}], & \mathfrak{g}^{j+1} &= [\mathfrak{g}^j, \mathfrak{g}^j]. \\ \mathfrak{g}_0 &= \mathfrak{g}, & \mathfrak{g}_1 &= [\mathfrak{g}, \mathfrak{g}], & \mathfrak{g}_{j+1} &= [\mathfrak{g}, \mathfrak{g}_j].\end{aligned}$$

1. Show that each $\mathfrak{g}^j$ and each $\mathfrak{g}_j$ is an ideal in $\mathfrak{g}$, and that $\mathfrak{g}^j \subseteq \mathfrak{g}_j$ for all j .
2. Show that the following are equivalent:
 - (i) $\mathfrak{g}^k = 0$ for some k ;
 - (ii) $\mathfrak{g}$ has a sequence of Lie subalgebras $\mathfrak{g} = \mathfrak{h}_0 \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_k = 0$ such that $\mathfrak{h}_{i+1}$ is an ideal in $\mathfrak{h}_i$ and $\mathfrak{h}_i/\mathfrak{h}_{i+1}$ is abelian.
 - (iii) $\mathfrak{g}$ has a sequence of ideals $\mathfrak{g} = \mathfrak{h}_0 \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_k = 0$ such that $\mathfrak{h}_i/\mathfrak{h}_{i+1}$ is an abelian Lie algebra.
3. Show that the following are equivalent:
 - (i) $\mathfrak{g}_k = 0$ for some k .
 - (ii) $\mathfrak{g}$ has a sequence of ideals $\mathfrak{g} = \mathfrak{h}_0 \supset \mathfrak{h}_1 \supset \cdots \supset \mathfrak{h}_k = 0$ such that $\mathfrak{h}_i/\mathfrak{h}_{i+1} \subset Z(\mathfrak{g}/\mathfrak{h}_{i+1})$
 - (iii) $\operatorname{ad}_{X_1} \circ \operatorname{ad}_{X_2} \circ \cdots \circ \operatorname{ad}_{X_k} Y = [X_1, [X_2, \dots, [X_k, Y] \dots]] = 0$ for some integer k and all $X_1, \dots, X_k, Y \in \mathfrak{g}$

Proof. For (1). Show $\mathfrak{g}^j$ is an ideal by induction on j as an example. Clearly it's an ideal when $j = 0, 1$. If $\mathfrak{g}^k$ is an ideal, then take any $g \in \mathfrak{g}, [u, v] \in \mathfrak{g}^{k+1}, u, v \in \mathfrak{g}^k$, we have

$$[g, [u, v]] = [[g, u], v] + [u, [g, v]]$$

Since $\mathfrak{g}^k$ is an ideal, $[g, u], [g, v] \in \mathfrak{g}^k$. This shows $\mathfrak{g}^{k+1}$ is an ideal. We also prove $\mathfrak{g}^j \subseteq \mathfrak{g}_j$ by induction on j . It's clear $\mathfrak{g}^j = \mathfrak{g}_j$ if $j = 0, 1$. But for $j = 2$, then

$$\mathfrak{g}^2 = [\mathfrak{g}^1, \mathfrak{g}^1] \subseteq [\mathfrak{g}, \mathfrak{g}^1] = [\mathfrak{g}, \mathfrak{g}_1] = \mathfrak{g}_2$$

For higher order, $\mathfrak{g}^{j+1} \subseteq \mathfrak{g}_{j+1}$ since $\mathfrak{g}^j \subseteq \mathfrak{g}_j$ and $\mathfrak{g}^j \subseteq \mathfrak{g}$.

For (2). It's a problem for equivalent conditions of solvable Lie algebras. It's clear from (i) to (iii), since we can take $\mathfrak{h}_i = \mathfrak{g}^i$. (iii) to (ii) is also trivial, since if $\mathfrak{h}_{i+1}$ is an ideal of $\mathfrak{g}$, it's also an ideal of $\mathfrak{h}_i$. It suffices to show (iii) to (i), we claim that $\mathfrak{g}^k \subseteq \mathfrak{h}_k$. Indeed, take $k = 1$ for example, since $\mathfrak{g}/\mathfrak{g}^1$ is its abelianization, thus if $\mathfrak{g}/\mathfrak{h}_1$ is abelian, then $\mathfrak{g}/\mathfrak{h}_1 \subseteq \mathfrak{g}/\mathfrak{g}^1$, which implies $\mathfrak{g}^1 \subseteq \mathfrak{h}_1$. The argument for higher order is same.

For (3). It's a problem for equivalent conditions of nilpotent Lie algebras. (i) to (ii) is clear since $\mathfrak{h}_i = \mathfrak{g}_i$ satisfies the desired properties. It suffices to show (ii) to (i), since the equivalence between (i) and (iii) we will prove in the first problem in the next problem set. Here we also need to show $\mathfrak{g}_i \subseteq \mathfrak{h}_i$. Indeed, take $k = 0$ for an example, if

$$\mathfrak{g}/\mathfrak{h}_1 \subset Z(\mathfrak{g}/\mathfrak{h}_1)$$

Then $\mathfrak{g}/\mathfrak{h}_1$ must be abelian, then $\mathfrak{g}_1 \subseteq \mathfrak{h}_1$. If $\mathfrak{g}_i \subseteq \mathfrak{h}_i$ holds for $i < k$, and assume $\mathfrak{g}_k \subsetneq \mathfrak{h}_k$, then there exists $g \in \mathfrak{g}_k = [\mathfrak{g}, \mathfrak{g}_{k-1}] \subseteq [\mathfrak{g}, \mathfrak{h}_{k-1}]$, $g \notin \mathfrak{h}_k$. But $\mathfrak{h}_{k-1}/\mathfrak{h}_k \subseteq Z(\mathfrak{g}/\mathfrak{h}_k)$ implies that

$$[\mathfrak{g}, \mathfrak{h}_{k-1}/\mathfrak{h}_k] \subseteq \mathfrak{h}_k$$

A contradiction to the existence of g . □

25. PROBLEM SET 13: SOLVABLE, NILPOTENT AND SEMISIMPLE LIE ALGEBRA

Exercise 25.0.1 (Criteria for nilpotency and solvability). Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. Show that

1. $\mathfrak{g}$ is nilpotent if and only if the Lie algebra $\text{ad } \mathfrak{g}$ is nilpotent.
2. $\mathfrak{g}$ is solvable if and only if $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

Proof. For (1). If $\mathfrak{g}$ is nilpotent, then $\text{ad } \mathfrak{g}$ is nilpotent clearly, since $\text{ad } \mathfrak{g} = \mathfrak{g}/Z(\mathfrak{g})$; Conversely, if $\text{ad } \mathfrak{g}$ is nilpotent, then there exists integer n such that

$$\underbrace{[\text{ad } \mathfrak{g}, [\dots, \text{ad } \mathfrak{g}]]}_{n \text{ times}} = 0$$

That's equivalent to

$$\underbrace{[\mathfrak{g}, [\dots, \mathfrak{g}]]}_{n \text{ times}} \subset Z(\mathfrak{g})$$

And that's equivalent to

$$\underbrace{[\mathfrak{g}, [\dots, \mathfrak{g}]]}_{n+1 \text{ times}} = 0$$

So $\mathfrak{g}$ is nilpotent.

For (2). If $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent, then $[\mathfrak{g}, \mathfrak{g}]$ is solvable, so is $\mathfrak{g}$; Conversely, if $\mathfrak{g}$ is solvable. From (1) it suffices to show $\text{ad}[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. Since $\mathfrak{g}$ is solvable, by Lie's theorem, over an algebraic closure $\bar{k}$ of the ground field the elements of $\text{ad } \mathfrak{g}$ are simultaneously upper triangularizable. It follows that over $\bar{k}$ the elements of $\text{ad}[\mathfrak{g}, \mathfrak{g}]$ can be represented by strictly upper triangular matrices, and hence $\text{ad}[\mathfrak{g}, \mathfrak{g}]$ is nilpotent over $\bar{k}$. But it doesn't depend on whether we extend our ground field or not. This completes the proof. $\square$

Exercise 25.0.2. Let $V \neq 0$ be a finite-dimensional complex vector space, $\mathfrak{g} \subset \mathfrak{gl}(V)$ be a complex Lie algebra. Show that

1. If $\mathfrak{g}$ is nilpotent, then there exists a basis for V such that the matrix representative of each $X \in \mathfrak{g}$ is strictly upper triangular.
2. If $\mathfrak{g}$ is solvable, then there exists a basis for V such that the matrix representative of each $X \in \mathfrak{g}$ is upper triangular.

Proof. Just Engel/Lie's theorem together with induction on dimension of V . $\square$

Exercise 25.0.3 (Simple Lie algebras). Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. Show that the following are equivalent:

1. $\mathfrak{g}$ is simple, i.e. $\mathfrak{g}$ has no nontrivial ideals and $\dim \mathfrak{g} > 1$.
2. $\mathfrak{g}$ has no nontrivial ideals and $\mathfrak{g}$ is nonabelian.
3. $\mathfrak{g}$ has no nontrivial ideals and is semisimple.

Proof. From (1) to (2): It suffices to show $\mathfrak{g}$ is nonabelian, if $\mathfrak{g}$ is abelian and $\dim \mathfrak{g} > 1$, then any subspace $0 \subsetneq \mathfrak{h} \subsetneq \mathfrak{g}$ is an ideal. A contradiction.

The equivalence between (2) and (3): From Proposition 11.18, we know that for any finite-dimensional Lie algebra, $\mathfrak{g}$ is semisimple if and only if $\mathfrak{g}$ has no abelian ideals. Since here $\mathfrak{g}$ has no nontrivial ideals, so $\mathfrak{g}$ is semisimple if and only if $\mathfrak{g}$ is nonabelian.

From (2) to (1) is trivial, since any Lie algebra of dimension 1 must be abelian. $\square$

Exercise 25.0.4. Semisimplicity of $\mathfrak{sl}(n, \mathbb{C})$

1. Show that $Z(\mathfrak{gl}(n, \mathbb{C})) = \mathfrak{s}(n, \mathbb{C})$, where $\mathfrak{s}(n, \mathbb{C}) = \{\lambda I_n \mid \lambda \in \mathbb{C}\}$ and I_n is the $n \times n$ identity matrix. Show that $Z(\mathfrak{sl}(n, \mathbb{C})) = 0$.
2. Show that $\text{rad}(\mathfrak{sl}(n, \mathbb{C})) = Z(\mathfrak{sl}(n, \mathbb{C}))$, hence $\mathfrak{sl}(n, \mathbb{C})$ is semisimple. Hint: Show that $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ is contained in every maximal solvable subalgebra $\mathfrak{b} \subset \mathfrak{sl}(n, \mathbb{C})$. Use Lie's theorem to show that $\text{rad}(\mathfrak{sl}(n, \mathbb{C})) \subset \mathfrak{sl}(n, \mathbb{C}) \cap \mathfrak{o}(n, \mathbb{C})$, where $\mathfrak{o}(n, \mathbb{C})$ denotes the Lie algebra of diagonal $n \times n$ matrices.

Proof. For the first part of (1), it's a basic result in linear algebra: Clearly $\mathfrak{s}(n, \mathbb{C}) \subseteq Z(\mathfrak{gl}(n, \mathbb{C}))$; Conversely, if $A \in Z(\mathfrak{gl}(n, \mathbb{C}))$, consider elementary matrix E_{ij} , then for any $1 \leq i, j \leq n$

$$E_{ij}A = AE_{ij} \implies \begin{cases} a_{ij} = 0, & i \neq j \\ a_{ii} = \lambda, & \forall 1 \leq i \leq n \end{cases}$$

That is $A \in \mathfrak{s}(n, \mathbb{C})$.

Remark 25.1. It suffices to use $E_{ij}, i \neq j$ to complete the proof, since $AE_{ii} = E_{ii}A$ gives nothing new.

For the second part of (2), note that

$$\mathfrak{sl}(n, \mathbb{C}) = \{A \in \mathfrak{gl}(n, \mathbb{C}) \mid \text{tr } A = 0\} \subset \mathfrak{gl}(n, \mathbb{C})$$

And use the remark we mentioned above, we have

$$Z(\mathfrak{sl}(n, \mathbb{C})) = Z(\mathfrak{gl}(n, \mathbb{C})) \cap \mathfrak{sl}(n, \mathbb{C}) = 0$$

For (2). Recall that $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ is the maximal solvable ideal of $\mathfrak{sl}(n, \mathbb{C})$. If $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ isn't contained in some maximal solvable subalgebra $\mathfrak{b}$, and if we can prove $\text{rad}(\mathfrak{sl}(n, \mathbb{C})) + \mathfrak{b}$ is still a solvable subalgebra, then it contradicts to the maximality of $\mathfrak{b}$. So it suffices to show a solvable ideal plus a solvable subalgebra is still a solvable subalgebra. Assume $\mathfrak{h}$ is a solvable ideal and $\mathfrak{b}$ is a solvable subalgebra, clearly $\mathfrak{h} + \mathfrak{b}$ is still a subalgebra. If we use second isomorphism, we have

$$\mathfrak{h} + \mathfrak{b}/\mathfrak{h} \cong \mathfrak{h}/\mathfrak{h} \cap \mathfrak{b}$$

Use the fact $\mathfrak{h}$ and $\mathfrak{h}/\mathfrak{h} \cap \mathfrak{b}$ are solvable to get desired result.

Applying Lie's theorem to $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$, we can think $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ consists of upper triangular matrices. Use the fact that $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ is an ideal, we can show

$$\text{rad}(\mathfrak{sl}(n, \mathbb{C})) \subset \mathfrak{sl}(n, \mathbb{C}) \cap \mathfrak{o}(n, \mathbb{C})$$

It suffices to show for any $A \in \text{rad}(\mathfrak{sl}(n, \mathbb{C}))$, all entries in the first row are zero except a_{11} , then we can use induction to get desired result. Since $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$

is an ideal, then first take E_{1n} and consider $[E_{1n}, A]$, which is still upper triangular, then we must have $a_{1i} = 0, i \neq 1, n$. Then use E_{2n} and consider $[E_{2n}, A]$ to get $a_{1n} = 0$, since we have already proved matrices in $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ has the property $a_{12} = 0$ ⁸.

Finally let's show $\text{rad}(\mathfrak{sl}(n, \mathbb{C})) \subseteq Z(\mathfrak{sl}(n, \mathbb{C}))$ which still depends heavily on the fact $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ is an ideal. Since $\mathfrak{sl}(n, \mathbb{C})$ is spanned by $E_{ij}, i \neq j$ and $E_{ii} - E_{jj}, 1 \leq i < j \leq n$, we just need to show for any $A \in \text{rad}(\mathfrak{sl}(n, \mathbb{C}))$, we have $[A, E_{ij}]$ or $[A, E_{ii} - E_{jj}] = 0$. It's easy to see $[A, E_{ii} - E_{jj}] = 0$ always holds for any diagonal matrix A , here we don't need the fact $\text{rad}(\mathfrak{sl}(n, \mathbb{C}))$ is an ideal. But for $[A, E_{ij}], i \neq j$, it is compelled to be zero, since it's diagonal. This completes the proof. $\square$

⁸In fact, here we can use more E_{ij} to test, then we can conclude $\text{rad}(\mathfrak{sl}(n, \mathbb{C})) = \{A \in \mathfrak{sl}(n, \mathbb{C}) \mid A = \lambda I_n\}$

26. PROBLEM SET 14: REPRESENTATIONS OF $\mathfrak{sl}(2, \mathbb{C})$

Exercise 26.0.1 (The Casimir operator for $\mathfrak{sl}(2, \mathbb{C})$). Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V)$, $\rho' : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V')$ be representations. Define $Z = \frac{1}{2}\rho(H)^2 + \rho(H) + 2\rho(Y)\rho(X)$

1. Show that $[Z, \rho(A)] = 0$ for all $A \in \mathfrak{sl}(2, \mathbb{C})$.
2. Assume that V and V' are irreducible. Let $L \in \text{Hom}(V, V')$ such that $L \circ \rho = \rho' \circ L$. Show that either $L = 0$ or L is invertible. Furthermore, if $V' = V$, then L is multiplication by a scalar.
3. Let $V^{(n)}$ be an irreducible representation of dimension $n + 1$. Show that $Zv = (\frac{1}{2}n^2 + n)v$ for $v \in V^{(n)}$.

Proof. For (1). It suffices to check Z commutes with $\rho(X), \rho(Y), \rho(Z)$. Directly compute as follows:

$$\begin{aligned}
 [\rho(X), \frac{1}{2}\rho(H)^2 + \rho(H) + 2\rho(Y)\rho(X)] &= \frac{1}{2}\rho(H)\rho([X, H]) + \frac{1}{2}\rho([X, H])\rho(H) \\
 &\quad + \rho([X, H]) + 2\rho(Y)\rho([X, X]) + 2\rho([X, Y])\rho(X) \\
 &= -\rho(H)\rho(X) - \rho(X)\rho(H) - 2\rho(X) + 2\rho(H)\rho(X) \\
 &= [\rho(H), \rho(X)] - 2\rho(X) \\
 &= 2\rho(X) - 2\rho(X) \\
 &= 0
 \end{aligned}$$

Similar for $\rho(Y)$ and $\rho(Z)$.

For (2). Parallel to Schur lemma.

For (3). From (1) and (2) we know that Z acts on $V^{(n)}$ as multiplication by a constant, thus we can choose a special $v \in V^{(n)}$ to figure it out, that is, choose v to be the eigenvector of $\rho(H)$ with respect to eigenvalue n . So

$$(\frac{1}{2}\rho(H)^2 + \rho(H) + \rho(Y)\rho(X))v = (\frac{1}{2}n^2 + n)v$$

As desired. □

Exercise 26.0.2 (Complementary invariant subspace for representations of $\mathfrak{sl}(2, \mathbb{C})$). Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V)$ be a finite dimensional representation. Let $W \subset V$ be an invariant subspace of codimension 1. Show that there is a complementary invariant subspace W' .

Hints: Consider first the case $\dim W = 1$ and analyze the quotient representation V/W . Then consider the case $\dim W > 1$ and $\rho|_W$ is irreducible, and study $\ker Z$ where Z is the Casimir operator in above problem. Finally, consider the case $\dim W \geq 1$ and $\rho|_W$ is not necessarily irreducible. Apply induction on $\dim W$.

Proof. □

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